

Towards Machine Learning Foundation Models for Materials Chemistry



Janosh Riebesell

University of Cambridge
St. Catherine's College
Department of Physics

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Supervisors	Alpha Lee, Ulrich Keyser
Advisors	Anubhav Jain, Rhys Goodall, Kristin Persson
Examiners	Chris Pickard, Aaron Walsh
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Declaration of Authorship

This thesis is the result of my own work and includes nothing which is the outcome of work done in collaboration except as declared in the preface and specified in the text. It is not substantially the same as any work that has already been submitted, or, is being concurrently submitted, for any degree, diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the preface and specified in the text. It does not exceed the prescribed word limit for the relevant Degree Committee.

Abstract

This thesis demonstrates how recent advances in machine learning (ML) for materials can accelerate our search for new stable inorganic crystals. We show how best to measure and compare the utility of different models, what range of applications a foundational ML force field can be expected to cover, what remaining shortcomings current ML potentials exhibit and how some of them can be overcome. Specifically, we present Matbench Discovery, an evaluation framework that closely mimics a real-world materials discovery campaign and establishes universal interatomic potentials (UIPs) as the state-of-the-art ML method for accelerating the discovery of thermodynamically stable inorganic crystals.

Next, we design and execute an ML-guided dielectric discovery workflow that integrates rapid ML screening, targeted crystal generation and high-throughput ab initio validation all feeding into informed experimental characterization which culminated in the synthesis of two novel dielectric materials, CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$, with CsTaTeO_6 generated by our workflow.

Finally, we comprehensively analyze MACE-MP, the best-performing model we trained for Matbench Discovery, which has since proven to be a highly versatile foundation model for atomistic simulations. While pre-trained purely on inorganic bulk crystals, it exhibits unexpected extrapolation to diverse chemistries and material classes far beyond its training distribution, as evidenced by its qualitative and often even quantitative agreement with density functional theory (DFT) in 36 diverse test cases, including phonon spectra, ammonia/borane, amorphous carbon, aqueous interfaces, batteries, carburane, cathode materials, combustion, dichalcogenides, dislocation, dissolution, heterogeneous catalysis, hydrogen, ice & water, MOFs, molten salts, multi-component alloys, nanoparticles, Perovskites, polymerization, Pt surface, Si interstitial, solvent mixtures, zeolites and more.

Taken together, these projects showcase how graph neural network (GNN) force fields can form a central pillar of computational materials science, inhabiting a different point on the cost-accuracy Pareto front than DFT, not much worse in accuracy yet orders of magnitude cheaper thanks to linear instead of cubic scaling with system size. Machine learning force fields

have thus unlocked the study of complex phenomena over length and time scales previously inaccessible to numerical simulation.

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No large language models (LLMs) were harmed in the making of this thesis but any passage I wasn’t happy with after rewriting more than once benefited from their editing assistance for the task of increasing clarity and conciseness. Models used for this purpose include GPT4(-Turbo) and later Claude 3 Opus. Figure 1 was converted from a screenshot to a TikZ figure with the help of Claude 3.5 Sonnet.

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Outline

Chapter 1 gives a brief introduction to the new field of materials informatics and the promise it holds by combining advances in machine learning with the growing wealth of materials data in open experimental and first-principles databases. We highlight some challenges holding back progress that motivate the projects presented in later chapters.

Chapter 2 provides a brief overview of core concepts in density functional theory and machine learning, which underpin all work presented in this thesis.

The actual research starts in **Chapter 3** which explores the utility of machine learning (ML) models for materials discovery. At the start of my Ph.D., the ability of ML to accelerate the search for new stable crystals remained questionable [10]. We sought to address open questions around measuring discovery acceleration factors enabled by ML guidance in the context of a realistic high-throughput materials discovery effort. This work resulted in **Matbench Discovery**, a benchmark task designed to resemble a real-world materials discovery campaign, published in

[11] J. Riebesell *et al.*, “Matbench Discovery – A Framework to Evaluate Machine Learning Crystal Stability Predictions” <http://arxiv.org/abs/2308.14920>. 2023–08–28
[Currently under review at Nature Machine Intelligence]

Using Materials Project [12] as training data and the Wang-Botti-Marques (WBM) dataset [1] as an out-of-distribution test set, we compared a variety of ML models to predict the thermodynamic stability of inorganic crystals from unrelaxed structures.

Chapter 4 presents work aimed at demonstrating the real-world benefits of ML-guided materials discovery in the context of a challenging multi-objective search for novel dielectrics. This project was published in

[13] J. Riebesell, T. W. Surta, R. Goodall, M. Gaultois, and A. A. Lee, “Pushing the Pareto Front of Band Gap and Permittivity: ML-guided Search for Dielectric Materials” <https://arxiv.org/abs/2401.05848v1>. 2024–01–11 [Invited submission to Digital Discovery]

and culminated in the successful de novo synthesis of CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$, showcasing the real-world viability of ML-guided discovery efforts, especially in a multi-objective optimization setting with a concave Pareto front.

The training of MACE-MP in **Chapter 3** directly led to **Chapter 5** which presents an extensive analysis of MACE-MP as a foundation model for inorganic materials published in

[8] J. Riebesell *et al.*, “A Foundation Model for Atomistic Materials Chemistry” <https://arxiv.org/abs/2401.00096v1>. 2023–12–29 [Currently under review at Nature]

Although trained purely on bulk crystals, MACE-MP is surprisingly capable of accurately simulating the physics of highly diverse chemistries as demonstrated in over 35 test cases.

Finally, [Chapter 6](#) summarizes results, and [Chapter 7](#) explores directions for future work.

Chapter 1

Introduction

1.1 Importance of Materials Science for Technology

Eras of human development are named for the materials we knew how to leverage at the time into tools and increasingly complex technology, reflecting the close link between our ability to manipulate new classes of materials and the progress of civilization. The rough chronology goes *stone* $\rightarrow$ *bronze* $\rightarrow$ *iron* $\rightarrow$ *steam* $\rightarrow$ *nuclear* $\rightarrow$ *silicon*.

Despite our growing ability to control matter on the atomic scale, synthesizing and characterizing new materials remains laborious, expensive and failure-prone. Designing atomic configurations with desired properties that enable new technologies and then devising working recipes for synthesizing them requires decades of work in the best case. More often, such efforts encounter unforeseen roadblocks and remain outside our reach entirely. Thus, prioritizing our efforts towards candidate materials of greatest promise for critical applications is paramount. However, conserving time and resources requires the ability to computationally predict how specific atomic arrangements attain stability and give rise to target properties. Over recent decades, computational materials science has made great strides in this direction. Functional materials design is an inherently interdisciplinary subject at the intersection of physics, chemistry, engineering, and computer science which has become a driving force behind innovations in every sector of modern civilization, from electronics and energy to medicine and transportation. Prominent examples include:

- Decades of research into **semiconductors** led to incredible transistor miniaturization and exponential gains in computing power.
- A dramatic drop in price per watt even outpacing optimistic projections has allowed **solar cells** to undercut many historical price projections to become one of the cheapest means of electricity generation [14], [15], [16], [17]. Energy used in cell production has similarly fallen with concomitant reductions in time to energy and cost amortization, making solar the most scalable and perhaps cleanest energy source going forward.
- **Li-ion batteries** store energy in a safe, rechargeable, portable, energy-dense medium, enabling ubiquitous modern technologies like mobile computing and electric vehicles.

To accelerate such developments, the vastness of materials space requires faster and more efficient discovery methods. The complexity of the underlying physics makes it difficult, if not impossible, for a single tool to accurately model all relevant processes while remaining compu-

tationally tractable. Many problems cannot be solved with *any* existing tool, inviting further method development to enable systematic chemical design across a wider range of materials applications.

1.1.1 Need for New Tools to Navigate Materials Space

Multi-decade delays between discovery and deployment into functional applications are common in materials science. A large part of the intervening time is spent by human experts iterating on synthesis recipes and optimizing process conditions to produce the desired material in sufficient quantity and purity at acceptable energy, time and precursor cost.

Traditional materials discovery and design have proceeded iteratively, often guided by heuristic design rules and accumulated human expert knowledge confined to narrow material subspaces. This reduces time and resource expenditure, helps de-risk the research and allows navigating local materials space with some degree of intent as opposed to pure random walk. At the same time, it biases efforts towards minor variations of known materials, thereby limiting discovery to close relatives and curtailing far-flung exploration.

1.1.2 Autonomous Labs and High-Throughput Synthesis

While iterative research improves over a purely Edisonian trial-and-error approach, urgency around global warming and the need for cheaper mass-producible sustainable technology demands we accelerate all aspects of the multi-stage new-chemistry-to-device pipeline. The immense scale of materials space and the time-consuming nature of human-driven experimentation mean comprehensive experimental exploration will remain permanently infeasible. Thus, a strong focus should be placed on aggressively pursuing lab automation [18], [19], [20], combinatorial synthesis [21], [22], [23], [24], and rational synthesis route efforts [25], [26], [27], [28].

While no fully autonomous inorganic chemistry lab exists today and the path to such a future is long and fraught with challenges, the combined expertise and ingenuity of experimental chemists, roboticists, software engineers and computational materials scientists can be brought to bear on this problem to achieve a step-change reduction in the time and cost of future experiments. Fully self-driving labs running human-out-of-the-loop exploration with integrated actively learning simulations and synthesis planning will likely prove the most scalable solution in the long run, thus promising to alleviate the growing bottleneck of validating even a fraction of computational property predictions from ever faster and more accurate simulation methods. However, the reality of the challenge is that even automated labs cannot possibly scale to the full space of possible materials spanned by arbitrary combinations of the 98 naturally occurring elements and the sheer infinite ways of arranging atoms in 3D space. To elucidate the scale of the challenge, as of April 2024 the Inorganic Crystal Structure Database (ICSD) [29], [30] holds 299 k [31] experimentally determined structures. Davies et al. [32] estimated that taking

into account stoichiometric and polymorphic degrees of freedom in the 3.6M possible element combinations of the 98 naturally occurring elements up to quaternary order [33] expands the search space to 10^{10} distinct and plausible materials that comply with valency and charge balancing rules. Thus, the extensive coverage of known materials in ICSD represents just 0.003% of the available design space up to quaternary order. Such a vast space is impossible to fully explore with first-principles simulation, let alone high-throughput experiments. Hence there is a persistent need for highly scalable computational tools to guide and prioritize experimental efforts.

While the slow pace of synthesis and characterization is perhaps the most pressing present bottleneck impeding progress in materials science today, it cannot be tackled by individual research groups but instead requires major governmental research initiatives and large industry investments. This thesis will not delve deeper into the experimental side of the problem and instead focus on the problem of optimally directing what experimental throughput we have towards the highest-reward candidate materials.

1.2 Accelerating Discovery through Virtual Screening

Virtual screening efforts for inorganic materials have relied primarily on ab initio calculations, especially density functional theory (DFT). DFT has been the workhorse of computational materials science for several decades and has led to the discovery of numerous novel functional materials [34], [35], [36], [37], [38].

A crucial concept for making computational stability prediction is the convex hull of stability illustrated in Figure 1. The convex hull, or just hull, is spanned by the lowest energy surface in composition space for a given chemical system (all possible ratios for a given set of elements). Materials whose energy lies on this hull are stable against decomposition into other phases. Materials above the hull are metastable or unstable depending on the height of stabilizing kinetic energy barriers. Larger hull distances correlate strongly with greater instability. Whenever a new polymorph with lower energy than all previously known phases made from the same set of elements is discovered, it lowers the convex hull in that chemical system. In a well-studied chemical system, new low-energy discoveries are rare. However, in unexplored chemical spaces, even high-energy structures may initially be on the known hull until true ground-state structures are found. Only thorough exploration can increase confidence that the known hull approximates the true hull and that phases predicted to be on the hull are truly stable.

However, the current silicon age (a.k.a. information age) has brought the momentous development that existing knowledge in the form of structured, computationally-generated data significantly boosts our ability to further advance our understanding of materials on the atomic scale, creating the possibility for a positive feedback loop of accelerating progress. A new field

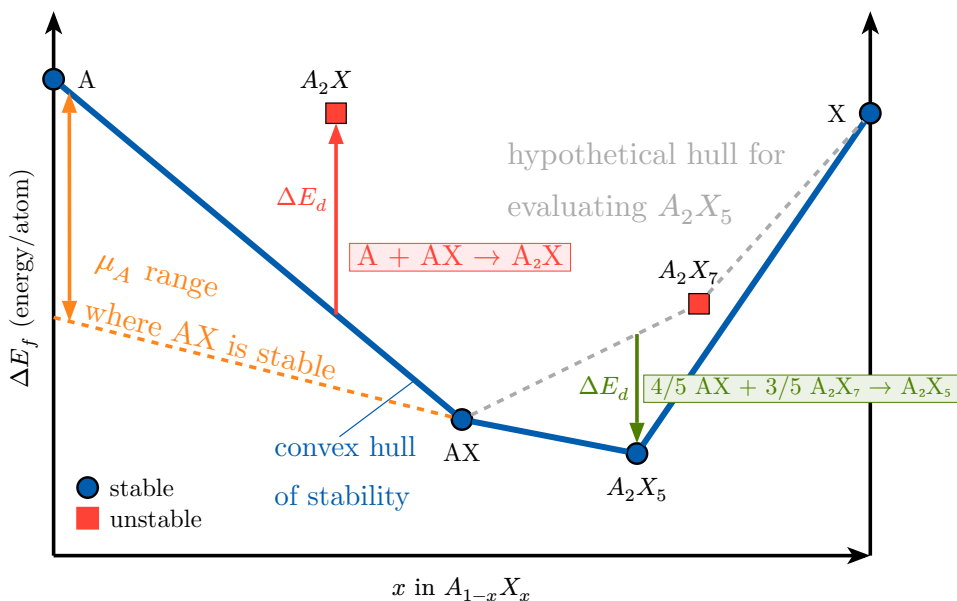


Figure 1: The convex hull of stability for a hypothetical A–X chemical system. The hull is spanned by the lowest energy surface in composition space. Materials on the hull are thermodynamically stable against decomposition into competing phases. For visual clarity, we show only ground states in this phase diagram. Usually, every observed stable phase has many higher-energy experimentally inaccessible polymorphs which would quickly phase-separate into the stable phases if synthesized. ML-predicted hull distance ΔE_d , a.k.a. $E_{\text{hull dist}}$ hereafter, is the central property of interest in many parts of this thesis. The boxed reactions are what define the decomposition energies ΔE_d for A_2X and A_2X_5 . The dashed orange line is tangent to the convex hull at composition AX and provides a lower bound on A’s chemical potential μ_A where AX would still be stable. Figure reproduced from [39], available on GitHub [40].

of materials informatics has emerged to put this flywheel into motion by developing automated workflows that perform intricate multi-step simulations to quickly predict relevant material properties in large quantities and algorithms that can leverage our growing databases generated by such workflows into statistical models with unprecedented cost-accuracy tradeoffs. Such models can in turn power said workflows at an even lower cost albeit with a significant but shrinking reduction in reliability and extrapolability. Physically-agnostic data-driven models can thus serve a vital function in aiding the systematic exploration of much larger regions of unknown chemical space. They may become a similar workhorse for materials simulation as DFT over the next decade.

Database	Year	N_{tot}	Database	Year	N_{tot}
NOMAD [41]	2018	12.1 M	Alexandria [2], [3]	2021	5.1 M
Materials Cloud [42]	2020	4.5 M	AFLOW [43]	2012	3.5 M
GNoME [44]	2023	2.2 M	OQMD [45], [46]	2013	1.2 M
odbx [47]	2024	523.2 k	COD [48]	2023	512.3 k
MPDS [49]	2020	507.2 k	Materials Project [12]	2013	154.4 k
JARVIS-DFT [50]	2020	77.1 k	omdb [51]	2020	68.6 k
TCOD [52]	2017	7.5 k	2DMatpedia [53]	2019	6.4 k
CMR [54]	2012	1.5 k	MPOD [55]	2012	401

Table 1: Widely used material databases sorted by their number of entries N_{tot} . All data (except GNoME) taken from the 2024 OPTIMADE paper [47, tab. 1]. The year refers to the publication date of the paper introducing the database. While size matters, an equally important yet difficult-to-quantify metric for database utility is homogeneity, especially when training ML models. Diverging data for the same or similar compounds renders the learning task ill-defined and confuses the model.

1.2.1 Role of Standardization and Databases

As the number of known materials and their properties grows it has become impossible for any one human to assimilate and formulate hypotheses connecting all relevant information. The severe limits of human memory mean efforts to standardize and digitize our collective knowledge are essential, both for managing this complexity and to enable the development of algorithmic tools that can fully leverage our compounding data. Preeminent experimental databases include the ICSD [29], [30] and the Crystallography Open Database (COD) [56] from which some of

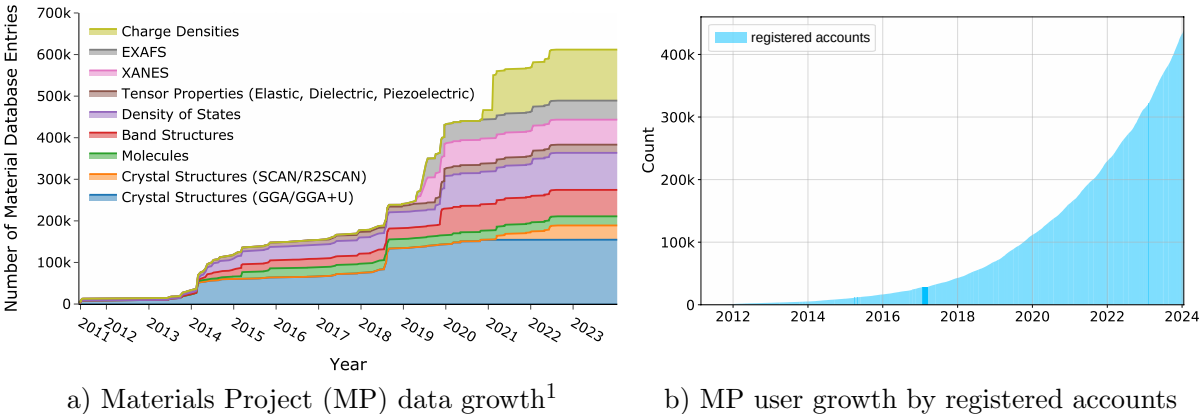


Figure 2: Materials databases like those listed in Table 1 form the backbone of modern data-driven materials research. They have seen compounding growth and use over the last decade and combined now contain many millions of entries. MP [12] for example saw roughly linear data growth (b) and exponential user growth (c) since its inception in 2012. In early 2024, it contained 154k materials with over 600k computed properties and serves over 400k users. (b) and (c) taken from the unpublished *The Materials Project: A Decade of Materials Discovery*.

¹XANES: X-ray absorption near-edge structure, EXAFS: extended X-ray absorption fine structure

the ab initio databases in [Table 1](#) were seeded. As the amount of existing data informing new research grows, it also becomes increasingly crucial to ensure that data is trustworthy, has clear provenance and is disseminated in a standardized, machine-readable format. Databases like [Materials Project \(MP\)](#) [12], [Open Quantum Materials Database \(OQMD\)](#) [45], [Cambridge Structural Database \(CSD\)](#) [57], [Materials Cloud](#) [42], [NOvel MAterials Discovery \(NOMAD\)](#) [41] have been instrumental in pushing in this direction for over a decade as can be seen in [Figure 2](#) by example of MP’s roughly linear data ([Figure 2a](#)) and exponential user growth ([Figure 2b](#)) since its inception in 2011.

However, the proliferation of databases also brings challenges. There is now an urgent need for standardization and interoperability between databases to ensure data can be easily shared, compared and combined across different platforms. If the many new rows in [Table 1](#) since 2020 are any indication, this need is only going to increase in importance as the number of datasets appears to grow super linearly. The FAIR principles [58] provide a framework for Findable, Accessible, Interoperable and Reusable data. While each database aids findability and accessibility (FA), the siloed nature of databases and the lack of common data formats and application programming interfaces (APIs) hinder interoperability and reusability (IR).

The OPTIMADE consortium [47] champions consolidation by providing a standardized API specification for any materials database to implement. OPTIMADE enables uniform client access and data integration across providers, amplifying the utility of the existing conglomerate of databases beyond the sum of its parts and fostering federated interoperability that frees the end user from needing to know the wildly incompatible API details from each data provider. As the volume and diversity of materials data grow rapidly ([Table 1](#)), OPTIMADE’s vendor-neutral integration is uniquely positioned to uphold the IR in FAIR materials data, allowing stakeholders to harness more of our collective materials knowledge at constant user effort.

1.2.2 Limitations of DFT

The cubic scaling of DFT’s computational cost with system size severely limits the length and time scales that can be simulated, even on modern supercomputers. It remains challenging to study complex dynamical processes such as catalytic reactions, phase transitions, and ion transport, which often involve thousands of atoms over nanoseconds to microseconds. Furthermore, the accuracy of semi-local DFT is limited by its simple form of the approximate exchange-correlation functional. More accurate hybrid functionals exist but come at such high computational cost as to prohibit their use in high-throughput searches.

While DFT remains indispensable for high-fidelity calculations and is sufficiently cheap to generate reference data for small systems, there is a growing need for faster and more scalable methods to navigate the vast chemical space in search of promising materials. ML has emerged as a powerful complementary approach that can leverage the wealth of DFT data to build faster

methods that scale to larger systems and longer times yet approach DFT’s accuracy for in-domain predictions.

1.2.3 Emergence of ML as a Complementary Tool

The rapid progress of ML for materials in recent years has led to property predictors that are sufficiently accurate and reliable to act as useful early-stage filters in many virtual screening tasks such as:

- molecular electronic properties [59]
- direct property prediction [60], [61]
- IS2RE² stability [11], [44], [62], [63], [64]
- driving diverse simulations [8], [65], [66], [67], [68]

Materials simulations historically driven by physics-based models (either semi-empirical or fully ab initio) are increasingly making use of purely data-driven (physics-agnostic) models. Hybrid workflows that combine the two (where ML often serves as a low-cost, lower-fidelity filter) are also quickly gaining traction [7], [69]. Geometry optimization and molecular dynamics (MD) in particular benefit immensely from ML force fields that predict energies and forces at a fraction of the cost of DFT by circumventing the need to update atom positions via Hellmann-Feynman forces. The Hellmann-Feynman theorem [70], [71] requires solving the electronic Schrödinger equation anew at each step in the trajectory which constitutes a large computational overhead if the sole goal is to update ion positions. Pre-relaxing with ML force fields has been shown to significantly reduce the number of ionic steps needed to reach the DFT ground state. DFT calculations can then be appended at the end to reach the true DFT ground state or compute electronic structure where necessary e.g. to obtain band gaps, conductivity, optical properties, Born charges to estimate long-range interactions [72] and resulting non-analytical corrections (NAC) to the dynamical matrix [73], etc.

Significant progress was made recently in applying inverse design methods to inorganic crystals [74], [75], [76], enabled by fruitful cross-pollination between the pure ML and cheminformatics communities. The idea is to avoid the arduous task of checking every one of the $\mathcal{O}(10^{10})$ possible materials with individual forward calculations and instead ‘jump’ straight to the proverbial needle in the haystack by generating crystal structures conditioned on desired properties. Conditional sampling takes place in a generative model’s latent space. A separate surrogate model for the functional property of interest is constructed that operates in the latent space and identifies embeddings corresponding to structures with a high likelihood of exhibiting the desired functionality [77]. Samples corresponding to promising embedding vectors are then decoded into crystal structures by the generative model. Such methods were largely pioneered by the organic materials community, where chemical diversity is lower (usually focussed on C, H, N, O, S, P and a varying handful of heavy elements), symmetry constraints are simpler,

²IS2RE = Initial Structure to Relaxed Energy. Terminology introduced by [62]

thermodynamic stability concerns play a smaller role and industry interest and funding is higher. However, equivalent methods for inorganic materials have emerged that efficiently handle the increased complexity of more variable chemistries and more stringent symmetry constraints imposed by the periodicity of bulk crystals [76], [78]. Generative models for inverse design may be on the cusp of becoming trusted tools in the wider materials community. However, synthesizing and measuring a generated material experimentally that satisfies the initial target properties has yet to be demonstrated to prove viability.

1.3 Machine Learning for Materials Simulation

1.3.1 Molecular Dynamics

MD simulations are an essential tool for gaining atomic-scale insights into many-body interactions over time. While DFT provides a computationally tractable method for approximately solving the electronic Schrödinger equation, its time-independent formulation only yields the potential energy surface (PES) for a given set of fixed atomic positions at zero temperature. To gain insights into dynamical processes such as catalytic reactions on surfaces or ion migration through an electrolyte requires atomic simulations over time at non-zero temperature. MD provides that by time-integrating Newton’s equations of motion:

$$\mathbf{F} = -\nabla V(\mathbf{r}) = m\mathbf{a} = m\ddot{\mathbf{r}}, \quad (1)$$

for a potential energy surface $V(\mathbf{r})$ with resulting forces $\mathbf{F}$ from its negative gradient $-\nabla V(\mathbf{r})$.

However, the short femtosecond timesteps needed to accurately simulate chemical reactions that occur over nano- or even microseconds require millions to billions of integration steps, each one having to compute the forces acting on all particles.

1.3.2 Classical Force Fields and Ab-Initio MD

Classical force fields can efficiently simulate up to millions of atoms and nanosecond timescales. However, their simplified functional forms limit accuracy and transferability, especially for complex reactions with changing chemical environments as encountered e.g. in catalysis, phase transitions, ion transport, and solid-electrolyte interphase formation. Conversely, quantum-mechanical methods like DFT-driven ab initio MD (AIMD) provide high accuracy but are limited to hundreds or a few thousand atoms and sub-nanosecond timescales due to the cubic scaling of compute cost with system size.

1.3.3 Machine Learning Interatomic Potentials

MLIPs, also known as ML force fields, have emerged as a promising third paradigm on a new previously unreachable point of the cost-accuracy Pareto front. They are statistical, physically agnostic models that learn energies and forces from reference data, often generated in large quantities with DFT. Many have the appealing property of scaling *linearly* with system size, enabling simulations of millions of atoms over nanosecond to microsecond timescales on current supercomputers. However, until recently, MLIPs have been severely limited in transferability, often fitted to only single- or few-element chemical systems due to insufficient training data and/or scaling constraints in the model architecture [79].

The latest generation of high-body-order message-passing MLIPs [67], [68], [80], [81], have demonstrated a step-change in the ability to benefit from training on the largest and most diverse ab initio databases available [3], [12], [44], [45], [46], [62], [82], [83], yielding unprecedented applicability across diverse chemical spaces³. By learning rich, generalizable representations of atomic environments, these so-called *foundation models* pre-trained on millions of crystal structures and their energies, forces, stresses, and more recently magnetic moments [67] and response to external fields [84], have the potential to run a wide variety of materials simulation at near-DFT accuracy but orders of magnitude cheaper, or, conversely, at orders of magnitude larger length and time scales. They also serve as excellent starting points for both small and large-data fine-tuning in cases where out-of-the-box performance proves insufficient. This ‘pre-train then fine-tune’ paradigm marks the defining feature of a foundation model and is an advantage of deep learning not applicable to prior ML methods that lack transferable end-to-end-learned internal representations. R. Bommasani *et al.*, [85] who coined the term ‘foundation model’ in the context of natural language processing (NLP) define them as models broadly applicable across a wide range of tasks and datasets, transferable to new tasks with minimal fine-tuning, and a strong baseline for further model development. In direct analogy to chemistry, we define **foundational ML force fields** in this thesis as models that are

- trained on millions of chemical spaces covering much of the periodic table, thereby becoming
- applicable to various tasks in computational materials science, from crystal structure prediction to molecular dynamics,
- effective starting points for data-efficient fine-tuning on account of having learned rich, generalizable representations of millions of atomic environments.

Developing foundational force fields that can accurately simulate the full diversity of inorganic materials presents an exciting avenue for achieving the trifecta of large-system, long-timescale high-accuracy simulations. Such simulations greatly facilitate understanding and designing materials for applications like catalysis, energy storage and CO₂ capture. While encouraging

³How much of this transferability is true extrapolation as opposed to interpolation in a high dimensional space is debatable but also somewhat moot as utility is unlocked either way.

progress has been made in training single large models on multiple modalities beyond bulk crystals such as amorphous materials, surfaces, interfaces, alloys and defects [65], [66], [86], challenges remain regarding:

1. Developing **good metrics** to compare foundation models going forward. We partially address this in [Chapters 3](#) and [5](#). In particular, [Chapter 3](#) presents results on thermodynamic stability prediction from unrelaxed hypothetical crystals under realistic conditions to filter out likely unstable ones, demonstrating that stability can be predicted to useful accuracy before conducting costly DFT relaxation.
2. Ensuring model development remains **open and reproducible**. We contribute to this by open-sourcing CHGNet [67] and MACE-MP [8], two models co-developed in [Chapter 3](#) and thoroughly tested in [Chapter 5](#). Both models established a new state of the art (SOTA) at time of publication among 10 tested open-source, open-data models⁴.
3. **Commoditizing** foundation models so they become usable by non-experts. By academic standards, [CHGNet](#) and [MACE-MP](#) have well-maintained code bases and versatile easy-to-use and install APIs, public weights, public training data and public training scripts to reproduce our results and finetune the models on new data.

While scaling efforts from industry are likely to drive rapid progress that academia will likely struggle to keep up with (as recently demonstrated by large language models [87], [88], [89], [90]), the academic and open source communities must step up to champion reproducibility and FAIRness in model development and training data generation. Overcoming these challenges will enable the rapid deployment of GNN foundation models to the wider materials community and help unlock their full scientific utility.

⁴The proprietary GNoME of NequIP architecture and trained on a much bigger proprietary dataset was the only model known to perform better on Matbench Discovery.

Chapter 2

Methods in Computational Materials Science

[Section 2.1](#) provides an overview of core concepts in first-principles simulation with a focus on density functional theory (DFT). [Section 2.2](#) gives a primer on data-driven modeling with a focus on machine learning force fields (MLFF). These two widely used methods in computational materials science underpin all work in this thesis.

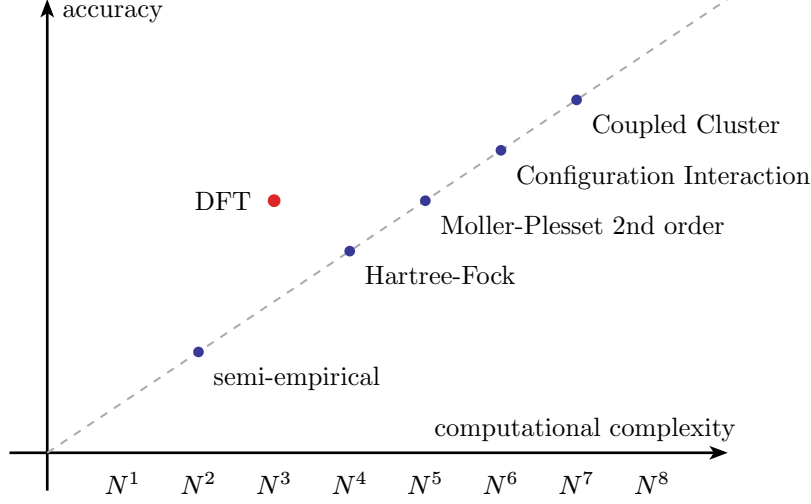
2.1 Density Functional Theory

The Kohn-Sham DFT self consistency cycle depicted in [Figure 3c](#)) at the PBE level of theory is the exclusive source of reference data for all models and benchmarks presented in this thesis ([Chapter 3](#), [Chapter 5](#)). It also serves as the highest-fidelity computational filter in a high-throughput funnel-type discovery workflow ([Chapter 4](#)). The first two sections of the following high-level introduction closely follow [\[91\]](#).

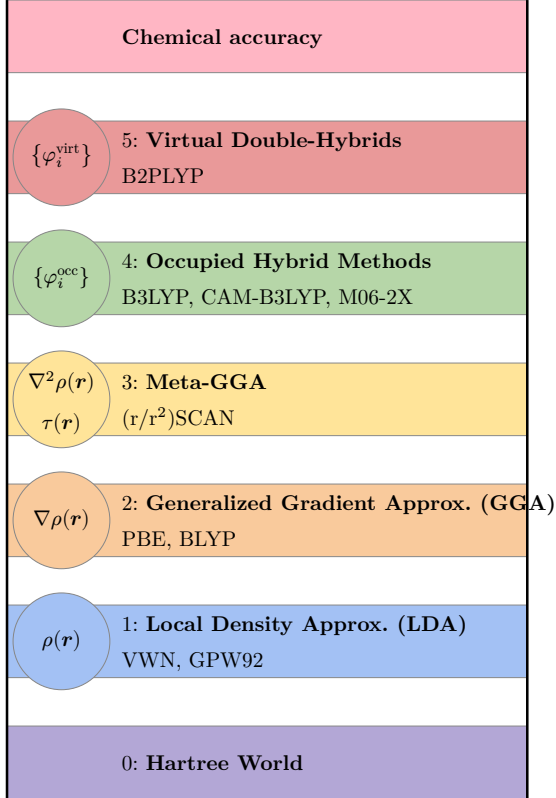
DFT has been the workhorse of computational materials science for decades, providing a first-principles quantum mechanical approach to calculating the electronic structure and resulting properties of materials up to system sizes of $\sim 10^4$ atoms with today’s supercomputers. Its prevalence is due in large part to its prominent placement on the cost-accuracy Pareto front shown in [Figure 3a](#)), offering an attractive combination of unusually high accuracy at acceptable computational cost.

The central quantity in DFT is the electron density $n(\mathbf{r})$, a function of just three spatial coordinates $\mathbf{r} = (x, y, z)$. It replaces the much more complex full electronic many-body quantum wave function $\Psi(\mathbf{R}, \mathbf{r})$ of $3N + 3M$ spatial coordinates for a system of N electrons and M nuclei. Despite being a much simpler object, $n(\mathbf{r})$ nonetheless uniquely determines the system’s ground state. This crucial insight is due to Hohenberg and Kohn, who showed that the ground-state energy $E[n(\mathbf{r})]$ of a system is a unique functional of the electron density. Moreover, the ground-state electron density is the minimizer of this functional, thus offering variational access to the ground state of any quantum system. An iteratively-derived self-consistent solution [\[93\]](#) of the Kohn-Sham equations [\[94\]](#) as shown in [Figure 3c](#)) therefore yields both the ground-state electron density and the total energy of the system from which many downstream properties can be derived.

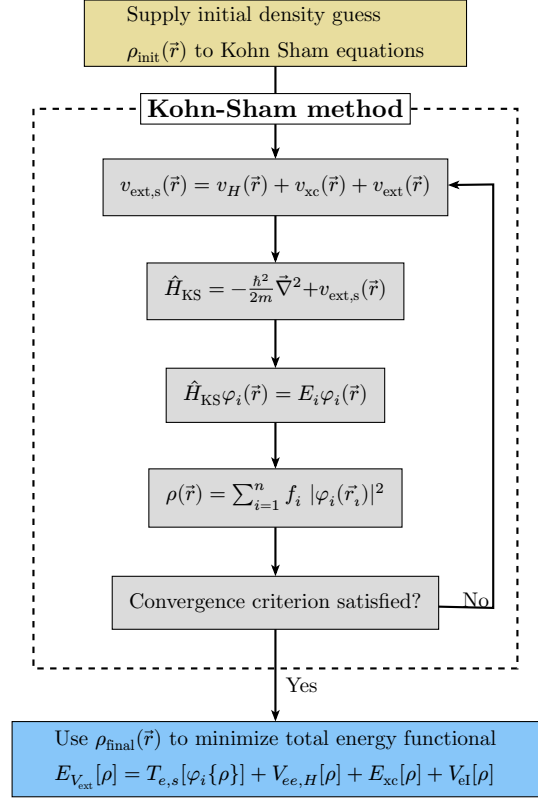
The time-independent many-body Schrödinger equation describes the quantum behavior of a system of N electrons and M nuclei [\[95\]](#):



a) Cost-accuracy tradeoff of different quantum mechanical approximation methods.



b) Jacob's ladder of E_{xc} approximations [92].



c) Kohn-Sham cycle.

Figure 3: **(a)** The cost-accuracy tradeoff of different quantum mechanical methods. DFT is the most widely used method in materials science due to its exceptional accuracy given its favorable cubic cost scaling. **(b)** Jacob's ladder of DFT according to Perdew represents a hierarchy of increasingly sophisticated, accurate and expensive approximations to the exchange-correlation energy. While displayed as a single dot in (a), the ladder shows that even within DFT a wide range of techniques with different cost-accuracy tradeoffs exist to estimate E_{xc} , from cheap local and semi-local functionals to accurate hybrid and double-hybrid/range-separated functionals. **(c)** The Kohn-Sham cycle enables iteratively solving for the ground state electron density, which in turn uniquely determines the total energy of the system. All 3 diagrams available on GitHub (a), (b), (c) [40].

$$\hat{H} \Psi(\mathbf{R}, \mathbf{r}) = E \Psi(\mathbf{R}, \mathbf{r}). \quad (2)$$

with $\Psi(\mathbf{R}, \mathbf{r})$ the many-body wave function of M nuclear coordinates $\mathbf{R} = \{\mathbf{R}_1, \dots, \mathbf{R}_M\}$ and N electronic coordinates $\mathbf{r} = \{\mathbf{r}_1, \dots, \mathbf{r}_N\}$, E the energy, and $\hat{H}$ the many-body Hamiltonian operator that splits into

$$\begin{aligned} \hat{H} = & -\overbrace{\frac{\hbar^2}{2m_e} \sum_i \nabla_i^2}^{\hat{T}_e} - \overbrace{\frac{\hbar^2}{2} \sum_k \frac{\nabla_k^2}{M_k}}^{\hat{T}_n} \\ & + \underbrace{\frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}}_{\hat{V}_{ee}} + \underbrace{\frac{1}{2} \sum_{k \neq l} \frac{Z_k Z_l}{|\mathbf{r}_k^n - \mathbf{r}_l^n|}}_{\hat{V}_{nn}} - \underbrace{\sum_{i,k} \frac{e Z_k}{|\mathbf{r}_i - \mathbf{r}_k^n|}}_{\hat{V}_{ne}} \end{aligned} \quad (3)$$

where $\hat{T}_{e/n}$ are the electronic/nuclear kinetic energy operators, $\hat{V}_{ee}$ is the electron-electron interaction energy, $\hat{V}_{en}$ is the electron-nucleus interaction (also referred to as external potential $\hat{V}_{\text{ext}}$), E_{nn} denotes the classical nucleus-nucleus interactions. $\hbar$ is the reduced Planck constant, m_e the electron mass, M_k/Z_k the mass/nuclear charge of nucleus k and e the elementary charge.

The Born-Oppenheimer approximation [96] makes use of the fact that nuclei are 3 orders of magnitude heavier than electrons, allowing us to treat the nuclear wave function as approximately stationary and solving for the electronic wave function independently at a given nuclear configuration defined by the external potential $\hat{V}_{en}$. The resulting Hamiltonian with purely-nuclear terms removed reads:

$$\hat{H}_{\text{BO}} = -\hat{T}_e + \hat{V}_{en} + \hat{V}_{ee}. \quad (4)$$

Although straightforward to formulate, solving this equation scales exponentially with system size, rendering it intractable for more than $\mathcal{O}(10)$ particles [97]. To circumvent this, DFT relies on a reformulation of the many-body problem onto a system of non-interacting electrons in an external potential. This transformation is predicated on two theorems, known as the Hohenberg-Kohn theorems [98]. The first theorem states that for a given system $\hat{V}_{en}$, its total energy E is a functional of the ground state charge density $n(\mathbf{r})$ given by:

$$E[n(\mathbf{r})] = \int d^3r V_{en}(\mathbf{r}) n(\mathbf{r}) + F[n(\mathbf{r})] + E_{nn}, \quad (5)$$

where the functional $F[n(\mathbf{r})]$ maps electron density to kinetic energy and electron-electron interactions. The second theorem asserts that the density minimizing the total energy corresponds to the ground state charge density. Together, the two theorems establish the profound insight that the ground state total energy of a system of interacting electrons and nuclei is determined by the ground state electron density. Thus, variational minimization of the energy functional $E[n(\mathbf{r})]$ yields the ground state charge density and the total energy of the system. In contrast to the $3N$ variables of an N -particle wavefunction, the charge density $n(\mathbf{r})$ with $\mathbf{r} = (x, y, z)$ has just three variables.

In practice, the self-consistency cycle for a given configuration of nuclei is often initialized to a charge density resulting from the superposition of atomic orbitals centered at each nucleus. From this charge density, we calculate the effective potential $\hat{V}_{en}$ and the resulting electronic wavefunction $\Psi_{i(\mathbf{r})}$. We then repeatedly solve the Kohn-Sham equations to get the total energy, each time modifying the charge density in the direction of decreasing energy until the input charge density guess and resulting wavefunction converge to a desired degree of self-consistency.

While the Hohenberg-Kohn theorem proves the existence of the functional [eq. \(5\)](#), it says nothing about how to derive it nor how to make it computable. Thankfully, as mentioned earlier, Kohn and Sham devised a method to map the many-body problem onto a non-interacting electron system in an effective potential produced by the surrounding nuclei and electrons. Solving for the ground state charge density of the non-interacting approximation yields the same ground state as the true interacting many-body system. Specifically, the Kohn-Sham approach reformulates the Hohenberg-Kohn expression for the ground state functional as:

$$E_{KS}[n] = T_{S[n]} + \int d^3\mathbf{r} V_{en}(\mathbf{r}) n(\mathbf{r}) + E_{\text{Hartree}}[n] + E_{nn} + E_{xc}[n], \quad (6)$$

with $T_{S[n]}$ the kinetic energy of non-interacting electrons, $E_{\text{Hartree}}[n]$ the mean-field Coulomb interaction energy of the electron density, and $E_{xc}[n]$ the exchange-correlation functional. This is a useful restating of the problem as it packages all of the difficult-to-compute quantum exchange and correlation contributions into the last term $E_{xc}[n]$. This so-called exchange-correlation energy is where most of the DFT development effort since its 1965 inception has been focused. By rewriting [eq. \(6\)](#), we see that $E_{xc}[n]$ captures the difference in kinetic and potential energy between the interacting and non-interacting electron systems [\[95\]](#):

$$E_{xc}[n] = \langle \hat{T} \rangle - T_{S[n]} + \langle \hat{V}_{en} \rangle - E_{\text{Hartree}}[n]. \quad (7)$$

2.1.1 Density Functionals

DFT would not have reached its modern utility and resulting ubiquity without the development of highly effective approximations of $E_{xc}[n]$. Some of the most successful are ranked by increasing accuracy and cost in [Figure 3b](#)), Jacob’s ladder of E_{xc} approximations [\[92\]](#). The local density approximation (LDA) and the generalized gradient approximation (GGA) are the first two rungs of this ladder and the most popular choices for inorganic crystals.

LDA is a strictly local functional, using only the value of the electron density $n(\mathbf{r})$ at each point to estimate $E_{xc}^{\text{LDA}}[n]$. It is based on the exact exchange of a homogeneous electron gas $\epsilon_{xc}^{\text{homogen}}$ as calculated by quantum Monte Carlo [\[94\]](#), [\[99\]](#), [\[100\]](#):

$$E_{xc}^{\text{LDA}}[n] = \int d^3\mathbf{r} n(\mathbf{r}) \epsilon_{xc}^{\text{homogen}}(n(\mathbf{r})), \quad (8)$$

Despite its strictly local dependence on the charge density and its assumption of a uniform electron gas, seemingly unjustified for highly non-uniform systems like molecules and crystals, LDA has been remarkably successful in describing the properties of condensed matter systems. It contributed in no small part to the tremendous early success of DFT, motivating many of the resources subsequently directed toward the development of more sophisticated functionals.

GGA comprises a class of functionals one step up on the complexity ladder that approximate $E_{\text{xc}}[n]$ as a functional of both the charge density $n(\mathbf{r})$ and its spatial gradient $\nabla n(\mathbf{r})$:

$$E_{\text{xc}}^{\text{GGA}}[n] = \int d^3\mathbf{r} n(\mathbf{r}) \varepsilon_{\text{xc}}^{\text{homogen}}(n(\mathbf{r}), \nabla n(\mathbf{r})). \quad (9)$$

By far the most widely used GGA variant is the Perdew-Burke-Ernzerhof (PBE) functional [101]. It generated all ML reference data used in this thesis and was the functional of choice for high-throughput screening in Chapter 4. GGAs improve on LDA results for many types of systems. However, both LDA and GGA fail to reproduce long-range van der Waals dispersion interactions. These are non-local correlations that require distant information about the charge density not captured by its local value and gradient. While LDA exhibits a systematic overbinding error, PBE is biased towards underbinding atoms. That is, LDA (GGA) lattice parameters are typically smaller (larger) than experiment. Both are well known to band gaps (the energy difference between highest occupied and lowest unoccupied electronic states in a material) by on average 10–40% [102], [103], [104]. For more accurate band gaps, a hybrid functional like Heyd-Scuseria-Ernzerhof (HSE) [105], [106] or an excited-state formalism beyond DFT such as many-body perturbation theory must be used.

2.1.2 Plane waves and pseudopotentials

To represent wavefunctions, a (preferably orthonormal) basis set is required. According to Bloch’s theorem [107], solutions to the Schrödinger equation in a periodic potential can be expressed as plane waves modulated by periodic functions. This makes plane waves a natural choice for basis expansion of the electron wave functions in simple metals where ionic cores only apply small perturbations to the nearly free periodic electron potential. In fact, plane waves apply well to general extended systems such as infinitely periodic crystals. For computational efficiency, the infinite series of plane waves must be truncated at a finite number of terms by imposing a cutoff E_{cut} on the maximum electronic kinetic energy. Depending on the unit cell’s lattice parameters and a given choice of E_{cut} , all plane waves $\exp(i\mathbf{G} \cdot \mathbf{r})$ satisfying the following equation are used in the wavefunction expansion:

$$\frac{\hbar^2}{2m_e} |\mathbf{G} + \mathbf{k}|^2 < E_{\text{cut}}, \quad (10)$$

where $\mathbf{G}$ is a reciprocal lattice vector and $\mathbf{k}$ is an electron wave vector. For example, for $E_{\text{cut}} = 520$ eV, the basis set resolution d is:

$$d = |G_{\text{cut}}|^{-1} = \frac{\hbar}{\sqrt{2m_e E_{\text{cut}}}} = \frac{\hbar}{\sqrt{2m_e \cdot 520 \text{ eV}}} \approx 8.6 \text{ pm} = 0.086 \text{ \AA}. \quad (11)$$

Core electrons are densely packed and highly localized near the nucleus. The need for fermionic wavefunctions to be orthogonal to one another results in highly oscillatory wavefunctions for core electrons. Meanwhile, valence electrons with more slowly varying wavefunctions in the outermost shells are much more important for chemical bonding and the resulting material properties. Thus, all-electron DFT calculations incur a large, often unjustified (in terms of their chemical accuracy) computational cost due to the need to include high-energy plane waves to accurately describe the oscillating core electrons. To accurately describe core wavefunctions would require a basis resolution comparable to these $\sim 10^{-2} \text{ \AA}$ oscillations, which would require E_{cut} to grow ~ 100 -fold due to appearing in the square root. A very common and often accurate approximation is to freeze core electrons into an effective nucleus whose positive charge is reduced by the number of neglected core electrons. This so-called pseudopotential formalism creates a smoother external potential and reduces the number of plane waves needed to accurately electronic wavefunctions. This approach is extremely common and was used for all DFT simulations pertaining to this thesis.

2.1.3 Self-Interaction Errors

When representing a many-body quantum system as an effective potential in the Kohn-Sham formalism, electrons neglecting exact exchange experience spurious self-interaction with their own contribution to the charge density [100]. This error leads to issues such as over-delocalization of electrons, underestimation of bandgaps, and incorrect description of charge transfer and dissociation [108]. It is particularly problematic in the context of redox reactions where an electron migrates between atoms with strongly differing localization environments [109], [110]. In such cases, the self-interaction errors between initial and final electron environment do not cancel, leading to inaccurate results. To mitigate self-interaction errors, various techniques have been developed including hybrid functionals, [111], the Perdew-Zunger one-electron self-interaction correction scheme [100], and the DFT+ U method [112].

Hybrid functionals such as HSE include a fraction $a \in (0, 1)$ (e.g. $a = 0.25$ in HSE06 and PBE0 [101]) of exact Hartree Fock exchange [111],

$$E_x^{\text{HF}} = -\frac{1}{2} \sum_{i \neq j} \sum_{\sigma=\uparrow\downarrow} \iint d^3r_1 d^3r_2 \frac{\Psi_i^*(\mathbf{r}_1) \Psi_j^*(\mathbf{r}_2) \Psi_i(\mathbf{r}_1) \Psi_j(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}, \quad (12)$$

where i and j sum over electron pairs and Ψ_i are the Kohn-Sham orbitals. This term is added to the hybrid exchange-correlation functional:

$$E_{\text{xc}}^{\text{hybrid}} = a E_x^{\text{HF}} + (1 - a) E_x^{\text{PBE}} + E_c^{\text{PBE}}, \quad (13)$$

with E_x^{PBE} and E_c^{PBE} the PBE exchange and correlation functionals.⁵

However, while hybrid functionals avoid the need for Hubbard U corrections and do so with much less empirical fitting, they also incur a significantly higher computational cost, making them as yet unsuitable for high-throughput calculations. Existing high-throughput DFT databases instead largely opted for PBE(+ U) for its computational efficiency.

2.1.4 Hubbard U Corrections

The Hubbard U correction is primarily designed to address the problem of strongly correlated electron systems, where standard (semi-)local density functionals like LDA/GGA fail to accurately describe the electronic structure [112]. The Hubbard U correction introduces an additional term to the DFT Hamiltonian that accounts for the on-site Coulomb interaction (the U term) between electrons in localized orbitals, such as d or f orbitals in transition metal oxides or rare-earth compounds [113]. This correction helps to improve the description of localized electronic states, opens up bandgaps, and better describes magnetic properties in strongly correlated systems [114], as well as mitigate the lack of self-interaction error cancellation among environments with contrasting localization in redox reactions [109], [110]. The Materials Project uses U corrections for all compounds containing Co, Cr, Fe, Mn, Mo, Ni, V, W in combination with O or F [115]. U values are calibrated specifically for the VASP PBE_52 pseudo-potentials and phase stability analyses of oxides, and the same oxide U values are also used for sulfides (presumably due to a lack of experimental sulfide fitting data). However, MP U corrections are not structure-site specific (a potential source of inaccuracy), instead applying to all sites with one of the above elements, and in those, only to the d -orbitals. MP values were fitted solely with phase stability in mind. As a result, they may harm just as easily as help in the calculation of properties such as band structures. Crucially when training ML models on a wide range of chemical systems, the use of U corrections causes inconsistent energies and forces between corrected and non-corrected DFT calculations, presenting a potential source of model (as well as user) confusion. This establishes a lower bound on the error with which even a perfect ML model can be fit to MP reference data if U -corrected energies are not carefully filtered out or otherwise accounted for during training [115], [116]. We consider this a dangerous pitfall to newcomers and even experts, especially those entering materials informatics from the computer science domain (see e.g. spuriously elevated errors for MACE-MP and M3GNet in [117, fig. 5] from not applying [MaterialsProject2020Compatibility](#) [116] GGA/GGA+ U mixing corrections). For an extensive discussion of U corrections in MP, we refer to [109], [110], [115].

⁵Incidentally, fortuitous error cancellation between E_x^{PBE} and E_c^{PBE} contributed in no small part to the success of semi-local functionals.

2.2 Machine Learning

Machine learning (ML) is a powerful technique for encoding carefully designed inductive biases (task-specific constraints and symmetries) into optimization algorithms that can make machines “write” or rather converge on useful programs given the right loss function to optimize. These “programs” consist largely of conceptually simple linear algebra operations, but at a scale and level of fine-tuning that would be impossible to achieve manually. The viability of current ML methods relies on the massive increase in compute power over recent decades and the compounding growth of digital data, be it in the form of natural language, images and video, or, in the case of materials science, high-throughput ab initio databases. This potent combination of large data and large compute has allowed artificial neural networks in particular to significantly outperform many other ML methods. A challenge with applying ML to materials is designing models capable of manipulating internal representations for 3D atomic configurations in a way that adheres to physical symmetries, most importantly rotation, translation and inversion symmetry, invariance to permutation of identical atom species and, specific to crystals, respecting periodic boundary conditions.

Similar to the development of machine learning in other domains, computational chemists have approached this issue using two different paradigms: shallow and later deep learning.

1. **Shallow Learning** involves the design of hand-crafted input features for representing atomic systems as fixed-length numerical vectors to use with existing ML architectures [118], [119], [120], [121].
2. **Deep Learning** for materials, by contrast, developed ML architectures tailor-made for end-to-end learning of internal representations of chemical systems [59], [67], [68], [122], [123], [124], [125], taking lower-level data like atomic coordinates and element types as input.

The next two sections (2.2.1 and 2.2.2) provide a high-level overview of these two approaches following [126]. Unsupervised and self-supervised learning – while important subfields of ML in other domains – have had comparatively little impact in materials science, and will be skipped in this introduction for brevity. We touch on the future prospects of generative modeling for materials in the outlook (sec. 7.6), another major subfield of ML omitted in this introduction.

2.2.1 Shallow Learning

The first appearances of ML in materials science were off-the-shelf architectures such as linear regression, support vector machines, random forests and gradient boosting. These were trained in a supervised manner on labeled data and input features crafted by domain experts to encode local atomic environments based on existing knowledge of physical properties relevant to the target problem.

Ensemble methods like random forests [127] and gradient boosting [128] have been particularly successful in materials informatics due to their ability to capture complex, non-linear relationships between descriptors and properties from relatively small amounts of data [126].

The performance of shallow models in materials science is limited by the task relevance and information content of their descriptors. Various hand-crafted atomic structure-based descriptors have been developed to represent local atomic environments [120], [129], [130], [131], [132], [133], [134]. To improve data efficiency and guarantee physically consistent models, these descriptors must reflect the symmetries of the underlying systems. That means a model’s internal representation must be equivariant under $E(3)$, the group of rotations, translations, reflections in 3d space, as well as permutations of same-species atoms [129].

Two featurization schemes to highlight since they are the only shallow learning techniques used in this thesis (see Voronoi Random Forest in Section 3.3) are the Magpie set [135] of general-purpose compositional descriptors and structural attributes derived from Voronoi tessellations [136]. Magpie introduced a 145-dimensional feature vector derived from element-amount-weighted statistics (mean, min, max, ...) of atomic, stoichiometric and electronic properties. Unlike some earlier compositional descriptors, Magpie can be applied to arbitrary chemical systems. It found use in discovering metallic glasses [137] and predicting band gaps of diverse inorganic compounds [135]. The second technique, Voronoi tessellation, has the advantage of not relying on knowledge of the DFT-relaxed lattice parameters and being invariant to dilations or contractions of the lattice, making it a particularly applicable shallow learning method for incorporating structural information into ML-guided materials discovery where only the unrelaxed crystal structure is known. The concatenated Magpie + Voronoi tessellation feature set is fed into a Random Forest in Section 3.3 to establish a strong shallow learning baseline and measure the scaling advantages of end-to-end-learned representations.

2.2.2 Deep Learning

Unlike shallow learning, deep learning models are trained to construct information-dense internal features directly from raw low-level data. This sidesteps the time-consuming process of handcrafting bespoke features for every new problem of interest. Since deep learning representations are not limited by domain knowledge and human intuition, they simultaneously lift the ceiling on feature fidelity and expressiveness. Empirically, end-to-end learned representations generally outperform hand-crafted features once training data exceeds a domain-dependent threshold, often on the order of $\gtrsim \mathcal{O}(10^4)$ samples [138], [139], [140], [141].

}

The first and still widely used deep learning model is the multi-layer perceptron (MLP) [146] dating back to 1958. It stacks layers of simple linear transformations combined with non-linear

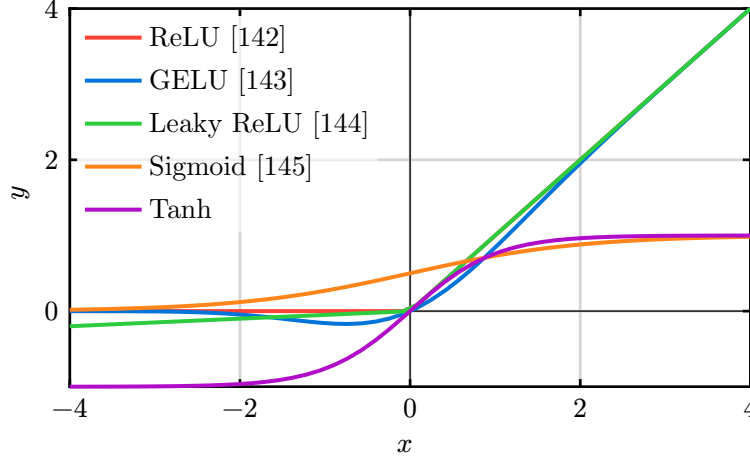


Figure 4: Popular ML activation functions. $\text{ReLU}(\mathbf{x}) = \mathbf{x}^+ = \max(\mathbf{x}, 0)$ is the most widely used due to its simplicity and computational efficiency. $\text{GELU}(\mathbf{x}, \mu = 0, \sigma = 1) = \frac{\mathbf{x}}{2} \left(1 + \text{erf}(\mathbf{x}/\sqrt{2}) \right)$ is a differentiable variant of ReLU. $\text{Leaky ReLU}(\mathbf{x}) = \max(0, \mathbf{x}) + \alpha \cdot \min(0, \mathbf{x})$ with $\alpha < 0$ adds a small gradient for negative activations. $\text{Sigmoid}(\mathbf{x}) = (1 + \exp(-\mathbf{x}))^{-1}$ smoothly squashes the input to the range $(0, 1)$. $\text{Tanh}(\mathbf{x}) = \frac{\exp(\mathbf{x}) + \exp(-\mathbf{x})}{\exp(\mathbf{x}) - \exp(-\mathbf{x})}$ is a scaled and shifted cousin of the sigmoid function. [Diagram available on GitHub \[40\]](#).

activation functions to gradually extract hierarchical representations of the input features $\mathbf{x}$ to predict some target property $\mathbf{y}$ [126]:

$$\hat{\mathbf{y}} = f(\mathbf{x}) = \mathbf{W}_l \sigma(\mathbf{W}_{l-1} \dots \sigma(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) \dots + \mathbf{b}_{l-1}) + \mathbf{b}_l, \quad (14)$$

where $\mathbf{W}_i \in \mathbb{R}^{M_i \times M_{i-1}}$ is the matrix storing the parameters of layer i which are tuned from data, M_i is the number of units (a.k.a. neurons) in the layer, and $\sigma(\mathbf{x})$ is a non-linear activation function, often one of the popular choices shown in Figure 4. These have largely replaced sigmoidal activations [145] due to their susceptibility to vanishing gradients [147].

Gradient Descent and Automatic Differentiation

A central ingredient to the success of deep learning is the ability to train deeply layered models with simple first-order gradient optimization algorithms like stochastic gradient descent [148]. This feature relies on the ability to automatically and efficiently differentiate through the model via backpropagation (i.e. repeatedly applying the chain rule of calculus to obtain gradients of the loss function with respect to the model parameters). Given sufficient data and compute power, this allows deep learning models to learn extremely complex functions impossible to write by hand. Modern deep learning frameworks [149], [150], [151] provide a plethora of modular building blocks including custom layers tailored for certain application domains with out-of-the-box automatic differentiation, abstracting away the technical complexities of implementing backpropagation [152]. This allows researchers to focus on innovating the model forward pass and combining existing layers in novel ways, contributing to the rapid pace of progress in the field. These frameworks also enable hardware acceleration such as graphical and tensor processing units (GPU and TPU) support to significantly reduce training times.

Among the many choices of optimizers, gradient descent is the simplest, simply updating the model parameters θ at each step according to the gradient of the loss function $L(\theta_t)$:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t). \quad (15)$$

The learning rate η is a hyperparameter (i.e. not optimized during training) and determines the size of each parameter update. In practice, the full gradient on the complete data set is replaced by a stochastic approximation calculated from (usually randomly sampled) mini-batches of data samples drawn without replacement until the training set is exhausted, marking the completion of one training epoch. Stochastic gradient descent tends to speed up convergence and provide regularization effects. Training can often be further sped up by choosing more sophisticated optimizers that include momentum, learning rate schedules or adaptive learning rates [153], as well as weight decay, early-stopping and dropout regularization methods [154].

2.2.3 Graph Neural Networks

A common and widely used strategy for infusing domain knowledge into deep learning revolves around designing custom layers with inductive biases whose moving parts often naturally map onto core concepts in the data being modeled. Notable examples include recurrent neural networks (RNNs) for series data [155], [156], transformers for language [87], [157] and convolutional neural networks (CNNs) for vision [158], [159], [160]. In computational chemistry, this paradigm has led to the adoption of graph neural networks (GNNs), also known as message-passing neural networks (MPNNs), that operate on graph representations of molecules and crystals where nodes map onto atoms and edges model interactions between atoms as illustrated in Figure 5. GNNs are not chemistry-specific but have emerged as a powerful approach in various domains to learn from arbitrary graph-structured data including reconstruction, tagging, and end-to-end analysis of collider events in high-energy physics [161], [162], modeling dynamics of online social networks [163], recommendation systems [164] and predicting time-of-travel in navigation tools like Google Maps [165]. The one commonality in all cases is that nodes naturally map onto entities of the modeled data and edges represent relationships and interactions between them, providing a highly versatile framework for inductive biases.

The key idea behind GNNs is to iteratively update each node representation by aggregating information from its neighboring nodes and edges. This allows GNNs to learn node embeddings that infuse node and edge features with the local graph structure directly from raw data.

Mathematically, a GNN layer can be formulated as a two-step process of message passing and node update. Let $\mathbf{h}_i^{(t)}$ denote the hidden state (a.k.a. learned representation or embedding) of node i at layer t , and $\mathbf{e}_{ij}^{(t)}$ denote the state of the edge connecting nodes i and j . In the message passing step t , each node i aggregates the messages $\mathbf{m}_{ij}^{(t)}$ from its neighbors $j \in \mathcal{N}(i)$:

$$\mathbf{m}_i^{(t)} = \bigoplus_{j \in \mathcal{N}(i)} M_t(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}) \quad (16)$$

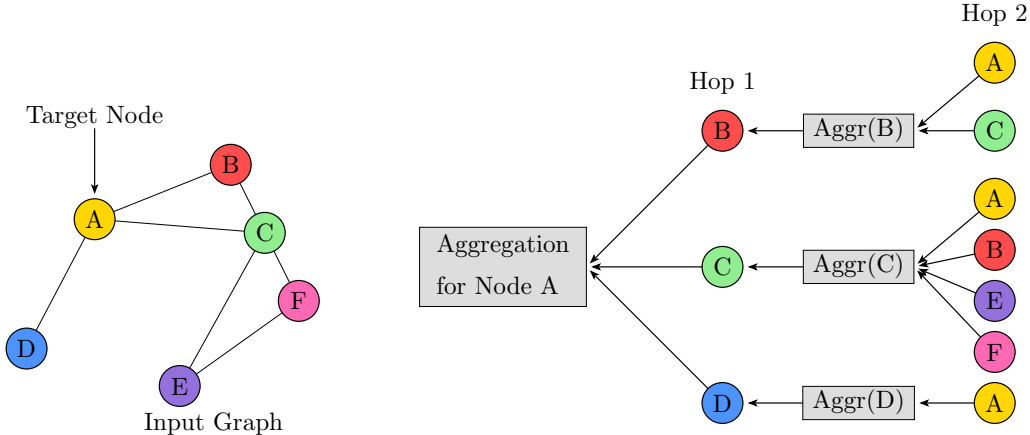


Figure 5: Diagram of the node aggregation process common to all message-passing GNNs. It shows the information flow from other nodes to the target node A from up to two hops away corresponding to a two-layer GNN. The first layer aggregates information from neighboring nodes B , C and D by combining the node states into a message function U whose output is used to update node A 's state before it enters the next layer. The second layer then aggregates (attenuated) information up to two hops away including nodes E and F , illustrating the quickly growing receptive field of each node with layer count, especially in highly connected graphs. [Diagram available on GitHub \[40\]](#), adapted from [166, p. 55].

where M_t is a differentiable and thus trainable message function (usually a small feed-forward neural network) and $\oplus$ is a permutation-invariant aggregation operator (e.g. sum, mean, min, max). The node state $\mathbf{h}_i^t$ is then updated by combining the aggregated message $\mathbf{m}_i^{(t)}$ with the node's previous state:

$$\mathbf{h}_i^{(t+1)} = U_t(\mathbf{h}_i^{(t)}, \mathbf{m}_i^{(t)}) \quad (17)$$

where U_t is again a learnable update function. Likewise, the edge state $\mathbf{e}_{ij}^{(t)}$ is updated by a learnable E_t based on the hidden states of the connected nodes i and j :

$$\mathbf{e}_{ij}^{(t+1)} = E_t(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}) \quad (18)$$

By stacking T message-passing layers (often $2 \leq T \leq 5$), the node representations can (in principle⁶) capture increasingly higher-order interactions and longer-range dependencies in the graph from up to T hops away. In chemistry, this makes a powerful inductive bias as it allows atoms to exchange information about their local bonding environment, thus setting up the model to recognize recurring structural motifs and what changes to properties of interest such as the energy they correlate with. For example, a model might learn that certain pairs of elements, when nearby, enter into tight bonds and thus lower the system's energy more than other less attracted pairs of atoms, thus allowing the model to rank different atomic configurations by stability. The aggregation behavior of GNNs is exactly analogous to the kernel in convolutional

⁶In practice, GNNs often overly attenuate long-range interactions, e.g. capturing only $\sim 1\%$ of expected van der Waals forces in unpublished testing.

neural networks (CNNs), except rather than operating on spatially-defined pixels, GNNs operate on the relational structure of a graph, making CNNs a special case of GNNs. For this reason, GNN layers are also referred to as graph convolutional networks (GCN) [167].

Several key architectural design choices in GNNs determine how exactly information is aggregated and propagated through the graph:

The **initial node and edge features** $\mathbf{h}_i^{(0)}$ and $\mathbf{e}_{ij}^{(0)}$ are the starting point for learning graph-dependent representations. Initial node features can depend on site properties like element type, charge, and coordination number. Edge features usually encode interatomic distances r_{ij} [123]. Later models also include bond angles θ_{ijk} [67], [168]. Such atom and bond attributes are often not directly used as features but are first mapped to continuous vectors via learnable embeddings:

$$\mathbf{h}_i^{(0)} = \text{Embed}(Z_i, q_i, C_i, \dots), \quad \mathbf{e}_{ij}^{(0)} = \text{Expand}(r_{ij}, \theta_{ijk}) \quad (19)$$

where Embed is a lookup table for atomic embeddings that returns initial node features for the first message-passing iteration (with atomic number Z_i , charge q_i , coordination number C_i) and Expand projects the scalar distances r_{ij} and angles θ_{ijk} from node i to its neighbors into a higher-dimensional space using e.g. a radial basis function and spherical harmonics expansion [124], [169].

The **message function** M_t is a crucial component that determines how information is exchanged between nodes. SchNet [124] uses a simple element-wise multiplication (denoted as $\odot$) between the edge embeddings and the neighboring node features:

$$M_t(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}) = \mathbf{h}_j^{(t)} \odot \mathbf{e}_{ij}^{(t)} \quad (20)$$

DimeNet [169] incorporates angular information by using a spherical Bessel function j_n and spherical harmonics Y_ℓ^m expansion of the distances and angles:

$$M_t(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}) = \mathbf{h}_j^{(t)} \odot \left(\sum_{n,\ell} a_{n\ell}^{(t)} \cdot j_n(r_{ij}) \cdot \sum_{m=-\ell}^{\ell} Y_\ell^m(r_{ij}) \cdot Y_\ell^{m*}(r_{ik}) \right) \quad (21)$$

where $a_{n\ell}^{(t)}$ are learnable parameters and k ranges over the neighbors of i .

NequIP [80] uses Clebsch-Gordan tensor products to construct rotationally equivariant message functions:

$$\left[M_t(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}) \right]_{k'q'} = \sum_{k,q,k'',q''} C_{kqk''q''}^{k'q'} \cdot \left[\mathbf{h}_j^{(t)} \right]_{kq} \cdot \left[\mathbf{e}_{ij}^{(t)} \right]_{k''q''} \quad (22)$$

where $[\cdot]_{kq}$ denotes the component of a feature vector transforming as the irreducible representation of $SO(3)$ with angular frequency k and phase q , and $C_{kqk''q''}^{k'q'}$ are the Clebsch-Gordan coefficients.

The **node and edge update functions** U_t and E_t are typically implemented as MLPs, sometimes with additional non-linearities like the Swish activation [170] and layer normalization [171]:

$$\mathbf{h}_i^{(t+1)} = \text{MLP}\left(\mathbf{h}_i^{(t)}, \mathbf{m}_i^{(t)}\right), \quad \mathbf{e}_{ij}^{(t+1)} = \text{MLP}\left(\mathbf{h}_i^{(t)}, \mathbf{h}_j^{(t)}, \mathbf{e}_{ij}^{(t)}\right) \quad (23)$$

To improve the flow of information across layers, models such as Schnet [172], TorchMD-Net [173], CGCNN [123] and CHGNet [67] among others use gated update functions, inspired by the gating mechanisms in LSTMs [156] and gated recurrent units (GRUs) [155]. Residual connections [160] between layers can also be used, especially to facilitate the training of deep GNNs:

$$\mathbf{h}_i^{(t+1)} = \mathbf{h}_i^{(t)} + U_t\left(\mathbf{h}_i^{(t)}, \mathbf{m}_i^{(t)}\right) \quad (24)$$

In chemistry, we are usually interested in making graph-level predictions such as the system’s total energy. Thus, the final readout of convolved node features consists of a **global pooling** operation in which final node embeddings after the last message-passing layer are pooled into a single vector:

$$\mathbf{h}_G = \text{Pool}\left(\left\{\mathbf{h}_i^{(T)} \mid i \in G\right\}\right). \quad (25)$$

Pool is a permutation-invariant pooling function (e.g. sum, mean, min, max) and T is the total number of layers. More sophisticated schemes such as differentiable [174], [175] and attention-based pooling [176] are being explored in pure ML research but have not made their way into GNNs for chemistry yet.

The pooled graph representation $\mathbf{h}_G$ can then be passed through a final MLP to make atomic-environment-aware property predictions for the entire molecule or material:

$$\hat{\mathbf{y}} = \text{MLP}(\mathbf{h}_G), \quad (26)$$

with $\hat{\mathbf{y}}$ commonly being the system’s total energy E such that forces $\mathbf{F}_i$ and stress $\boldsymbol{\sigma}$ follow from it via automatic differentiation w.r.t. the atomic positions $\mathbf{r}_i$ and lattice strain tensor $\boldsymbol{\epsilon}$, respectively:

$$\mathbf{F}_i = -\nabla_{\mathbf{r}_i} E \quad \boldsymbol{\sigma} = \frac{1}{V} \nabla_{\boldsymbol{\epsilon}} E. \quad (27)$$

Symmetry plays another crucial role. Since all physical properties are either *invariant* (scalars like energy) or *equivariant* (vectors and higher-order tensors like forces, magnetic moments, stress) to $E(3)$ symmetry transformations, all GNNs must be designed to preserve these symmetries by using invariant or equivariant message-passing operations. Notably, invariant models can only predict invariant quantities like total energy (from which equivariant properties like forces can nonetheless be derived), while equivariant models can directly predict equivariant quantities such as forces or magnetic moments. Examples of equivariant models include Tensor Field Networks [177], Cormorant [178], PaiNN [172], E3NN [179], NequIP [80], TorchMD-Net [173] and MACE [81]. Architectural variations have proliferated to the point that a compre-

hensive list would be lengthy and quickly outdated. Many other models such as Schnet [124], DimeNet [169], CGCNN [123], MEGNet [125], M3GNet [68] and CHGNet [67] use only invariant messages, or $L = 0$ equivariance, meaning the message transforms like spherical harmonic $Y_{L=0}^M$ under rotation.

Which GNN architecture to use depends on various factors including how much training data is available, the chemical diversity and types of systems being studied, the desired balance between accuracy and efficiency and which properties need to be predicted. As the field evolves, further innovations in GNN architecture often originating in pure ML research will likely make some of the above obsolete. However, given the generality and extensibility of the GNN framework, it is likely to remain a key asset in the future computational chemist toolbox. For further details on the full design space of $E(3)$ equivariant interatomic potentials, we refer to [180].

Early efforts to incorporate additional degrees of freedom beyond ionic positions include the Charge Equilibration Neural Network Technique (CENT) [181], [182] and 4G-BPNN [183] which feature a partial charge equilibration scheme that minimizes electrostatic energy to better handle ionic systems. SpookyNet [184] goes further and directly takes the number of electrons and their spin state as inputs, allowing it to differentiate the PES of systems with different electronic structures. It also integrates a distance-independent interaction term allowing it to learn electron delocalization effects and non-local atomic interactions. However, it still relies on semiempirical corrections for long-ranged electrostatic and dispersion interactions, making it an MLIP-classical force field hybrid. More recently, CHGNet [67] has incorporated magnetic moments as a proxy for oxidation states to more accurately describe heterovalent ions whose interactions can depend strongly on their valence state. Closer adherence to the full physics provides additional latent-space regularization to the model and often allows CHGNet to infer the absolute value of magnetic moments from ionic structure, enabling the study of charge distribution and its effects on ionic dynamics in large-scale simulations. Similarly named but more explicit in its treatment of electronic structure is ChargeE3Net [185], an $E(3)$ -equivariant GNN that directly predicts electron density at arbitrary points in atomic systems. The authors show starting the Kohn-Sham self-consistent field (SCF) cycle from ChargeE3Net-predicted electron densities requires fewer iterations to converge than VASP’s default initialization which uses a superposition of atomic densities (SAD) [186]. Incorporating electronic degrees of freedom into MLIPs at higher fidelity is an active area of research with many innovations likely to come. If done efficiently and with appropriate inductive biases, it has the potential to enable fully data-driven electronic structure simulations.

GNNs have become established tools for driving MD simulations and various property prediction tasks, often outperforming other ML methods by wide margins, especially on data-rich tasks. The ability to (pre-)train GNNs on multiple datasets, possibly in an active learning cycle that generates new training data on the fly as needed based on simulation accuracy

metrics, is pushing the boundaries of the cost-accuracy Pareto front set by previous, fixed approximations. This approach maintains linear scaling of simulation cost with system size while quickly increasing in accuracy, with more advances from better data, better models and better training techniques likely to come.

2.2.4 From Classical to ML Force Fields: Evolving Fixed into Flexible Functional Forms

Force field development aims to expand the capabilities of (semi-)empirical materials modeling which revolves around increasing accuracy and transferability while minimizing cost. This progression started with classical force fields (CFF), also known as empirical potentials, which typically model interactions as the sum of bonded and non-bonded energy terms.

$$E_{\text{total}} = E_{\text{bonded}} + E_{\text{non-bonded}} \quad (28)$$

where E_{bonded} includes terms for bond stretching, angle bending, and bond torsions. In their most simplistic approximation using Hooke’s law, these terms can be written as

$$\begin{aligned} E_{\text{bonded}} &= E_{\text{bonds}} + E_{\text{angles}} + E_{\text{torsions}} \\ &= \sum_{\text{bonds}} k_r (r - r_0)^2 + \sum_{\text{angles}} k_\theta (\theta - \theta_0)^2 + \sum_{\text{torsions}} k_\varphi [1 + \cos(n\varphi - \delta)], \end{aligned} \quad (29)$$

where r , θ , and φ are bond lengths, angles, and dihedral angles with r_0 , θ_0 , and φ_0 their equilibrium values, while k_r , k_θ , k_φ , are force constants and n and δ are phase angles.

$E_{\text{non-bonded}}$ accounts for electrostatics and van der Waals interactions, the most simplistic approximation here being Lennard-Jones potentials.

$$\begin{aligned} E_{\text{non-bonded}} &= E_{\text{electrostatics}} + E_{\text{van der Waals}} \\ &= \sum_{i < j} \left(\frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} + 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] \right), \end{aligned} \quad (30)$$

where q_i and q_j are the partial charges on atoms i and j , r_{ij} is the distance between them, ϵ_0 is the vacuum permittivity, ϵ_{ij} and σ_{ij} are the Lennard-Jones parameters, and k_r , k_θ , k_φ , n , and δ are force constants and phase angles for the bonded terms.

CFFs enabled simulations of specific classes of (bio-)molecules and materials which for which they had to be specifically designed for. As a result, every new material class or even chemical system required fitting a new bespoke force field, potentially requiring architectural changes, and thus placing a high barrier to entry for studying new systems, making classical force fields a suboptimal choice exploring new chemical spaces. While computationally efficient, thus allowing simulation of up to millions of atoms over nanosecond timescales, their simple functional forms place inherent limits on accuracy and transferability, especially for chemical reactions or many-

component chemical environments [187], [188], [189]. Key limitations of classical force fields include:

1. Unable to describe bond breaking/formation
2. Limited accuracy for polarization and many-body interactions
3. Struggle to capture complex electronic effects
4. Challenges in determining when to use more complex functional forms (e.g., polarizable force fields)
5. Reliance on predefined bonding patterns

The progression from classical force fields to modern MLFFs can be broadly characterized by several key steps:

1. Reactive Force Fields (1990s – 2000s): Introduced bond order terms and allowed for dynamic formation and breaking of chemical bonds [190].
2. Ab Initio Molecular Dynamics (1980s – present): Directly use quantum mechanical calculations to determine forces, providing high accuracy but at huge computational cost [191].
3. Neural Network Potentials (2000s – 2010s): Used artificial neural networks to represent potential energy surfaces, trained on quantum mechanical data [120].
4. Deep Learning Potentials (2010s – present): Employ graph neural networks and end-to-end training to learn representations directly from atomic coordinates and charges [192], [193].

MLFFs have emerged as a promising avenue to address these limitations by learning the potential energy surface directly from high-quality quantum mechanical reference data. Human engineering occurs only at the high-level of inductive biases (e.g. designing a suitable graph representation and adherence to $E(3)$ equivariance) but otherwise trained end-to-end from raw data with little or no tailoring of the model to specific systems. They typically employ flexible functional forms such as neural networks or kernel methods. The universal approximation theorem states that feedforward networks with non-polynomial activations are dense in continuous function space between two Euclidean spaces [194]. As a result, MLIPs offer essentially infinite functional flexibility able to model atomic interactions of all kinds. A simple feed-forward neural network potential might take the form:

$$E = \sum_i \text{NN}(G_i) \quad (31)$$

where NN is a neural network and G_i is a descriptor of the local atomic environment of atom i used to predict each atom’s contribution to the system’s total energy E . Knowing E , many other system properties can be derived. More sophisticated graph neural network architectures, as described in the previous section, offer increased expressivity and data efficiency due to equivariance to physical symmetries.

Key advantages of MLIPs include:

1. Automatic capture of complex many-body interactions without predefined functional forms

2. Highly flexible end-to-end learned descriptors for arbitrary numbers of diverse chemical environments and reactions
3. Potential for ab initio accuracy with classical force field efficiency
4. Ability to systematically improve towards ab initio accuracy with more training data while maintaining linear scaling with system size
5. Controllable increase in model inference cost with growing data set size (increased parameter count to keep up with growing datasets often leads to sublinear increase in inference cost due to efficient hardware acceleration)

The flexibility of MLIPs comes at the cost of requiring large, high-quality datasets and careful training procedures. Generating such datasets and training large models on them both require highly specialized skill sets that are in high demand but draw from largely disjoint talent pools. However, once trained, they enable simulations at near-DFT accuracy but orders of magnitude faster [195]. While hard to predict where the rapid pace of current improvements might stall, MLFFs currently appear as the most promising method for bridging the gap between classical force fields and DFT for accurate yet large-scale materials simulations. They approach both the speed and favorable scaling of classical force fields and the accuracy of ab initio quantum simulations, thus pushing the cost-accuracy Pareto boundaries in materials modeling as illustrated in Figure 6.

2.2.5 ML in DFT

While this thesis is exclusively focused on training full-PES ML models from DFT reference data, it is worth noting that ML can also be integrated into the DFT formalism itself by

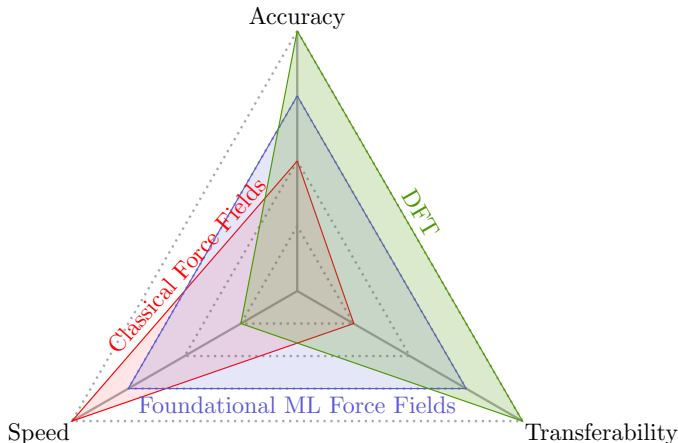


Figure 6: Speed-accuracy-transferability trade-off of DFT, foundational MLFFs, and CFFs. DFT provides the highest accuracy and versatility but is slow and scales poorly with system size. CFFs are very computationally efficient at decent accuracy but are tailored to specific chemical systems and interaction types and so offer poor transferability. MLFFs offer near-DFT accuracy at a fraction of the cost and scale linearly with system size, thus bridging the gap between the speed of classical force fields and the accuracy and versatility of DFT. [Diagram available on GitHub \[40\]](#).

only learning the most challenging contribution to the energy of a quantum system: an ML functional for the Kohn-Sham exchange-correlation $E_{\text{xc}}^{\text{ML}}[n(\mathbf{r})]$ [196], [197], [198], [199], [200]. This approach is particularly compelling as it keeps all the easily calculable physics, such as electronic and ionic kinetic energy, ion-ion and electron-ion interactions while learning only $E_{\text{xc}}^{\text{ML}}[n(\mathbf{r})]$ from high-quality reference data. This reduces the problem space needed to solve with statistical methods and provides new avenues for regularization as exchange-correlation $E_{\text{xc}}[n(\mathbf{r})]$ is well-known to adhere to 17 exact physical constraints [201] which if satisfied greatly enhance the functional’s transferability and accuracy. By using ML only to learn a correction, the combination of DFT and ML acts in a similar fashion to a gradient-boosting ensemble. DFT becomes the first tree that models most of the physical variation while ML as the second tree learns to account for the residuals. Similar to linear-scaling GNNs, this line of research also provides a path to pushing the cost-accuracy Pareto front, albeit at a higher point on the cost spectrum. In the best case, it has the potential to yield an orbital-dependent and non-local functional with the accuracy of hybrid DFT at the cost of semi-local functionals [200].

Chapter 3

Matbench Discovery

This chapter is based on

[11] J. Riebesell *et al.*, “Matbench Discovery – A Framework to Evaluate Machine Learning Crystal Stability Predictions” <http://arxiv.org/abs/2308.14920>. 2023–08–28.

Under review at Nature Machine Intelligence. I performed the benchmark implementation and the test set curation, all of the model testing and ensuing data analysis, and trained some of the models. All the universal force fields were trained in collaborations as their training can easily reach the scope of a PhD project. This project turned out to be a major curation and iterative benchmark design effort that took much more time and care than initially anticipated. It would not have been feasible without significant help of many incredible collaborators. Rhys Goodall was instrumental in initial benchmark design and conceptualization, asking the right questions, and help with editing the manuscript. Anubhav Jain supervised the project by meeting with me every 2 weeks while I was at Berkeley and Kristin Persson enabled me to work on it as a Materials Project staff member. Bowen Deng curated the MPtrj dataset and trained CHGNet. Philipp Benner and Yuan Chiang took on model training of ALIGNN [168] (Philipp) and MACE [81] (Philipp and Yuan) which were both sizable projects in their own right and initially started with the sole intent of submitting these models to Matbench Discovery. MACE-MP later turned out to be much more widely applicable. Philipp also invested months of work into training and testing ALIGNN-FF [202] and Equiformer v2 [203] on MPtrj which both had to be aborted after weeks of trial and error because the models proved unable to meet the task due to lack of robustness and excessive memory requirements. We left a paper trail of training attempts made and issues encountered for ALIGNN-FF and hope to release the same for Equiformer v2 if time permits. We believe this helps the community learn just as much as new SOTA results and avoids giving a false impression of the state of the field from showing only the success stories.

3.1 Introduction

Materials science can be viewed as a combinatorial problem of mixing and arranging different atoms to leverage the complex range of properties that emerge. To date, $\sim 10^5$ combinations have been tested experimentally [29], [204] and $\sim 10^7$ have been simulated [12], [41], [43], [45]. However, an estimated $\sim 10^{10}$ quaternary materials are allowed by electronegativity and charge-balancing rules [32]. The space of quaternaries and higher is even larger but much less explored, leaving vast numbers of potentially useful materials to be discovered. The discovery of new materials is a key driver of technological progress and lies on the path to more efficient solar cells, lighter and longer-lived batteries, and smaller and more efficient transistor gates just to

name a few. In light of our sustainability goals, these advances cannot come fast enough. Any speed-up novel discovery methods might yield should be leveraged to their fullest extent.

Despite significant advances in all areas of materials science, discovering new compounds still requires complex calculations, and labor-intensive trial-and-error experimentation, and often happens fortuitously rather than through rational design. Machine learning (ML) methods efficiently extract and distill trends from huge datasets, and can handle high dimensionality, multiple objectives [13], uncertainty [205], [206], [207], [208], and noisy or sparse data [10], [209], making them powerful additions to the computational materials science tool set.

ML models are less accurate and reliable but orders of magnitude faster than *ab initio* simulation. This makes them most suitable for use in high-throughput (HT) searches to triage more expensive, higher-fidelity simulation methods. The use of neural networks for learning the Kohn-Sham density-functional theory (DFT) potential energy surface (PES) can be traced as far back as 2007, [120]. This work kicked off rapid advances and significant efforts to fit ever more sophisticated ML models to known samples of the PES. Initially, most of these models were trained and deployed as interatomic potentials (also known as force fields) to study known materials of interest, a workflow that requires curating custom training data for each new system of interest [210], [211]. As larger and more diverse datasets have emerged from initiatives like the Materials Project (MP) [3], [12], [44], [45], [46], [62], [82], [83], researchers have begun to train so-called universal models that cover 90 or more of the most-application relevant elements in the periodic table. This enables new ways of ML-guided materials discovery across the periodic table in hopes of increasing the hit rate of stable crystals and speeding up DFT- and expert-driven searches.

Progress in ML for materials is often measured according to performance on standard benchmark data sets. As ML models have grown in complexity and applicability, benchmark datasets need to grow with them to accurately measure their usefulness. However, due to the rapid pace of the field and the variety of possible approaches for framing the discovery problem, no large-scale benchmark yet exists for measuring the ability of ML to accelerate materials discovery. As a result, it is unclear which methodologies or models are best suited for this task. Recent works focusing on prospective computational materials discovery have proposed strategies based on one-shot coordinate free predictors [206], iterative Bayesian optimizers [212], and universal interatomic potentials (UIP) [67], [68], [81]. These papers deploy their respective models on specific systems and custom datasets to highlight certain strengths but have yet to be compared in a systematic standardized manner that allows them to be ranked by their ability to accelerate materials discovery. Our work aims to identify the state-of-the-art (SOTA) model by proposing a novel evaluation framework that closely simulates a real-world discovery campaign guided by ML models.

Specifically, we designed a task that addresses three central challenges. We believe meeting these challenges is a requirement when seeking to justify experimental validation of ML predictions:

1. **Mimic Real-world Deployment:** Idealized and overly simplified benchmarks can fail to reflect the real-world challenges a model faces when used for actual prospective discovery. This can result in a disconnect between benchmark metrics and real-world performance. Possible reasons for this include choosing the wrong target [10] or picking an unrepresentative train/test split [213], [214]. In the case of materials discovery, formation energies - although widely used as regression targets - do not indicate thermodynamic stability. (Free) energy distance to the convex hull spanned by competing phases in the same chemical system is a much more informative indicator. Moreover, ML models relying on relaxed crystal structures as input render any discovery pipeline circular since obtaining relaxed structures requires computationally expensive DFT simulations, thus depending on the very process we intend to accelerate.
2. **Opportunity cost:** Accurate regressors are susceptible to unexpectedly high false-positive rates if those accurate predictions lie close to the decision boundary. Looking purely at global metrics like MAE, RMSE and R^2 can give practitioners a false sense of security about their model’s reliability. Failed experiments incur a high opportunity cost by wasting lab time and resources. Thus false positive rate (FPR) is a pivotal metric.
3. **Scalability:** Future discovery efforts are likely to encounter large data regimes. Small benchmarks can lack chemical diversity, obfuscate poor scaling relations or poor out-of-distribution (OOD) performance (see Figure 49 for a UMAP projection of our train and test sets). For instance, random forests achieve excellent performance on small datasets but are prone to overfit and are typically outperformed by neural networks on datasets exceeding $\mathcal{O}(10^4)$ samples due to the benefits of deep representation learning [139].

Two prior efforts worth highlighting that have partially addressed the above challenges are Matbench [60] and the Open Catalyst Project (OCP) [62].

By providing a standardized collection of 13 datasets ranging in size from ~ 300 to ~ 132 k samples from both DFT and experimental sources, Matbench addresses the scalability challenge, highlighting how model performance changes as a function of data size. It also helped direct the field, increase comparability across papers and provide a quantitative measure of progress over time. However, all tasks were exclusively concerned with the properties of known materials. We believe a task that simulates a materials discovery campaign by requiring materials stability prediction from unrelaxed structures to be a missing piece for ranking ML model utility.

OCP is a large-scale initiative to discover substrate-adsorbate combinations that catalyze key industrial reactions to convert said adsorbates into more useful products. OCP has released two data sets for training and benchmarking ML models thus far, OCP20 [62] and OCP22

[215]. OCP certainly addressed challenge 1 of closely mimicking a real-world problem. However, the metrics are not task-specific, only reflect regression performance and fail to account for opportunity cost. They have recently shown that despite not reaching the target accuracy they deem necessary to entirely replace DFT, using ML in conjunction with confirmatory DFT calculations dramatically speeds up their combinatorial screening workflow [216].

We believe that addressing all three challenges in a single benchmark will enable future ML-guided discovery efforts to expand materials databases to confidently select appropriate models and methodology. Our findings show that UIPs outperform all other methodologies we tested both in accuracy and robustness.

3.2 Evaluation Framework for Materials Discovery

Our proposed evaluation framework places no constraints on the type of data a model trains on as long as it would be available to a practitioner conducting a real materials discovery campaign. We only enforce that at test time, all models must make predictions on the convex hull distance of the relaxed structure with only the unrelaxed structure as input. This setup avoids circularity in the discovery process, as unrelaxed structures can be cheaply enumerated through element substitution methodologies and do not contain information inaccessible in a prospective discovery campaign. Eschewing the frequently used regression target of formation energy, we instead choose to focus on the relaxed structure’s zero Kelvin convex hull distance as the highest fidelity proxy of true crystal stability available in large quantities $>\mathcal{O}(10^5)$. We recognize that our ground truth is ignorant of entropic stabilization and metastable states. While these factors influence experimental synthesizability, both data availability and current model restrictions force us to neglect them. Future benchmarks should aim to use the free energy hull distance instead as an even more realistic stability proxy once a sufficiently comprehensive free energy hull becomes available, preferably at multiple temperatures. The decision to use hull distance over formation energy was also motivated by the analysis in [10] who found even composition-only models capable of predicting DFT formation energies with useful accuracy. However, when tasking those same models with predicting decomposition enthalpy, performance deteriorated sharply due to the smaller value spread of formation energies and less systematic cancellation of errors among similar compounds. This insight meant that ML models are much less useful than DFT for discovering new inorganic solids. Moreover, the qualitative leap in performance from Roost [139], the best compositional model benchmarked, to CGCNN [123], the single structural model they tested, reaffirms the well-known fact that structure plays a crucial role in determining material stability. Hence our decision to restrict test-time model input to unrelaxed structures to avoid the above-mentioned circularity and measure true prospective utility while still allowing for more precise structure-based energy predictions.

Standard practice in ML benchmarks is to hold as many variables as possible fixed – most importantly the training data – and vary only the model architecture to isolate architectural effects on the performance. We deliberately deviate from this practice due to diverging objectives from common ML benchmarks. Our goal is to identify the best methodology for accelerating materials discovery. An ML potential that can learn a higher fidelity model of the physical interactions between ions from the forces and stresses recorded in DFT relaxations has a genuine advantage in training data size over another model that trains on energies only which we want our benchmark to reflect. For that reason, valid model submissions are permitted to train on the entirety of the Materials Project (MP) [12] database release [v2022.10.28](#). Any subset of the energies, forces, stresses, magnetic moments or any other recorded DFT data are valid training targets. All of these would be available to a practitioner performing a real materials discovery campaign and hence are fair game in our benchmark. Models may train on every single relaxation frame, or diversity maximizing subsets thereof such as initial and final relaxed structures only. Our test set consists of the unrelaxed structures generated via element substitution in the WBM dataset [1]. Their target values are the PBE formation energies of the corresponding DFT-relaxed structures.

3.2.1 Materials Project Training Set

The Materials Project is a well-known effort to calculate the properties of all inorganic materials using high-throughput ab initio methods. Seeded from a subset of the Inorganic Crystal Structure Database (ICSD) [217], the initial release of the database consisted of ~ 9 k crystals. At the time of writing, the Materials Project database [12] has grown to ~ 154.4 k crystals, covering diverse chemistries and providing the relaxation trajectory and VASP charge densities among other things for every entry.

Our benchmark defines the training set as all data available from the [v2022.10.28 MP release](#). We recorded a snapshot of energies, forces, stresses and magnetic moments for all MP ionic steps on 2023-03-15 as the canonical training set for Matbench Discovery, and provide convenience functions through our [Python package](#) for easily feeding that data into future model submissions.

The flexibility in specifying the maximum allowed dataset is intended to allow authors to experiment with and exploit the large variety of available data. This choice is motivated by two factors. First, whilst it may appear that models that train on multiple snapshots containing energies, forces and stresses are getting more training data than models that are only trained or able to be trained on the energies of the relaxed structures the critical factor is that all this additional data was generated as a byproduct of the workflow to produce the relaxed structures i.e. all models have been trained from the same total cost of training data acquisition. If some architectures or approaches can leverage more of this byproduct data to make improved predic-

tions this is a fair comparison between the two models. This approach diverges philosophically from other benchmarks such as the OCP and Matbench where it has been more common to subcategorize different models and look at multiple alternative tasks (e.g. composition-only vs structure-available in Matbench or IS2RS, IS2RE, S2EF in OCP) and do not make direct comparisons of this manner. Second, recent work in the space from [218], [219] has claimed that much of the data in large databases like MP are redundant and that models can be trained more effectively by taking a subset of these large data pools. We believe that from a systems-level perspective identification of novel cleaning and or active-learning strategies that do not leak signal from the test set are equally as important as architectural improvements as they can lead to similar improvements in performance, particularly given the prevalence of errors in high-throughput DFT that can disrupt learning of the PES. Consequently, such strategies where they lead to improved performance should be able to be recognized within the benchmark. We encourage submissions to submit ablation studies showing how different system-level choices affect performance.

We highlight for example data sets that are valid within the rules of the benchmark that take advantage of these freedoms. The first is the [MPF.2021.2.8](#) dataset ref. [68] curated to train the M3GNet model which takes a subset of materials from the [v2021.02.08](#) MP release. Rather than taking every ionic step from the relaxation trajectory this dataset opts to select only the initial, final and one intermediate structure for each material to avoid biasing the dataset towards examples where more ionic steps were needed to relax the structure. Additionally, the [MPF.2021.2.8](#) dataset filters the [v2021.02.08](#) release significantly to select a subset of calculations for increased self-consistency. That is, keeping only DFT calculations that ran prior to minor changes in the `MPRelaxSet` VASP settings prior to 2019 to increase dataset homogeneity. [MPF.2021.2.8](#) is a proper subset of the training data as no materials were deprecated between database releases [v2021.02.08](#) and [v2022.10.28](#). The second dataset we highlight is `MPtrj` [67]. It was introduced alongside `CHGNet` [67] but has since been used to train other universal force fields including `MACE-MP` and `SevenNet`. `MPtrj` was curated from the [v2021.11.10](#) MP release and contains 1.6 M frames from 154 k DFT relaxations, i.e. nearly all of MP. It removed only a small number of suspect and potentially confusing energy, force, stress and magnetic moment labels which required designing carefully tuned consistency filters before the frames were subsampled using the `pymatgen` [4] `StructureMatcher` to remove nearly identical structures with no additive training value. It therefore covers nearly all of MP’s structural and chemical variety, thus making it an excellent training set for ML potentials for its time.

Importantly, `MPtrj` is also a proper subset of the canonical Matbench Discovery training set, making any model trained on it a valid submission. The [v2022.10.28](#) release contains a small number of additional Perovskite structures not found in `MPtrj` that could be added to the training set while staying within the scope of the benchmark.

We also note that [v2023.11.1](#) deprecated ~ 5 k calculations so data queried from subsequent database releases is not considered valid for Matbench Discovery.

3.2.2 WBM Test Set

The WBM dataset [\[1\]](#) consists of ~ 257 k structures generated via chemical similarity-based element substitution of MP source structures followed by DFT relaxation and calculating each crystal’s convex hull distance. Which element replaces an existing one in a given source structure was determined by random sampling according to the weights in a chemical similarity matrix data-mined from the ICSD [\[220\]](#).

The WBM authors performed five iterations of this substitution process (we refer to these iterations as batches). After each step, the newly generated structures found to be thermodynamically stable via PBE relaxation flow back into the source pool to partake in the next round of substitution. This split of the data into batches of increasing substitution count is a unique and compelling feature of the test set as it allows out-of-distribution (OOD) testing by seeing if model performance degrades with substitution count. A higher number of element substitutions on average carries the structure further away from the region of material space covered by the MP training set (see [Figs. 14](#) and [49](#) for details). While this batch information makes the WBM dataset a very useful data source for examining the extrapolation performance of ML models, except for [Figure 14](#) our evaluation primarily looks at metrics that consider all batches as a single test set.

Throughout this work, we define stability as being on or below the convex hull of the MP training set ($E_{\text{MP hull dist}} \leq 0$). ~ 42.8 k out of ~ 257 k materials (16.7%) in WBM satisfy this criterion. Of these, ~ 33 k are unique prototypes, meaning they have no matching structure prototype in MP nor another higher-energy duplicate prototype in WBM. Our code treats the stability threshold as a dynamic parameter, allowing for future model comparisons at different thresholds. This may reveal systematic differences in stability calibration, as hinted by the different TPR/TNR tradeoffs observed for CHGNet and M3GNet in [Table 4](#). A looser stability threshold (e.g. $E_{\text{MP hull dist}} \leq 0.05$ eV/atom) may benefit a model like M3GNet which is more conservative in its stability predictions, whereas a stricter threshold should favor a model like CHGNet that is more optimistic in its stability predictions. In [Figure 9](#) (or the [interactive web version](#) which shows the stability threshold corresponding to each point on hover), the ROC curves of CHGNet and M3GNet indeed exhibit the largest gap for a stricter stability threshold of $E_{\text{MP hull dist}} \leq -10$ meV/atom.

As WBM explores regions of materials space not well sampled by MP, many of the discovered materials that lie below MP’s convex hull are not stable relative to each other. Of the ~ 42.8 k that lie below the MP convex hull less than half, or around ~ 20 k, remain on the joint MP+WBM convex hull. This observation suggests that many WBM structures are repeated

samples in the same new chemical spaces. It also highlights a critical aspect of this benchmark in that we purposely operate with an incomplete convex hull. Only current knowledge of competing points on the PES is accessible when doing real discovery and our metrics are designed to reflect this.

3.3 Models

To test a wide variety of methodologies proposed for learning the potential energy surface, our initial benchmark release includes 10 models.

1. **MACE** [81] (UIP-GNN) - MACE builds upon recent advances [80], [177] in equivariant neural network architectures by proposing an approach to computing high-body-order features efficiently via Atomic Cluster Expansion [221]. Unlike the other UIP models considered, MACE was primarily developed for molecular dynamics of single-material systems, not the universal use case studied here. Submitting it to this benchmark required significant upfront engineering effort including migrating the edge embeddings of interatomic distances to Gaussian radial basis functions, tuning the loss function by moving to Huber loss which deweights outliers and stabilizes the training process, and implementing multi-GPU training required to scale to the this larger training set than what MACE has traditionally been trained on. MACE is the only open $E(3)$ -equivariant model we tested.⁷
2. **CHGNet** [67] (UIP-GNN) - CHGNet is an invariant ML potential for charge-informed atomistic modeling. Its distinguishing feature is that it was trained to predict magnetic moments on top of energies, forces and stresses in the MPtrj dataset consisting of relaxation trajectories for ~ 1.5 M MP structures (see Section A.2 for a detailed analysis of this dataset). By modeling magnetic moments, CHGNet learns to accurately represent the orbital occupancy of electrons which allows it to predict both atomic and electronic degrees of freedom.
3. **M3GNet** [68] (UIP-GNN) - M3GNet is another $E(3)$ -invariant GNN-based universal interatomic potential for materials trained on up to 3-body interactions that significantly inspired CHGNet’s development. The version submitted to this benchmark was trained on the initial, middle and final frame of pre-2019 MP DFT relaxations. Like CHGNet and MACE-MP, model was tested in relaxation mode, meaning force predictions were used to move atoms to the input structures ML-PES ground state before predicting its energy.
4. **ALIGNN** [168] (GNN) - The Atomistic Line Graph Neural Network (ALIGNN) is a message-passing GNN architecture that takes as input both the interatomic bond graph and a line graph corresponding to 3-body bond angles. The ALIGNN architecture involves

⁷GNoME shown in Table 11 using a NequIP architecture is also $E(3)$ -equivariant but trained on a much larger incompatible dataset, allowing no direct architectural comparisons to the other invariant UIPs CHGNet and M3GNet.

a global pooling operation which means that it is ill-suited to force-field applications. To address this the ALIGNN-FF model was later introduced without global pooling [202]. We also attempted to test that model variant in relaxation mode but encountered severe [technical hurdles](#) which forced us to abort.

5. **MEGNet** [125] (GNN) - MatErials Graph Network is a GNN inspired by and similar to CGCNN originally intended mainly for property prediction of relaxed structures.

It can incorporate global features (like pressure, temperature and entropy) in its message-passing operation. This work showed that learned element embeddings encode periodic chemical trends and can be transfer-learned from large data sets (formation energies) to predictions on small data properties (band gaps, elastic moduli).

1. **CGCNN** [123] (GNN) - The Crystal Graph Convolutional Neural Network (CGCNN) was the first neural network model to directly learn eight different DFT-computed material properties from a graph representing the atoms and bonds in a periodic crystal. CGCNN was among the first to show that just like in other areas of ML, given large enough training sets, neural networks can learn embeddings that outperform human-engineered structure features directly from the data.
2. **CGCNN+P** [222] (GNN) - This work proposes simple structure perturbations of relaxed structures as a means of training data augmentation to supplement the data set with further from equilibrium configurations but mapped to the same DFT final energy. The authors introduced this data augmentation method applied to CGCNN but in principle, it could be used with any structure-based GNN. We chose $P = 5$ for our CGCNN-P submission, meaning the training set is augmented with 5 random perturbations of each relaxed MP structure mapped to the same target energy. In contrast to all other structure-based GNNs considered in this benchmark, CGCNN+P is not attempting to learn the Born-Oppenheimer potential energy surface. The model is instead taught the PES as a step-function that maps each valley to its local minimum. The idea is that during testing on unrelaxed structures, the model will predict the energy of the nearest basin in the PES. The authors confirm this by demonstrating a lowering of the relaxed energy error when given unrelaxed structures as input.
3. **Wrenformer** (Transformer) - For this benchmark, we introduce Wrenformer which is a variation on the coordinate-free Wren model [206] constructed using standard QKV-self-attention blocks [87], [157] in place of message-passing layers. This architectural adaptation reduces the memory usage allowing the architecture to scale to structures with greater than 16 Wyckoff positions on common hardware. Like its predecessor, Wrenformer is a fast coordinate-free model aimed at accelerating screening campaigns where even the unrelaxed structure is a priori unknown. The key idea is that by training on the coordinate anonymized Wyckoff positions (symmetry-related positions in the crystal structure), the model learns to

distinguish polymorphs while maintaining discrete and computationally enumerable inputs. The benefit of an enumerable input is that it allows users to predict the energy of all possible combinations of spacegroup and Wyckoff positions for a given composition and maximum unit cell size. The lowest-ranked prototypes can then be fed into downstream analysis for further validation, thus potentially making the task of materials discovery a countable exercise.

4. **BOWSR** [212] (BO-GNN) - BOWSR combines a symmetry-constrained Bayesian optimizer (BO) with a surrogate energy model to perform an iterative exploration-exploitation-guided search of the potential energy surface. Here we use MEGNet [125] for the energy model as proposed in the original work. The high sample count needed to explore the PES with BO makes this by far the most expensive model tested.
5. **Voronoi RF** [136] (Structure Fingerprint) - A random forest trained to map a combination of composition-based Magpie features [135] and structure-based relaxation-robust Voronoi tessellation features [136] (effective coordination numbers, structural heterogeneity, local environment properties, etc.) to DFT formation energies. This fingerprint-based model predates most deep learning for materials but significantly improved over earlier fingerprint-based methods such as the Coulomb matrix [129] and partial radial distribution function features [223]. It serves as a baseline model to see how much value the learned featurization of deep learning models can extract from the increasingly large corpus of available training data.

Except for M3GNet and MACE-MP which are trained to predict the total PBE electronic free energy, all models predict the formation energy E_{form} including MP2020 GGA/GGA+U and anion energy corrections [116].

3.4 Results

Table 2 shows performance metrics for all models included in the initial release of Matbench Discovery. Regression metrics for ML vs PBE convex hull distance have units of eV/atom. MACE takes the top spot and emerges as the current SOTA for ML-guided materials discovery. It outperforms the nine other models on all six classification metrics but is slightly overtaken by CHGNet in 2 of 3 regression metrics, R^2 and RMSE, which are sensitive to outliers. The discovery acceleration factor (DAF) is the ratio of the precision to the prevalence. It measures how many more stable structures a model found compared to the dummy discovery rate achieved by randomly selecting test set crystals. The maximum possible DAF is the inverse of the prevalence or dummy discovery rate which on our dataset is $(32.9 \text{ k} / 215.5 \text{ k})^{-1} \approx 6.5$. Thus, the current SOTA of 3.85 achieved by MACE-MP leaves room for improvement. However, evaluating each model on the much smaller subset of its 10k most stable predictions as shown in Table 4, CHGNet takes the lead in all nine metrics, in particular achieving an impressive

	F1 $\uparrow$	DAF $\uparrow$	Prec $\uparrow$	Acc $\uparrow$	TPR $\uparrow$	TNR $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	R^2 $\uparrow$	Training Size	Model Params	Model Type	Targets
MACE	0.67	3.78	0.58	0.88	0.80	0.89	0.06	0.10	0.70	1.6M (145.9K)	4.7M	UIP	EFS
CHGNet	0.61	3.36	0.51	0.85	0.76	0.87	0.06	0.10	0.69	1.6M (145.9K)	412.5K	UIP	EFSM
M3GNet	0.57	2.88	0.44	0.81	0.80	0.81	0.07	0.12	0.58	188.3K (62.8K)	227.5K	UIP	EFS
ALIGNN	0.57	3.21	0.49	0.84	0.67	0.87	0.09	0.15	0.30	154.7K	4.0M	GNN	E
MEGNet	0.51	2.96	0.45	0.83	0.58	0.87	0.13	0.21	-0.25	133.4K	167.8K	GNN	E
CGCNN	0.51	2.85	0.44	0.82	0.60	0.86	0.14	0.23	-0.60	154.7K	128.4K	GNN	E
CGCNN+P	0.50	2.56	0.39	0.79	0.69	0.80	0.11	0.18	0.02	154.7K	128.4K	GNN	E
Wrenformer	0.47	2.26	0.34	0.74	0.72	0.75	0.11	0.19	-0.04	154.7K	5.2M	Transformer	E
BOWSR	0.42	1.96	0.30	0.71	0.72	0.69	0.12	0.17	0.15	133.4K	167.8K	BO-GNN	E
Voronoi RF	0.33	1.58	0.24	0.67	0.54	0.69	0.15	0.21	-0.33	154.7K	0.0	Fingerprint	E
Dummy	0.18	1.00	0.15	0.69	0.23	0.77	0.12	0.18	0.00				

Table 2: Classification and regression metrics for all models tested on our benchmark ranked by F1 score. All metrics are calculated from ML-predicted $E_{\text{above MP hull}}$ in units of eV/atom. For classification metrics, only $\text{sign}(E_{\text{above MP hull}} - t_{\text{stab}})$ matters where the stability threshold $t_{\text{stab}} = 0$ is a benchmark parameter we set to zero. Thus, classification metrics measure whether a model’s predictions for compounds to be above or below the convex hull agree with PBE DFT. The heat map ranges from yellow (best) to blue (worst) performance. DAF = discovery acceleration factor is the ratio of model precision to percentage of stable structures in the test set, TPR = true positive rate = $\frac{\text{TP}}{\text{TP} + \text{FN}}$, TNR = true negative rate = $\frac{\text{TN}}{\text{TN} + \text{FP}}$, MAE = mean absolute error, RMSE = root mean squared error. The dummy classifier uses the `scikit-learn stratified` strategy of randomly assigning stable/unstable labels according to the training set prevalence. The dummy regression metrics MAE, RMSE and R^2 are attained by always predicting the test set mean. The three universal interatomic potentials MACE, CHGNet and M3GNet take the top 3 spots by F1 score and achieve an impressive performance gap especially in regression metrics to the other seven force-free models, highlighting the high information content of forces and stresses for learning a more accurate PES. Voronoi RF, CGCNN and MEGNet perform worse than dummy in regression metrics but better than dummy on some classification metrics which are more indicative of model utility for guiding discovery, demonstrating that regression metrics alone can be misleading. To increase ‘OOD-ness’ of WBM, this table is restricted to the subset of materials with no matching structure prototype in MP nor another higher-energy duplicate prototype in WBM, reducing the test set size to 257k $\rightarrow$ 215.5k. See Table 3 for metrics on the whole WBM test set and details on prototype matching.

DAF of 5.29 which approaches optimal performance, indicating that the most confident stable crystal predictions from SOTA models are becoming reliable.

We find a large performance gap between models that make one-shot predictions directly from unrelaxed inputs (MEGNet, Wrenformer, CGCNN, CGCNN+P, ALIGNN, Voronoi RF) compared to UIPs that learn from force and stress labels to emulate DFT relaxation (CHGNet, M3GNet, MACE). While the F1 scores and DAFs of force-free models are surprisingly less affected, their regression metrics such as R^2 and RMSE are significantly worse. Of the force-free models, only ALIGNN, BOWSR and CGCNN+P achieve positive predictive power R^2 . Negative R^2 means model predictions explain the observed variation in the data less than simply predicting the test set mean. In other words, these models are not predictive in a global

sense (across the full dataset range). However, even models with negative R^2 can be locally good in the positive and negative tails of the test set distribution. They suffer most in the mode of the distribution near the stability threshold of 0 eV/atom above the hull. This reveals an important shortcoming of R^2 as a metric for stability prediction which ultimately comes down to a yes or no decision on whether to validate a prediction and hence reduces to a binary classification task.

The results for M3GNet depart from the trend that F1 is broadly rank-correlated with the other classification metrics. M3GNet achieves a high true positive rate (TPR = 0.8) but a notably lower true negative rate (TNR = 0.81) than CHGNet and ALIGNN which have similar F1 scores. Figure 14 provides a visual explanation for this observation. M3GNet has a lower rolling mean of the absolute errors (rolling MAE) as a function of hull distance for materials above the convex hull (right half of plot) but incurs comparably large errors for materials below the hull (left half of plot). Since $\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}$, lower error for energies above the hull increases TN and decreases FP, resulting in M3GNet’s high TPR.

The reason CGCNN+P achieves better regression metrics than CGCNN but performs worse as a classifier is apparent from Figure 13. The CGCNN+P histogram is more sharply peaked at the 0 hull distance decision boundary, causing even small errors in the predicted convex hull distance to be large enough to invert a stability classification. Again, this is evidence to choose carefully which metrics to optimize for materials discovery. Regression metrics are far more prevalent when evaluating energy predictions. However, our benchmark treats energy predictions as merely a means to the end of stability classification. Improvements in regression accuracy are of limited use to materials discovery in their own right unless they also improve classification accuracy. Our results demonstrate that this is not a given.

Figure 7 has models rank materials by model-predicted hull distance from most to least stable; materials furthest below the known hull first, materials on the hull last. For each model, we iterate through that list and calculate at each step the cumulative precision and recall of correctly identified stable materials. This simulates exactly how these models would be used in a prospective materials discovery campaign and reveals how a model’s performance changes as a function of the number of materials screened for stability. Practitioners have a certain amount of resources available to validate model predictions. These curves allow them to identify the best model given their search pool and validation budget as well as the optimal trade-off between fewer false positives (precision) or fewer negatives (recall) for the discovery task at hand. In this case, it so happens that CHGNet achieves the highest precision *and* recall initially, but is overtaken by MACE-MP after ~ 25 k+ materials are screened.

A line terminates when a model believes there are no more materials in the WBM test set below the MP convex hull. The dashed vertical line shows the actual number of stable structures in our test set. All models are biased towards stability to some degree as they all overestimate

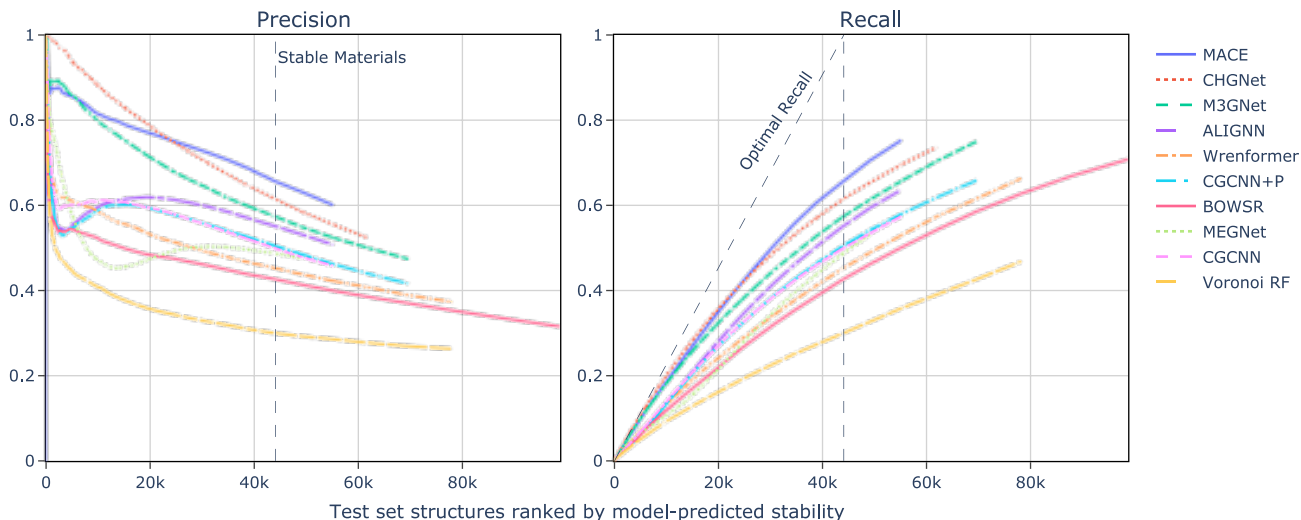


Figure 7: This figure measures model utility for discovery efforts of varying sizes by plotting the precision and recall as a function of the number of model predictions validated. CHGNet initially achieves the highest cumulative precision and recall. However, for a campaign that validates $\sim 25\text{k}+$ model predictions, MACE-MP overtakes CHGNet in both precision and recall and maintains that lead till the end. A typical discovery campaign will rank hypothetical materials by model-predicted hull distance from most to least stable and validate the most stable predictions first. A higher fraction of correct stable predictions corresponds to higher precision. The reverse, fewer stable materials overlooked corresponds to higher recall. This figure highlights how different models perform better or worse depending on the length of the discovery campaign. The UIPs (CHGNet, M3GNet, MACE) are seen to offer significantly improved precision on shorter campaigns of $\sim 20\text{k}$ or less materials validated as they are much less prone to false positive predictions among highly stable materials.

this number, most of all BOWSR by 133%, least of all MEGNet by 30%. This is only a problem in practice for exhaustive discovery campaigns that validate *all* stable predictions from a model. More frequently, model predictions will be ranked most-to-least stable and validation stops after some pre-determined compute budget is spent, say, 10 k DFT relaxations. In that case, the concentration of false positive predictions that naturally accumulates near the less stable end of the candidate list can be avoided with no harm to the campaign’s overall discovery rate (see Table 4 for a similar metrics table considering only the 10 k materials predicted by each model to be furthest below the known convex hull).

The diagonal Optimal Recall line would be achieved if a model never made a false negative prediction (classifying a stable structure as unstable) and stopped predicting crystals as stable exactly when the true number of stable materials is reached. Zooming in on the top-left corner of the precision plot, we observe that CHGNet is the only model without a sudden drop in precision right at the start of the discovery campaign, meaning its first few hundred most stable materials are largely actually stable. MACE in particular suffers from several initial failure cases. MACE underpredicts the energy of these unstable materials by several eV/atom (see Figure 10), resulting in MACE’s initial precision starting at 0 before abruptly climbing to $\sim$

0.8. We experienced this behavior with multiple independently trained MACE variants; even more so with a checkpoint we received directly from the MACE authors which was trained on the M3GNet training set. The results shown here are from a superior MACE trained on the $\sim 10\times$ larger (though much more redundant) MPtrj training set which we trained specifically for Matbench Discovery. M3GNet likewise exhibits the early-on precision drop. As a result, CHGNet has a strong lead over M3GNet until reaching ~ 3 k screened materials. From there, CHGNet and M3GNet slowly converge until they almost tie at a precision of 0.52 after ~ 56 k screened materials. At that point, CHGNet’s list of stable predictions is exhausted while M3GNet continues, dropping in precision to 0.45 at 76 k, attributable to many false positives near the convex hull.

All force-free models exhibit much stronger early-on precision drop, falling to 0.6 or less in the first 5 k screened materials. Many of these models (all except BOWSR, Wrenformer and Voronoi RF) display an interesting hook shape in their cumulative precision, recovering again slightly between ~ 5 k and up to ~ 30 k screened materials before dropping again towards the end.

Figure 8 provides a visual representation of the reliability of different models as a function of a material’s DFT distance to the MP convex hull. The lines show the rolling MAE of model-predicted hull distances vs DFT. The red-shaded area, which we coin the ‘triangle of peril’,

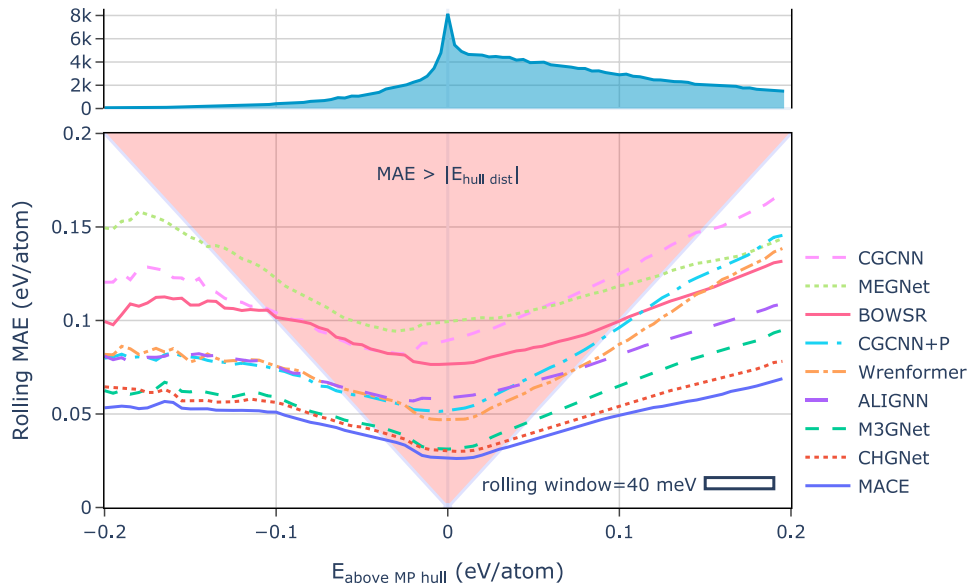


Figure 8: Universal potentials are more reliable classifiers because they exit the red triangle earliest. These lines show the rolling MAE on the WBM test set as the energy to the convex hull of the MP training set is varied. Lower is better. Inside the large red ‘triangle of peril’, models are most likely to misclassify structures. As long as a model’s rolling MAE remains inside the triangle, its mean error is larger than the distance to the convex hull. If the model’s error for a given prediction happens to point towards the stability threshold at $E_{\text{above MP hull}} = 0$, its average error will change the stability classification from true positive/negative to false negative/positive. The width of the ‘rolling window’ box indicates the width over which prediction errors were averaged.

emphasizes the zone where the average model error surpasses the distance to the stability threshold at 0 eV. As long as the rolling MAE remains within this triangle, the model is most susceptible to misclassifying structures. Because the average error is larger than the distance to the classification threshold at 0, it is large enough on average to flip a correct classification into an incorrect one (if the error happens to point toward the decision boundary).

The sooner a model exits the triangle on the left side (negative DFT hull distance), the less likely it is to incorrectly predict stable structures as unstable, thereby reducing false negatives. Exiting the triangle early on the right side (positive DFT hull distance) results in a lower likelihood of predicting unstable structures as stable, leading to fewer false positives.

On both sides, MACE exits the triangle first at a remarkably low ± 30 meV/atom hull distance followed closely by CHGNet, consistent with the high accuracy of these models in Table 2. M3GNet performs similarly to CHGNet for stable structures but has a widening performance gap for increasingly unstable structures on the right side of the plot. This leads to M3GNet’s relatively low TNR of 0.81 compared to CHGNet’s 0.87 in Table 2.

All models tend to have lower rolling error towards the plot’s left edge compared to the right edge. This imbalance shows that models are more prone to false positive than false negative predictions. In other words, all models are less likely to predict a material at -0.2 eV/atom DFT hull distance as unstable than a material at 0.2 eV/atom DFT hull distance as stable. From a practitioner’s standpoint, this is undesirable as attempting to validate an unstable material usually has a much higher opportunity cost than missing a stable one. We hypothesize this error imbalance is due to the MP training set having an uncharacteristically high share of stable materials, causing models trained on it to be biased towards low-energy predictions even for high-energy structures.

As the convex hull becomes more thoroughly sampled by future discovery, the fraction of unknown stable structures decreases, naturally leading to less enriched future test sets. This has several implications, including allowing for higher maximum DAFs but also skewing the hull distance distribution towards positive values (i.e. the right half of Figure 8) as there are fewer undersampled chemical spaces. Consequently, just to retain equally accurate stability classifications, model accuracy will need to improve concomitantly. This is particularly the case for the left half of Figure 8, where models will most need to become more accurate to still be able to exit the shaded triangle.

3.4.1 Metrics on full test set and for 10k materials predicted most stable

While Table 2 evaluates model performance on the subset of 215.5k WBM structures with no matching structure prototype in MP and no higher-energy duplicate prototype in WBM, Table 3 includes the full WBM test set of 257k unrelaxed structures. The 44k removed materials with

non-unique prototypes comprise 11,175 discarded due to having a matching structure prototype in MP and 32,784 materials which are prototype duplicates of another WBM material with lower energy. The motivation to remove identical structure prototypes is that when correctly relaxed, both will result in the same ground state structure. This has two downsides: it increases the chance of leakage between train and test set and screening materials already in the database does not constitute discovery. For the most part, however, we find metrics to be insensitive to testing on the full WBM test set or the subset of unique prototypes. The most noteworthy difference between the two tables is a drop in DAF for all models in Table 3 compared to Table 2. MACE for example achieves a DAF of 3.5 on the full test set (Table 3) compared to 3.85 on the subset of 215.5 k materials with unique prototypes (Table 2). This 10 % increase is largely due to an 8.3 % decrease in the fraction of materials below the MP convex hull: 15.3 % (32,942 out of 215,488) in Table 2 vs 16.7 % (42,825 out of 256,963) in the full dataset (Table 3). Since DAF is the ratio of the model’s precision for stability prediction to the prevalence of stable structures, a lower prevalence results in a higher DAF if model performance remains constant.

Although matching prototypes is non-trivial and could potentially introduce bias into the test set during deduplication, we still opted to feature Table 2 as the main metrics table. Even if it made little difference here, we expect the removal of prototypes overlapping with MP to better reflect a model’s true ability to extrapolate to out-of-domain materials. Prototypes were matched based on an Aflow-style [224] Wyckoff representation as implemented in `aviary.wren.utils.get_protostructure_label_from_spglib` [206]. This string encodes the crystal’s prototype and is invariant to relaxation (unless the system changes spacegroup during relaxation, usually by gaining symmetry). It will therefore count structures with different lattice parameters as duplicates if both are expected to relax to the same ground state.

	F1 ↑	DAF ↑	Prec ↑	Acc ↑	TPR ↑	TNR ↑	MAE ↓	RMSE ↓	R² ↑
MACE	0.67	3.40	0.58	0.87	0.78	0.89	0.06	0.10	0.70
CHGNet	0.61	3.04	0.52	0.84	0.74	0.86	0.06	0.10	0.69
M3GNet	0.58	2.65	0.45	0.80	0.79	0.80	0.07	0.12	0.58
ALIGNN	0.56	2.92	0.50	0.83	0.65	0.87	0.09	0.15	0.27
MEGNet	0.51	2.70	0.46	0.81	0.57	0.86	0.13	0.20	-0.28
CGCNN	0.51	2.63	0.45	0.81	0.59	0.85	0.14	0.23	-0.62
CGCNN+P	0.51	2.40	0.41	0.78	0.67	0.80	0.11	0.18	0.03
Wrenformer	0.48	2.13	0.36	0.74	0.69	0.75	0.10	0.18	-0.04
BOWSR	0.44	1.84	0.32	0.70	0.71	0.68	0.12	0.16	0.14
Voronoi RF	0.34	1.51	0.26	0.67	0.51	0.70	0.14	0.21	-0.31
Dummy	0.19	1.00	0.17	0.68	0.23	0.77	0.12	0.18	0.00

Table 3: Same as Table 2 but computed for all 257k structures in the WBM test set, no duplicate prototypes excluded. The prevalence of stable structures is 16.7 % (42,825 out of 256,963), placing a ceiling of 6 on the maximum possible DAF.

A real-world discovery campaign is unlikely to validate all stable predictions from a given model as we did in Table 2. Presumably, it will rank model predictions from most to least stable and screen that list as far as time and compute budget permits. Assuming that increases in compute resources will allow average future discovery campaigns to grow in scope, we believe 10k model validations to be a reasonable scope for a representative discovery effort. Table 4 measures performance for this scenario by calculating classification and regression metrics for the 10k test set materials predicted to be most stable by each model. We again show dummy performance in the bottom row. Note that each model is now evaluated on a different slice of the data. However, the bottom row still shows dummy performance across the whole dataset. CHGNet and M3GNet achieve an impressive precision of 0.83 and 0.8, respectively. In concrete terms, a discovery campaign that validates 10k model predictions from a search pool of 257k crystals that are chemically dissimilar from the training set and of which 16.7% are stable, would get four stable structures for every five predictions from CHGNet and M3GNet. We emphasize that in light of the significant resulting increase in stability hit rate, these models are well worth integrating into future materials searches.

	F1 $\uparrow$	DAF $\uparrow$	Prec $\uparrow$	Acc $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	R² $\uparrow$
CHGNet	0.92	5.57	0.85	0.85	0.06	0.10	0.82
MACE	0.89	5.22	0.80	0.80	0.09	0.17	0.51
M3GNet	0.87	5.02	0.77	0.77	0.10	0.16	0.55
ALIGNN	0.75	3.90	0.60	0.60	0.18	0.25	0.08
CGCNN	0.74	3.88	0.59	0.59	0.17	0.23	0.18
CGCNN+P	0.74	3.81	0.58	0.58	0.20	0.28	-0.08
Wrenformer	0.72	3.69	0.56	0.56	0.18	0.24	0.14
BOWSR	0.66	3.25	0.50	0.50	0.26	0.32	-1.17
MEGNet	0.63	3.02	0.46	0.46	0.30	0.34	-0.91
Voronoi RF	0.55	2.49	0.38	0.38	0.35	0.42	-1.01
Dummy	0.19	1.00	0.17	0.68	0.12	0.18	0.00

Table 4: Stability prediction metrics for the 10k materials predicted to be most stable by each model. Every model picks out a different slice of the test set as the 10k materials it believes to be furthest below the known convex hull. CHGNet achieves an impressive F1 score of 0.91 and a discovery acceleration factor (DAF) of 5.29, both approaching the optimal values of 1 and 6, respectively. CHGNet notably outperforms MACE on its first 10k predictions sorted by stability which are the most important ones when guiding a discovery effort as these are usually the first materials to be validated by higher fidelity methods. This is reflected in the left panel of Figure 7 and the top row of Figure 10 where MACE, unlike CHGNet, suffers a sudden drop in precision right at the start of the discovery campaign from severe underpredictions of a small number of compounds. These can be seen as points along the negative y axis in the MACE parity plot in Figure 10 which are not present for CHGNet.

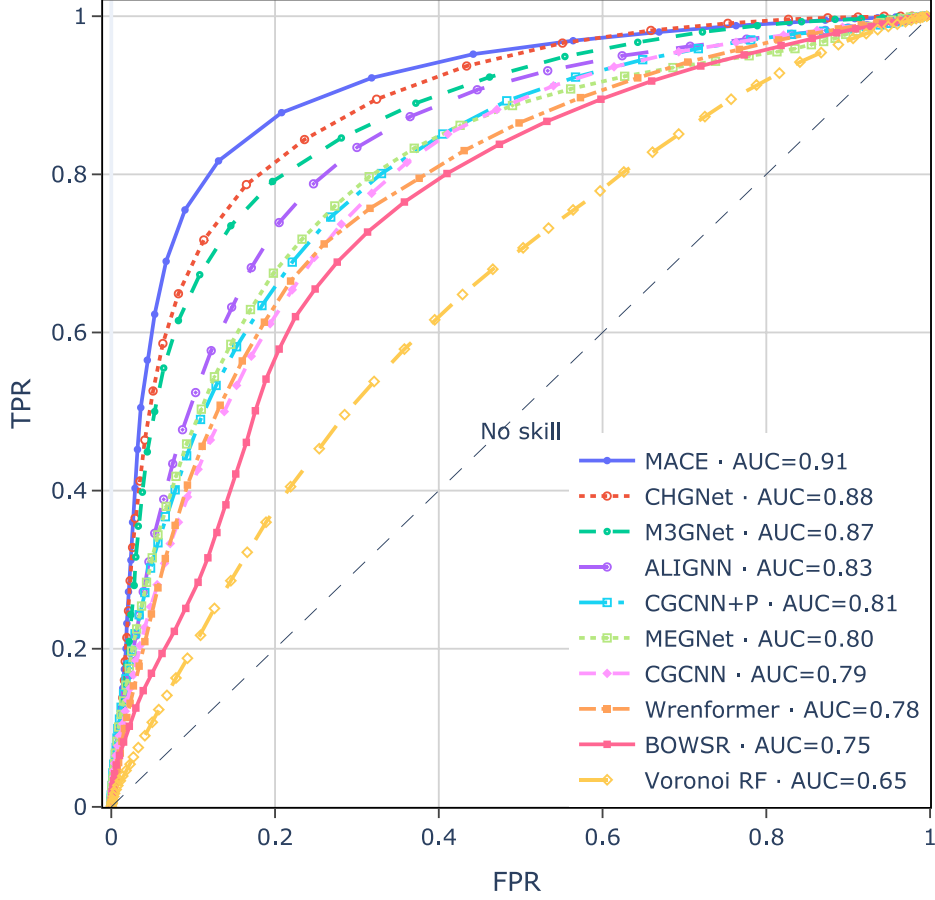


Figure 9: **Receiver operating characteristic (ROC) curve for each model as a function of varying stability threshold $-1 \text{ eV/atom} < E_{\text{MP hull dist}} < 1 \text{ eV/atom}$.** TPR/FPR = true/false positive rate are the fraction of unstable structures classified as stable and the fraction of stable structures classified as stable, respectively. The diagonal **No skill** line shows the performance of a dummy model that randomly ranks material stability. We refer to the [interactive web version](#) for a hover tooltip showing each point’s stability threshold.

3.4.2 ROC Curves

A material is classified as stable if the predicted $E_{\text{above hull}}$ is below the stability threshold which we set to $t_{\text{stab}} = 0 \text{ eV/atom}$ hull distance. To test the sensitivity of different models to this threshold, we plot the receiver operating characteristic (ROC) curve for each model in [Figure 9](#). MACE-MP achieves the highest area under curve (AUC) of 0.91, coming in 1.4x higher than the worst model, the Voronoi Random Forest.

3.4.3 Parity Plots

[Figure 10](#) shows that all models do well for materials far below the convex hull (left side of the plot). Performance for materials far above the convex hull is more varied with occasional

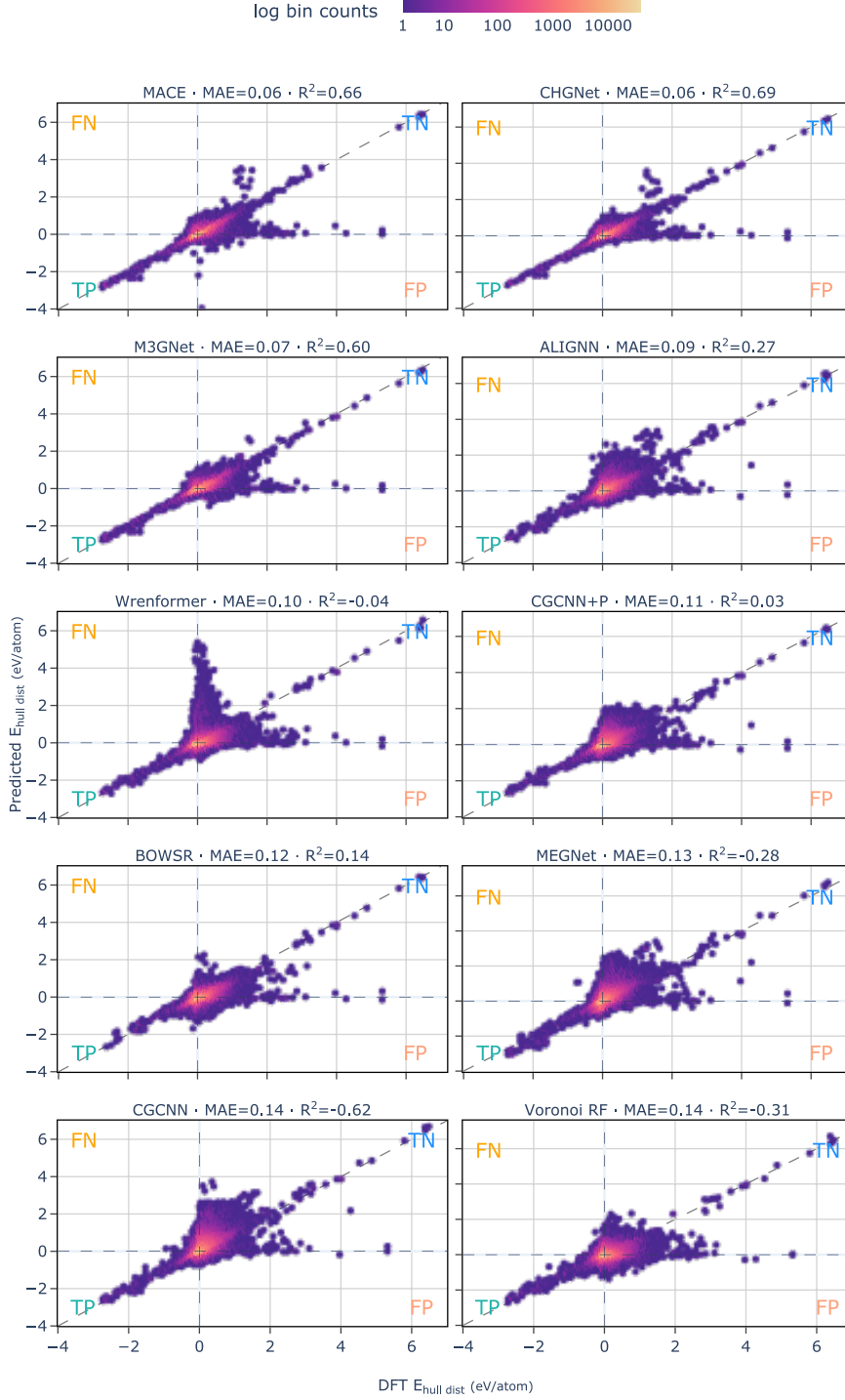


Figure 10: Parity plots of model-predicted energy distance to the convex hull (based on their formation energy predictions) vs DFT ground truth, color-coded by log density of points. Models are sorted left to right and top to bottom by MAE. For parity plots of formation energy predictions, see [Figure 16](#).

underpredictions of the energy of materials far above the convex hull (right side). All models suffer most in the mode of the distribution at $x = 0$.

Two models stand out as anomalous to the general trends.

Wrenformer is the only model with a large number of severe energy overpredictions at 0 DFT hull distance scattered along the positive y axis. We investigate these failure cases in more detail in Figure 11 and find these overpredictions to be dominated by spacegroup 71. This appears to be a data rather than a model issue caused by 2.1 k materials in MP having relaxed to a saddle point rather than the true ground state. The Alexandria dataset [2], [3] reached a lower energy state for these same materials. So the model seems to have learned to predict saddle point rather than ground state energies for materials similar to the ill-converged spacegroup-71 structures found in its MP training data.

The other anomalous model is MACE with several severe underpredictions at $x = 0$ along the negative y axis. We investigated these points for common traits in composition or crystal symmetry but noticed no pattern.

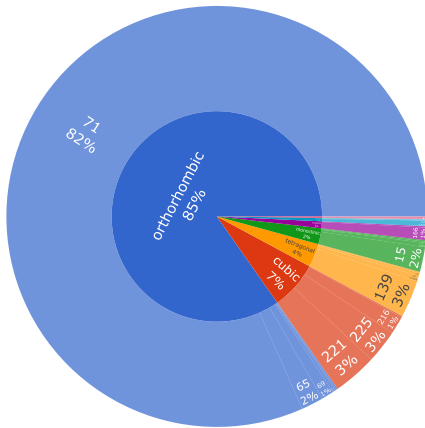
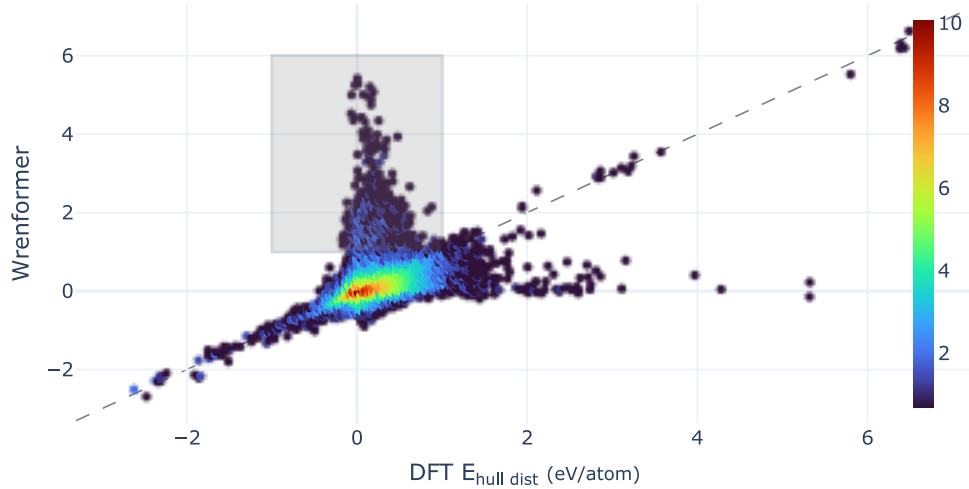
Beyond these MACE outliers visible in the plot, MACE exhibits another rare but reproducible type of failure case, in which the final predicted energy after relaxation is off by several orders of magnitude. The largest ‘derailed’ prediction was -10^{22} eV/atom for wbm-3-31970 (formula H_2Ir). In each case, the MACE relaxation exhausted the maximum number of ionic steps set to 500 and caused a volume implosion from initial cell volumes of hundreds to relaxed cell volumes of tens of cubic Angstrom. Using the checkpoint trained on the M3GNet dataset which we received from the MACE authors, this failure mode occurred for several hundred of the 250 k test set crystals. Using the checkpoint we trained ourselves on the MPtrj dataset, it affects only 44 test crystals, suggesting that these holes in the MACE PES can perhaps be fully plugged by further increases in training data. Further analysis in this direction is ongoing. Since these derailed values are easily identified and not harmful in practice when actually performing a prospective discovery campaign, we excluded them from the MACE parity plot and all other downstream analyses.

3.4.4 Wrenformer High-energy Predictions

The table in Figure 11 shows 456 + 194 (69 %) of the failure cases inside the shaded rectangle in the hull distance parity plot come from just two prototypes in spacegroup 71: AB2C2-oI10-71-a-f-h and ABC2-oI8-71-a-b-f. The occurrence of those same prototypes in the MP training set shows little data support for the failing prototypes. Moreover, the relatively few examples MP does provide in this regime appear to be ill-converged saddle-point structures that should be deprecated from future database releases, likely biasing the model towards high-energy predictions for these prototypes.

3.4.5 Hull Distance Box plot

Figure 12 shows the interquartile ranges (IQR) of hull distance errors for each model. BOWSR has the highest median error of 0.054, while Voronoi RF has the highest IQR of 0.19. MACE-



	Isopointal Prototypes	count	MP occurrences
0	AB2C2-ol10-71-a-f-h	456	75
1	ABC2-ol8-71-a-b-f	194	0
2	AB2-ol6-71-a-e	68	16
3	ABC3-cP5-221-a-b-c	30	1429
4	ABC2-cF16-225-a-b-c	24	4395
5	AB3-ol8-71-a-bf	19	2
6	AB2-mC12-15-e-2f	18	0
7	ABC2-tl8-139-a-b-e	17	5
8	AB2C5-ol16-71-c-e-am	16	3
9	A2B3-ol10-71-e-cg	13	3

Figure 11: **Symmetry analysis of the 941 Wrenformer high-predicted-energy failure cases.** Included are all points in the shaded rectangle defined by $E_{\text{DFT hull dist}} < 1$ and $E_{\text{ML hull dist}} > 1$. The spacegroup sunburst (bottom left) plot shows that 85% of severe energy overestimations are orthorhombic with 82% from spacegroup 71. The table on the right shows occurrence counts of exact structure prototypes for each material in the sunburst plot as well as their corresponding prevalence in the training set.

MP by contrast achieves the lowest median error of 0.014 and the lowest IQR of 0.062. MEGNet and CGCNN are the only models with a positive median, meaning they are more likely to overestimate rather than underestimate the DFT hull distance. This explains their ability in the cumulative precision/recall curves (Figure 7) to get the overall number of stable structures in the test set nearly right.

3.4.6 Classification Histograms using Model-Predicted Energies

The thickness of the red and yellow bands in Figure 13 shows how often models misclassify as a function of how far away from the convex hull they predict a material. While CHGNet and M3GNet perform almost equally well overall, they do so via different trade-offs. M3GNet has a thinner yellow band than CHGNet, meaning it commits fewer false negatives, but a thicker red

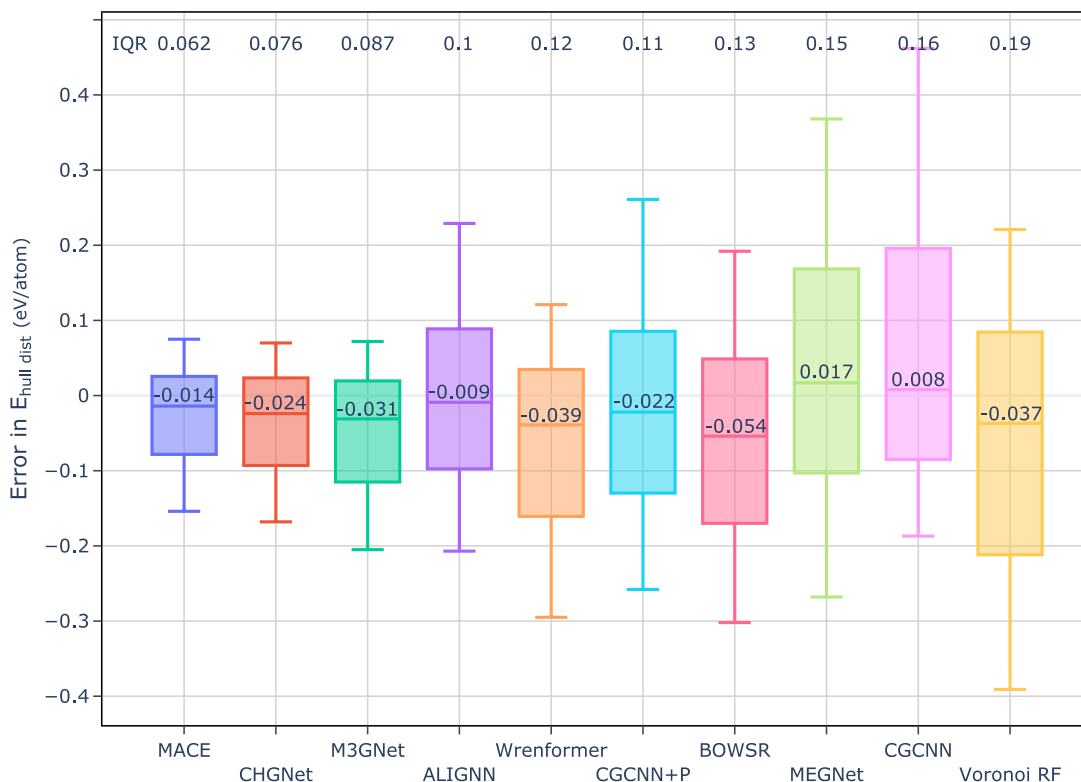


Figure 12: **Hull distance error interquartile ranges (IQR) for each model show almost universal bias towards stability.** The whiskers extend to the 5th and 95th percentiles. The horizontal line inside the box shows the median.

band corresponding to more false positives. In a real discovery campaign, false positives (red) have a higher opportunity cost than false negatives (yellow), since they result in wasted DFT relaxations or even wasted synthesis time in the lab. A false negative by contrast is merely one missed opportunity out of many. Thus, a practitioner might prefer a model like CHGNet over M3GNet in order to trade more false negatives (missed stable materials) in exchange for fewer false positives. This observation is also reflected in the higher TPR and lower TNR of M3GNet vs CHGNet in Table 2, as well as the lower rolling MAE for CHGNet vs M3GNet on the stable side (left half) of Figure 8 and vice-versa on the unstable side (right half).

Note the CGCNN+P histogram in Figure 13 is more strongly peaked than CGCNN’s which agrees better with the actual DFT ground truth distribution of hull distances in our test set (see Figure 48b)). This explains why CGCNN+P performs better as a regressor, but also reveals how it can simultaneously be a worse classifier. By moving predictions closer to the stability threshold at 0 eV/atom above the hull, even small errors can tip a stability prediction over the decision boundary.

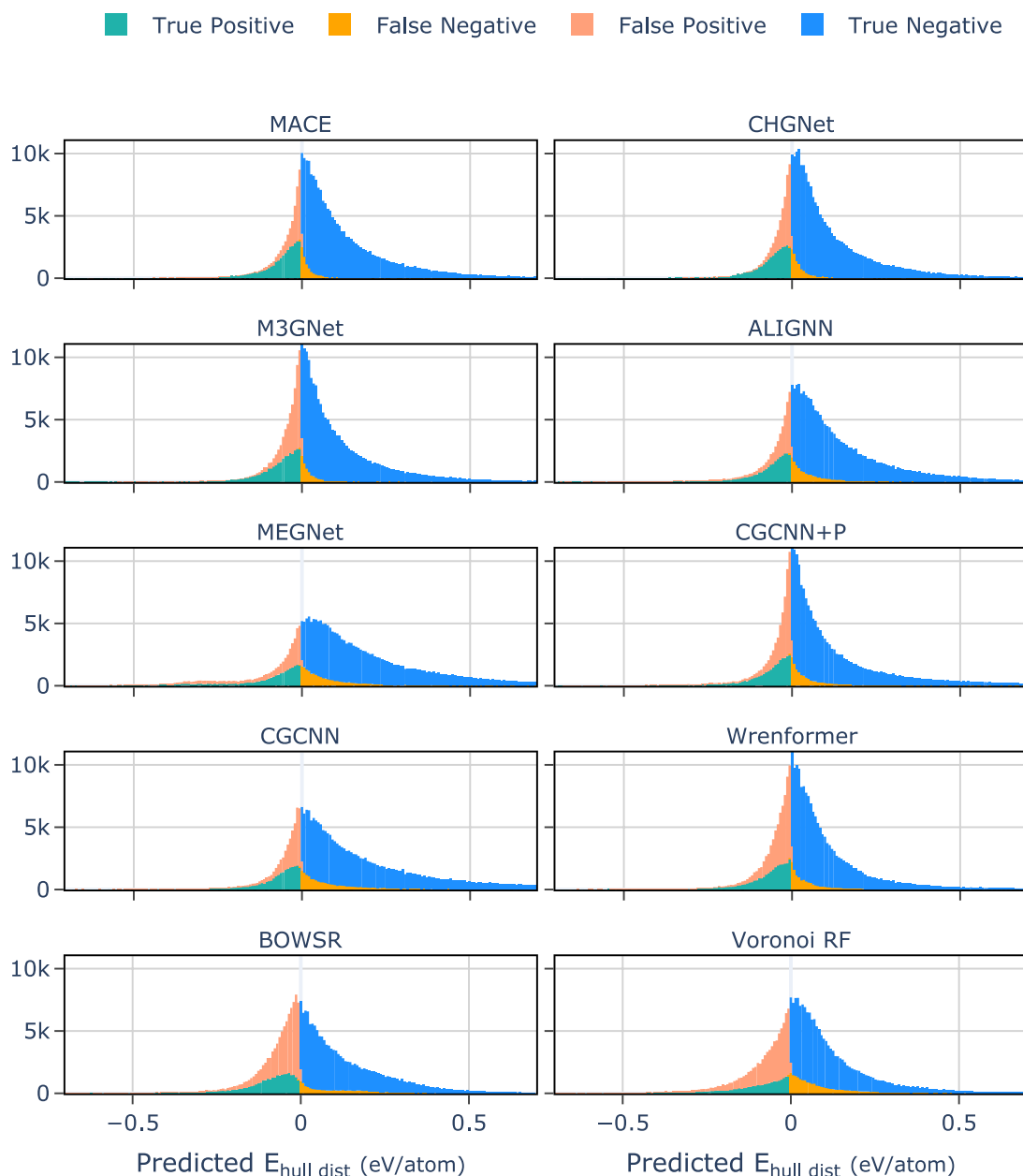


Figure 13: **Model-predicted hull distance distributions colored by stability classification.**

Models are sorted from top to bottom by F1 score.

3.4.7 Measuring extrapolation performance from WBM batch robustness

The WBM test set was generated in 5 successive batches, each step applying another element replacement to an MP source structure or a new stable crystal generated in one of the previous replacement rounds. The likelihood by which one element replaces another is governed by ISCD-mined chemical similarity scores for each pair of elements. Naively, one would expect model performance to degrade with increasing batch count, as repeated substitutions should on average

‘diffuse’ deeper into uncharted regions of material space, requiring the model to extrapolate more. We observe this effect more for the energy-only models than the ML potentials.

Figure 14 shows the rolling MAE of model-predicted convex hull distance as a function of DFT hull distance for each of the 5 WBM rounds of element substitution. These plots show a stronger performance decrease for Wrenformer and Voronoi RF than for UIPs like M3GNet, CHGNet, MACE-MP and even force-less GNNs with larger errors like MEGNet and CGCNN.

On WBM batch 1, Wrenformer performs comparably to the SOTA UIPs M3GNet and CHGNet. However, Wrenformer exhibits a pronounced increase in MAE with batch count. We view these plots as a strong indicator that Matbench Discovery is indeed testing out-of-distribution extrapolation performance as it appears to be the simplest (Occam’s razor) explanation for the observed model performance drop with increasing batch count. The effect is larger for some models than others but batches 4 and 5 consistently incur the highest convex hull distance errors for all models.

MEGNet and CGCNN both incur a higher rolling MAE than Wrenformer across all five batches and across most or all of the hull distance range visible in these plots. However, similar to the UIPs, MEGNet and CGCNN exhibit less degradation for higher batch counts. A stronger increase in error for later WBM batches appears to be specific to Wrenformer and Voronoi RF which suggests training only on composition and coarse-grained structural features (spacegroup and Wyckoff positions in the case of Wrenformer coordination numbers, local environment properties, etc. in the case of Voronoi RF) alone is insufficient to learn an extrapolatable map of the PES.

Considering its strong performance on batch 1, given sufficiently diverse training data Wrenformer could potentially become similarly accurate as the UIPs (at least for energies) across the entire PES while being substantially faster at inference time. However, in light of the difficulty of creating high-quality high-volume training sets, data efficiency is a key consideration for any ML model and Matbench Discovery shows ML potentials have an edge over other architectures when it comes to data efficiency. Moreover, the lack of force and stress predictions makes energy-only models unattractive for simulations.

3.4.8 Largest Errors vs DFT Hull Distance

Given the large variety of models tested, we asked whether any additional insight into the errors can be gained from looking at how the predictions vary between different models. In Figure 15 we see two distinct clusters emerge when looking at the 200 structures with the largest errors. This clustering is particularly apparent when points are colored by model disagreement. There is a strong color separation between the mostly dark blue low-energy bias predictions and green/yellow/red high-error predictions. Blue means models are in good agreement, i.e. all models are “wrong” together. Green/yellow/red means large-error predictions with little model agreement,

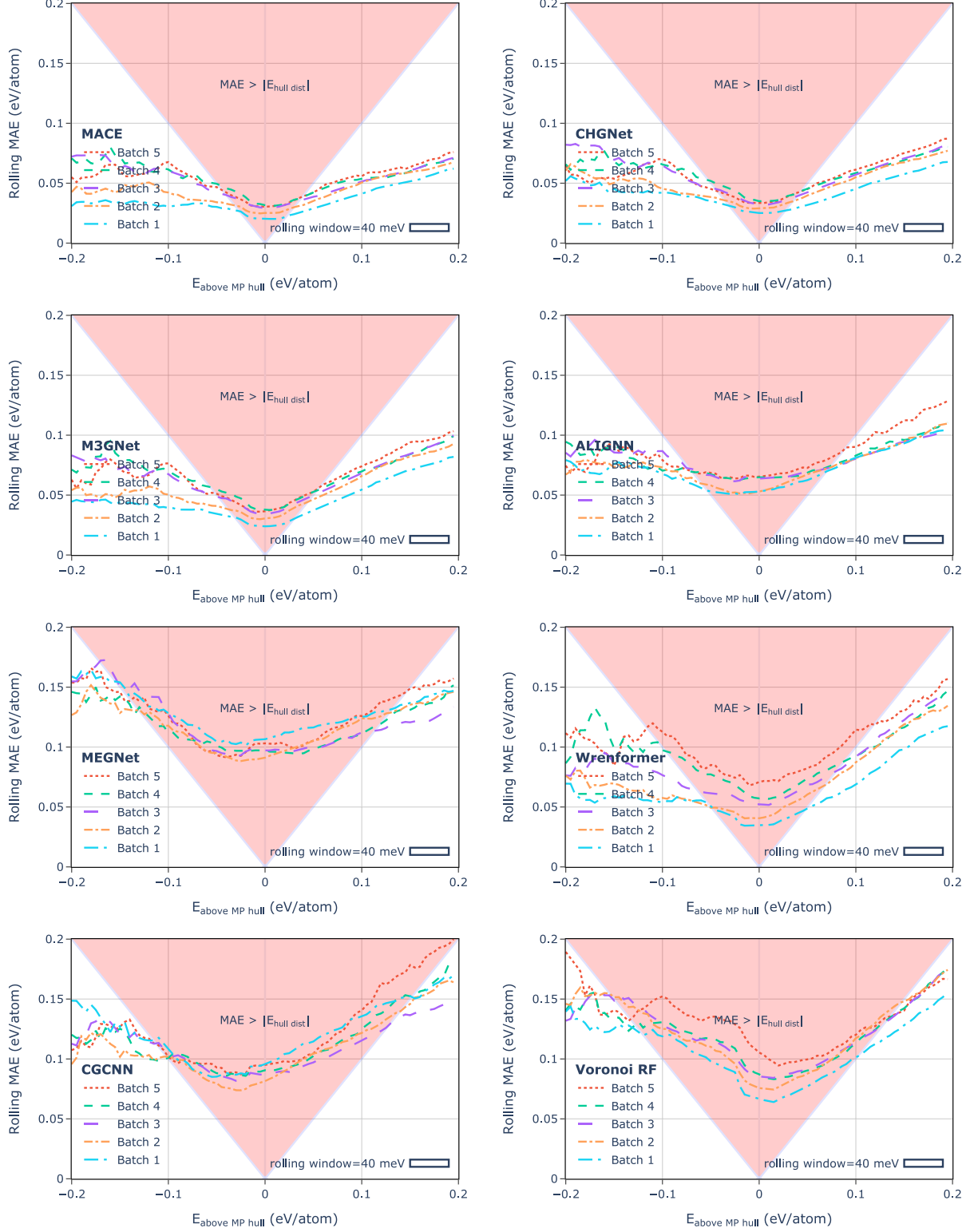


Figure 14: **UIP energy errors remain consistently low even for increased out-of-distribution (OOD) structures in later WBM substitution batches.** Each model’s rolling MAE as a function of DFT convex hull distance split by WBM substitution batch. In general, later batches (esp. 4 and 5) show higher errors than earlier ones, providing strong empirical evidence in support of the claim that the WBM test set constitutes OOD testing. In particular, MACE-MP, CHGNet and M3GNet show strong extrapolation performance as they exhibit low error across the whole DFT stability spectrum and only minor deterioration in performance on later batches.

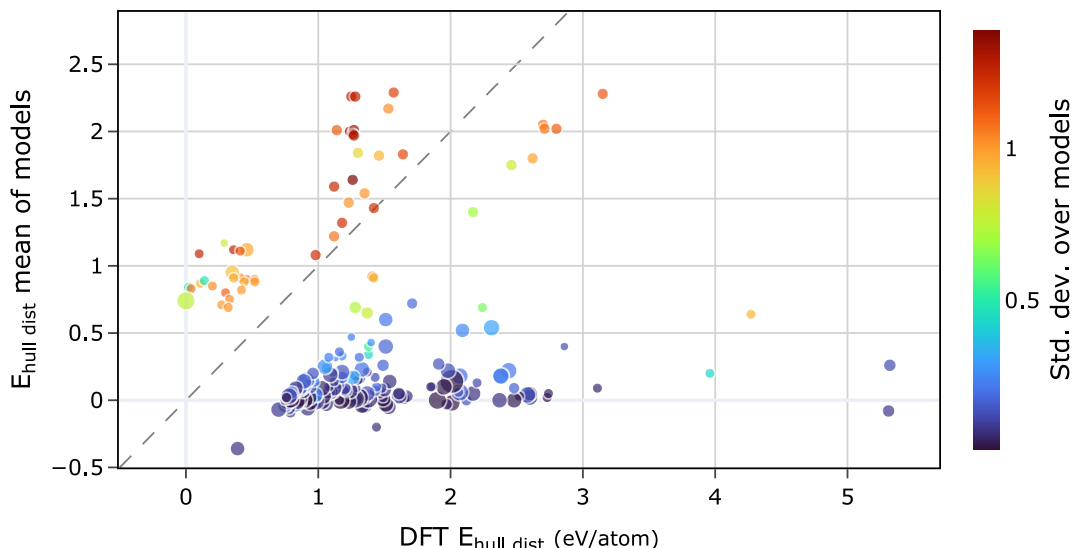


Figure 15: **Model disagreement (red) is a strong indicator of high-error predictions. Conversely, some high-agreement large-error points (blue) could indicate DFT issues.** ML-predicted vs DFT hull distance parity plot (averaged over all models) for the 200 largest error structures in WBM. Points are colored by model disagreement as measured by the standard deviation in hull distance predictions from different models. Marker size scales with the number of atoms in the structures.

i.e. all models are wrong in different ways. Some of the blue points with large errors yet good agreement among models may be accurate ML predictions for ill-converged DFT relaxations as was the case for the Wrenformer errors analyzed in [Section 3.4.4](#). The dark blue points also tend to be slightly larger corresponding to larger structures where DFT issues may be less surprising. ML model committees could potentially be used to cheaply screen large databases for DFT errors in a high-throughput manner.

The lack of points near the parity line ($x = y$) is due to the nature of the plot only including high-error predictions. The bias towards underestimating hull distance apparent from the larger share of points below the parity line is unsurprising considering MP training data mainly consists of low-energy structures, leading to a bias towards low-energy predictions.

3.4.9 MEGNet formation energies from UIP-relaxed structures

[Table 5](#) shows that MEGNet predicts less accurate energies than CHGNet and M3GNet even when it has access to either the CHGNet or M3GNet-relaxed structure. While all regression metrics improve for the model combos M3GNet→MEGNet and CHGNet→MEGNet over standalone MEGNet, the classification metrics even deteriorate except for TPR which markedly increases. However, [Table 5](#) shows that CHGNet→MEGNet and M3GNet→MEGNet both perform worse than either CHGNet or M3GNet by themselves. There are too many confounding effects at play to draw firm conclusions from this but this tentatively suggests having a separate specifically trained for the purpose of formation energy prediction offers no benefits over a

Model	F1 $\uparrow$	DAF $\uparrow$	Prec $\uparrow$	Acc $\uparrow$	TPR $\uparrow$	TNR $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	R ² $\uparrow$	Training Size	Model Type
CHGNet	0.61	3.04	0.52	0.84	0.74	0.86	0.06	0.10	0.69	1.6M (145.9K)	UIP-GNN
M3GNet	0.58	2.65	0.45	0.80	0.79	0.80	0.07	0.12	0.58	188.3K (62.8K)	UIP-GNN
M3GNet→MEGNet	0.55	2.54	0.43	0.79	0.75	0.80	0.09	0.14	0.36	188.3K (62.8K)	UIP-GNN
CHGNet→MEGNet	0.55	2.56	0.43	0.79	0.74	0.80	0.09	0.15	0.34	1.6M (145.9K)	UIP-GNN
MEGNet	0.51	2.70	0.46	0.81	0.57	0.86	0.13	0.20	-0.28	133.4K	GNN

Table 5: **MEGNet, an energy-only model, does not make better final energy predictions than UIPs when given UIP-relaxed structures as input.** All rows in this table show metrics for the task of predicting the relaxed energy given the unrelaxed structure (IS2RE). Metrics in rows labeled M3GNet→MEGNet and CHGNet→MEGNet are the result of passing M3GNet/CHGNet-relaxed structures into MEGNet for formation energy prediction. Both model combos perform worse than using the respective UIPs on their own with a more pronounced performance drop from CHGNet to CHGNet→MEGNet than M3GNet to M3GNet→MEGNet. This suggests that MEGNet has no additional knowledge of the PES that is not already encoded in the UIPs. However, both combos perform better than MEGNet on its own, demonstrating that ML relaxation imparts some of the latent knowledge of the PES into the relaxed structure from which a more informed final energy prediction is possible. This underscores the utility of ML relaxation at a very low cost for any downstream structure-dependent analysis.

model simultaneously trained on energies, forces and stresses, even when allowing the energy-only model to make its prediction from the same ML-relaxed structure as the UIP’s final energy.

While CHGNet was trained to predict formation energies per atom including [MP2020 energy corrections](#) [116] and MEGNet to predict formation energies with the [legacy MP corrections](#) (which we swapped for the [MP2020](#) corrections in our analysis), M3GNet was trained directly on PBE electronic free energies.

Changing the target from raw DFT energies to formation energies in principle merely constitutes a change of gauge since the difference is just a linear transformation of subtracting the elemental reference energies weighted by composition. But in practice and in the absence of z-scoring the targets, one might expect the problem of stability prediction to be slightly easier for the model when training targets are formation energies, as the data range is much smaller, requiring model parameters to shift less during training from their random initialization.

3.4.10 Predicted Formation Energy Analysis

[Figure 16](#) shows the distribution of model-predicted formation energies vs PBE ground truth for all 257k WBM materials. While similar to the parity plots in [Figure 10](#) which shows the predicted convex hull distance vs PBE hull distance, [Figure 16](#) better visualizes the density of points due to the inherently wider spread of formation energies compared to convex hull distances. We observe broadly the same failure modes with a handful of high-DFT-energy outliers predicted as near 0 formation energy by all models.

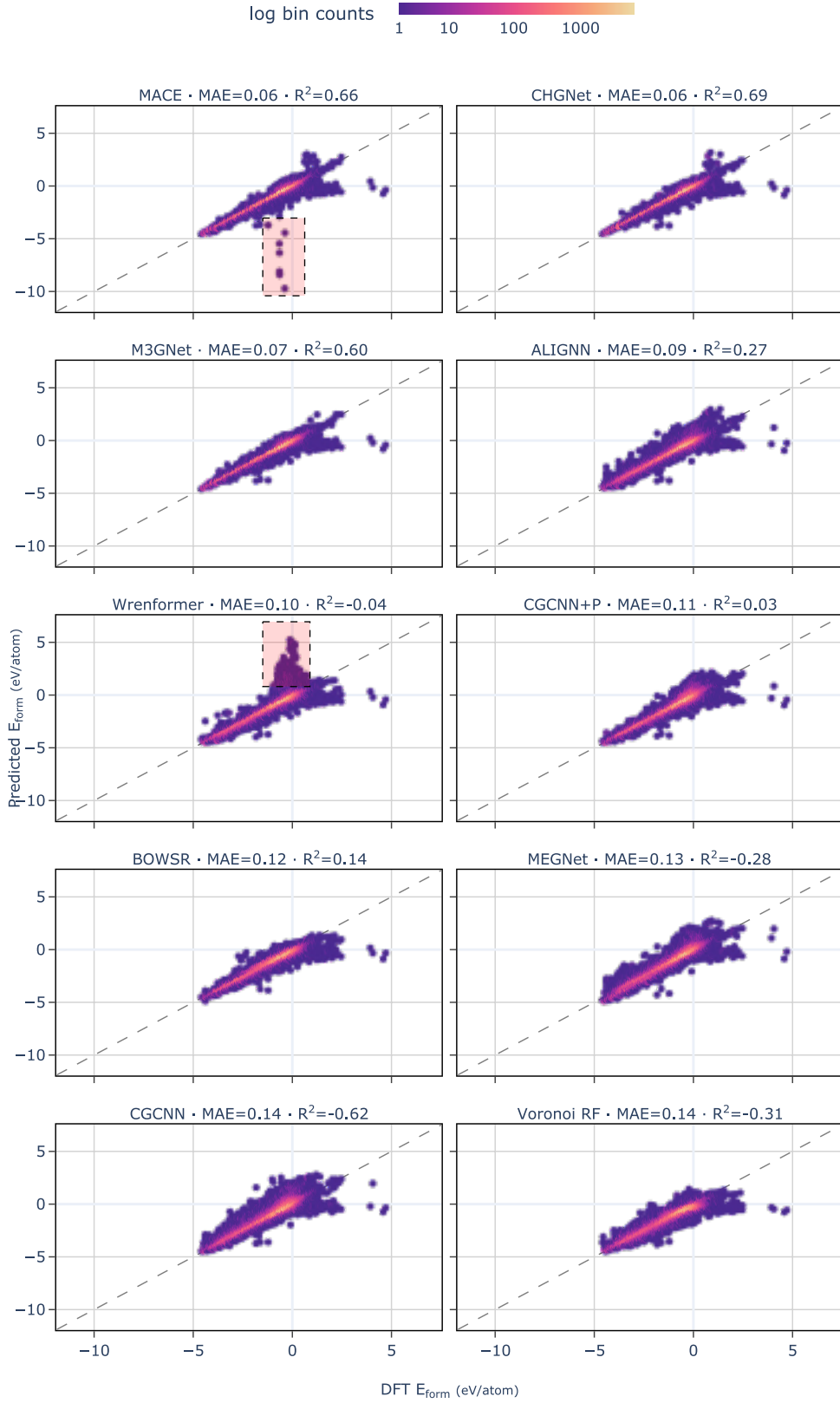


Figure 16: **A handful of high-DFT-energy outliers repeat across models, indicating potential data issues while other outliers are model-specific, most notably MACE-MP and Wrenformer (red-shaded rectangles).** Parity plots of model-predicted formation energies vs DFT ground truth, color-coded by log density of points. The models are sorted from left to right and top to bottom by MAE. MACE-MP has ~ 10 outliers at $E_{\text{form}}^{\text{DFT}} \approx 0$ and $-5 < E_{\text{form}}^{\text{MACE}} < -10$ which may be due to a model failure. The $\sim 2\text{k}$ highlighted Wrenformer outliers with $0 < E_{\text{form}}^{\text{Wren}} < 5$ are likely due to ill-converged training structures in Materials Project misleading the model as discussed in [Section 3.4.4](#).

Figure 17 plots the distribution of convex hull distance errors projected onto elements by composition for MACE and CHGNet as periodic tables. We observe very similar error distributions for MACE-MP and CHGNet, suggesting both models learn a similar map of the PES. There’s a weak trend of distributions more sharply peaked at 0 error for CHGNet than MACE-MP. We also observe slightly skewed error distributions from MACE-MP on the halides which accumulate more density on negative than positive errors. That is, MACE-MP tends to underestimate the convex hull distance and therefore overestimate the stability of halide-containing materials.

3.5 Discussion

We have demonstrated the effectiveness of ML-based triage in high-throughput materials discovery and posit that the benefits of including ML in discovery workflows now clearly outweigh the costs and risks. In a realistic benchmark scenario, several models achieve a discovery acceleration greater than 3.5 across the whole dataset (Table 2) and up to 5.6 when considering only the 10 k most stable predictions from each model (Table 4), approaching the highest possible DAF of 6.5 on our benchmark. When starting this project, we were unsure which is the most promising ML methodology for high-throughput discovery. Our findings demonstrate a clear superiority in the accuracy and extrapolation performance of UIPs like CHGNet, M3GNet and MACE-MP. Modeling forces enables these models to chart a path through atomic configuration space closer to the DFT-relaxed structure from where a more informed final energy prediction is possible. The consistently linear log-log learning curves observed in related literature [44, fig. 1e], [225, fig. 2] suggest that further decreases in the error of UIPs can be readily unlocked with increased training data, even without improvements in model architecture or training techniques.

In summary, we found a substantial increase in the predictive power of UIPs for crystal stability since the sobering 2020 analysis in ref. [10]. Despite its exciting pace of progress, the field of universal ML potentials is still in its early stages. Many shortcomings remain and invite further research. All models are biased towards stability, overestimating the correct number of stable materials in our test set by 30% (MEGNet) to 133% (BOWSR). As a result, even the best models in our analysis still exhibit a significant fraction of false positive predictions including for materials as high as 50 meV/atom above the hull (see Figs. 8 and 13), inviting further research into model calibration. By contrast, both DeepMind’s proprietary GNoME and Microsoft’s MatterSim trained on $\sim 55x$ and $\sim 11x$ larger dataset, respectively, get the true number of 42.8 k materials stable with respect to the MP convex hull almost right. GNoME predicts 39 k as stable while MatterSim nails it with 43.8 k, in both cases contributing to fewer false positives from these models and a resulting increase in DAF as shown in Table 11. We opted to relegate the metrics table Table 11 that includes proprietary models trained on larger

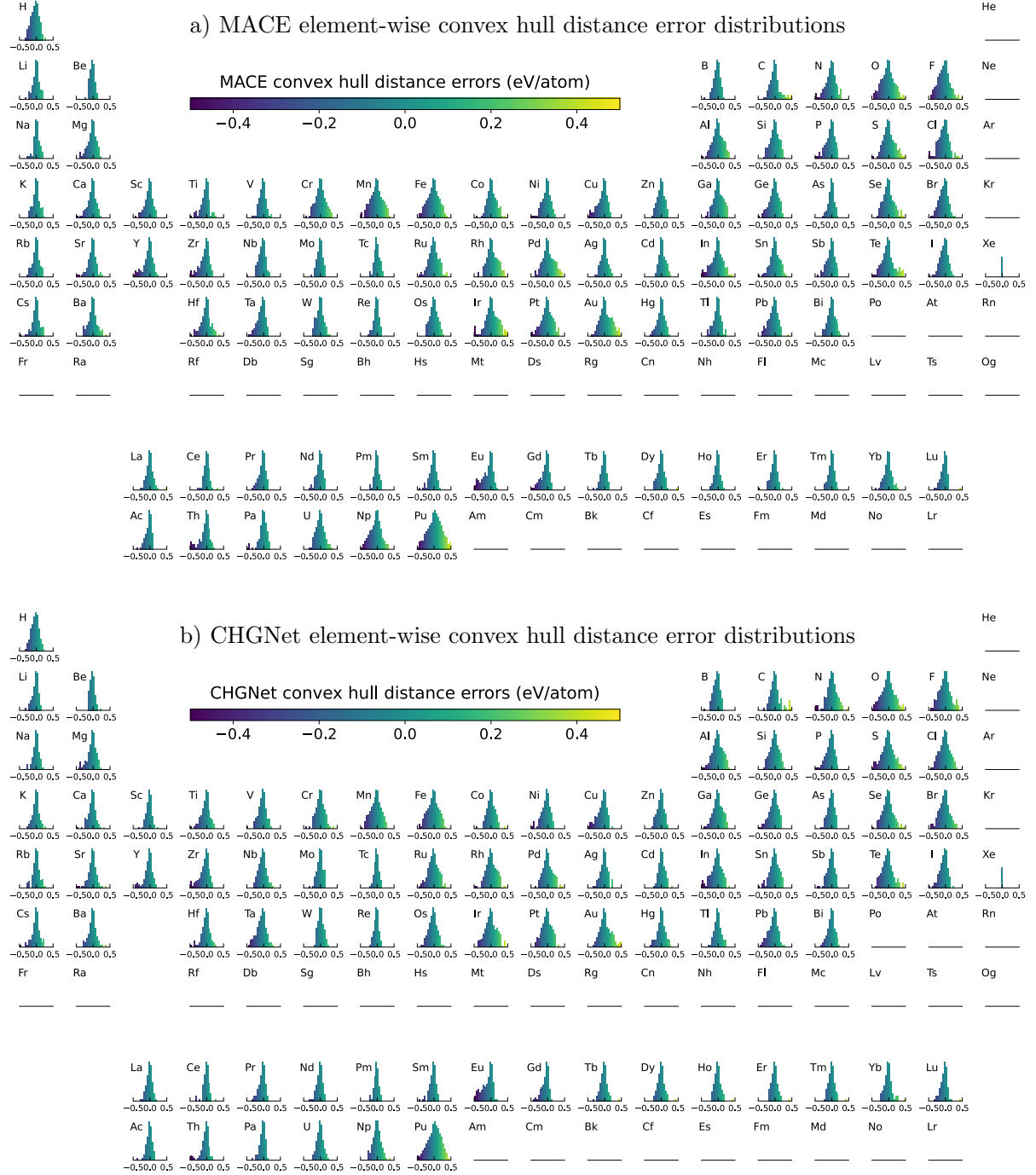


Figure 17: **Convex hull distance error distributions projected onto elements weighted by composition for all WBM test materials.** We show the two best open models submitted to Matbench Discovery, MACE-MP [81] and CHGNet [67]. The way these plots project model errors onto elements is best explained by example: assume Fe_2O_3 has a convex hull distance error of 50 meV/atom, $\frac{2}{3}$ or 20 meV/atom of that error would be attributed to Fe and $\frac{3}{5}$ or 30 meV/atom to O. Fe_2O_3 would show up in the above histograms as one count of 20 for Fe and one count of 30 for O.

incompatible datasets (otherwise identical to Table 2) to the appendix as the results are not

comparable and strictly speaking not admissible by the rules of the benchmark, yet are likely of interest to a wider audience as demonstrations of how far this technology can be pushed. As such, proprietary closed-source model submissions, while irreproducible and thus impossible to verify by outsiders given the closed data, trained weights and/or code, may act as guideposts for future community efforts. In particular, looking beyond the obvious advantage of having seen $\sim 55\times$ the number of structures during training than the open UIPs trained on MPtrj, we speculate that GNoME’s excellent performance is in no small part due to the larger share of far-equilibrium configurations in GNoME’s 89 M training set. By comparison, MPtrj contains only 1.6 M structures of which $\sim 154\text{ k}$ are relaxation end points. Specifically, they used a batch size of 32 structures, 16 of which were randomly sampled from across the whole relaxation trajectory while the other 16 were drawn only from initial relaxation frames, increasing the representation of far-equilibrium structures during training to maximize structure diversity. They report this sampling strategy to “substantially improve performance on downstream tasks” [44, p. 10]. We still caution that no firm conclusions can be drawn from this $N = 1$ sample size, especially since many details of the GNoME dataset are unbeknownst to us which could have a confounding effect on this result. Gaining a detailed understanding of the chemical, structural, energy and force distributions even of the publicly shared subset of the GNoME training data and how that affects ML potential predictions will require a large concerted effort by a diverse committee of expert chemists and ML researchers [226]. However, our results in Section 5.9 investigating the origins of PES softening in MLFFs likewise indicate datasets less biased towards stable structures will alleviate if not fully remedy the observed tendency for models trained on MP to underpredict energy.

This reinforces the need for future models to train on more diverse datasets less biased towards near-equilibrium structures than MPtrj. The Alexandria dataset [2], [3] offers a compelling choice for a highly diverse open dataset specifically engineered for training foundational GNN potentials. It contains PBE, PBEsol and SCAN data with a more even distribution of elements and crystal symmetries than MP which is heavily oxide-dominated (see Figure 45a)). Alexandria’s impressive size of over 3 M PBE geometry optimizations ($\sim 20\times$ the size of MP), 150k of which resulted in relaxed structures less than 50 meV/atom above the convex hull, provide good coverage of both ground-state and metastable regions of the PES. The MatPES collaboration will likewise help fill this data gap, which we launched in the second half of 2023 to generate an even higher-fidelity diverse MLFF training set at the r2SCAN level of theory with $\mathcal{O}(10^5)$ structures specifically sampled from high-temperature MLMD trajectories to cover far-from-equilibrium regions of the PES. At time of writing, MatPES has reached 130 k r2SCAN completed static calculations of 300 K M3GNet-MD snapshots with a similar number at 600 K and 900 K hopefully to be finished by the end of 2024. MatPES uses the latest PBE_64 VASP pseudo-potentials which offer increased accuracy for several rare-earth elements as well as finer convergence settings than MPRelaxSet which generated all data in MPtrj.

We believe our work (e.g. [Figure 14](#)) and others [\[216\]](#) have caught early glimpses that sufficiently large training sets allow us to narrow the performance gap between in-domain and out-of-distribution performance. Closing this gap will further increase the discovery acceleration factor, ideally resulting in a virtuous cycle between ML models accelerating new training data generation while becoming more accurate and even stronger accelerators after retraining. This is a typical active learning cycle but at the scale of the entire ML potential community. Meanwhile, the setup cost and learning curve of ML potentials are steadily decreasing, likely with a much lower floor on the entry barrier than DFT given the intricacies of high-throughput ab initio simulation. Given their already strong performance, this suggests that UIPs combined with data accumulation over the next few years may pave the way to a more comprehensive yet cheap-to-evaluate map of the PES that helps commoditize materials discovery.

Looking beyond finding new stable materials via thermodynamic stability prediction at zero Kelvin, materials research also requires understanding and predicting properties under varying conditions of positive temperature and pressure. This is where interatomic potentials provide significantly increased utility over energy-only models. Force and stress predictions enable insights into dynamical properties. One open question of particular relevance is the extent to which such models may aid research into the computational prediction of synthesis pathways which has received growing attention in recent years [\[28\]](#), [\[63\]](#), [\[227\]](#), [\[228\]](#), [\[229\]](#), [\[230\]](#), [\[231\]](#), [\[232\]](#). Many existing approaches for reaction pathway prediction involve the use of heuristic rules for dealing with the significant added complexity of metastability alongside traditional ground state ab initio data [\[25\]](#), [\[233\]](#), [\[234\]](#). These algorithms stand to benefit immensely from more efficient estimates of the reaction energy barriers that future UIPs can provide and may enable much more specific guidance to experimentalists including specific synthesis steps to try in order to reach a desired configuration of atoms that was identified as a promising candidate material. If sufficient reaction barrier accuracy is reached, this will open up a the field of synthesis pathway prediction to a more impactful high-throughput mode.

Chapter 4

ML-Guided Dielectric Materials Discovery

This chapter is based on

[13] J. Riebesell, T. W. Surta, R. Goodall, M. Gaultois, and A. A. Lee, “Pushing the Pareto Front of Band Gap and Permittivity: ML-guided Search for Dielectric Materials” <https://arxiv.org/abs/2401.05848> v1. 2024–01–11.

Invited submission to Digital Discovery. Rhys and I designed the workflow, I implemented it and performed the data curation and analysis, model training and testing, selected promising candidates for DFT validation based on ML predictions and ran the high-throughput DFPT simulations to test them. Michael Gaultois, Wesley Surta and I had many online meetings for the second selection round of promising candidates for experimental synthesis based on the DFPT results. Wesley performed all synthesis and characterization experiments. I drafted and edited the manuscript and made the figures. Rhys Goodall and Alpha Lee helped with conceptualization and manuscript editing. This work is currently under review at Digital Discovery.

4.1 Introduction

Dielectric materials are indispensable in numerous modern electronic devices including central processing units (CPUs), random access memory (RAM), solid-state disks (SSDs), high-frequency (5G) antennas, photovoltaics, and light-emitting diodes (LEDs) [235], [236]. Their utility hinges on balancing dielectric constant and band gap, two anti-correlated properties that rarely both assume large values in a single material. High band gaps are crucial for reducing leakage current and preventing dielectric breakdown when subjected to high voltage. Conversely, a large dielectric constant is desirable for minimizing the energy required for polarization, which is especially important in applications like transistor gates. As transistors continue to shrink, the need for high-performance dielectrics that can serve as nanometer-thin gate dielectrics while withstanding operating voltages and suppressing leakage currents from quantum tunneling becomes more pressing.

Historically, the discovery of dielectric materials has often relied on time-consuming trial-and-error experimentation. Recent advances, particularly in automated workflows for computational screening using density functional perturbation theory (DFPT) have shown promise in systematically searching for high-performance dielectrics, e.g. mapping the bandgap-dielectric Pareto front of binary and ternary oxides [237]. Increasing computing power and workflow robustness have since enabled scaling simulations to thousands of candidate materials [238], [239], [240].

However, the sheer size of the space of $\sim 10^5$ known, let alone the $\sim 10^{10}$ hypothesized materials (up to quaternary order) [32], prohibits sampling without inductive bias and presents a daunting challenge for existing computational methods. Consequently, the dielectric properties of the vast majority of the $\sim 10^7$ simulated inorganic crystals remain unknown, making a more comprehensive exploration of the space likely to yield novel high-performance materials. To screen even a small fraction of the full space requires orders of magnitude cheaper methods. Moreover, going beyond the $\sim 10^5$ experimentally known materials introduces the added challenge of predicting thermodynamic stability in addition to estimating band gap and dielectric constant.

To address this, we propose a new dielectric discovery workflow that judiciously integrates machine learning (ML) as the first filter in a multi-step funnel. ML, while less reliable than ab initio methods like DFPT, is orders of magnitude faster and continues to improve with new data. Our ML filtering of the vast search space uses surrogate models for band gaps, dielectric constants, and formation energies. Instead of exact Cartesian coordinates, we employ crystal symmetries and Wyckoff positions in a coordinate-free, coarse-grained structure representation. The motivation behind this choice is to substitutionally generate large numbers of structures ($\sim 10^5$) and rapidly estimate their thermodynamic stability while sidestepping expensive DFT relaxations. Following DFPT validation of the most promising candidates according to our surrogate model ranking, the last selection step is a committee of human experts to incorporate domain knowledge when weighing the risks, precursor availability and ease of experimental synthesis of all high-expected-reward materials. We validate the entire workflow by deploying it from start to finish which culminated in making and characterizing two new metastable materials: CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$ which partially and fully satisfy our target metrics, respectively.

Finding exceptional materials that extremize a single property such as hardness [212], [241] necessarily requires extrapolation from the training data. This is fundamentally at odds with the statistical nature of ML, leading to increased error and less reliable predictions. Our approach diverges from previous efforts by choosing a target class of materials where the path to application relevance requires balancing two (or more) conflicting properties. Instead of needing to maximize both properties, this allows our ML models to operate in regions of good training support while still identifying materials with exceptional figures of merit. This type of tradeoff between competing properties is ubiquitous across many classes of materials such as thermoelectrics (need high low thermal but high electrical conductivity) [242], [243], catalysts (need high activity for fast reactions which tends to lower selectivity, increasing unwanted side reactions) [244], [245], high-strength and shape-memory alloys (need high strength and high ductility) [246], [247], and many more. While multi-objective optimization is often seen as compounding the discovery challenge, we propose that concave Pareto fronts such as the above examples may instead facilitate ML-guided discovery by reducing the need for extrapolation.

Despite a nascent but growing body of work on automated and high-throughput synthesis [18], [20], [248], [249], [250], [251], experimental validation remains a key bottleneck in the design of materials. The process of manually developing experimental synthesis recipes for theoretical materials is very time-consuming, often taking months to a year per material. The central claim of ML-guided screening and related efforts in rational materials design is that we can reduce the downside risk of attempting novel synthesis procedures by increasing the hit rate of desired properties. To test the performance of our ML-guided approach we develop synthesis procedures for two materials predicted to be high-performing: $\text{Bi}_2\text{Zr}_2\text{O}_7$ (mp-756175) and CsTaTeO_6 (a substitutionally-generated structure mp-1225854:W $\rightarrow$ Te). Both materials were measured to exhibit dielectric response with experimental permittivities in the 43rd and 81st percentile, respectively, of 136 experimental reference results for high dielectric materials reported in [238], [252], supporting our hypothesis that ML surrogates in a concave Pareto optimization problem allow widening the search space to completely uncharted regions of property space while maintaining a good hit rate of high-performance materials.

We first present our workflow in Section 4.2, report its computational results in Section 4.3 and then present experimental validation of CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$ in Section 4.4.

4.1.1 Related Work

While previous studies have made significant strides in automating high-throughput DFPT to uncover new dielectrics, our work diverges in 3 important regards. We prefix DFPT with generative and pre-filtering ML which allows us to consider a much larger initial candidate pool as well as venture into uncharted regions of material space in our search for high dielectrics. Using ML-preselection and biasing the structure generation to crystals similar in chemistry to known high dielectric materials in MP allows us to nonetheless maintain a higher hit rate of materials with high $\Phi_M > 240$ than previous works as shown in Table 6. Third, we built a web UI that enabled effective collaboration with experimentalists to select two promising candidates which we successfully synthesized and characterized.

To our knowledge, ref. [237] were the first to develop codes that fully automate ab initio calculation of band gaps and dielectric permittivities. They calculated 1,800 structures of binary and ternary oxides from the ICSD to generate a dielectric property map which confirmed the inverse correlation between band gap and permittivity for most oxides, with occasional outliers that exhibit both large permittivity despite large band gaps.

Ref. [252] calculated electronic and ionic dielectric tensors for 88 compounds to test the predictive power of DFPT against experiment for total dielectric constant and refractive index. While they observed a Mean Average Deviation (MARD) of 16.2% when using PBE as compared to LDA, they noted that DFPT is less accurate for compounds with complex structural effects

or strong anharmonicity. Their results, however, showed a high Spearman correlation factor of 0.92, demonstrating the utility of DFPT in identifying promising materials by ranking.

The following year, ref. [238] extended their previous work [252] by running high-throughput DFPT on 1,056 inorganic compounds. The resulting database of dielectric tensors was integrated into the Materials Project for public access. While this greatly improved data FAIRness and likely may have helped expand the search pool for experimentalists seeking synthesis candidates, the scale of the data remained too limited to cover more than a small fraction of compositional and even less of the configurational space of potential high dielectrics.

While the above works resulted in novel and promising candidate materials, they relied exclusively on expensive DFPT calculations, making truly high-throughput screening of hundreds of thousands of materials cost-prohibitive. Yet they produced a sizeable pool of DFT dielectric properties with which we are now able to train ML models to accelerate and amortize the high cost of DFPT in the search for dielectrics, allowing screening of a much more expansive chemical space.

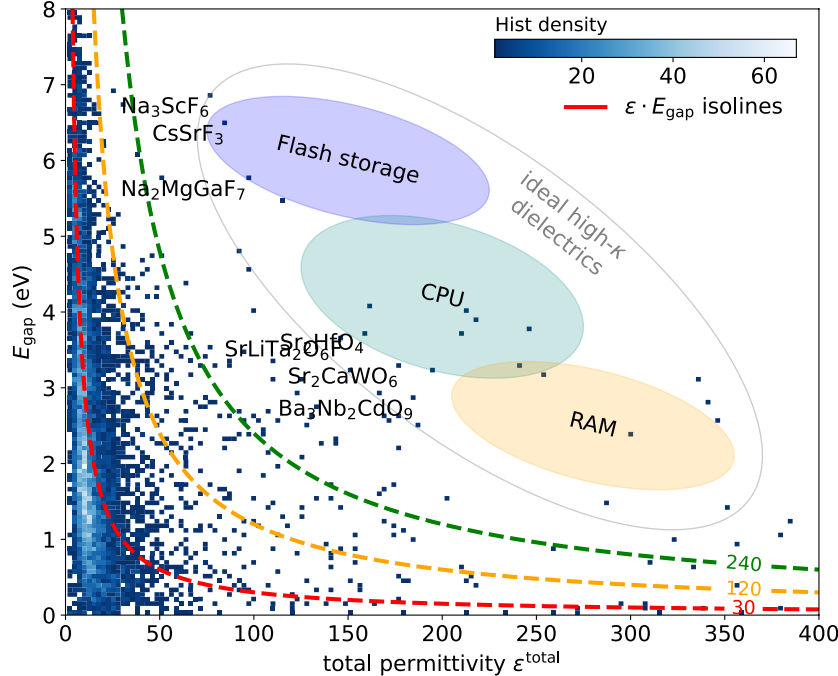


Figure 18: **Pareto front of dielectric constant and band gap for computing applications.** 2d histogram showing the $\frac{1}{x}$ relationship between E_{gap} and total dielectric constant ϵ_{tot} for 7.2k MP materials. The dashed isolines indicate levels of constant figure of merit ($\epsilon_{\text{tot}} \cdot E_{\text{gap}}$). The colored ellipses highlight desirable trade-offs between band gap and permittivity for specific device applications (CPU, RAM, SSD).

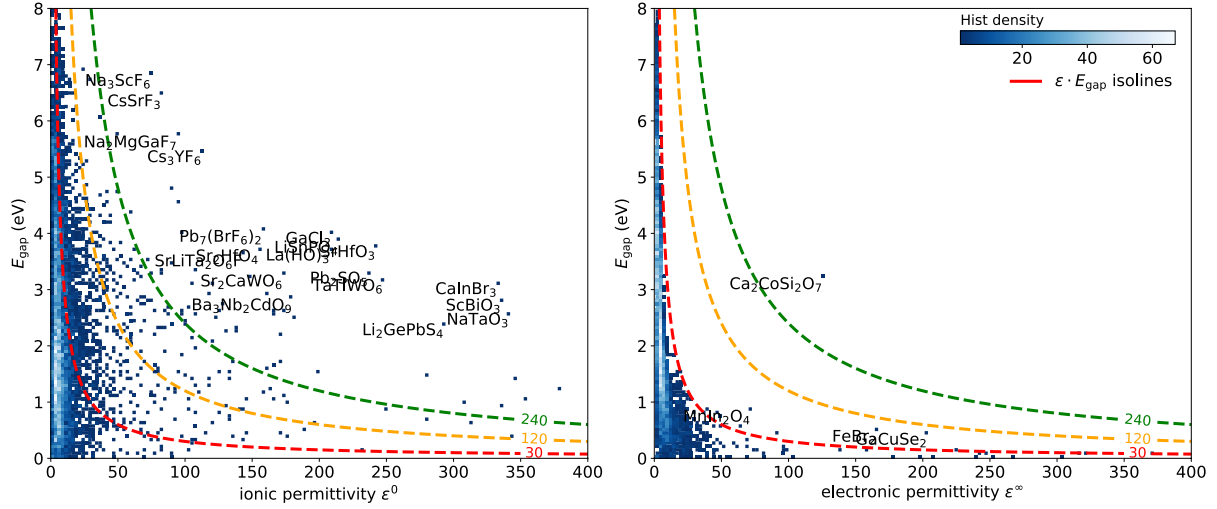


Figure 19: **Pareto front split by ionic (left) and electronic (right) dielectric contributions.** Ionic permittivity tends to be larger and is more likely to break the $\frac{1}{x}$ relationship with E_{gap} , making it the more promising property to optimize for computing device applications.

4.1.2 Dielectrics for Computing Applications

We plot the PBE total dielectric constant vs PBE band gap of all 7.2 k MP materials with DFPT results in Figure 18. As indicated by the blue, green and yellow shaded ellipses, ideal high-band-gap-high-permittivity dielectric materials lie in the top right of this plot. However, different computing applications have different requirements with different priorities for balancing band gap and permittivity. Materials for flash storage in particular require large band gaps to minimize leakage current and maintain polarization over extended time periods. CPU gate dielectrics tradeoff lower band gaps in exchange for increased permittivity which lowers the gate voltage required to achieve polarization and hence decreases power consumption. For random access memory (RAM) applications, increased leakage current resulting from a lower band gap is acceptable since RAM is memory-refreshed hundreds of times a second (stored data is read and immediately rewritten unmodified to preserve integrity; this prevents polarization sapping over time). Instead, optimal RAM performance relies on exceptionally high permittivity so that each repolarization costs minimal energy. Our goal is to discover materials in any of these regions beyond the dashed green isoline of constant figure of merit $\Phi_M = 240$ eV). Figure 19 reveals the principal permittivity contribution in most materials is ionic rather than electronic and that ϵ_{ionic} is more likely to overcome the $\frac{1}{x}$ relationship with E_{gap} and allow breaching the current Pareto front.

4.2 Scalable Dielectric Discovery Workflow

The large search space and high cost of experimental validation demand a funnel approach. To maximize the size of the initial candidate pool and still retain tractable computational cost,

less auspicious materials must be discarded by a hierarchy of successively more expensive but higher-fidelity computational filters. Such an approach maximizes return on invested effort by allotting more resources to candidates which accumulated evidence of expected utility in earlier filters. [Figure 20](#) gives an overview of the components and design decisions that go into our dielectric discovery workflow. Our proposed implementation precedes high-throughput DFPT with 5 to 6 orders of magnitude cheaper ML pre-screening which in this instance reduced 131,685 generated candidate materials down to 2,522 with computed dielectric properties.

We pre-screen based on 4 ML-predicted quantities: thermodynamic stability (derived from predicted formation energy E_{form}), band gap E_{gap} , ionic permittivity ϵ_{ionic} and electronic permittivity ϵ_{elec} . Each property is predicted by a separate ensemble of 10 Wren models [\[206\]](#) independently trained from random initializations to enable epistemic uncertainty estimation. This allows us to both massively expand the search pool of initial candidates and waste fewer resources on unpromising compounds.

The total dielectric response of a material to a given electric field is largely determined by two mechanisms:

1. Electronic contributions ϵ_{elec} due to **electron cloud distortion**, also frequently denoted ϵ_{∞} since electrons have little mass and thus polarize quickly even in high-frequency electric fields ($\sim 10^{15}$ Hz)
2. Ionic contributions ϵ_{ionic} due to **atomic displacements**, also frequently denoted ϵ_0 since ions have thousands of times greater mass, making atomic displacements more sluggish and only allowing their polarization to manifest in lower-frequency electric fields ($\sim 10^{12}$ Hz)

We found two separate ML ensembles to estimate the sum $\epsilon_{\text{tot}} = \epsilon_{\text{ionic}} + \epsilon_{\text{elec}}$ in [eq. \(32\)](#) to perform better than a single model predicting ϵ_{tot} directly.

The formation energy and band gap training sets each contain 319,601 data points, the combination of 98,850 Materials Project (MP) [\[12\]](#) calculations and 220,751 from the WBM dataset [\[1\]](#) which was generated with MP-compatible VASP settings. The ϵ_{ionic} and ϵ_{elec} ensembles are trained on the much smaller dataset of [7,172 DFPT calculations](#) in MP (2020-09-08) due to the lack of additional MP-compatible dielectric datasets (as far as we know).

While simply screening materials within large ab initio databases for which properties of interest have yet to be calculated is a viable strategy, it is also important to demonstrate the generative capabilities of ML-based workflows. To this end, we identify the top 1 k MP structures by figure of merit $\Phi_M^{\text{PBE}} = \epsilon_{\text{tot}}^{\text{PBE}} \cdot E_{\text{gap}}^{\text{PBE}}$ and use them as seed crystals for element substitution. The expectation is that this generates novel structures with increased likelihood of high Φ_M . The substitution process involves replacing all sites of one element in the structure with a chemically similar element (e.g. $\text{NaCl} \xrightarrow{\text{Na} \rightarrow \text{K}} \text{KCl}$), as determined by a similarity matrix mined from the ICSD [\[204\]](#). After filtering out duplicates (compositions that already exist in MP or

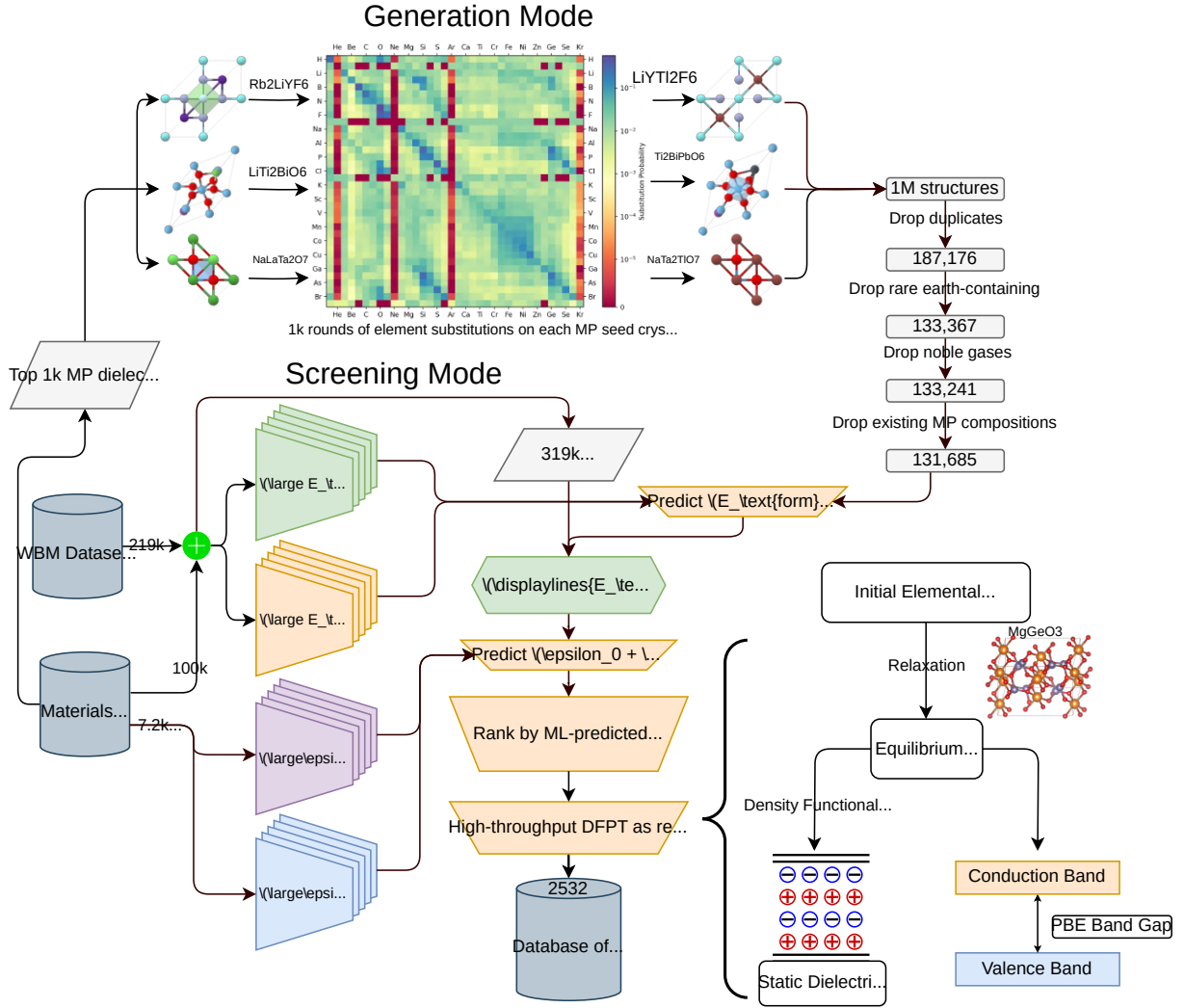


Figure 20: **Dielectric material discovery workflow integrating ML pre-screening, substitutionally-generated crystals and high-throughput DFPT validation.** The discovery pipeline can operate in two modes: screening and generation. Screening mode searches for large permittivity among known materials. In generation mode, we feed the top 1 k MP structures by figure of merit (Φ_M) into a chemical-similarity-guided element substitution process.

WBM, i.e. we do not consider structural degrees of freedom) as well as compounds containing noble gases, lanthanides or actinides, we are left with 131,685 potential new dielectric materials.

Using the trained Wren ensembles, we predict ϵ_{ionic} and ϵ_{elec} for all candidates, both those sourced from high-throughput databases and those produced using our generative methodology, and E_{form} and E_{gap} for the latter where DFT values are not available. We estimate the convex hull distance for each crystal from these predicted energies and discard those more than 0.1 eV/atom above the hull. This is motivated by the observation that 90% of crystals in ICSD are predicted to be less than 67 meV/atom above the convex hull [253]. This tolerance towards instability accounts for errors in DFT energies and the fact that some thermodynamically unstable materials are kinetically or entropically meta-stable and hence synthesizable.

The remaining candidates are ranked by their ML-predicted figure of merit Φ_M^{Wren} and subjected to a high-throughput DFPT workflow as our computational budget permits, resulting in a database of 2,522 dielectric properties.

4.3 Finding Dielectrics on the Computational Pareto Front

The violin plot in Figure 21 shows Gaussian kernel density estimates (KDE) of all 2,522 DFPT-computed electronic and ionic dielectric constants split by crystal system. Unlike the electronic contribution which is lower-bounded by the vacuum permittivity, the ionic dielectric constant can be zero in all crystal systems. We observe a general trend of higher dielectric constant the higher the crystal symmetry, especially for the ionic contribution. Only cubic crystals reach significant electronic permittivity with a median of 10.

Figure 22 compares the results from our methodology against those published in Petousis et al. [238] and Qu et al. [102] by plotting the PBE band gap on the y -axis against the total dielectric constant on the x -axis on a log-log scale. The blue circles show the 2,522 DFPT results we computed. The 441 orange diamonds show data generated by [102] while the 139 green squares are from [238]. The dark blue dashed isolines indicate constant figure of merit at values $\Phi_M = E_{\text{gap}} \cdot \epsilon_{\text{tot}} = c \in \{30, 60, 120, 240\}$ for band gap E_{gap} and total dielectric constant

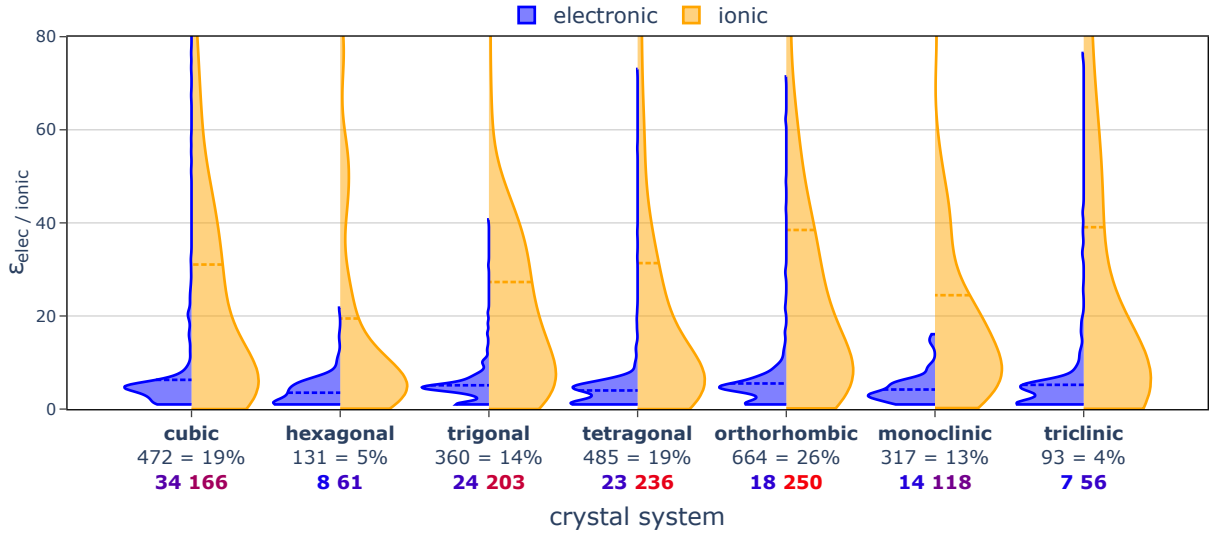


Figure 21: **Distribution of DFPT-computed electronic and ionic dielectric constants split by crystal system.** Violin plots show Gaussian KDEs of electronic (blue left halves) and ionic (orange right halves) dielectric constants. The dashed horizontal lines in each violin show the median. Below each crystal system is the number of materials we have for it and its percent share of the DFPT dataset. The colored bold numbers (blue = low, red = high) show the mean of the top 30 electronic/ionic dielectric constants for each crystal system. Tetragonal and orthorhombic systems have the highest outliers, top-30 ionic dielectric constants of 236 and 261, respectively.

Study	Number of Hits / Total	Hit Rate
Petousis et al. [238]	7 / 139	5 %
Qu et al. [102]	15 / 441	3.4 %
This work [13]	146 / 2,522	5.8 %
This work (with $E_{\text{gap}} > 0.1$ eV)	146 / 1,911	7.6 %

Table 6: **ML-guidance increases the hit rate of high- Φ_M materials.** Comparison of absolute and relative number of materials with $\Phi_M > 240$ eV with previous works. Excluding metals misclassified as insulators by our band gap models (which did not enter the workflows of other studies in the first place due to not generating novel structures with unknown band gaps), we achieve a $\Phi_M > 240$ eV hit rate of 5.8 %. This validates our approach of substitutionally generating candidate structures from known dielectrics and pre-screening with ML.

ϵ_{tot} . Our results achieve a larger number of materials beyond the highest Φ_M isoline of 240 than both previous works combined. We also achieve a higher hit rate per DFPT calculation of such high-merit materials as shown in Table 6. For ref. [102], $15/441 = 3.4\%$ of materials achieve $\Phi_M > 240$, while ref. [238] reaches $7/139 = 5\%$ and our data has $146/2,522 = 5.8\%$ materials with $\Phi_M > 240$. Note that our hit rate increases even further when posthoc excluding metals, i.e. filtering the hit rate analysis for materials with a band gap of at least 0.1 eV. While the other works started from DFT structures with known band gaps and hence were able to filter out metals from the outset, the same is not possible when generating novel crystals with unknown band gaps. Our workflow instead relies on ML band gap prediction to filter out metals. This step unfortunately suffers from a high false positive rate (metals misclassified as semiconductors/insulators). By upgrading to a better band gap model, a future realization of our workflow could achieve a high-merit hit rate over $146/1,911 = 7.6\%$.

Figure 22 compares our DFPT data to the results of [238] and [102]. Our workflow generates more high- Φ_M materials than both previous works combined and at a higher hit rate per expensive DFPT calculation than either [238] or [102]. We believe this hit rate increase is attributable to ML pre-screening and substituting elements into known dielectric materials.

4.4 Prospective Experimental Validation

To validate our workflow’s ability to yield viable dielectric materials in practice, we selected CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$ for experimental synthesis and characterization. Our selection criteria incorporated DFPT results, prior literature or related materials appearing in the ICSD [254], as well as precursor availability and expected ease of synthesis. The selection process was facilitated by a custom web interface to visualize DFPT results on the Pareto front hooked up to a shared database for note-taking and collecting prior literature appearances on individual candidate materials detailed in Section 4.5.5. Even so, making CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$ required several trial-and-error iterations to optimize the synthesis conditions which we detail in this section.

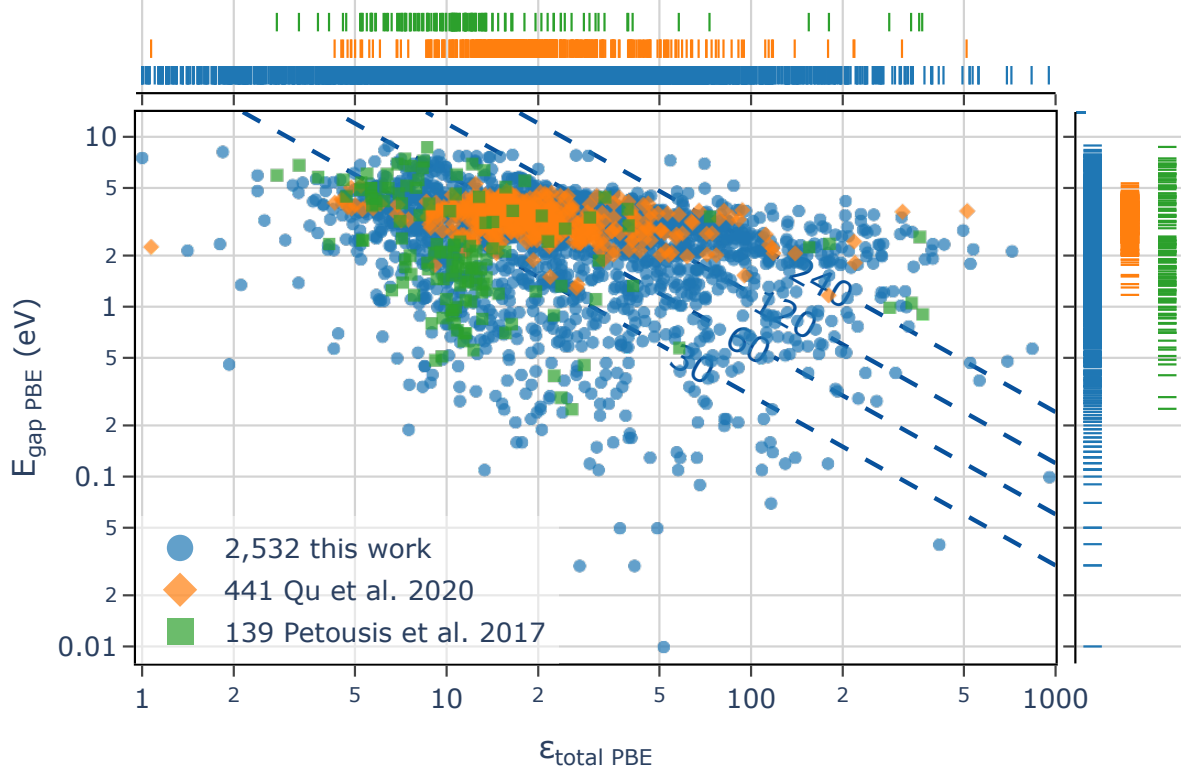


Figure 22: **Comparing hit rates for high- Φ_M materials from different studies.** Log-log plot of PBE band gap E_{gap} vs total dielectric constant ϵ_{tot} . 146 (5.8%) of our DFPT results (blue circles) lie beyond the $\Phi_M > 240$ eV isoline. [102] (orange diamonds) and [238] (green squares) produce fewer $\Phi_M > 240$ eV materials, both in absolute numbers and relative to dataset size (see Table 6). The dark blue lines indicate constant figure of merit $\Phi_M = E_{\text{gap}} \cdot \epsilon_{\text{tot}}$. The stacked marginal rugs along the top and right show the distribution of band gaps and dielectric constants in each dataset.

Optimization and Purity

We use X-ray diffraction (XRD) data and Rietveld fits to test our structural models for CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$. Results are shown in Figure 23.

The measured XRD pattern for our CsTaTeO_6 sample at 80% of the theoretical weight density readily fits a pyrochlore model. The atomic displacement parameters were small but positive within error. Even though multiple disordered models were explored, the simplest pyrochlore provided the best fit. We detected Ta_2O_5 impurities constituting $4.20 \pm 0.12\%$ of total weight that are highlighted in Figure 23a).

For $\text{Bi}_2\text{Zr}_2\text{O}_7$, we explored optimal synthesis temperatures between 550 °C – 750 °C. An extensive 8-hour XRD scan of $\text{Bi}_2\text{Zr}_2\text{O}_7$ after 48 h of heating at 650 °C confirmed the absence of Bi_2O_3 and ZrO_2 impurities in the sample, which significantly surpasses existing literature in terms of purity [255]. After sintering, we obtained a ceramic sample with 92+% of the theoretical density of a single crystal. Contrary to literature reports that typically describe an impure pyrochlore with a noticeable (111) reflection [255], our samples exhibit no such peaks. Prolonged heating did result in a broad (111) peak but was accompanied by undesired

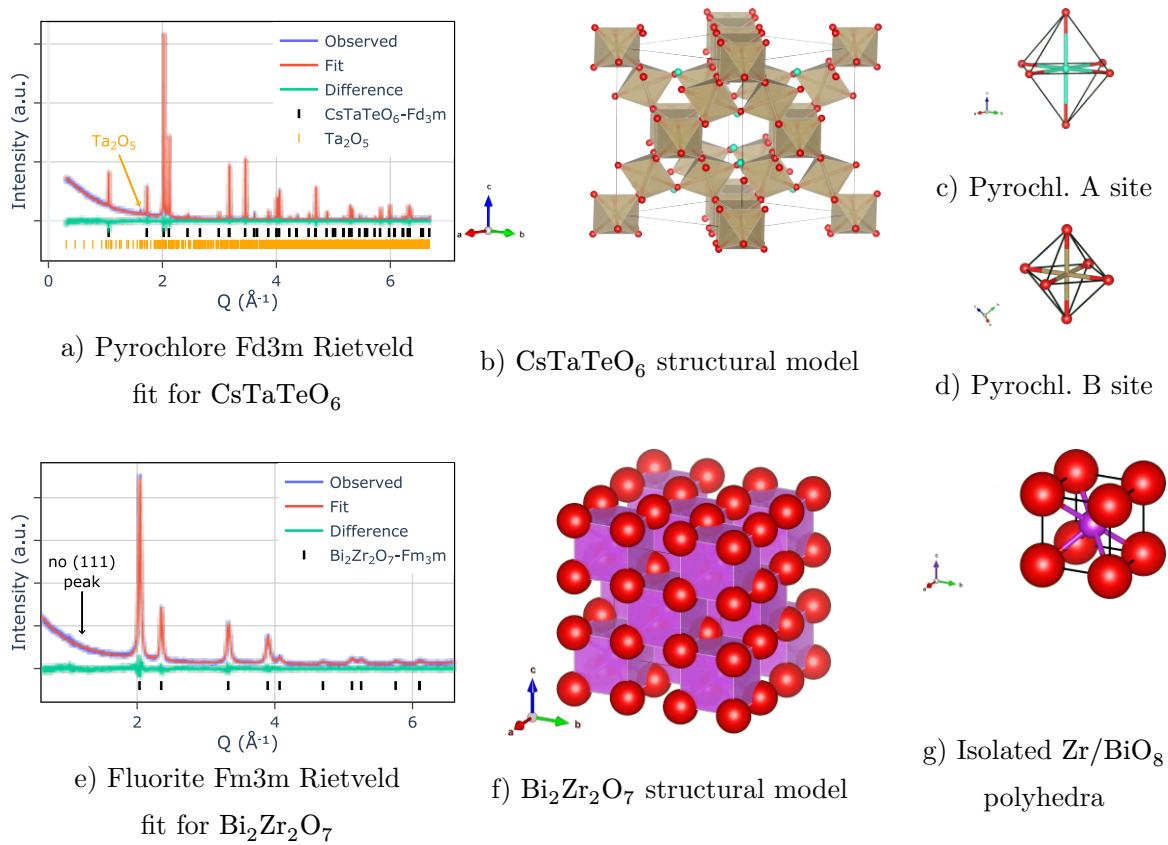


Figure 23: **Structural determination of CsTaTeO₆ (a) and Bi₂Zr₂O₇ (mp-756175) (e) using XRD and Rietveld refinement.** $Q = 2\pi \cdot d^{-1} [\text{\AA}^{-1}]$ is the scattering vector. (b) Crystal structure of the best Rietveld fit for CsTaTeO₆, with (c) and (d) showing the pyrochlore A and B site octahedra. (e) Crystal structure of the best Rietveld fit for Bi₂Zr₂O₇ with (f) showing the isolated Zr/BiO₈ polyhedra. Notable Ta₂O₅ impurities were detected in the CsTaTeO₆ XRD scan (e). Ta₂O₅ has many hkl reflections, most of which are not distinguishable from the background noise. The most prominent observable Ta₂O₅ peak at $Q = 1.7$ is marked by the orange arrow. The absence of a (111) peak in the Bi₂Zr₂O₇ Rietveld fit (g) suggests a fluorite structure, in contrast to the literature-proposed pyrochlore model.

Bi₂O₃ and ZrO₂ impurities. Avoiding prolonged heat, the Rietveld analysis in Figure 23e) shows the (111) peak to be absent, favoring a fluorite model for Bi₂Zr₂O₇, in contrast to the literature-proposed pyrochlore models. The compound exhibited large atomic displacement parameters (B_{iso}) which may arise from two superimposed crystallographic positions or due to off-stoichiometry (occupancy). Both commonly result in models with large atomic displacement parameters that simulate the distribution of electron density from these sites. However, attempts to reduce atomic displacement using site splitting and occupancy refinement for disordered materials did not yield better fits. Higher-quality diffraction data, e.g. from neutron scattering, would likely be required for more accurate modeling.

Further details on synthesis development, equipment used and XRD fitting for both Bi₂Zr₂O₇ and CsTaTeO₆ are provided in methods Section 4.5.6.

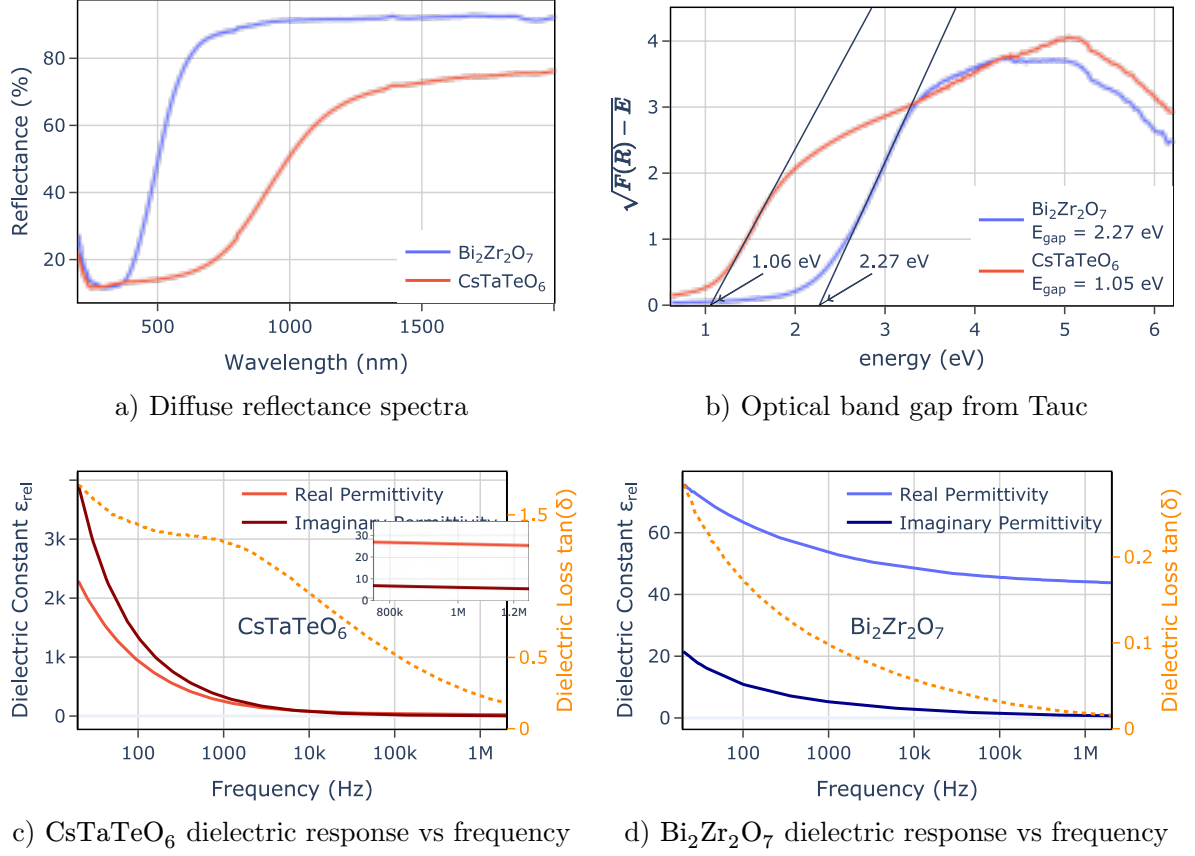


Figure 24: **Dielectric measurements of Bi₂Zr₂O₇ and CsTaTeO₆.** (a) Diffuse reflectance spectra for both compounds exhibit distinctive absorption edges, indicating ordered crystalline structures. (b) Tauc plot measuring absorption coefficient $\alpha(E_{ph})$ vs photon energy $E_{ph} = h\nu$ for both compounds. The extracted optical band gaps are $E_{gap}^{BZO} = 2.27$ eV for Bi₂Zr₂O₇ and $E_{gap}^{CTTO} = 1.05$ eV for CsTaTeO₆. (c) Dielectric response of CsTaTeO₆ as a function of frequency. We measure $\epsilon_{tot}^{CTTO} = 26$ at 1 MHz electric field (compared to 67 from DFPT). Its unwelcome high dielectric loss of $\tan(\delta) = 0.23$ at 1 MHz confirms the semiconducting nature observed in the Tauc plot's spectroscopic data. (d) Dielectric response of Bi₂Zr₂O₇ as a function of frequency yields $\epsilon_{tot}^{BZO} = 20.5$ at 1 MHz (compared to 206 from DFPT). We highlight Bi₂Zr₂O₇'s dielectric loss of less than 0.1 above 1 kHz, a sufficiently low value for many practical applications.

Dielectric Characterization

Having selected, synthesized at high purity, and confirmed the structures of CsTaTeO₆ and Bi₂Zr₂O₇, we investigate their physical properties in Figure 24. The band gaps of both materials were identified using UV-vis impedance spectroscopy on powders using diffuse reflectance and an integrating sphere. These data can be seen in Figure 24a) and they were modified and fit using the Kubelka Munk [256] equation to extract the bandgap, seen in Figure 24b). Figure 24a) shows diffuse reflectance measurements for CsTaTeO₆ and Bi₂Zr₂O₇ exhibiting distinctive absorption edges. The extracted band gaps are 2.27 eV for Bi₂Zr₂O₇ and 1.05 eV for CsTaTeO₆. It is worth noting that the measurements for both CsTaTeO₆ and Bi₂Zr₂O₇ turned out much lower than the DFT-calculated values of 2.09 eV and 2.96 eV respectively. This is surprising

given PBE’s tendency to underestimate experimental band gaps. For CsTaTeO_6 this may be due to complex defect effects not captured by DFT arising from Cs or Te volatility [257]. A more accurate ML band gap model that provides a more specific filter for metals and semiconductors would save future implementations of our workflow from spending unwarranted compute and lab time on semiconducting compounds like CsTaTeO_6 . However, given the limitations of PBE observed for these materials, it would be advisable to train the model on reference data obtained from higher levels of theory.

The low value of $E_{\text{gap}}^{\text{CTTO}} = 1.05 \text{ eV}$ for CsTaTeO_6 is consistent with its observed black color and unfortunately renders it unusable as a dielectric material. The dielectric measurements in Figure 24c) confirm a band gap-related high dielectric loss⁸. It is worth noting that despite its low band gap, CsTaTeO_6 exhibits high polarizability of $\epsilon_{\text{real}} = 26$ at 1 MHz up to its low breakdown voltage. However, its high dielectric loss of $\tan(\delta) = 0.23$ at 1 MHz confirms the semiconducting behavior observed in the spectroscopic data. We also caveat the measured dielectric constant with the fact that 23% loss makes the extracted ϵ_{real} value less reliable.

The $\text{Bi}_2\text{Zr}_2\text{O}_7$ compound has an observed band gap of $E_{\text{gap}}^{\text{BZO}} = 2.27 \text{ eV}$, making it a useful dielectric. Importantly, the observed band gap is 0.27 eV (13.5 %) higher than the previously reported mixed phase [255] who report $E_{\text{gap}}^{\text{BZO}} = 2 \text{ eV}$. This suggests reduced defect states and further substantiates the high purity and distinct phase of our synthesized materials. Dense ceramics were only accessible using spark plasma sintering, due to the metastable nature of the compound. Room temperature dielectric properties as a function of frequency can be seen in Figure 24d). Dielectric properties arise from a variety of different mechanisms: space charge, dipolar, ionic, and electronic polarization. Measuring as a function of frequency allows mechanisms with slower response times, such as space charge polarization arising from ionic conductivity, to be isolated from more meaningful mechanisms. At high frequency (1 MHz) the dielectric response shows a dielectric permittivity (ϵ_{real}) of 44 and a dielectric loss of $\tan(\delta) = 0.018$. The low dielectric loss (<0.1) indicates that the value of ϵ_{real} is free from conductive contributions. The permittivity of 44 is similar to doped Bi_2O_3 with fluorite-related structures, such as a 10% Ta^{5+} -doped Bi_2O_3 with a ϵ_{real} of 42 [258]. However, $\text{Bi}_2\text{Zr}_2\text{O}_7$ has a higher ϵ_{real} than HfO_2 or ZrO_2 (ϵ_{real} between 22-25) fluorites which are used as high-k dielectrics industrially [259], making it a worthwhile material to consider for real-world application. Furthermore, the aqueous-based synthesis with low calcination temperature of $\text{Bi}_2\text{Zr}_2\text{O}_7$ presents promising opportunities for solution processing of dielectrics which are compatible with existing industrial MOSFET processing technologies.

⁸The dielectric loss measures dissipation of electromagnetic energy propagating inside a dielectric material to heat. It is defined as the phasor in the complex plane between the real resistive (lossy) and imaginary reactive (lossless) components of the relative permittivity $\epsilon_{\text{rel}} = \epsilon_{\text{real}} + i\epsilon_{\text{imag}}$ and is commonly given as the tangent of that angle, $\tan(\delta) = \epsilon_{\text{imag}}/\epsilon_{\text{real}}$.

4.5 Methods

4.5.1 Derivation of Φ_M

Since dielectric constant and band gap are both crucial factors when considering electronic device applications, we measure materials by a figure of merit defined as

$$\Phi_M = E_{\text{gap}} \cdot \epsilon_{\text{tot}} \quad \text{where} \quad \epsilon_{\text{tot}} = \epsilon_{\text{ionic}} + \epsilon_{\text{elec}}. \quad (32)$$

A product ensures materials exhibit at least intermediate levels of band gap *and* permittivity. This follows ref. [260] who define this semi-empirical expression for the leakage current through a MOSFET gate dielectric:

$$J_G \propto \exp \left\{ -\frac{4\pi\sqrt{2}q}{h} \cdot (m_{\text{eff}} \Phi_b)^{1/2} \epsilon_{\text{tot}} \cdot t_{\text{ox,eq}} \right\} \quad (33)$$

with charge q , effective tunneling mass m_{eff} of the electron or hole, injection barrier of the gate dielectric Φ_b , and the SiO₂-equivalent-capacitance oxide thickness $t_{\text{ox,eq}} = (\epsilon_{\text{SiO}_2}/\epsilon_{\text{tot}}) \cdot t_{\text{phys}}$. Increasing $(m_{\text{eff}} \Phi_b)^{1/2} \epsilon_{\text{tot}}$ exponentially suppresses the tunneling current. Thus MOSFET device miniaturization requires materials that maximize this quantity. The effective tunneling mass m_{eff} and the carrier injection barrier Φ_b are expensive to compute from first principles and out of reach for high throughput workflows. Ref. [261], therefore, approximate their product as proportional to the band gap, $E_{\text{gap}} \propto (m_{\text{eff}} \Phi_b)^{1/2}$. Increasing $\Phi_M = E_{\text{gap}} \cdot \epsilon_{\text{tot}}$ should thus result in exponentially suppressed tunneling current.

In generation mode, the candidate materials are ranked for synthesis selection not by Φ_M but by an ensemble-risk-adjusted figure of merit $\Phi_M^{\text{std-adj}}$ which derives from Φ_M via uncertainty propagation rules. Expanding the variance σ_Φ^2 of $\Phi_M(\epsilon_{\text{tot}}, E_{\text{gap}})$, henceforth written without subscripts as $\Phi(\epsilon, E)$ for brevity, yields:

$$\sigma_\Phi^2 = \left| \frac{\partial \Phi}{\partial \epsilon} \right|^2 \sigma_\epsilon^2 + \left| \frac{\partial \Phi}{\partial E} \right|^2 \sigma_E^2 + 2 \frac{\partial \Phi}{\partial \epsilon} \frac{\partial \Phi}{\partial E} \sigma_{\epsilon E} \quad (34)$$

$\sigma_{\epsilon E} = \sigma_\epsilon \sigma_E \rho_{\epsilon E}$ is the covariance between ϵ and E and $\rho_{\epsilon E}$ their correlation. Since $\Phi = \epsilon \cdot E$ is just the product, $\frac{\partial \Phi}{\partial \epsilon} = E$ and $\frac{\partial \Phi}{\partial E} = \epsilon$ so that eq. (34) simplifies to:

$$\sigma_\Phi^2 = E^2 \sigma_\epsilon^2 + \epsilon^2 \sigma_E^2 + 2\epsilon E \sigma_{\epsilon E} \quad (35)$$

Since the band gap and dielectric ensembles are uncorrelated, $\rho_{\epsilon E} = 0$ and so the covariance term $2\epsilon E \sigma_{\epsilon E}$ vanishes.

$$\sigma_\Phi = \sqrt{E^2 \sigma_\epsilon^2 + \epsilon^2 \sigma_E^2} \quad (36)$$

Finally, we define the risk-adjusted figure of merit $\Phi_M^{\text{std-adj}}$ as the ensemble mean of Φ_M minus one standard deviation σ_Φ (with subscripts reinstated):

$$\Phi_M^{\text{std-adj}} = \Phi_M - \sigma_{\Phi_M} = \epsilon_{\text{tot}} \cdot E_{\text{gap}} - \sqrt{\epsilon_{\text{tot}}^2 \sigma_{E_{\text{gap}}}^2 + E_{\text{gap}}^2 \sigma_{\epsilon_{\text{tot}}}^2} \quad (37)$$

where $\epsilon_{\text{tot}} = \epsilon_{\text{tot}}^{\text{Wren}}$ and $E_{\text{gap}} = E_{\text{gap}}^{\text{Wren}}$ are the Wren ensemble means of predicted permittivity and band gap, respectively, and $\sigma_{\epsilon_{\text{tot}}}$ and $\sigma_{E_{\text{gap}}}$ the epistemic uncertainties given by the ensemble standard deviations. We use $\Phi_M^{\text{std-adj}}$ (rather than Φ_M) to rank substitutionally-generated structures for priority when allocating DFPT compute budget. Being a conservative estimate of merit that penalizes structures with high predictive uncertainty, Φ_M should yield fewer false positives at the cost of increased false negatives.

4.5.2 Initial Candidate Generation

As shown in Figure 20, we begin our discovery campaign by generating a large set of initial candidates. The Materials Project currently holds 7,172 materials with DFPT-calculated permittivity. Starting with the 1000 highest FOM MP dielectric materials, we perform 1000 rounds of element substitution on each source structure. Substitutions are guided by a chemical similarity matrix [1] mined from the Inorganic Crystal Structure Database (ICSD) [204], resulting in 1 million potential new structures.

The chemical similarity matrix offers a likelihood score for element substitution, based on their co-occurrence in the same space group in ICSD. This approach is inspired by previous works [206], [220]. During substitution, we swap out one element for another across the entire structure and limit ourselves to the 89 elements present in the Materials Project. This process yields 187,176 potential candidates, which we then filter as follows:

1. Remove duplicates: $10^6 \rightarrow 187,176$
2. Exclude structures containing rare earths (lanthanides and actinides): $187,176 \rightarrow 133,367$
3. Exclude structures containing noble gases: $133,367 \rightarrow 133,241$
4. Remove existing Materials Project compositions: $133,241 \rightarrow 131,685$

We remove rare earths because DFT is well-known to struggle with the 4f electrons [262], making any DFPT on such compounds less reliable. We filter noble gases because they are chemically inert and hence unlikely to occur in stable compounds. We remove structures with matching compositions in the Materials Project since many MP structures are sourced from the ICSD and hence the experimentally observed ground state. As such, any structures we generate with polymorphs in MP have an increased risk of being metastable at best.

4.5.3 Model Training

We trained Wren [206] deep ensembles [263] with ten members each on 319,601 PBE-level formation energies and band gaps obtained by combining the Materials Project and WBM datasets.

- The **Materials Project** (MP) database [12] is a well-curated database of high-throughput DFT calculations. We used database version 2020-09-08 powered by pymatgen v2022.0.8 containing 146,323 crystal structures [264].
- Wang et al. [1] calculated energies and properties for a large number of crystal structures substitutionally generated from MP source structures with chemically similar elements as pioneered in [220]. After substitution, the structures were relaxed using MP-compatible workflows. Using the author’s initials, we refer to this as the **WBM dataset**. After de-duplication and cleaning, WBM contains ~ 221 k structures.

MP also contains dielectric properties for 7,172 materials which we used to train two more Wren ensembles for predicting ionic and electronic permittivity. In screening mode, PBE formation energies of band gaps of candidate materials are already known and so only the permittivity needs to be predicted (as shown in Figure 20). In generative mode, all four ensembles are deployed to predict formation energy, band gap and permittivity for each of the 131,685 generated candidate materials as part of the ML pre-filtering step (Section 4.5.2).

The Wren architecture uses a coarse-grained Wyckoff position-based material representation that discards exact atomic coordinates in favor of discrete, enumerable symmetry labels identifying groups of sites that map onto each other under the crystal’s symmetry operations. Each Wyckoff position is embedded into a vector space and concatenated with the crystal site’s Matscholar element embeddings [265] before being placed in a fully connected graph with all other Wyckoff sites. Each node in the graph is then allowed to contextualize to its neighbors via multiple message-passing layers and finally mean-pooled to get a permutation- and relaxation-invariant, fixed-length, symmetry-aware crystal descriptor which is much cheaper to obtain than PBE or even MLFF-relaxed atom positions. A simple feed-forward network with skip connections [266] and ReLU [267] activations then maps the Wren crystal embedding onto one or multiple target variables. This featurization becomes more informative with higher symmetry in the structure. For our use case of filtering out substitutionally-generated unrelaxed structures, its distinct advantage is invariance under structure relaxations as long as the relaxation does not affect the structure’s symmetry (many DFT relaxations enforce keeping the initial symmetry throughout the relaxation, e.g. by setting `ISYM > 0` for VASP).

Training each ensemble member on identical data but with different random initialization offers two advantages:

- The ensemble average yields more reliable point estimates compared to single models.
- Ensemble variance allows us to assess epistemic model uncertainty, which we incorporate into the risk-aware figure of merit $\Phi_M^{\text{std-adj}}$ via error propagation (see eq. (37)). This reduces false positives at the cost of increased false negatives.

Moreover, for the formation energy ensemble, we also estimate aleatoric uncertainty, i.e. uncertainty that is inherent to the data, by using a “robust” loss function. This loss requires

changing the final output layer of each model to predict two numbers per sample. The loss function interprets the first number as the predicted mean and the second as predicted aleatoric uncertainty. This uncertainty enters the loss function as an attenuation term on the L^p norm. This allows the model to deweight the loss on predictions it attributes higher uncertainty to at the cost of incurring a higher regularization penalty.

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \frac{1}{2\sigma(\mathbf{x}_i)^2} \|\mathbf{y}_i - \mathbf{f}(\mathbf{x}_i)\|^2 + \frac{1}{2} \log \sigma(\mathbf{x}_i)^2 \quad (38)$$

where $\mathbf{x}_i$ are the model inputs, $\mathbf{y}_i$ the corresponding target values, $\mathbf{f}(\mathbf{x}_i)$ the model predictions and $\sigma(\mathbf{x}_i)$ is the model-predicted data noise (aleatoric uncertainty). $\sigma(\mathbf{x}_i)$ is learned by the model as a function of the input, making the loss heteroscedastic (i.e. sample-dependent). Thus the model can learn to deweight the standard L^2 loss in the first term by increasing its predicted $\sigma(\mathbf{x}_i)$. We do not set a robust loss on the other 3 ensembles due to an increase in validation error which we did not observe for the formation energy ensemble.

For all 4 Wren ensembles (E_{form} , E_{gap} , ϵ_{ionic} , ϵ_{elec}), we adopt the same hyperparameters as [206] to which we refer for details on the model architecture. Briefly, each ensemble member consists of 3 message passing layers, each with a single attention head. Both parts of the soft-attention mechanism use single-hidden layers with 256 hidden units and LeakyReLU activation functions [144]. The output network following the message-passing layers is a simple feed-forward network with skip connections and ReLU activation functions [142]. Its 4 hidden layers have sizes 64, 256, 256, and 1, respectively.

4.5.4 Dielectric Properties from DFPT

Candidates that pass our ML filters are fed into high-throughput density functional perturbation theory (DFPT). This stage offers more accurate property estimates at 5 – 6 orders of magnitude increase in computational cost.

Structure Relaxation

We used the Vienna Ab Initio Simulation Package (VASP) [93], [268] in projector augmented wave (PAW) mode [269] for all DFT calculations. The exchange-correlation energy was computed in the generalized gradient approximation (GGA) [270] using the Perdew-Burke-Ernzerhof (PBE) functional [101]. Input files were auto-generated by pymatgen [4]. To perform high-throughput DFT, we used the Materials Project workflow library atomate [6], the job launcher, queue manager and progress monitor Fireworks [5], and the automatic error handler Custodian [4]. Structures were relaxed until all interatomic forces fell below 10^{-2} eV/Å and the total energy change between self-consistent field (SCF) cycles fell below 10^{-7} eV. We compute the permittivity using linear response theory [271] at the generalized gradient approximation (GGA) level of density functional perturbation theory as implemented in VASP 6.2.0.

High-throughput calculations were orchestrated on the Cambridge CSD3 cluster using the `wf_dielectric_constant` workflow in `atomate` v1.1.0 [6]. This yields Born effective charges and phonon modes at the Γ point [272].

DFPT Settings

We depart from standard MP dielectric settings in several respects. First, by using the (at the time) most recent `PBE_54` release of VASP POTPAW pseudopotentials (MP uses `PBE`). We used the default structure-dependent k -point grid as implemented in `pymatgen` which constructs Gamma-centered meshes for hexagonal and face-centered cells, and Monkhorst-Pack grids otherwise. However, we increased the grid density to 3000 k -points per atom despite the significant cost increase for a high-throughput workflow to accommodate the sensitivity of linear-response calculations to k -point sampling. We also set tight convergence criteria of `EDIFF` = 10^{-7} eV (default = 10^{-5} eV) and a high kinetic energy cutoff for the plane wave basis set of `ENCUT` = 700 eV (default = 520 eV). We expect these changes to increase the fidelity of our results, or at worst, increase compute cost at no detriment to accuracy.

The total dielectric tensor splits into ionic (ϵ^0) and electronic (ϵ^∞) contributions:

$$\epsilon_{ij}^{\text{tot}} = \epsilon_{ij}^0 + \epsilon_{ij}^\infty \quad (39)$$

with $i, j \in x, y, z$ the 3 spatial dimensions and 0 and ∞ representing the electric field frequency. The ionic contribution ϵ^0 is computed from the Born effective charges Z^* and the phonon modes ω [273],

$$\epsilon_{ij}^0 = \frac{4\pi}{\Omega} \sum_m \frac{Z_{m,i}^* Z_{m,j}^*}{\omega_m^2} \quad (40)$$

with Ω the unit cell volume, m the phonon mode index, ω_m the infrared phonon frequency of mode m and $Z_{m,i}^*$ the i th component of the Born effective charge of mode m .

The scalar dielectric constant that enters our figure of merit equation eq. (32) is the mean of the eigenvalues ϵ_i of the total dielectric tensor:

$$\epsilon_{\text{tot}} = \frac{1}{3} \text{tr}(\epsilon_{\text{tot}}) = \frac{1}{3} \sum_{i=1}^3 \epsilon_i^{\text{tot}} \quad (41)$$

The ionic contribution ϵ_{ionic} to the permittivity is known to be sensitive to low-frequency phonon modes which are incorrectly softened by the lattice parameter overestimation due to GGA's tendency to underbind [237], resulting in higher mean error than LDA. We chose to run GGA DFPT despite this known shortcoming for better compatibility with existing MP dielectric data [238], [252].

DFPT Validation

To validate our DFPT results, [Figure 25](#) compares our ab initio results generated using the `wf_dielectric_constant` workflow in `atomate` [6] against available experimental dielectric constants collected in [238], [252]. While we achieve better agreement with experiment than Petousis as indicated by the lower MAE of 16.5 (vs 20.4) and higher R^2 of 0.41 (vs 0.0), and similar performance to MP (MAE = 14.9, R^2 = 0.12), we incur a slightly larger fraction of outliers than either of them at 14% (vs 9% and 10%, respectively). We define outliers as points with absolute relative deviation greater than $\pm 50\%$ relative to experiment. The reason we nonetheless achieve higher R^2 is due to the lack of extreme outliers; we have more but they are less severe. This is advantageous in high-throughput settings where the goal is to guide experiment. Even rare cases of extreme outliers will show up given sufficient throughput and extreme permittivity overpredictions are more likely to result in wasted experimental effort.

We note that while the data in MP was generated with the same `atomate` workflow as designed and benchmarked by ref. [252], our data is expected to deviate from MP/Petousis due to our departure in choice of VASP parameters described in [Section 4.5.4](#), most notably the use of `PBE_54` POTCARs, increased k -point density of 3000 points per atom, increased `ENCUT` = 700 eV plane wave energy cutoff and decreased `EDIFF` = 10^{-7} eV SCF convergence criterion. All of the above, though most notably the newer pseudopotentials may explain the less extreme outliers with respect to experiment. Overall, the variations with respect to MP/Petousis are within reason for run-to-run variability using slightly modified settings.

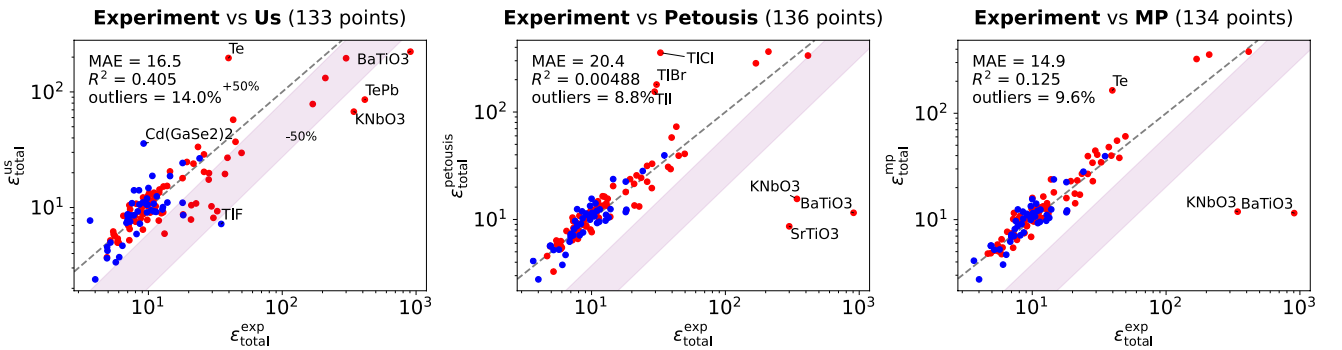


Figure 25: **Comparing experimental and DFPT-computed values for total permittivity ϵ_{tot} from different sources.** Our data shows lower MAE and higher R^2 but more outliers (defined as points with $> 50\%$ error) compared to Petousis et al. Comparing our DFPT dielectric constants with experimental values, we achieve an MAE of 16.5 and R^2 of 0.4 while MP results attain a slightly lower MAE of 14.9 and R^2 of 0.125. A CSV file with the plotted experimental data is available on [GitHub](#).

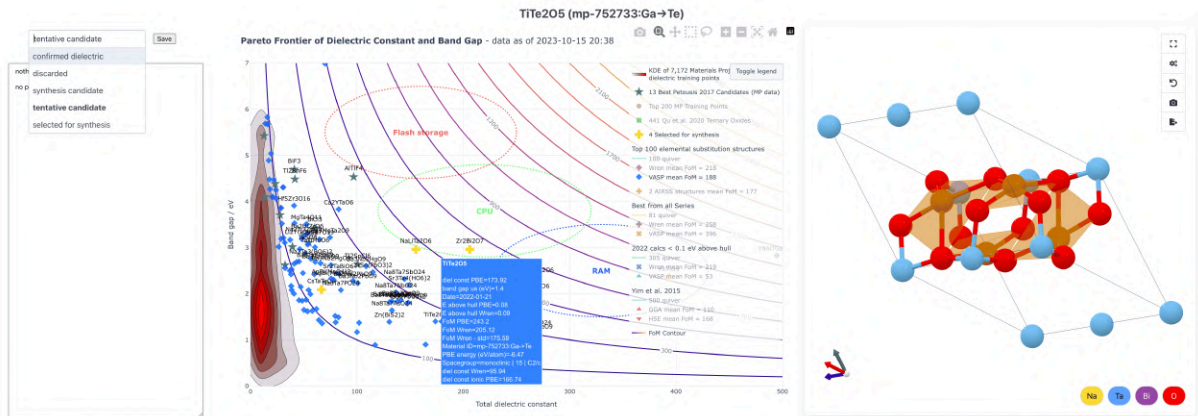


Figure 26: **Screenshot of the web app that aids with collaborative synthesis selection.** The centerpiece of the app is an interactive scatter plot similar to Figure 22 showing the Pareto front of band gap and total dielectric constant. The legend on the right enables toggling between our dataset and data from prior works discussed in Section 4.1.1. For our data, we have legend groups for different calculation batches and Φ_M -based subsets of the data that allow switching between viewing ML predictions and/or their corresponding DFPT results with a third option to show a quiver plot that renders arrows between these points. This makes it easy to visualize how ML predictions differ from DFPT results and to look for trends in ML errors w.r.t. chemistry. Ellipses again indicate regions of particular interest for specific device applications (CPU, RAM, Flash storage). The density contours show lines of constant figure of merit.

4.5.5 Web App for Collaborative Synthesis Selection

The last step in our discovery workflow before experimental synthesis involves a custom-built web interface shown in Figure 26 powered by a MongoDB Atlas M2 instance on the backend. This database is automatically updated when new *atomate* [6] DFPT workflows finish. To implement the frontend, we leveraged multiple open-source technologies:

pymongo [274] for fetching the latest calculation results from our *atomate* tasks collection.

plotly powers the interactive scatter plot used to show computed and/or ML-predicted band gaps and dielectric constants, switch between different calculation series or best-of subslices of the data as well as clicking points to select individual materials for closer inspection.

CrystalToolkit [275] renders the 3d structure of the material selected in the scatter plot with pan and zoom functionality.

dash [276] stitches the above three components together with callback functions. The two main ones are updating the structure viewer whenever a new point is selected from the scatter plot and updating the database when new free-form notes or categorizations are recorded in the text area and dropdown menu on the left.

In our case, selecting individual points from the scatter plot and annotating them with free-form text took place during remote live discussions between theorists and experimentalists while screen sharing. Since these meetings took place over months, the web app massively

helped with keeping track of reasons for categorizing a given material as discarded or tentative/firm synthesis candidate or recording links to prior art for materials categorized as already confirmed dielectrics. Since this web app effectively facilitated remote collaboration between computational and experimental groups, we emphasize the importance of developing more custom tools to improve information flow and data visualization. Moreover, we found our tool significantly facilitates the process of keeping provenance, recording planned next steps during meetings and being able to retrace the reasoning process behind why a given candidate was selected or discarded. Ideally, this process should be automated entirely in the future as this area is extremely prone to human error.

The final verdict of these discussions results in a classification as one of

confirmed dielectric prior experimental literature exists confirming our candidate material to be a dielectric. No point in synthesizing and re-characterizing, but increases trust in our workflow.

selected for synthesis Promising in every way, i.e. high calculated band gap and permittivity, cheap and easily accessible precursors, synthesis procedure matches our experimentalists' area of expertise and has ideally been demonstrated in earlier experimental works but without dielectric characterization.

strong candidate promising in some ways, i.e. high calculated band gap and permittivity but perhaps no existing literature reporting successful prior synthesis or compound looks challenging to make (e.g. might require aerobic environment)

weak candidate less promising in terms of simulated properties but potentially easier to make than other materials with superior expected properties

discarded failures of our screening method, usually due to existing literature indicating properties are not as we predict such as when a material was previously synthesized but reported as black, indicating a small band gap.

This interactive selection tool proved invaluable for extracting maximum utility and insight from the data we generated and resulted in identifying two candidates for final selection as suitable candidates for experimental synthesis.

We use GitHub Pages to host a figure-only version of this web interface at janosh.github.io/dielectrics. It is set up with continuous integration to update automatically as new data is generated. It has no write access to the database and hence cannot be used to annotate or categorize candidate materials but serves as a user-friendly public entry point to the most promising results in our database that requires no setup nor technical knowledge to use.

4.5.6 Synthesis Details

CsTaTeO_6 was synthesized using standard solid-state synthesis techniques. Stoichiometric amounts of Cs_2CO_3 (Alfa Aesar, 99.95), Ta_2O_5 (Afla Asear, 99.999), and $\text{Te}(\text{OH})_6$ (Aldrich, 99.5) were added to an agate pestle and mortar and ground to homogenize the precursors before calcining at 400 °C for 24 h in an Al_2O_3 crucible. After calcining, samples were reground and pressed into a 10 mm disk and annealed at 750 °C for 48 h in a covered Al_2O_3 crucible to form the final product.

$\text{Bi}_2\text{Zr}_2\text{O}_7$ was synthesized using an ethylenediaminetetraacetic acid (EDTA) and nitrate chelation and combustion, similar to a sol-gel process, a reaction modified from [255]. Equal molar quantities of Bi_2O_3 and $\text{ZrO}(\text{NO}_3)_2$ were added to separate beakers and dissolved in minimal amounts of concentrated nitric acid by stirring with a magnetic stir bar. Once dissolved, these two solutions were mixed with a four times molar excess of EDTA to ensure chelation. The solution was then heated at 80 °C until all liquid evaporated, leaving a brownish-white powder that was amorphous to X-rays. The amorphous powder was then calcined as a loose powder in a furnace inside a Al_2O_3 crucible at temperatures from 550 °C – 750 °C in 50 °C increments for 1 h. The samples heated at 650 °C produced the sharpest XRD peaks, without any trace of impurities. Samples heated higher than 650 °C or for longer than 1 h resulted in the decomposition of the sample into Bi_2O_3 and ZrO_2 , indicating metastability.

Both samples were sintered using spark plasma sintering. Pure phase samples were loaded into 10 mm graphite dies in a Thermal Technology LLC DCS10 furnace. Samples (~ 0.75 g) were loaded into a 10 mm diameter graphite die lined with a graphite foil and loaded into a sample chamber which was evacuated and backfilled with He three times. The sample was pressed uniaxially at 60 MPa, heated to the desired temperature at a rate of 200 C/min, held for 1 min, and cooled at the same rate. The CsTaTeO_6 sample was heated to a maximum temperature of 750 °C and the $\text{Bi}_2\text{Zr}_2\text{O}_7$ sample was heated to 600 °C resulting in samples with 92% and 94% of theoretical densities, respectively.

Diffuse reflectance measurements were taken on powdered using a Cary 5000 UV-Vis-NIR Spectrometer. Dielectric permittivity data was collected on sintered samples that had been thinned to a thickness of 1 mm and sputtered with gold electrodes. Data was collected using an Agilent 4980A instrument with a home-built sample holder and a program created in LABVIEW. X-ray diffraction data was collected using a Paralytical X'pert Pro instrument with $\text{Co K}\alpha 1$ ($\lambda = 1.789 \text{ \AA}$) radiation. Rietveld analysis was carried out using Topas Academic on these X-ray data. Initial refinements started with parameters identified using the Pawley method. Final refinements included lattice parameters, atomic positions, atomic displacement parameters, profile parameters and the background.

4.5.7 Bi₂Zr₂O₇ Synthesis Development and Structure Fitting

Bi₂Zr₂O₇ is known and has seen research interest for its use as a photocatalyst [277], [278], [279]. In these reports, the compound has been said to have either a stoichiometric pyrochlore structure (A₂B₂O₇) [255], [278], [279], [280], [281], [282] or the structurally related defect fluorite structure [277], [283], [284], [285], [286]. Our results show that a pyrochlore could not be isolated without additional Bi₂O₃ or ZrO₂ impurities due to the metastable nature of this compound. Though the pyrochlore and fluorite structures yield similar XRD patterns, with the most intense peaks located in the same positions, the absence of the (111) peak at $Q = 1.01 \text{ \AA}^{-1}$ favors assignment of the fluorite structure (Figure 23e).

The XRD data was fit using Rietveld refinement. Attempts were made to fit the data with a pyrochlore structure. When using both a standard pyrochlore model and models with oxygen and Bi₃₊, displacive disorder produces calculated patterns that fail to fit the data properly. Intensity mismatch is observed for low-angle pyrochlore peaks, specifically the (111) reflection. No amount of disorder was sufficient to reduce the intensity of this peak in the model to noise levels in the data, further confirming that this compound does not crystallize as a pyrochlore.

Using a defect fluorite structure (Bi_{0.5}Zr_{0.5}O_{1.75}, Figure 23f) results in rapid model-to-data convergence with a good visual fit, Figure 23e). The resultant model shows atomic displacement parameters of 3.28(17) Å² for the cations and 8.4(4) Å² for the larger oxygen. Large atomic displacement parameters are commonly found in disordered compounds, and experimentally observable in the form of broad diffraction peaks, compared to Figure 23a). Attempts to account for the disorder in this compound in our structural model were not successful. Splitting the position of Zr and Bi to account for chemical displacements off their position in the center of the cubic polyhedra resulted in the cations refining back to their undisplaced positions. The same process was used for the oxygen positions but resulted in much larger atomic displacement parameters, leading us to discount this distortion. The occupancies of sites were also refined, resulting in the cations maintaining a 1:1 ratio, within error. This allowed us to conclude that a simple defect fluorite structure is the most sensible model. This final model can be seen in Figure 23f) and an isolated Zr/BiO₈ polyhedra can be seen in Figure 23g). This model produced sensible metal-oxygen bond lengths of 2.3160(5) Å, expectedly longer than ZrO₂ bond lengths of 2.25 Å.

4.5.8 CsTaTeO₆ Synthesis Development and Structure Fitting

The targeted CsTaTeO₆ pyrochlore compound was initially investigated as it both met the figure of merit criterion and had not been reported previously in the ICSD or MP. However, we did find mention of this compound and its crystallographic analysis in [287] after completing synthesis and characterization. Moreover, a related pyrochlore with composition CsNbTeO₆ had been reported in [257], [288] from which we extracted initial synthesis parameters. With

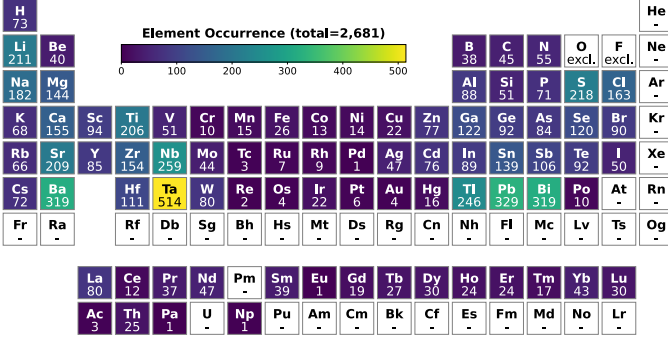
minor modifications of the synthetic procedure, the new compound was isolated in high purity, with only a 4.20 wt% Ta_2O_5 impurity.

Figure 23a) shows XRD data of the final CsTaTeO_6 product. This pattern indexes readily to the symmetry and lattice parameters of a cubic pyrochlore (Figure 23b)), consistent with both the Nb^{5+} -based analog and the computational predictions. Rietveld refinements were initiated using parameters taken from Pawley fitting and readily converged to a pyrochlore structural model. The structural model was taken from the refinement of CsNbTeO_6 which places Cs on the larger site 8b (Figure 23c)) site and the Ta^{5+} and Te^{6+} in equimolar amounts on the 16c site (Figure 23d)). This formulation is that of a defect pyrochlore (AB_2O_6), which is distinct from the traditional $\text{A}_2\text{B}_2\text{O}_7$ pyrochlore structure. Relative to a traditional pyrochlore, this structure has both cation and anion vacancies, while maintaining the same anion packing and BO_6 connectivity. After refining all parameters simultaneously, a good visual fit to the data is obtained with sensible atomic positions, sensible atomic displacement parameters ($0.087(15) - 0.78(2) \text{ \AA}^2$), a lattice parameter ($10.29894(5) \text{ \AA}$) close to that of the Nb^{5+} analog ($10.288 \text{ \AA}$), and a fit quality parameter ($\text{Rwp} = 8.095\%$) approaching that of the minimum set by the Pawley Fit ($\text{Rwp} = 7.399\%$) [289].

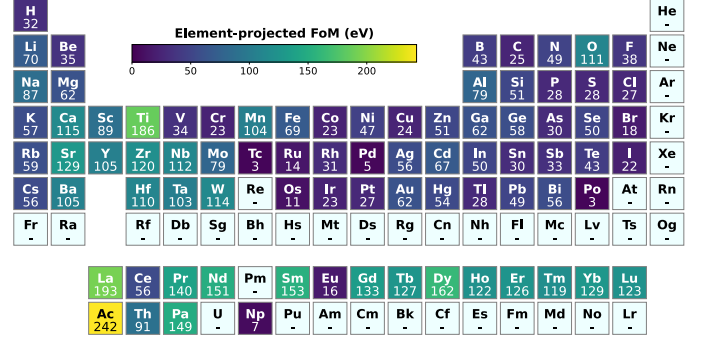
Due to the defect nature of this pyrochlore formulation, the Cs^+ (A-site) adopts an octahedral polyhedral environment (Figure 23c)) with six equal bond lengths of $3.183(6) \text{ \AA}$, instead of the cubic (AO8) environment found in stoichiometric $\text{A}_2\text{B}_2\text{O}_7$ pyrochlores. The observed bond lengths are consistent with AO_6 polyhedral environments seen in other Cs^+ pyrochlores such as CsNbTeO_6 or CsMoTeO_6 which range from $3.180 - 3.421 \text{ \AA}$ [289], [290]. The Ta_{5+} and Te_{6+} occupy the smaller octahedral environment (Figure 23d)) found in traditional and defect pyrochlores. This environment generates bond lengths of $1.9430(18) \text{ \AA}$, again falling within the expected range of related materials such as the Nb_{5+} and Mo_{5+} analogs previously mentioned, compounds that range from $1.941 - 2.013 \text{ \AA}$. This consistency of the structural environments found in CsTaTeO_6 with similar chemistries further validates the quality of our model.

4.6 Exploratory Data Analysis

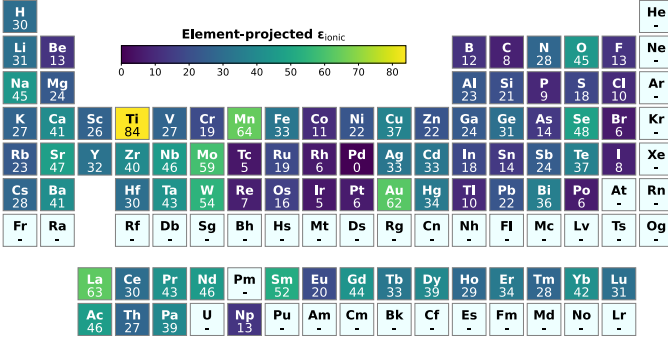
Figure 27 provides an exploratory view of the composition and element-projected properties of the 2522 structures we completed DFPT calculations for. The most prevalent elements are Ta (514 structures), Pb (329), Bi (319), Ba (319), Nb (259) (see Figure 27a)). We recover well-known trends for elements that tend to be present in high dielectrics. In particular, Figure 27c) shows titanium-containing structures on average have the highest ionic dielectric constants, consistent with the well-known excellent dielectric performance of alkaline earth metal titanates such as BaTiO_3 and SrTiO_3 perovskites. Interestingly, Figure 27d) reveals that late transition metals like Ru, Rh, Os, Ir and Pt yield the highest observed electronic permittivities.



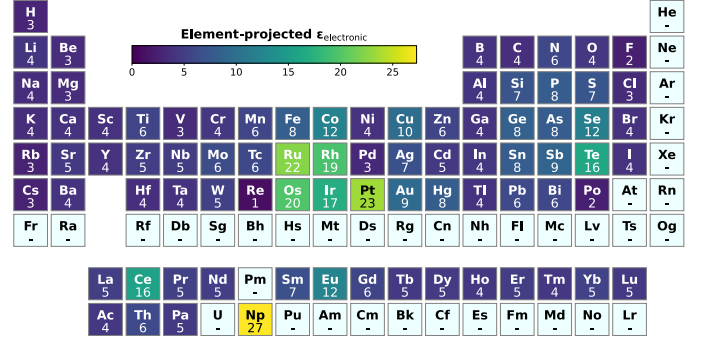
a) Element occurrences N



b) $\Phi_M = \epsilon_{\text{tot}} \cdot E_{\text{gap}}$



c) Ionic permittivity ϵ_{ionic}



d) Electronic permittivity ϵ_{elec}

Figure 27: **Element-projected heatmaps of ionic, electronic dielectric constant and Φ_M of our DFPT dataset** recover known trends for high dielectric materials but also reveal new ones. (a) Element occurrences, i.e. the number of structures among 2522 DFPT results containing each element. (b) Figure of merit $\Phi_M = \epsilon_{\text{tot}} \cdot E_{\text{gap}}$, (c) ionic ϵ_{ionic} and (d) electronic dielectric constants ϵ_{elec} projected onto elements by composition and averaged over all 2522 structures. E.g. Fe_2O_3 with a Φ_M of 100 would contribute a sample of 40 to the mean heatmap value of Fe and 60 to O. We filtered out ten untrustworthy calculations with $\epsilon_{\text{elec}} > 100$.

Table 7 lists all DFPT results in our dataset with $\Phi_M > 350$ sorted by Φ_M . The highest- Φ_M materials are almost exclusively oxides with only two fluorides and one selenide in the mix (AcF_3 , LiY_2F_7 and Sm_2CdSe_4). Some of the top materials, unfortunately, contain toxic or rare elements (e.g. Cd, Nd, Dy) which are undesirable for environmental, economic and lab-safety/logistic reasons. Others contain lanthanides and actinides, f -block elements which DFT is known to struggle with due to strong electron correlation effects in the atomic-like $4f$ orbitals near the valence band [262]. Both are strong arguments against attempting experimental synthesis, explaining why we did not simply select the top materials in this list.

4.7 Band Gap Prediction Issues

Substantial challenges remain concerning band gap prediction, due in part to a metal-heavy dataset imbalance and in part to band gaps being an inherently challenging-to-model non-local property of the electronic rather than ionic structure. This makes the prediction problem

	Material ID	Formula	Spacegroup	ϵ_{elec}	ϵ_{ionic}	ϵ_{total}	E_{gap} (eV)	Φ_M (eV)	n_{sites}	n_{elems}
1	mp-14550	TiCdO3	62	7.51	709	716	2.12	1,520	20	3
2	mp-997585:La->Y	Y8Al7GaO24	221	4.69	490	495	2.83	1,402	40	4
3	mp-32244:W->Mo	LiNbMoO6	113	5.76	528	534	2.19	1,172	18	4
4	mp-1097026	LaGaO3	221	5.23	305	311	3.42	1,062	5	3
5	mp-754225	YbTiO3	62	6.37	422	428	2.30	986	20	3
6	mp-3335	Sm2Ti2O7	227	6.13	331	337	2.82	951	22	3
7	wbm-1-40021	YbTiO3	62	6.36	387	393	2.33	916	20	3
8	mp-1226157	Cs2Ti(WO4)3	166	5.36	315	321	2.83	908	18	4
9	wbm-3-54931	Sr2LuTaO6	225	4.34	247	252	3.49	878	10	4
10	mvc-3783	MgTiO3	62	6.34	370	376	2.33	877	20	3
11	mp-556003	CaTiO3	74	6.39	384	390	2.16	843	10	3
12	mp-1202153	Na2Nb4O11	9	6.20	301	307	2.74	842	34	3
13	wbm-1-42539	La2MgZrO6	148	4.74	199	204	4.09	832	10	4
14	mp-1217978	SrPrScO4	107	4.64	224	228	3.52	804	7	4
15	mp-556925	TiMnO3	62	7.29	515	523	1.54	803	20	3
16	mp-979932:Sr->Pb	SiPb3O5	140	6.55	387	393	1.93	757	18	3
17	mp-1225952	CsNbWO6	74	5.43	248	254	2.92	741	18	4
18	wbm-1-25816	LiTaO3	167	5.41	223	228	3.21	733	10	3
19	mp-9890	TaAgO3	167	6.71	364	371	1.95	724	10	3
20	mp-754128:Ba->Ca	Ca3Hf2O7	139	4.24	199	203	3.47	704	12	3
21	mp-1219587	RbNbWO6	74	5.26	231	236	2.93	691	18	4
22	mp-1222804	LaY3Ti4O14	166	6.09	233	240	2.81	674	22	4
23	mp-1222933	LaTiNbO6	33	6.06	226	232	2.80	649	36	4
24	mp-755367:Cu->Rb	RbLiTa2O6	155	5.51	265	271	2.36	639	10	4
25	mp-4019	CaTiO3	62	6.39	257	263	2.31	606	20	3
26	wbm-1-38346	CaTiO3	62	6.39	254	261	2.32	605	20	3
27	mp-1218162	SrNd2Ti4O12	123	6.68	333	340	1.73	588	19	4
28	mp-756175	Zr2Bi2O7	227	6.53	200	206	2.80	578	22	3
29	mp-3858	NaTaO3	62	5.26	203	208	2.59	539	20	3
30	mp-755367:Cu->Ag	LiTa2AgO6	155	6.11	260	266	2.02	538	10	4
31	wbm-1-40022	YbTiO3	74	6.34	230	236	2.23	527	10	3
32	mp-1222976	LaTaBi2O7	74	6.39	262	268	1.96	526	22	4
33	mp-39511	LiCaTa2O6F	74	4.49	138	143	3.66	524	22	5
34	wbm-1-38356	CaZrO3	127	4.65	163	168	3.09	519	10	3
35	wbm-1-39241	NaTaO3	62	5.26	193	198	2.59	513	20	3
36	mp-6440	CaTiSiO5	15	4.51	171	175	2.91	509	16	4
37	wbm-3-51397	NdScO3	140	5.14	166	171	2.95	504	10	3
38	mp-1173711	Na3ErTi2Nb2O12	26	6.08	235	241	2.09	502	40	5
39	mp-559482	Ti2Bi2O7	227	8.42	180	189	2.62	495	22	3
40	mp-675778:Na->Li	LiY2F7	12	2.32	68	70	6.99	489	10	3
41	mp-1218781	Sr2La2MgTi3O12	1	5.67	233	239	1.97	471	20	5
42	mp-1220301	NbTiWO6	74	5.86	152	158	2.93	462	18	4
43	mp-755367:Cu->Na	NaLiTa2O6	155	5.32	150	155	2.96	460	10	4
44	mp-754936	DyAlO3	167	4.25	83	87	5.19	453	10	3
45	mp-1199037	NaNbO3	52	5.76	245	251	1.76	441	40	3
46	wbm-3-55347	Ba2ZrTiO6	225	5.68	192	198	2.23	441	10	4
47	wbm-2-35353	Ca2DyTaO6	87	4.46	111	116	3.79	439	10	4
48	mp-1218358	SrCaTi2O6	26	6.40	202	209	2.09	435	20	4
49	mp-38125	Sm2CdSe4	122	10.72	298	309	1.40	433	14	3
50	mp-1227686	Ca3Ta4(O6F)2	166	4.50	113	118	3.57	420	21	4
51	wbm-3-51131	Na2ZrO3	65	3.76	128	131	3.15	414	6	3
52	mp-1223520	KCa2Ta3O10	38	4.09	184	189	2.18	411	16	4
53	mp-5986	BaTiO3	99	5.63	218	223	1.78	397	5	3
54	mp-977360	AcF3	225	3.01	51	54	7.30	395	4	2
55	mp-759812	Li4Ti11O24	12	5.14	166	172	2.30	394	39	3
56	mp-769280	Ca5Ta4O15	164	4.67	127	131	2.98	391	24	3
57	mp-545665	WO3	191	6.14	313	319	1.19	381	12	2
58	mp-1518526	Sr2ZrTiO6	225	5.33	156	161	2.35	378	10	4
59	mp-1227336	BaSrTi2O6	123	6.65	206	212	1.75	371	10	4
60	mp-1227473	Ca2La2MgTi3O12	1	5.66	172	178	2.08	369	20	5
61	mp-1078457	Ba2ZrTiO6	225	5.68	157	162	2.22	359	10	4
62	wbm-1-40008	SrTiO3	99	5.96	190	196	1.80	354	5	3
63	wbm-1-38267	CaHfO3	127	4.39	91	96	3.67	352	10	3
64	mp-756214	YAlO3	167	4.25	66	70	5.00	351	10	3

Table 7: **Materials with DFPT-computed $\Phi_M > 350$, sorted by Φ_M .** While these are the highest-reward materials from a purely computational standpoint, synthesis of these high-arity compounds is made challenging by the proliferation of competing in higher dimensional chemical spaces. Many of the listed compounds therefore have a risk-reward profile of lower appeal than other materials in our dataset with lower-predicted Φ_M . A CSV file of this table is available on [GitHub](#).

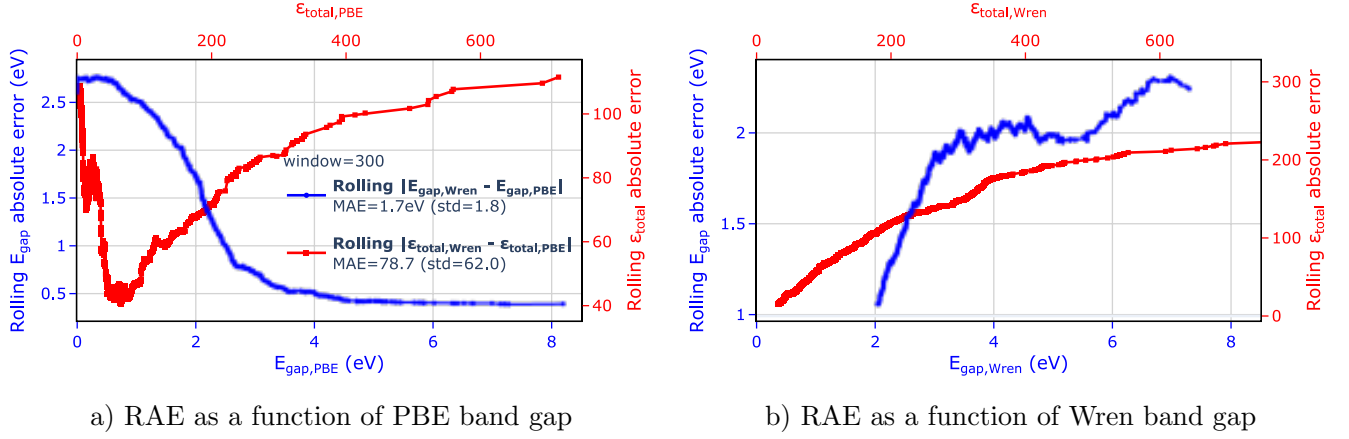


Figure 28: **ML predictions are low in the target region of $E_{\text{gap}}^{\text{PBE}} > 2 \text{ eV}$ and $\epsilon_{\text{total}}^{\text{PBE}} > 40$.** Rolling absolute error (RAE) of Wren band gap and dielectric constant predictions relative to DFPT.

poorly, if not ill-defined for a coarse-grained structural model with no concept of electronic degrees of freedom. We describe some attempts to mitigate this issue that achieved limited success but ultimately consider ML band gap prediction a high-impact but unsolved problem (see discussion [Section 4.9](#)).

Our ensemble of 10 Wren band gap models trained with L1 loss achieved a deceptively low $\text{MAE} = 0.2 \text{ eV}$ and a high coefficient of determination $R^2 = 0.969$. This is largely due to the aforementioned dataset imbalance. $243,095 / 319,601 = 76.1\%$ of the combined MP + WBM dataset are PBE metals. Not wanting to discard 3/4 of our training data, we attempted naive equal loss weighting across all samples as well as increased loss weighting of non-metals. Finally, we tried prepending a metal-nonmetal classifier to our band gap regressor to only predict the band gap for materials classified as non-metals. While the latter slightly decreased the false-positive rate, neither managed to significantly improve the overall performance of our band gap model nor fix this main failure mode in our discovery pipeline of metals classified as insulators/semiconductors. Many of the generated element substitution structures we predicted to have sizable band gaps turned out to be PBE metals. More recent efforts in training foundation models on giant datasets and then fine-tuning on smaller cognate datasets [86] have achieved impressive sub-100 meV/atom band gap MAEs and may be able to overcome this issue.

[Figure 28](#) plots the rolling absolute error of our Wren band gap and dielectric constant ensembles with respect to DFPT using a variable window size of 300 samples. In [Figure 28a](#)), the bottom x-axis spans the range of PBE-computed band gaps for which we also have Wren predictions. Similarly, the top x-axis spans the range of DFPT-computed dielectric constants for which we also have Wren predictions. In [Figure 28b](#)), we swap the x-axis values to PBE instead. That is Wren band gap predictions on the bottom x-axis and Wren dielectric constants on the top x-axis. The y-axis is identical in both subplots: the rolling Wren-vs-DFPT absolute error for band gaps on the left and dielectric constants on the right.

Figure 28a) reveals that the error in the dielectric constant shows a pronounced dip at intermediate ranges from about 40 to 120. This supports our initial argument for choosing dielectrics as the target material class for this discovery campaign. We hypothesized that by optimizing the trade-off between two opposing material properties, we can operate both the dielectric and band gap models in regions of good training support where ML models are most reliable and still discover materials with high Φ_M . In the case of the band gap model, this argument is less supported by the data. While the error in band gap prediction indeed drops significantly in our target region of $E_{\text{gap}} > 2 \text{ eV}$, the error does not increase again for extreme values but stays low even for outlier points beyond 5 eV. However, small errors on large band gaps do not negatively affect the chances of dielectric materials discovery and so are not in conflict with our objective. The issue with the band gap model is that its error for small band gaps is $> 2 \text{ eV}$ and therefore large enough to predict metals as insulators, thereby introducing false positives into our discovery pipeline.

Figure 28b) reveals that our workflow suffered from a negative feedback loop in that we purposely selected materials with large band gaps according to Wren which drew the bulk of our selection towards the lower end of the blue line. This line ends at a minimum band gap of 2 eV, indicating that no smaller Wren band gaps made it into our DFPT validation set. However, this is precisely the region where model error and its prediction are almost equal, resulting in a large number of false positive insulator predictions that turned out to be PBE metals.

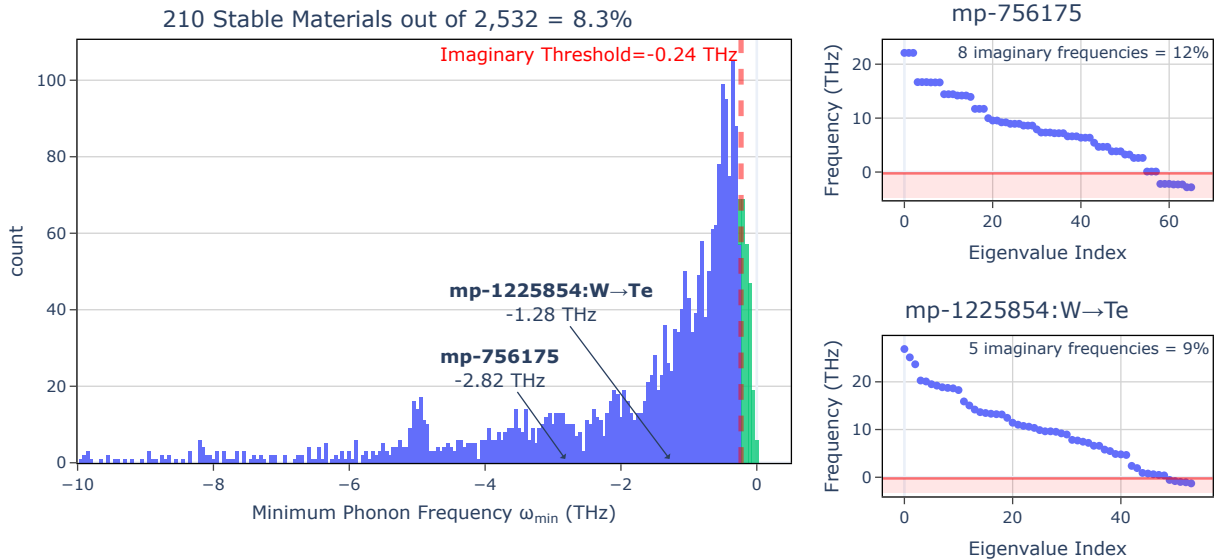


Figure 29: **Histogram of minimum phonon frequencies.** Negative frequencies should be interpreted as imaginary, i.e. dynamically unstable structures. The red dashed line indicates the dynamic stability threshold of $\omega_{\text{thresh}} = -1 \text{ meV} \triangleq -0.242 \text{ THz}$. Structures with minimum phonon frequency

4.8 Dynamic Stability Issues

Another significant failure mode in our workflow was the lack of dynamic instability filtering at the cheap ML stage. Due to using Wren [126], an energy-only model, we did not have access to harmonic phonon spectra from ML force predictions. Given Wren uses coarse-grained structural information for making energy predictions, its input features would either change discontinuously or, more often, remain unchanged when displacing atoms to obtain forces via finite differences. Even without this technical limitation, given the issues we observed with phonons predicted by CHGNet, M3GNet and MACE-MP in Section 5.4, the quality of ML phonons from a model not trained on forces would likely be much worse, possibly to the point of doing no better than a dummy dynamic stability classifier.

In the present implementation of the workflow, we only have access to the eigenspectrum of the dynamical matrix for a given structure after the DFPT calculation. At that point, the largest computational cost to screen a candidate material has already been incurred. As shown in Figure 29, the lack of dynamic instability filtering resulted in only 210 out of 2522 DFPT calculations (8.3 %) being performed on dynamically stable structures.

This is a major flaw of the present workflow implementation and could have been avoided by opting for a force model like M3GNet for the ML-filtering stage from the start. While CHGNet and MACE-MP came to be after the computational part of this project was completed, M3GNet pre-dates the start of this work (both 2022). In hindsight, M3GNet would have been a much better choice of model but was insufficiently considered since the performance gap between M3GNet and Wren demonstrated by Matbench Discovery was not yet known nor anticipated by us. In fact, part of the goal of this dielectric discovery workflow was to demonstrate the utility of coarse-grained structural models like Wren for accelerating materials discovery. This hypothesis was severely undermined by Matbench Discovery which found ML force fields to be both superior for PES exploration and more versatile than energy-only models.

This project outcome serves as a cautionary tale of embarking on long projects with false preconceptions poorly challenged due to insufficient methodological exploration. Favoring the familiar over the unknown may also have played a role. Wren was developed “in-house” by Rhys Goodall, myself and Rokas Elijosius whereas we had little experience with M3GNet at the time. The decision to use Wren was partially motivated by our false assumption at the time that ML potentials like M3GNet trained largely on the near-equilibrium structures in MP (where even the relaxation starting points, i.e. the highest-force configurations, are experimental ICSD ground state structures) would not be able to accurately extrapolate to unseen and unrelaxed structures generated by element substitution. The lesson learned is to perform more comprehensive method testing before committing significant computational resources to a new workflow. Discovery efforts should not be driven by the motivation to prove the benefits of a particular method but should commit significant effort to explore the best available tools for

the task at hand before entering the high-throughput phase. It also highlights the importance of Matbench Discovery which cured us of the misconception that coarse-grained structural energy-only models could be competitive with force models for PES exploration.

Fortunately, not much needs to change in the current workflow to incorporate a force model like MACE-MP which we showed in [Section 5.4](#) to achieve better than 80% dynamic stability classification accuracy (with respect to PhononDB PBE reference data [\[291\]](#), [\[292\]](#)). While outperforming CHGNet (65%) and M3GNet (70%), all 3 models outperform a random binary classifier and could potentially be combined into a committee classifier with even higher accuracy.

Assuming the 8.3 % dynamic stability ratio we observed is a base rate that holds generally when substitutionally-generating structures, and further assuming such a committee classifier would have a 10 % false stable rate (1 percentage point better than MACE-MP alone, see [Figure 34](#)), we could potentially reduce the number of DFPT calculations on dynamically unstable structures from 91.7 % to 9.17 %. Thus, we expect using universal MLFFs for dynamical instability filtering could almost invert the 210 / 2,312 stable-to-unstable ratio we obtained in the present work at the cost of a few extra minutes of compute time per material on a laptop. We are eager to test this in future work.

To what extent the imaginary frequencies of $\text{Bi}_2\text{Zr}_2\text{O}_7$ ([mp-756175](#)) and CsTaTeO_6 ([mp-1225854](#):W $\rightarrow$ Te) in [Figure 29](#) are indicative of ferroelectricity requires further investigation. The discrepancy between DFPT and synthesized structures may disappear under a displacive symmetry-lowering ferroelectric phase transition. However, given that imaginary modes associated with dynamic instabilities are anharmonic, a quantitative description would require moving beyond the harmonic approximation [\[293\]](#). Given neither structure appears to be polar, future work could search for unstable phonon modes that lead to a stable polar structure, a signature for ferroelectricity [\[294\]](#).

4.9 Discussion

We have implemented and run a high-throughput workflow for dielectric materials discovery that combines data-driven and first-principles methods to efficiently navigate the vast space of possible dielectric materials while balancing multiple material properties for optimal device performance. We show in [Table 6](#) that this combination achieves improved enrichment of materials that satisfy $\Phi_M = \epsilon_{\text{tot}} \cdot E_{\text{gap}} > 240 \text{ eV}$ than ab initio methods alone.

By deploying this workflow into practice, we identified and synthesized two candidate materials, CsTaTeO_6 and $\text{Bi}_2\text{Zr}_2\text{O}_7$. We performed Rietveld analysis to identify the synthesized structures, finding different polymorphs than the structures produced by the workflow, possibly due to the oversight of neglecting dynamic stability filtering, although entropic and defect effects

may also play a role which are much harder to account for in a high-throughput workflow. We nonetheless went on to measure band gaps and dielectric properties of both synthesized samples. $\text{Bi}_2\text{Zr}_2\text{O}_7$ shows strong promise for electronic applications given its measured band gap of 2.27 eV, dielectric constant of 20.5 and its relatively available constituent elements. CsTaTeO_6 is a black semiconductor with a low band gap of 1.05 eV and dielectric constant of 26, making it unsuitable for electronic applications. CsTaTeO_6 was substitutionally generated with no prior entries with this formula or even this chemical system in the ICSD or [MP](#). We thus demonstrated successful de novo synthesis on a challenging metastable phase and established a reference point for the dielectric properties of similar materials in this largely unexplored region of chemical space. Attempting metastable synthesis in an unknown chemical system is a high-risk undertaking, and the successful synthesis of CsTaTeO_6 demonstrates the benefits computational guidance can offer experiment. This experimental outcome serves as an early signal that ML-driven stability and property prediction has matured enough in reliability to warrant inclusion in complex multi-step workflows such as ours.

However, two major failure modes negatively impacted the hit rate of high- Φ_M materials in this work.

1. **Inaccurate band gaps** from the Wren ensemble caused a high false-positive insulator rate, predicting many generated structures as insulators which turned out to be PBE metals.
2. Lack of **dynamic instability filtering** at the ML stage due to unavailable phonon predictions from Wren.

Dynamic instability is straightforward to remedy. A future iteration of this workflow could use one of the top-performing force models from Matbench Discovery such as MACE-MP which not only provide more accurate energies but also allow filtering for unstable phonon modes at negligible compute cost (2 – 5 minutes on a laptop for most materials vs many hours with DFPT on a cluster). This would simultaneously reduce the number of false positives at an early stage of the workflow and provide a significant speedup, allowing for more materials to be screened for a given compute budget. The reason we did not do so here is largely chronological. This project started before we produced Matbench Discovery results. At the time, the performance gap between energy-only models and ML potentials was not yet clear to us (as detailed in [Section 4.8](#)).

The band gap prediction issue on the other hand is less straightforward to solve. We consider accurate band gap models to be an unsolved problem in materials informatics and encourage more efforts be directed at it. However, two recent works may offer significant improvements in this area.

- MatterGen [\[76, fig. 5b\]](#), a generative diffusion model, demonstrated an impressive ability to condition structure generation on band gaps much larger than the mode of its training

set. MatterGen’s training set is sharply peaked at $E_{\text{gap}}^{\text{PBE}} = 0$ with only long-tail, minuscule coverage of 3 eV and higher band gaps, yet its distribution of generated structures impressively achieved a mode at the band gap target of 3 eV.

- Ref. [86, table 4] demonstrates that pre-training on a large dataset of diverse materials prediction tasks can significantly improve band gap prediction accuracy on the Matbench $E_{\text{gap}}^{\text{MP}}$ task, reaching sub-100 meV/atom MAE after pre-training on many large but non-cognate tasks. This suggests current model architectures combined with large data curation and pre-training efforts can already achieve sufficiently accurate band gap predictions to significantly improve the hit rate of high- Φ_M materials in our workflow.

Models that predict a spectrum rather than a single scalar offer another promising avenue to pursue. Mat2Spec [295] proposed circumventing direct band gap prediction in favor of predicting the electronic density of states (eDOS). This allows inferring the band gap from the DOS and opens the door to more nuanced loss functions as well as increased regularization during training similar to how including forces in the loss function can regularize energy predictions. Sufficiently complex models with good inductive bias may learn more subtle features of the electronic structure from this approach than a direct band gap predictor. It should be noted, however, that Mat2Spec refrained from reporting the accuracy of band gaps inferred from their eDOS predictions, likely indicating more work is required to unlock such benefits. In summary, while it was not the main focus of this work, we believe there remains urgent need for improving ML band gap predictions.

However, the challenge of predicting band gaps in our workflow is not restricted to ML. It also applies to DFT. In the case of CsTaTeO₆, PBE exhibits an unusual severe overestimation $E_{\text{gap}}^{\text{PBE}} > 2.09$ eV of the experimental band gap of $E_{\text{gap}}^{\text{exp}} = 1.05$ eV. While defect chemistry may play a role in this effect, there are obvious computing limitations in a high-throughput workflow, making the simulation of defect effects cost-prohibitive. Possible improvements to narrow the gap between simulation and reality include incorporating a third computational filter to the funnel directly preceding experiment in the form of a hybrid functional such as HSE or HSE@GGA [237]. The latter only uses HSE to calculate energy levels at the band-edge k -points identified by GGA to lower cost. HSE@GGA assumes the band structure undergoes only a rigid shift from GGA to HSE which is often a good approximation. In either case, HSE would only be applied sparingly to compounds that have passed ML and PBE filters but before attempting experimental synthesis, thus keeping computational cost manageable while reducing false positives at the lab stage.

Given our workflow – despite its current limitations – nonetheless managed to result in the successful synthesis of materials with decent dielectric constants is cause for optimism about the potential for more meaningful discoveries in the future. In particular, combining MatterGen-like high-band-gap structure generation with MACE-MP dynamic stability filtering and HSE@GGA band gap prediction should make for a powerful dielectric materials discovery pipeline.

Chapter 5

Foundational Machine Learning Force Fields

This chapter is based on

[8] J. Riebesell *et al.*, “A Foundation Model for Atomistic Materials Chemistry” <https://arxiv.org/abs/2401.00096v1>. 2023–12–29 (under review at Nature)

except for [Section 5.9](#) which is based on [9] B. Deng *et al.*, “Overcoming Systematic Softening in Universal Machine Learning Interatomic Potentials by Fine-Tuning” <http://arxiv.org/abs/2405.07105>. 2024–05–11. Some analysis sections exclusively look at MACE-MP, while others compare it with CHGNet and M3GNet. The phonon analysis is more in-depth than either of the two papers and contains some results not previously published. In the MACE-MP project [8], I helped coordinate between the over 70 co-authors, directly overseeing several contributed analysis sections. I led the phonon and materials discovery analysis, and contributed to several others as detailed in the preamble of those sections ([5.5](#), [5.6](#) and [5.8](#)). I worked closely with Philipp Benner, Yuan Chiang and Ilyes Batatia on training multiple small, medium and large variants of the MACE-MP model (initially with the sole intent of submitting MACE to Matbench Discovery). I implemented the automated model checkpoint distribution in the [MACE repo](#) for easy sharing among contributors which sped up onboarding of the over 70 co-authors. I performed exploratory data analysis of the MPtrj training data reproduced in [Section A.2](#) to shed light on observed predictive biases in many of the 36 test cases we subjected the model to.

5.1 Introduction

DFT is perhaps the most successful tool for materials modeling to date [\[36\]](#), [\[94\]](#), [\[296\]](#), [\[297\]](#), [\[298\]](#), [\[299\]](#), [\[300\]](#) due to its attractive placement on the cost-accuracy Pareto front. However, the cubic scaling of compute cost with system size – while better than other ab initio methods with comparable or higher accuracy – still renders simulations of thousands of atoms prohibitively expensive. These include important use cases such as amorphous solids, condensed phase liquids, nanostructured materials, battery systems, and more.

Fast empirical interatomic potentials (force fields) have existed for years [\[301\]](#) and provide an alternative option with varying accuracy-generalization trade-offs. Yet they often fail to match DFT accuracy, especially around reactive events and phase transitions [\[302\]](#). Consequently, DFT has remained the main workhorse for materials simulation.

Recent years saw the rise of narrow ML interatomic potentials, custom-trained for a particular material or system, which compared to empirical force fields achieve significant accuracy gains at moderate increase in cost [\[80\]](#), [\[120\]](#), [\[124\]](#), [\[221\]](#), [\[303\]](#), [\[304\]](#), [\[305\]](#), [\[306\]](#). Their main drawback is the significant time and human effort needed to generate bespoke training sets (usually from

DFT reference data) when then used for model training and validation which are equally time-consuming and need to be repeated for each material class of interest [307]. This recurring upfront data curation cost that must be paid for every new chemical system is a major inhibitor to their widespread adoption and use by non-experts.

It would mark an important milestone in the lineage of work started nearly two decades ago by J. Behler and M. Parrinello [120] if we succeeded at training a single model whose compute cost scales linearly with system size yet accurately describes the potential energy surface (PES) across all chemical and structural spaces. This would enable reliable molecular dynamics (MD) simulations over unprecedented time scales and system sizes, opening up the immediate study of important physical processes that can only be studied in all-atom simulations of systems containing many thousands of atoms. These include vexing problems that have inhibited our understanding and resisted the rapid improvement of more performant sustainability-related materials such as metal-organic frameworks for direct air capture or solid-electrolyte interphases for increasing cycle life of batteries. Even though the physical laws that govern their behavior were discovered a century ago [308], actually simulating the outcome of these laws for large systems or over long-time scales needed for complex chemical reaction processes remains inaccessible to DFT even with today’s largest computers.

Hence there is an urgent need to develop computational tools that retain most of the accuracy and versatility of DFT yet ideally achieve linear scaling of compute cost with system size. The MEGNet model published in 2019 [125] marked an important first step in the direction of ML foundation models that could satisfy these lofty requirements. MEGNet enabled predicting various properties including band gaps, elastic moduli and most importantly formation energies of diverse inorganic crystals from having trained on the PBE [101] ground states of ~ 60 k materials covering all 89 elements present in the Materials Project (MP).

5.2 Related Work

M3GNet [68] (2022) and CHGNet [67] (2023) were important next milestones. These are $E(3)$ -invariant GNN potentials with the ability to predict not only energies but also forces from having trained on snapshots of MP PBE structure relaxations.

The proprietary GNoME [44] (NequIP architecture [80]) which currently ranks SOTA on Matbench Discovery (see Table 11) also starts from MP data (as well as OQMD [45], [46]), but uses a complex active learning workflow to generate and iteratively train on a dataset of inorganic crystals nearly two orders of magnitude larger than MPtrj (thus making the results more an indication of what’s possible with current architectures and data generation capabilities than something a regular research group can aspire to).

The above models were designed and trained primarily with materials discovery in mind, i.e. predicting thermodynamic stability of hypothetical crystals. In addition, the force fields are useful drivers for molecular dynamics. In fact, CHGNet and GNoME were both used to study alkali metal ion diffusion in battery materials.

Another contender for the most general and transferable force field currently in existence is the PFP model [66] from Matlantis (TeaNet architecture [65]) which is also proprietary. Its training set was recently expanded from 45 to 72 elements and at 20 M structures is more than an order of magnitude larger than MP, while also comprising large numbers of molecules and surfaces. PFP was shown to be able to simulate dynamics in solid-state ionic conductors, molecular adsorption and heterogeneous catalysis.

A largely separate lineage of ML force fields specialized for organic molecules (containing a much more limited set of elements comprising mainly C, H, N, O later extended to anions F, Cl, and S) is the ANI (and later AimNET) series of models [121], [309], [310].

However, the field has yet to deliver a comprehensive demonstration that a single ML potential can describe solid, liquid, and gaseous systems of materials and molecules across the periodic table and well beyond the training set distribution. This chapter presents MACE-MP, a new foundational interatomic potential that we thoroughly analyzed and benchmarked in a large collaborative effort involving over 70 co-authors and 36 test cases that we believe comes close to such a demonstration [8]. Built on the MACE force field architecture [81] and trained on the same MPtrj dataset curated for and released with CHGNet, we demonstrate its capabilities on an unprecedented range of qualitative and quantitative examples drawn from diverse areas of computational chemistry and materials science, including running stable molecular dynamics simulations in a wide variety of chemistries, predicting phonon spectra, calculating activation energies for point defect and dislocation motion, simulating solvent mixtures, combusting hydrogen gas, modeling a complete rechargeable battery cell, amorphous carbon, and much more; the sheer breadth of applications we found the model able to handle is illustrated in Figure 30. We find that this pre-trained *foundation model* exhibits remarkable zero-shot out-of-distribution performance, making it an all the more promising starting point for fine-tuning for specific applications needed by different modeling communities.

While the complete analysis of this model spanning more than 100 pages in [8] is too extensive to be reproduced here in its entirety, we instead focus on the parts of the analysis that I led or contributed to. These include the phonon spectra and elastic moduli benchmarks and the zero-shot OOD formation energy predictions. The medium-sized 4.7 M-parameter MACE model used throughout this paper cost ~ 2.6 k Nvidia A100 GPU hours to train. Compared to the smallest 2 M and largest 15.8 M parameter models we trained, medium offers the best cost-accuracy tradeoff.

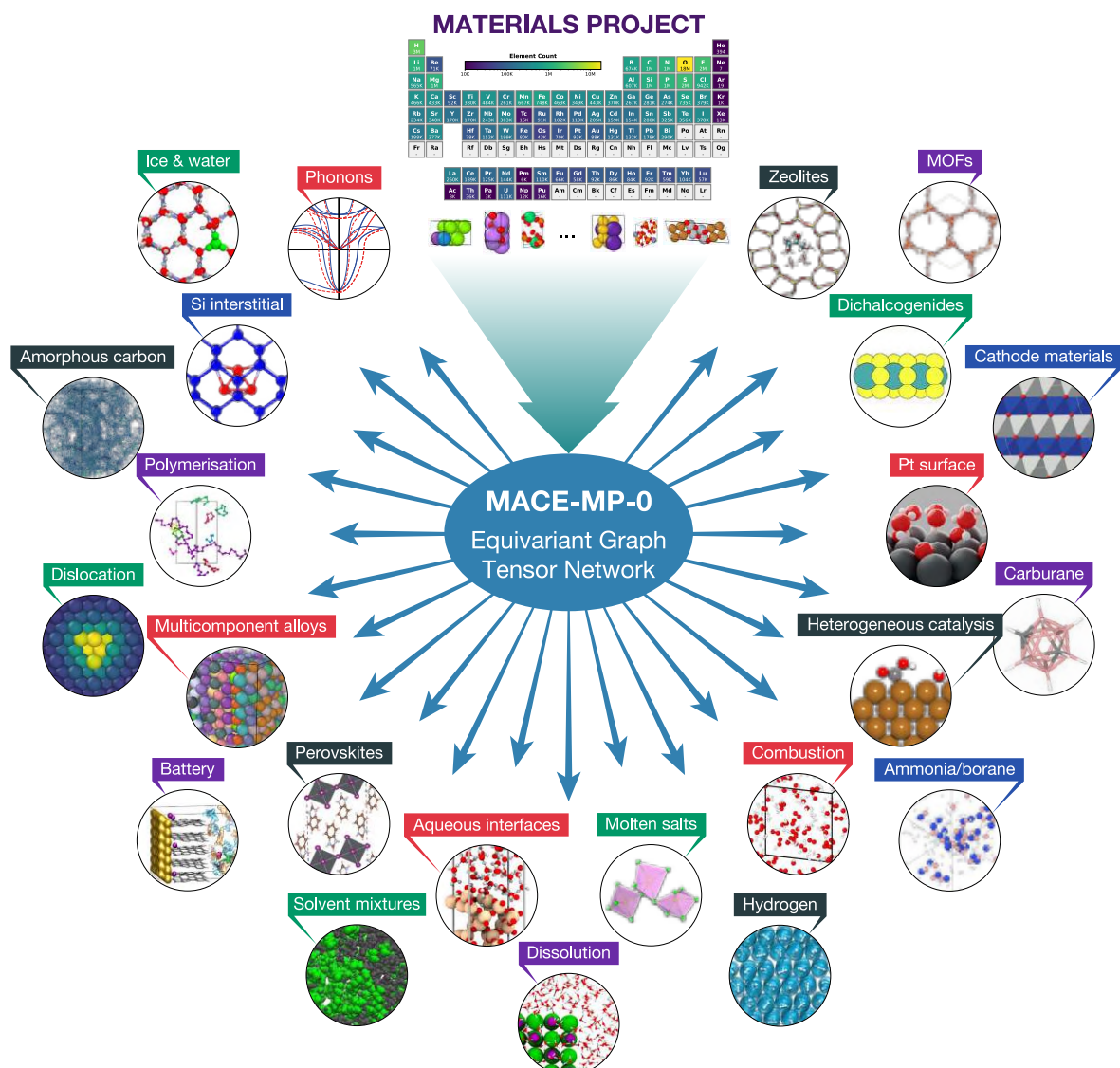


Figure 30: **MACE-MP** is an **unexpectedly versatile foundation model for materials simulation**. Trained only on the DFT relaxations of 154k Materials Project structures [12] which comprise almost entirely of inorganic crystals heavily dominated by oxides, MACE-MP can nonetheless drive molecular dynamics and other types of simulation across various chemistries in solid, liquid and gaseous phases. Reproduced from [8] with permission.

5.3 Training Protocol

The core piece of the MACE-MP pre-trained interatomic potentials are multi-body message passing (interaction) layers. In each layer, the message is encoded in an irreducible representation basis with C channels up to an angular frequency of order L . This is specified as $(128 \times 0e + 128 \times 1o + 128 \times 2e)$ for $C = 128$ channels and $L = 2$.

In each batch updating step, the weighted sum of Huber losses [311] of energy, forces, and stress incurred by all structures in a batch are averaged and back-propagated to update the

model parameters. For a given batch size N_b and number of atoms in each structure N_a , the full loss function reads

$$\begin{aligned} \mathcal{L} = & \frac{\lambda_E}{N_b} \sum_{b=1}^{N_b} \mathcal{L}_{\text{Huber}} \left(\frac{\hat{E}_b}{N_a}, \frac{E_b}{N_a}, \delta_E \right) + \frac{\lambda_F}{3 \sum_{b=1}^{N_b} N_a} \sum_{b=1}^{N_b} \sum_{a=1}^{N_a} \sum_{i=1}^3 \mathcal{L}_{\text{Huber}}^* \left(\frac{\partial \hat{E}_b}{\partial r_{b,a,i}}, F_{b,a,i}, \delta_F \right) \\ & + \frac{\lambda_\sigma}{9 N_b} \sum_{b=1}^{N_b} \sum_{i=1}^3 \sum_{j=1}^3 \mathcal{L}_{\text{Huber}} \left(\frac{1}{V_b} \frac{\partial \hat{E}_b}{\partial \epsilon_{b,ij}}, \sigma_{b,ij}, \delta_\sigma \right), \end{aligned} \quad (42)$$

where $\lambda_E, \lambda_F, \lambda_\sigma$ can be used to control the relative importance placed on energy, forces, and stress losses. We chose $(\lambda_E, \lambda_F, \lambda_\sigma) = (1, 10, 100)$ and trained for 200 epochs. The energy fine-tuned small model (“small-energy”) is subsequently trained for another 50 epochs with an increased emphasis on energy loss $(\lambda_E, \lambda_F, \lambda_\sigma) = (10, 1, 100)$. We used Huber deltas $\delta_E = \delta_F = \delta_\sigma = 0.01$. For forces, we use a conditional Huber loss $\mathcal{L}_{\text{Huber}}^*$ where the Huber delta δ_F is adaptive to the force magnitude on each atom. Specifically, δ_F decreases step-wise by a factor from 1.0 to 0.1 as the atomic force increases from 0 to 300 eV/Å. In other words, the force Huber loss is defined as:

$$\mathcal{L}_{\text{Huber}}^* \left(\frac{\partial \hat{E}_b}{\partial r_{b,a,i}}, F_{b,a,i}, \delta_F \right) = \begin{cases} \mathcal{L}_{\text{Huber}}(\dots, \delta_F) & \text{if } F_{b,a,i} < 100 \\ \mathcal{L}_{\text{Huber}}(\dots, 0.7\delta_F) & \text{if } 100 \leq F_{b,a,i} < 200 \\ \mathcal{L}_{\text{Huber}}(\dots, 0.4\delta_F) & \text{if } 200 \leq F_{b,a,i} < 300 \\ \mathcal{L}_{\text{Huber}}(\dots, 0.1\delta_F) & \text{if } F_{b,a,i} \geq 300 \end{cases} \quad (43)$$

Standardization of target variables (here energies, forces and stresses) with different scales has been proven to be important for weight initialization and training stability [312], in that a large spread of input or output will result in large and uneven weight values and cause model instability and slow convergence. After each message passing layer k , the node energies ϵ_a are scaled and shifted before sum pooling. The energy prediction of each structure therefore reads:

$$\hat{E} = \sum_{a=1}^N \left[\sigma \left(\sum_{k=1}^K \epsilon_a^{(k)} \right) + \mu_{Z_a} \right], \quad (44)$$

where K denotes the total number of message passing layers and $\epsilon_a^{(k)}$ is the node energy of atom a at layer k . μ_Z and σ are the mean of atomic energies and the root mean square (RMS) of the atomic forces computed over the training dataset. The predicted forces and stress are computed through PyTorch’s [151] automatic differentiation `torch.autograd` of total energy with respect to atomic positions and lattice strain tensor. The models are trained for 200 epochs with 40 – 80 NVIDIA A100 GPUs across 10 – 20 nodes with 4 GPUs each on the Perlmutter Cray EX supercomputer operated by the National Energy Research Scientific Computing Center (NERSC) at Lawrence Berkeley National Laboratory (LBNL). The change in validation metrics during training is shown in Figure 31. The total number of GPU hours used for training the medium model analyzed here was about ~ 2.6 k.

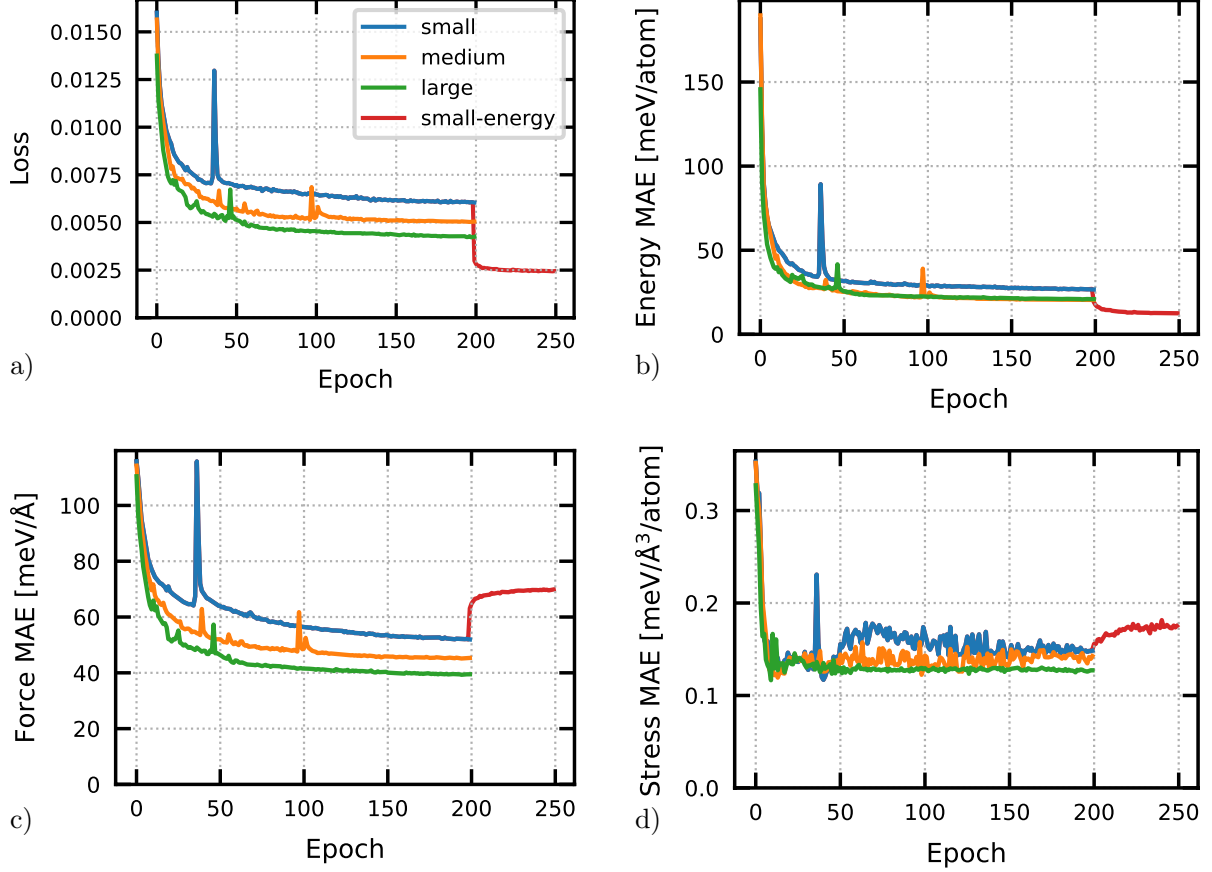


Figure 31: **Validation metrics during training of different sized MACE-MP models on MPtrj.** The medium model with 4.7 M trainable parameters offers the best tradeoff between cost and accuracy and was throughout the analysis in this section (Chapter 5). (a) Loss (eq. (42)). (b-d) Mean absolute error (MAE) of (b) energy per atom, (c) force, and (d) stress per atom.

5.4 Phonon Analysis

I led the work in this section, performed all the calculations, generated the figures (including interactive phonon band structure and DOS figures implemented with `plotly` in `pymatviz` [313]), drafted the initial text and extended the analysis to cover 3x more materials ($97 \rightarrow 299$) after MACE-MP [8] appeared as a preprint. I would like to thank Janine George and Aakash A. Naik at Federal Institute of Materials Research and Testing (BAM), Berlin, Germany for guidance and help with ingesting PhononDB PBE data as well as convergence testing of their VASP parameters in confirming reference data quality.

This section presents an in-depth comparison of phonon predictions from MACE-MP, CHGNet and M3GNet. Phonons (from the Greek word for sound) are quasiparticles that describe the collective vibrations of atoms in a periodic crystal. The phonon spectrum measures the energy of such vibrations as a function of atomic momenta. It gives a complete description of possible collective vibrations of atoms for a given crystal lattice. The discrete energy levels such sound waves can acquire turn into a multitude of continuous bands when plotted in reciprocal space as in Figure 32. This so-called phonon band structure is an information-

dense representation of the crystal’s vibrational energy spectrum. Lattice vibrations of crystals underpin many application-relevant material properties, including volumetric and constant-pressure heat capacities, electrical and thermal conductivity, thermal expansion and speed of sound. In addition, phonons are key to describing physical phenomena such as phase transitions [314], [315], superconductivity [316], [317] and ferroelectricity [318], [319] which all play critical roles in many materials applications. But most relevant to this work and crystal stability prediction more generally, the phonon spectrum also encodes the dynamic stability of crystals as well as entropic contributions S_{vib} to the free energy $A = U - TS$ [39], [320], [321] where $S = S_{\text{vib}} + S_{\text{config}}$ is the sum of vibrational and configurational entropy. Hypothetical crystal structures may exhibit lattice vibrations whereby atoms can lower their energy by gaining momentum. That would mean the crystal flows apart by displacing atoms in real space along the corresponding vibrational eigenvector, thereby transforming into a different crystal structure that inhabits a lower-energy saddle point or the true ground state of the free energy surface. Such dynamically unstable crystals cannot be realized experimentally and should be filtered out when attempting synthesis of novel structures to avoid wasted effort. Thus we view phonon analysis as a natural continuation of Matbench Discovery for measuring the ability of ML to accelerate materials stability prediction by accounting for previously neglected modes of crystal instability. In contrast to [Matbench Discovery](#) which considered only thermodynamic stability, benchmarking MLFFs on phonons allows progressing from a zero Kelvin energy surface to a free energy surface that enables a more holistic assessment of stability. Finally, finding that MLFFs can accurately model phonons would simultaneously open the door to directly validate ML against many new experimental results in various materials science domains by comparing ML-predicted thermal expansion, speed of sound, etc. to observed values.

5.4.1 Brief Primer on Phonons

Phonons are quanta of atomic vibrations in a crystal lattice and measure the curvature of the energy surface. The connection to curvature becomes apparent through the dynamical matrix ω which consists of the second derivative of the PES with respect to displacement of ionic positions in all possible directions:

$$\omega_{ij} = \frac{1}{\sqrt{m_i m_j}} \frac{\partial^2 E}{\partial \mathbf{u}_i \partial \mathbf{u}_j} \quad (45)$$

where m_i is the mass of atom i , $\mathbf{u}_i = (u_x, u_y, u_z)$ is its displacement and E is the potential energy and its second derivative gives the local PES curvature in different atom displacement directions. ω is a $3N \times 3N$ matrix (N is the number of atoms) whose eigenvalues $\omega_{\mathbf{q}\nu}^2$ are the squares of the phonon frequencies and whose eigenvectors $\mathbf{q}$ are called the phonon modes (or just phonons). As such, they may constitute a higher fidelity test for the quality of an ML force field than Matbench Discovery which restricted PES evaluation to forces (first derivative of E)

needed for relaxation and reading of the PES value at the relaxed geometry (zeroth derivative of E).

The eigenvalues $\omega_{\mathbf{q}\nu}$ can be positive or negative depending on whether the local energy surface curves upwards or downwards. Positive eigenvalues correspond to a quadratic increase in energy with atomic displacement, while negative eigenvalues indicate a decrease in energy when displacing atoms along the vibration’s corresponding eigenvector $\mathbf{q}$. At a local minimum of the PES, all eigenvalues are positive, whereas at a saddle point, most eigenvalues are positive, except for those associated with energy-lowering directions.

Since phonon frequencies are the square root of the eigenvalues of ω , they must be either positive real numbers or purely imaginary⁹.

We suspect phonons may offer a higher fidelity test for the quality of ML potentials compared to methods that restrict PES testing to forces (first derivative) needed for relaxation and the PES value at the relaxed geometry (zeroth derivative) such as Matbench Discovery.

5.4.2 Methodology

Harmonic (non-interacting) phonons can be calculated from the restoring force on atomic displacements and require highly accurate force predictions to be physical, making them a sensitive test for the fidelity of ML potentials. A model trained on PBE forces should be able to reproduce PBE lattice vibrations [322], [323]. To assess the accuracy of MACE-MP restoring forces, we compare phonon modes predicted via the finite-displacement method as implemented in Phonopy [291], [292], [324] and atomate2 [7] with the Togo phonon database [291], [292]. We restrict our analysis to materials in which magnetism and U-correction do not play a role since the Togo database was calculated at the PBE level of theory but without consideration of magnetism or U-corrections and so will be at odds with the model’s MP training data which uses said corrections for such systems. Additional inaccuracies in the comparison may stem from unconverged kinetic energy cutoffs of the plane-wave expansion or insufficient k -point density. Based on our analysis, Togo DB calculations used the VASP-recommended sufficiently high kinetic energy cutoff ($1.3 \cdot \text{ENMAX}$ of the highest ENMAX indicated in a compound’s pseudopotential files for the PAW method). In most cases, the reciprocal k -point density used matches that of MPRelaxSet for structure optimization and force calculations. However, some reciprocal k -point densities are slightly lower compared to MPRelaxSet by 29 ± 15 points for structure optimization and 35 ± 21 points for force calculations. These are small deviations, leading us to conclude PhononDB results serve as reasonable high-throughput reference data for an MLFF phonon analysis. To minimize the discrepancy between ML and Togo results, we simulate identical supercells. In addition, we remove the non-analytic corrections derived from Born charges which

⁹In band structure plots such as Figure 32b), imaginary frequencies are plotted as negative numbers for convenience but y -axis labels should be read as $-1 \rightarrow i, -2 \rightarrow 2i, \dots$

are unavailable from ML potentials with no concept of polarization.¹⁰ These corrections could be obtained even without changes to the model by a hybrid workflow that performs a DFT static calculation on the MLFF-relaxed structure to obtain Born effective charges and posthoc apply the resulting non-analytic corrections to the MLFF phonon spectrum. Since we are mainly interested in the part of the phonon spectrum the model **can** predict in order to gain an understanding of the fidelity of MACE-MP restorative forces, we did not pursue this more involved workflow. However, we hope to soon provide an off-the-shelf implementation of this hybrid workflow compatible with all force fields in **atomate2** [7] (currently CHGNet, MACE-MP, M3GNet, GAP, NequIP and SevenNet). Both MP [12] and PhononDB [291] provide phonons for several thousand compounds, calculated with density functional perturbation theory (DFPT) and frozen phonons, respectively. Unfortunately, both databases opted to use the PBEsol functional for their phonon calculations, introducing a systematic error when benchmarking models trained on PBE data. PBEsol phonon spectra tend to be slightly stretched compared to PBE, i.e. high frequencies in particular are slightly higher with PBEsol due to stiffer bonds. As such any underpredictions, especially of the highest frequency modes, cannot be purely attributed to model error. However, the error contribution from systematic differences between PBE forces seen during training and PBEsol forces giving rise to the reference phonon spectra is likely small. While DFPT and the frozen phonon method using the same flavor of DFT should in principle give comparable results, we chose PhononDB as our source of reference data to eliminate another potential source of systematic discrepancy, however minor, between reference data and model predictions.

5.4.3 Results

Figs. 32 and 33 show examples of phonon band structures and density of states (DOS) from MACE-MP and CHGNet compared to PBE data from PhononDB [291], [292]. Of the 299 materials we tested, we show 3 for each model where DFT and ML agree well (Figure 32a-c), and one of the compounds of largest qualitative disagreement (Figure 32d)). From mere visual inspection of many dozens of phonon band structures and DOS, we find MACE-MP generally achieves better agreement with PBE than CHGNet or M3GNet. While the band structures give a visual and intuitive understanding of the quality of MLFF phonons, the following analysis will focus on quantitative performance metrics to more rigorously assess the usefulness of MLFF phonons.

¹⁰The polarization and resulting Born effective charges can in principle be learned by current MLFFs by adding external perturbations such as electric and magnetic fields to the model input and including response properties such as electric/magnetic enthalpy, polarization and Born charges in the loss function. An impressive demonstration of this recently appeared in [84] but much more DFT training data at non-zero external fields needs to be generated before a model with the chemical scope of MACE-MP and an understanding of external perturbations can be trained.

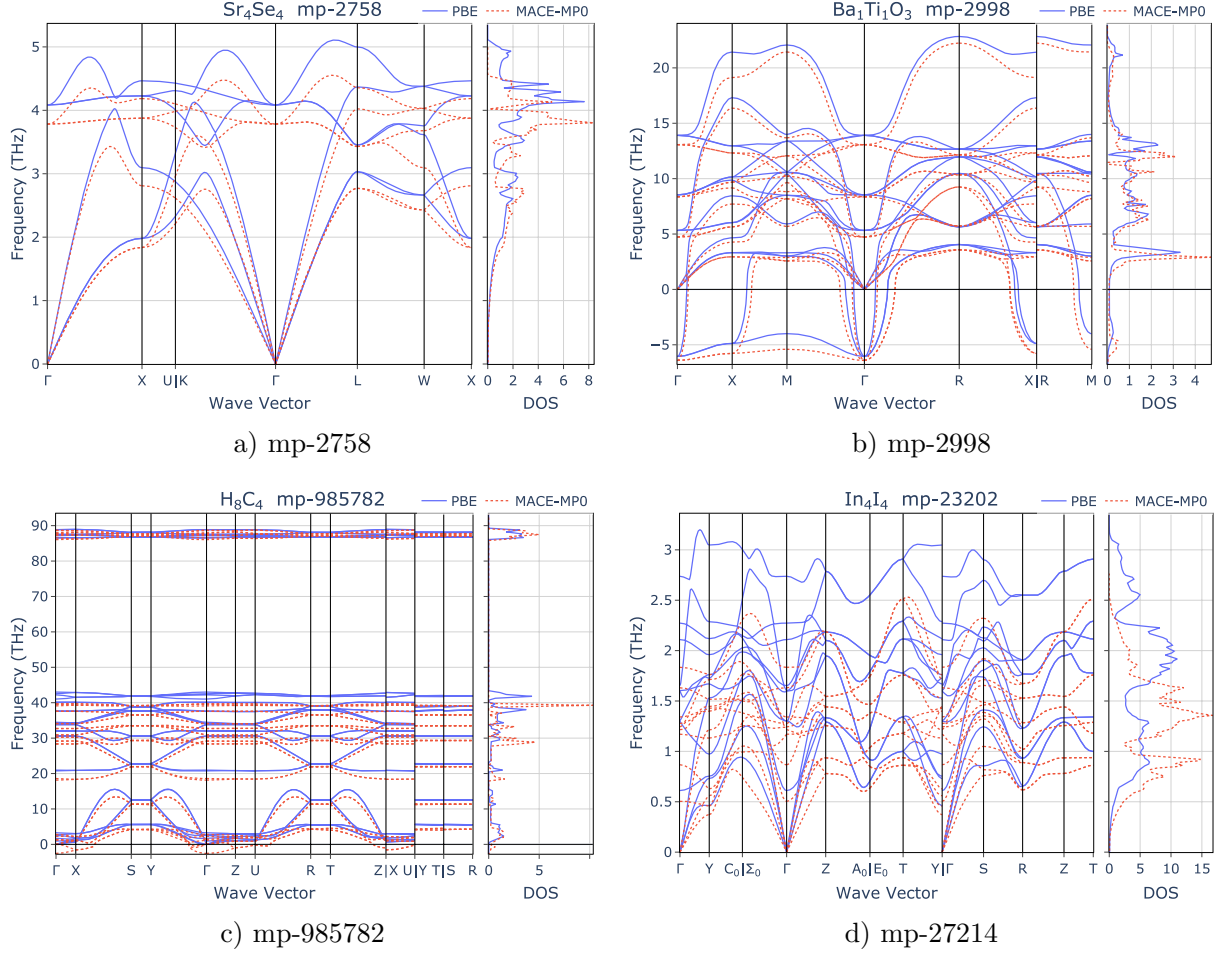


Figure 32: **Example phonon bands and DOS from MACE-MP.** Comparison of PBE and MACE-MP phonon band structures and density of states (DOS). PBE reference data was taken from PhononDB [291], [292]. [mp-2758](#) Sr_4Se_4 (Figure 32a)), [mp-2998](#) BaTiO_3 (Figure 32b)) and [mp-985782](#) H_8C_4 (Figure 32c)) show examples of particularly good phonon bands while [mp-23202](#) In_4I_4 (Figure 32d)) is a particularly bad example. On average across a diverse set of 299 randomly chosen MP materials, $\lesssim 50\%$ achieve what we would consider useful agreement. We present more quantitative analysis in Figure 34 below.

Figure 34 shows confusion matrices for the presence of imaginary frequencies at any q -point as predicted by MACE-MP, CHGNet and M3GNet compared with PhononDB PBE reference data. We use a tolerance of 0.01 THz, meaning any phonon modes above -0.01 THz are not considered imaginary to account for numerical imprecision. For example, phonon modes of -90 mTHz would not be considered dynamically unstable while -0.11 THz would. Using those settings yields an imbalanced classification task biased towards dynamic instability with 68% of PBE phonons containing imaginary modes and 62% at the Γ point. The confusion matrices are colored white for low values and dark blue for high values. Optimal performance corresponds to dark blue on the diagonals and white on the off-diagonals.

Surprisingly, there is a much larger performance gap in the accuracy of predicting the presence of imaginary modes between MACE-MP and CHGNet than what we observed in

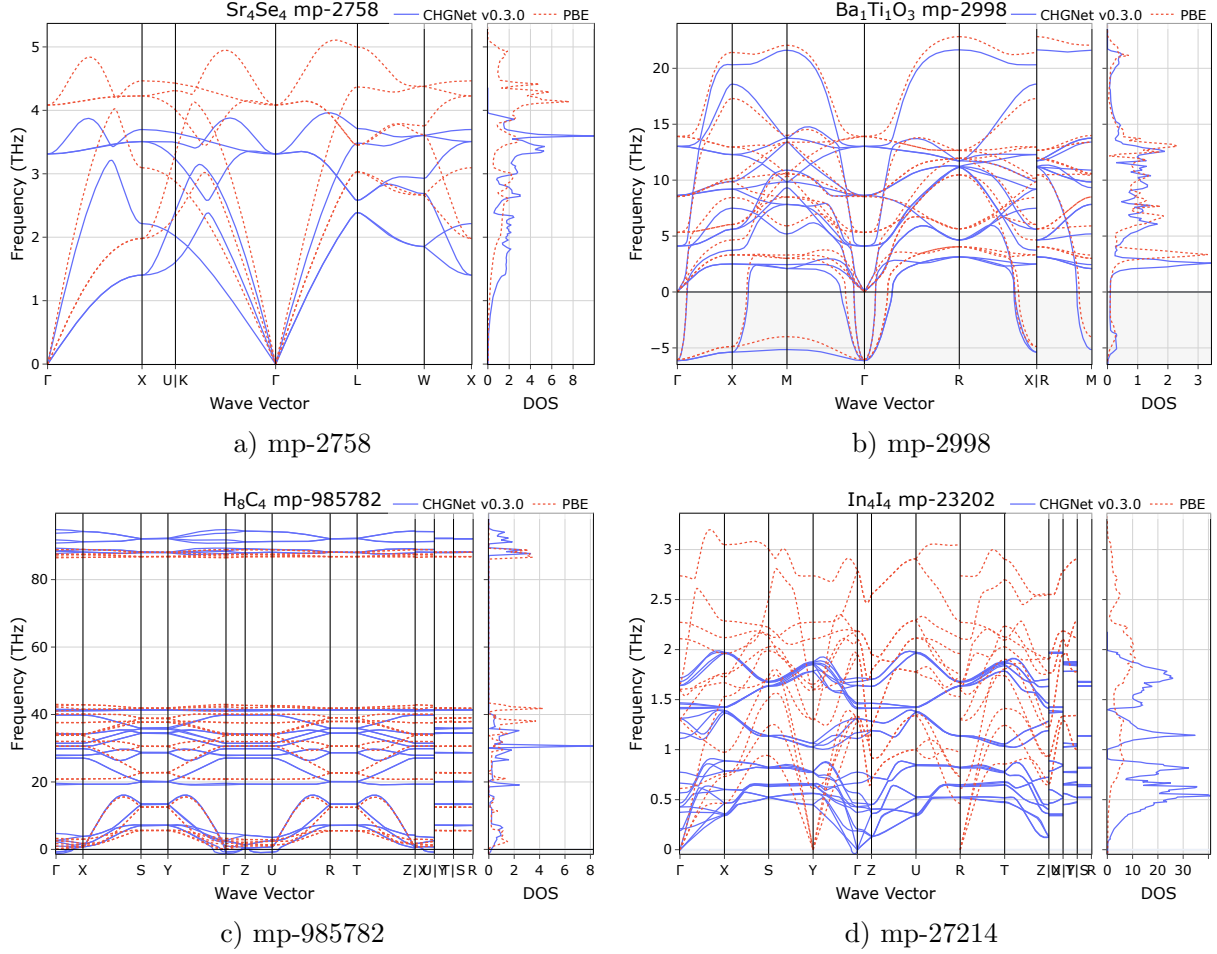


Figure 33: **Example phonon bands and DOS from CHGNet.** Same as Figure 32 except comparing CHGNet instead of MACE-MP with PBE phonon band structures and density of states. Across a diverse set of 242 randomly chosen MP materials, we generally observe MACE-MP achieve better agreement with PBE phonons than CHGNet. In all cases, the tolerance for imaginary frequencies was set to -0.01 THz.

Matbench Discovery for stability prediction (see Table 2). MACE-MP achieves an accuracy of 81% for predicting the presence of imaginary frequencies. This compares to 65% and 57% for CHGNet. Even M3GNet which performed noticeably worse in Table 2 outperforms CHGNet on imaginary phonon modes by a meaningful margin, achieving 70% and 67% accuracy for the same tasks. Looking at the top right quadrant in both CHGNet confusion matrices, we can localize CHGNet’s failure mode to a high false unstable rate of $\sim 40\%$. In other words, CHGNet predicts 34% of structures as dynamically unstable that are PBE-stable (Figure 34b)). This is a much higher false unstable rate than M3GNet (29% and 24%) and even more so compared to MACE-MP (8% and 8%). It is currently unclear why CHGNet is strongly biased towards false unstable phonons given that it was trained on a larger dataset and performed better at materials discovery than M3GNet. Work is ongoing on a more comprehensive phonon analysis that may shed light on this discrepancy.

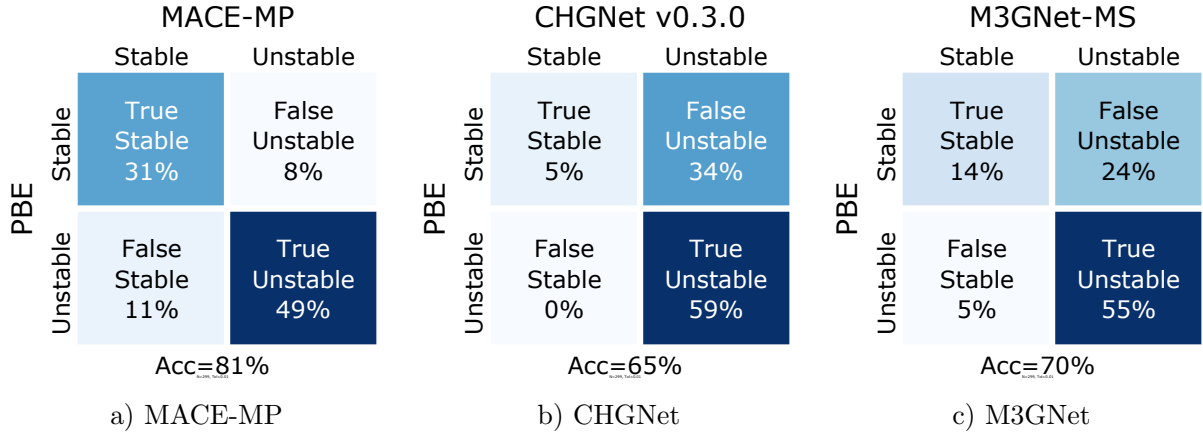


Figure 34: **ML foundation models trained purely for materials discovery can predict dynamic stability with useful accuracy.** Confusion matrices for the binary classification task of predicting the presence of imaginary frequencies at any q -point for MACE-MP, CHGNet and M3GNet on 299 randomly chosen MP materials. See Table 10 for a full list of MP IDs, chemical formulas and the supercells used for each material in this analysis. The reference data is skewed towards instability with a global PBE unstable rate of 68% and a Γ -unstable rate of 62%.

Table 8 presents a summary of the phonon band structure and density of states metrics averaged over the 299 randomly chosen MP materials listed in Table 10. MACE-MP outperforms CHGNet and M3GNet in all metrics except false negative rate (FNR) for binary classification of imaginary phonon modes, where MACE-MP’s FNR of 0.18 is much higher than CHGNet’s 0.01 and even M3GNet’s 0.09. That is, MACE-MP will predict a dynamically unstable structure as stable almost 20% of the time. In return, however, MACE-MP achieves a much lower false positive rate (FPR) which measures the fraction of PBE-dynamically stable structures that are

Model	MACE-MP	M3GNet-MS	CHGNet v0.3.0
Prec. $\uparrow$	0.79	0.59	0.53
Recall $\uparrow$	0.82	0.91	0.99
F1 $\uparrow$	0.81	0.72	0.69
ROC AUC $\uparrow$	0.80	0.64	0.56
Acc. $\uparrow$	0.80	0.64	0.56
FPR $\downarrow$	0.21	0.63	0.87
FNR $\downarrow$	0.18	0.09	0.01

Model	MACE-MP	CHGNet v0.3.0	M3GNet-MS
MAE _{ph DOS} (THz) $\downarrow$	3.87	4.83	5.44
$R^2_{\text{ph DOS}}$ $\uparrow$	0.03	-0.04	-0.02
MAE $_{\Omega^{\text{max}}}$ (THz) $\downarrow$	1.79	2.31	4.34
$R^2_{\Omega^{\text{max}}}$ (THz) $\uparrow$	0.99	0.98	0.94

a) Imaginary mode classification metrics

b) Ω_{max} and phonon DOS regression metrics

Table 8: **MACE-MP comes closer to being a universal ML phonon simulator and dynamic stability predictor than CHGNet and M3GNet.** Prediction metrics comparing MLFF with PhononDB PBEsol band structures and density of states averaged over 299 randomly chosen MP materials. Tolerance for the presence of imaginary frequencies was set to 0.01 THz. The color scale goes from white = poor to dark blue = good performance. With the notable exception of false negative rate (FNR) and recall for the presence of imaginary phonon modes at any q -point, MACE-MP outperforms CHGNet and M3GNet in all metrics. The performance gap is significant in all cases except for coefficients of determination $R^2_{\Omega^{\text{ph DOS}}}$ and $R^2_{\Omega^{\text{max}}}$.

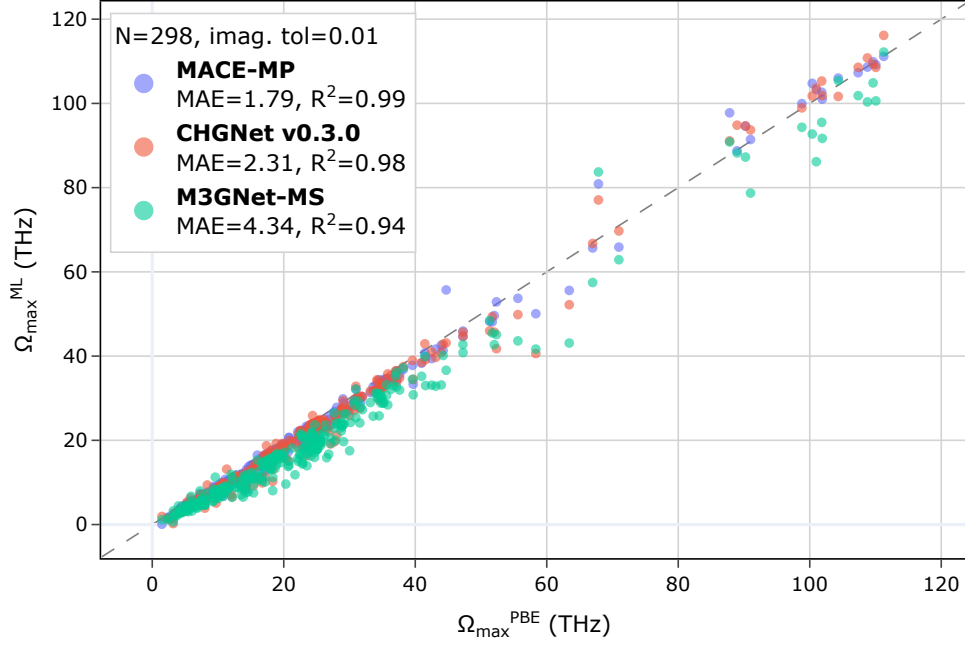


Figure 35: **MLFFs can predict the highest phonon band frequency with useful precision.** Parity plot of PBE vs ML foundation model predictions of the highest frequency present at any q -point. Despite the good accuracy of $\Omega_{\max}^{\text{ML}}$, the consistent trend of underestimated phonon frequencies stemming from lower-than-PBE predicted restorative forces is apparent.

predicted as unstable. MACE-MP’s $\text{FPR} = 0.12$ compares very favorably to CHGNet’s 0.87 and M3GNet’s 0.63.

In Table 8b), $R_{\Omega_{\text{ph DOS}}}^2$ and $R_{\Omega_{\max}}^2$ measure how much of the variation in the discretized PBE phonon DOS and the highest phonon frequency across the 299 test materials is captured by each model.

Figure 35 compares the highest band frequencies (a.k.a. bandwidth) predicted by ML force fields and DFT. The bandwidth correlates with the stiffness of the material’s strongest bond, making it a physically meaningful quantity in its own right. All three models demonstrate acceptable agreement with DFT, with MACE-MP ($R^2 = 0.99$, $\text{MAE} = 1.79$) slightly outperforming CHGNet ($R^2 = 0.98$, $\text{MAE} = 2.31$) which in turn beats M3GNet ($R^2 = 0.94$, $\text{MAE} = 4.35$) by a margin. The largest discrepancies between PBE and ML bandwidths are shown in Table 9. MACE-MP’s largest error of 13 THz are 27% and 26% lower than CHGNet’s 18 THz and M3GNet’s 20 THz.

However, while MACE-MP errors are generally smaller and more frequently include overpredictions of phonon frequencies, in general, all models exhibit the same systematic error direction of typically underestimating the bandwidth. This observation is not limited to phonon predictions but rather one example of many across several projects indicating a more general trend that ML force fields over-soften the PES (i.e. underestimate the forces they were trained on), highlighting a broader unresolved issue in the ML force field design and training methodology. For many more examples of PES softening, we refer the reader to [8, esp. secs. 2.1 on water, A.1

Model	MP ID	Formula	$\omega_{\max}^{\text{ML}}$ [THz]	$\omega_{\max}^{\text{PBE}}$ [THz]	$\omega_{\max}^{\text{err}}$ [THz]	$\omega_{\max}^{\text{err rel}}$
M3GNet	mp-570130	Al_6H_{18}	43.1	63.4	-20.3	-16.3 %
CHGNet	mp-642996	$\text{Li}_{24}\text{H}_{36}\text{Os}_6$	40.6	58.3	-17.7	-14.2 %
MACE-MP	mp-23803	$\text{Ga}_2\text{H}_2\text{O}_4$	80.9	67.9	13	10.4 %

Table 9: **Worst-case phonon band width predictions for each model out of 299 materials** (listed in [Table 10](#)). $\omega_{\max}^{\text{err}}$ is the signed error between ML-predicted and PBE phonon bandwidths $\omega_{\max}^{\text{ML}}$ and $\omega_{\max}^{\text{PBE}}$. $\omega_{\max}^{\text{err rel}}$ is the relative error in percent.

on self interstitials, A.2 on amorphous silicon, A.13 on metal defects, A.14 on bulk diffusion, A.25 on carborane barrier heights, A.26 on transition metal dichalcogenides, B.1 on phonons]. We suspect addressing this issue will lead to more reliable ML force fields with enhanced general applicability. In this case, the MAE can be lowered to between ~ 1 and ~ 2 THz by applying a constant ‘PES-stiffening’ shift that uniformly moves all predicted frequencies up by $\sim 1 - 3$ THz.

Despite this limitation, the acceptable MAEs and high R^2 in this zero-shot test are encouraging. They indicate significant potential for ML force fields to become accurate universal bandwidth predictors.

[Figure 36](#) shows the distribution of errors in the highest and lowest phonon frequencies comparing to PhononDB PBEsol results for 299 randomly chosen MP materials. $\Delta\Omega_{\max}$ strongly skews negative for all 3 models, indicating a systematic underestimation of the highest phonon frequency. MACE-MP has the most sharply peaked and closest-to-zero $\Delta\Omega_{\max}$ distribution. The green-shaded area indicating ± 1.5 THz $\Delta\Omega_{\max}$ contains the modes of $\Delta\Omega_{\max}$ for MACE-MP and CHGNet while M3GNet has a larger error and falls outside this range. In contrast, $\Delta\Omega_{\min}$ is almost uniformly distributed, showing no clear systematic bias for any model. [Section 5.9](#) explains this difference in (mis-)calibration between $\Omega_{\min}$ and $\Omega_{\max}$.

Highly specialized models for one element can achieve much lower errors of 0.1 – 0.2 THz over the whole band structure [\[322\]](#). MACE-MP, let alone CHGNet or M3GNet, in their current form are far from achieving such accuracy. However, the deviations are close to results from models that specifically train only on the highest phonon peak on MatBench [\[60\]](#). Here, the best models arrive at mean absolute errors of 0.8 – 1 THz for the last phonon DOS peak, which is a similar but *not* identical quantity to the bandwidth. The highest phonon peak is usually slightly but sometimes a lot smaller than the bandwidth. It’s worth reiterating that neither MACE-MP nor its training data was tailored in any way for phonon predictions. Even so, MLFF phonon band structures, especially those of MACE-MP, achieve at least qualitative agreement with PBE for most materials, which can be sufficient for many practical applications (e.g. models that screen for low or high thermal conductivity [\[325\]](#)).

While MACE-MP’s dynamic stability classification accuracy of 81% ([Figure 34](#)) outperforms dummy accuracy of 50%, it also leaves room for improvement. This is even more true for

MP ID	Formula	Supercell	Atoms	MP ID	Formula	Supercell	Atoms
mp-39	Tl ₂	4, 4, 4	2	mp-149	Si ₈	2, 2, 2	8
mp-422	Be ₄ S ₄	2, 2, 2	8	mp-661	Al ₂ N ₂	4, 4, 2	4
mp-886	Ga ₈ O ₁₂	1, 3, 2	20	mp-1022	K ₈ S ₄	2, 2, 2	12
mp-1105	Ba ₂ O ₄	3, 3, 2	6	mp-1224	Hg ₄ O ₄	2, 2, 4	8
mp-1253	Ba ₄ Se ₄	2, 2, 2	8	mp-1265	Mg ₄ O ₄	2, 2, 2	8
mp-1392	Zn ₈ P ₁₆	1, 2, 2	24	mp-1541	Be ₄ Se ₄	2, 2, 2	8
mp-1666	Th ₈ S ₂₀	2, 1, 2	28	mp-1784	Cs ₄ F ₄	2, 2, 2	8
mp-2657	Ti ₂ O ₄	2, 2, 4	6	mp-2659	Li ₂ N ₆	2, 3, 3	8
mp-2667	Cs ₁ Au ₁	4, 4, 4	2	mp-2672	K ₈ O ₈	2, 2, 2	16
mp-2691	Cd ₄ Se ₄	2, 2, 2	8	mp-2697	Sr ₂ O ₄	3, 3, 2	6
mp-2739	Te ₄ O ₈	2, 2, 2	12	mp-2741	Ca ₄ F ₈	2, 2, 2	12
mp-2758	Sr ₄ Se ₄	2, 2, 2	8	mp-2763	Nd ₂ O ₃	3, 3, 2	5
mp-2782	Zn ₈ P ₁₆	2, 2, 1	24	mp-2784	Na ₈ Te ₄	2, 2, 2	12
mp-2789	N ₁₂ O ₂₄	1, 1, 1	36	mp-2798	Si ₂₄ P ₂₄	3, 1, 1	48
mp-2892	Ba ₄ Nd ₈ O ₁₆	1, 3, 1	28	mp-2998	Ba ₁ Ti ₁ O ₃	3, 3, 3	5
mp-3107	Ga ₉ P ₃ O ₂₁	1, 1, 2	33	mp-3164	Ba ₄ S ₄ O ₁₆	1, 2, 2	24
mp-3193	Na ₈ Si ₈ O ₂₀	2, 2, 1	36	mp-3472	Pb ₄ S ₄ O ₁₆	1, 2, 2	24
mp-3491	Nb ₄ Sb ₄ O ₁₆	2, 3, 1	24	mp-3700	Li ₄ Y ₄ F ₁₆	2, 2, 1	24
mp-3707	Ca ₄ Ge ₈ O ₂₀	2, 1, 2	32	mp-3996	Ga ₃ As ₃ O ₁₂	2, 2, 1	18
mp-4056	Ca ₄ Sc ₈ O ₁₆	1, 4, 1	28	mp-4192	Ca ₁₂ As ₄ N ₄	2, 1, 2	20
mp-4473	Ba ₄ Cu ₈ Se ₈	2, 3, 1	20	mp-4511	La ₂ S ₁ O ₂	3, 3, 2	5
mp-4608	K ₃ P ₃ F ₁₈	1, 1, 2	24	mp-4651	Sr ₄ Ti ₄ O ₁₂	2, 2, 1	20
mp-4662	Sc ₄ O ₄ F ₄	2, 3, 2	12	mp-4718	Nd ₄ Ta ₄ O ₁₆	2, 1, 3	24
mp-4809	Ga ₄ Hg ₂ S ₈	2, 2, 2	14	mp-5285	Sr ₄ S ₄ O ₁₆	1, 2, 2	24
mp-5309	K ₄ Au ₄ F ₁₆	2, 2, 1	24	mp-5338	Zn ₈ Se ₈ O ₂₄	2, 1, 1	40
mp-5368	Li ₆ Ge ₃ F ₁₈	1, 1, 2	27	mp-5387	Y ₄ Nb ₄ O ₁₆	2, 1, 3	24
mp-5487	Sr ₁₂ Si ₄ O ₂₀	2, 2, 1	36	mp-5967	Ti ₄ Co ₄ Sb ₄	2, 2, 2	12
mp-5970	Ba ₄ Cu ₈ S ₈	2, 3, 1	20	mp-6037	Ba ₈ Sm ₄ Sb ₄ O ₂₄	1, 1, 1	40
mp-6065	Sr ₈ Ga ₄ Sb ₄ O ₂₄	1, 1, 1	40	mp-6169	Ba ₃ Ta ₆ Si ₄ O ₂₆	1, 1, 2	39
mp-6175	Ba ₈ Li ₁₆ P ₁₆ O ₅₆	1, 1, 1	96	mp-6211	Rb ₄ Li ₄ S ₄ O ₁₆	2, 2, 1	28

Table 10: **Materials analyzed in the phonon band structure comparison.** The first 60 materials sorted by ID in our test set of 299 randomly chosen MP materials. The full list of materials is available on [GitHub](#). Number of atoms is per unit cell.

CHGNet and M3GNet. However, given its low compute cost and general applicability, MACE-MP as-is can already serve as a useful pre-screening filter for dynamically unstable materials. Generating a complete phonon band structure on an Apple M2 Max CPU takes approximately 30 s. Running MACE-MP on a supercomputer could easily generate approximate phonon bands for every material in MP or other large databases. That said, MACE-MP harmonic phonons should not be used for more than pre-screening as current accuracy is far below what would be needed to reliably replace DFT.

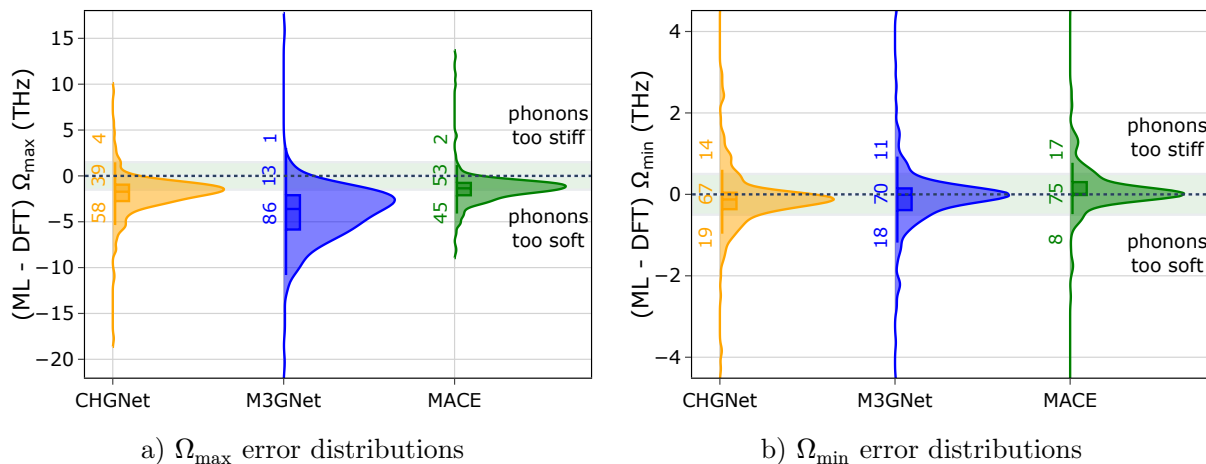


Figure 36: **ML force fields systematically underestimate $\Omega_{\max}$ but not $\Omega_{\min}$, the highest/lowest phonon frequencies in a material** (a) DFT vs ML error distributions of $\Delta\Omega_{\max} = \Omega_{\max}^{\text{ML}} - \Omega_{\max}^{\text{PBE}}$ for 299 randomly chosen MP materials. The vertical numbers colored yellow, blue and green indicate the fraction of materials with over-/underestimated highest phonon frequency for each model. (b) Same as (a) but for $\Omega_{\min}$ whose prediction error is approximately normally distributed with only minor bias for underestimation from CHGNet and M3GNet, while MACE-MP on average slightly overestimates $\Omega_{\min}$. The green-shaded area highlights ± 0.5 THz $\Omega_{\min}$ error which explains its better dynamic instability classification in Figure 34a).

A caveat to the above analysis is that the PBE unstable labels we use as ground truth may themselves contain false positives. That is, any unconverged values of ENCUT and k -point density may result in incorrect dynamically unstable PBE labels that would stabilize by reaching numerical convergence.

All data and code for this phonon analysis is available at <https://github.com/janosh/ffonons>. The code was written in a model-agnostic way and works with any predictor that conforms to the ASE Calculator API [326].

5.5 MACE-MP vs GNoME Relaxed Volume of Highly Coordinated Structures

Work in this section was led by Aaron D. Kaplan at Materials Sciences Division, Lawrence Berkeley National Laboratory, USA. I offered guidance, reproduced and checked the results and edited the text.

The recently published GNoME database [44] contains $\sim 381,000$ thermodynamically stable crystals which could not be matched to any experimental structure in the Inorganic Crystal Structure Database (ICSD) [254], nor any public repository of theoretical structures such as the Open Quantum Materials Database (OQMD) or Materials Project (MP).

The GNoME dataset was generated via SAPS [44] (i.e. partial Wyckoff-site) element substitution from MP + OQMD parent structures with deduplication and thus does not overlap

with MP by construction. That is not to say that GNoME and MP chemical space are entirely disjoint. For the 1,199 GNoME materials considered here, 131 (10.9 %) contain the same element combination as another MP structure. The chemical space of maximum overlap, Cu-La-Zn, contains four materials in GNoME (7ee54b7a37, 3422b9acb0, f36439bb45 and 68a5e7535f) and two in MP (mp-1223296 and mp-1093834).

An analysis of the local coordination environment of these structures using the `CrystalNN` algorithm [327] as implemented in `pymatgen` [4] reveals 21,300 structures with unusually high maximum coordination number $N_{\text{cnn}}^{\text{max}} > 16$.

To evaluate the performance of MACE-MP against GNoME which discovered these structures, we re-relaxed 1,199 of them with $N_{\text{cnn}}^{\text{max}} > 20$. The `FrechetCellFilter` class in the Atomic Simulation Environment (ASE) [326] was used to relax structures until the maximum (absolute) Cartesian component of any MACE-MP inter-atomic force was below 10^{-5} eV/Å. No dispersion correction was used to augment MACE-MP, consistent with the lack of dispersion correction in GNoME. While all relaxations completed successfully, two boron-dominated structures (GNoME identifiers 00edc694b5 and fc5cb3e024) required a much larger number of relaxation steps than the average structure to reach force convergence.

Figure 37 plots the volumes of these 1,199 structures after MACE-MP relaxation vs after GNoME relaxation, which are in excellent agreement. MACE-MP relaxes structures to near identical configurations as GNoME in nearly all cases. Only 19 structures (1.6 %) undergo energy lowering of more than 10 meV/atom during relaxation. 22 structures (1.8 %) change space group during relaxation, in all cases increasing in symmetry. The maximum calculated coordination number after MACE-MP relaxation changed for 135 structures (11.3 %) and decreased in all cases.

In short, MACE-MP performs comparably to GNoME for highly-coordinated structures and equilibrates to structures of equal or higher symmetry.

5.6 Homonuclear Diatomics

Work in this section was led by Tamas K. Stenczel at Engineering Laboratory, University of Cambridge, UK. I reproduced and checked the results, performed additional analysis of MACE-MP PES divergence issues for homonuclear dimers at close approach, edited the text and implemented the periodic table line plotting functions to generate Figure 38 in `pymatviz` [313].

When modeling high-temperature or high-pressure phases, a model must have strong core repulsion to prevent atoms from approaching too closely or even fusing. To test the physicality of all homonuclear 2-body interactions, Figure 38 shows diatomic energy-distance curves of MACE-MP for all elements. In a surprisingly large number of cases, we observe problematic fluctuations or erratic drops in energy at very close approach, e.g. for Li, B, Na, Al, P, Cl at

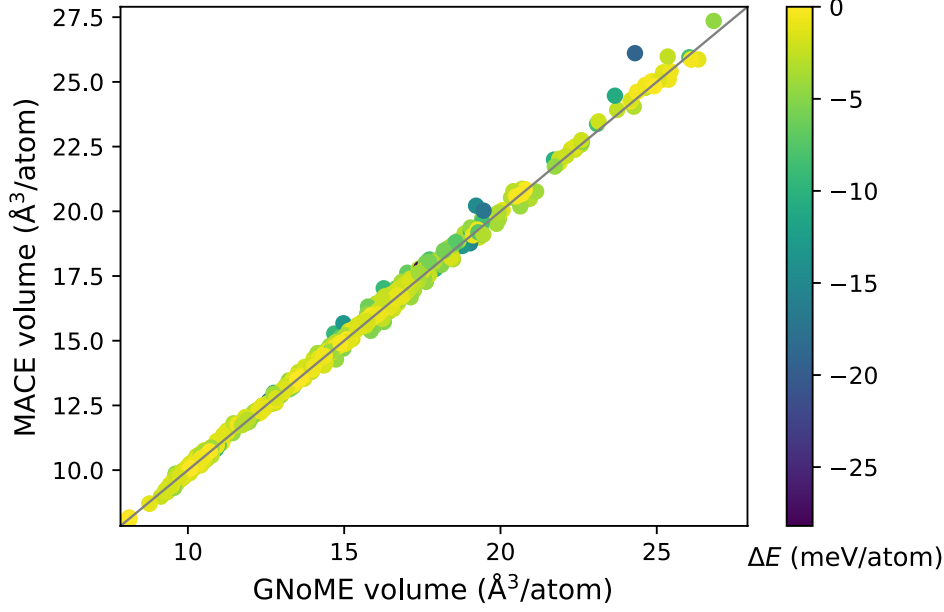


Figure 37: **MACE-MP relaxes highly-coordinated structures to near identical configurations as GNoME.** If $N_{\text{cnn}}^{\text{max}}$ changes, it always decreases, while symmetry always **increases**. Comparison of ML-relaxed volume for 1,199 highly-coordinated structures ($N_{\text{cnn}}^{\text{max}} > 20$) in the GNoME database [44] (x axis) vs the same structures relaxed with MACE-MP (y axis). Points are colored by the difference in the total MACE-MP potential energy (in meV/atom) before vs after relaxation. Darker points indicate larger energy lowering during relaxation. The line of best fit is

$$V_{\text{MLFF}} \approx 1.009 \times V_{\text{GNoME}} - 0.102, \text{ with } R^2 \approx 0.998.$$

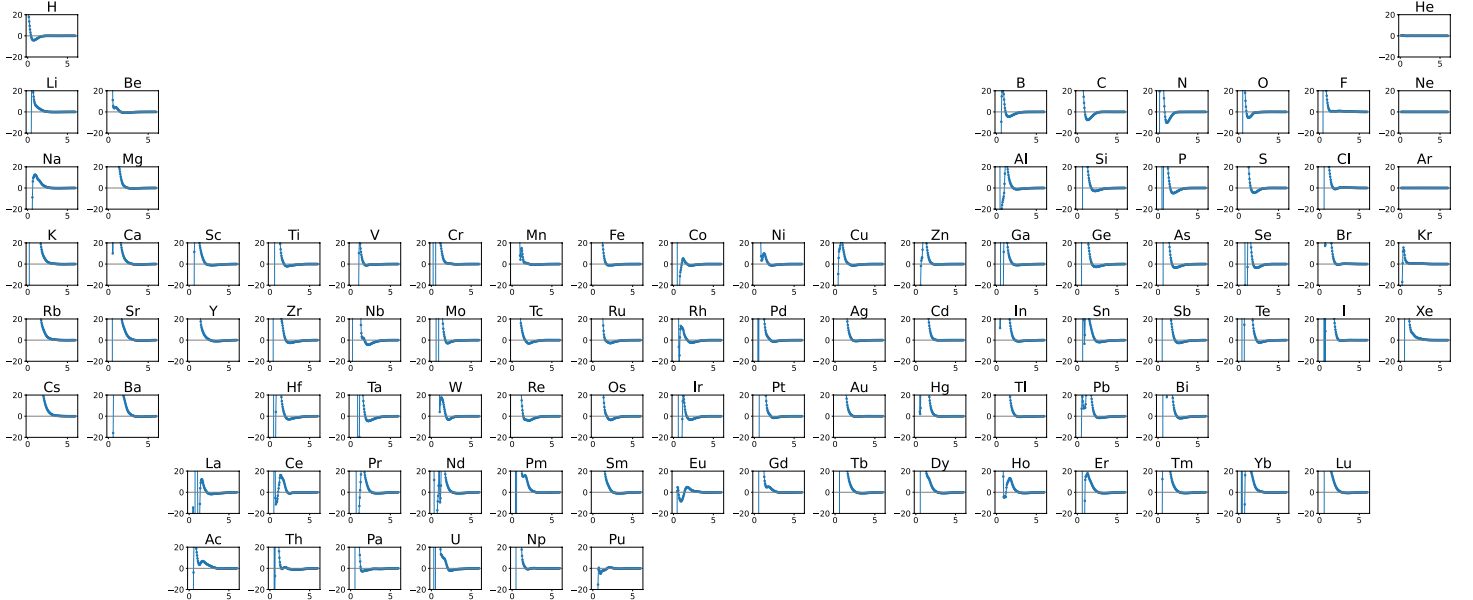


Figure 38: **Energy-distance curves of homonuclear diatomics in vacuum.** Units are eV for energy (y) and Å for distance (x). While many elements exhibit strong repulsion at small distances, an even larger number (60) show erratic behavior at sub-Angstrom separation with unphysical sharp drops in energy.

$\sim 0.5 \text{ \AA}$ and for O and F with higher training data representation at even smaller distances of $0.2 - 0.3 \text{ \AA}$. The noble gases He, Ne, Ar with negligible presence in the training set (Figure 45) are flat.

While Figure 38 shows the medium-sized MACE-MP with 4.7 M parameters as used throughout this chapter, the small model with 3.8 M parameters suffered from fewer elements with imploding negative energy at close approach (51 vs 60 elements going below -20 eV), possibly indicating a degree of overparameterization in the medium model.

One possible avenue for addressing this issue is to incorporate an analytical short-range repulsive term that is only active below a certain cutoff distance r_{cutoff} . The difficulty here would be to implement a switching mechanism that correctly untangles atoms from MACE’s many-body message passing as they enter the short-range regime, while also ensuring a smooth transition from MACE-MP at intermediate ranges to the short-range repulsion potential. An open question is how sensitive simulation results would be to the choice of r_{cutoff} and to what extent it would need to be element-dependent. A second option that avoids the switching mechanism altogether is to always include a short-range repulsive term. Simple 2-body terms like ZBL are inexpensive and only contribute when atoms are close. However, given the severe energy drops for some elements in Figure 38, even adding a positive ZBL term may not be sufficient to guarantee repulsion at close distances.

Another option that sidesteps complicating the model is to only use the analytical short-range potential such as Ziegler-Biersack-Littmark (ZBL) [328] to generate additional training labels for systems of densely packed atoms, or computing all dimers at various separation with exact DFT. This would leave the model architecture unchanged and simply teach the model to raise the energies of high-density configurations by compensating for the severe lack of sub-Angstrom separation data in MPtrj. We plan to explore these options, especially the data augmentation approach, in future work.

5.7 Materials Discovery via Element Substitution

Work in this section was led by Konstantin S. Jakob at Fritz-Haber-Institute, Max-Planck-Society, Berlin, Germany. I reviewed the results, edited the text and contributed/fixed parts of the code in `pymatgen` [4] used to perform the analysis.

We performed exhaustive element substitutions for 100 randomly selected MP compositions. The number of existing polymorphs in MP for each of these chemical systems varies widely, ranging from one (e.g. for NaNiIO_6 , NaSrSnBiO_6) to at most 423 (for Sr_2CoO_4). Specifically, MP was first filtered to remove all structures with more than 16 atoms in the reduced formula to ensure a sufficient number of possible substitutions. This set was randomly split into host and target compositions with a 60:40 ratio. 100 compositions were randomly drawn from the

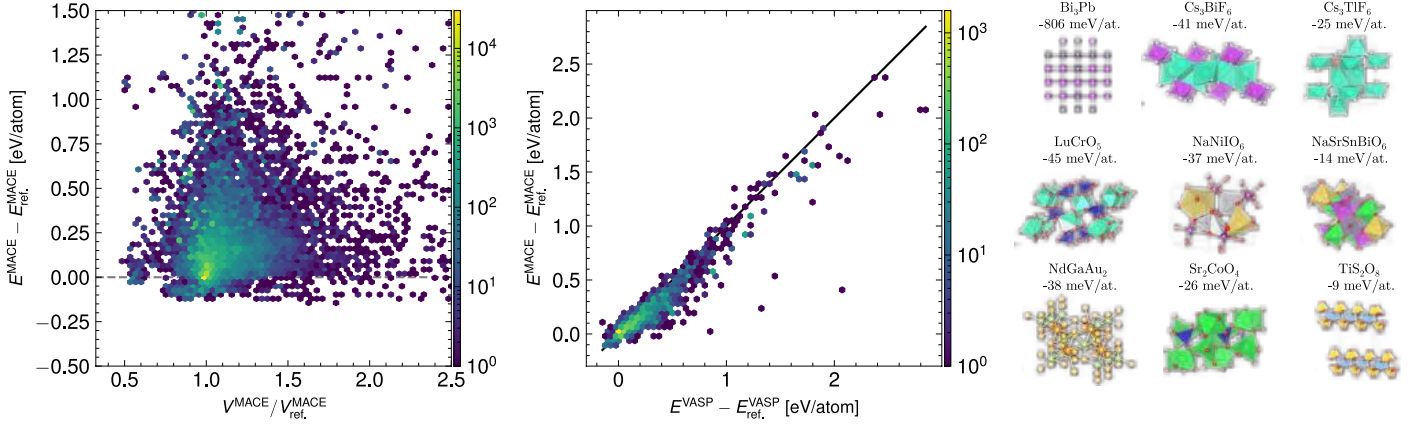


Figure 39: **MACE-MP discovers new stable phases.** Distribution of MACE-MP relative energies and volumes for 100 stoichiometric substitutions of MP compositions. The distribution is sharply peaked at $\frac{V}{V_{\text{ref}}} = 1$ and $E - E_{\text{ref}} = 0$, indicating that the substituted cells often relax back to the known ground state structure from MP.

40% target split and substituted into each stoichiometrically matching host, yielding 154,685 substituted structures. These were optimized with MACE-MP using full unit cell relaxations (without D3 dispersion corrections). Figure 39 (left) shows the results for relative energies and volumes compared to the most stable MP structure of that same composition. The distribution is sharply peaked at $\frac{V}{V_{\text{ref}}} = 1$ and $E - E_{\text{ref}} = 0$, indicating that the substituted cells often relax back to the known ground state structure from MP. This suggests MACE-MP’s suitability for finding the correct ground state of unknown materials. MACE-MP energies were validated with MP-compatible DFT calculations (using `pymatgen`’s `MPRelaxSet` to prepare VASP input files), yielding an RMSE of 97 meV/atom.

Beyond recovering the MP ground state, the wide range of substitutions tested also yields a large number of novel polymorphs for each composition. 2,518 of these have negative relative energies and are thus predicted to be more stable than the current MP ground state. To validate these predictions, DFT relaxations were performed for all structures with negative relative energies. The results confirm that 2,120 of them are indeed more stable than the corresponding MP reference. However, not all constitute new stable phases. Many relaxations converged to the same MP minimum configuration (with slightly lower energy likely due to numerical noise). In other cases, multiple lower-energy structures are found for a single composition, only one of which can be the true ground state. Nonetheless, for 9 of the original 100 MP compositions considered here, new structures with energies below the current MP hull were discovered (for [Bi₃Pb](#), [Cs₃BiF₆](#), [Cs₃TlF₆](#), [LuCrO₅](#), [NaNiO₆](#), [NaSrSnBiO₆](#), [NdGaAu₂](#), [Sr₂CoO₄](#), [TiS₂O₈](#)). We believe these to be genuinely new phases that are thermodynamically stable with respect to the MP convex hull. For each novel ground state discovered, the closest MP structure is: [mp-1106139](#) for Bi₃Pb, [mp-559695](#) for Cs₃BiF₆, [mp-561827](#) for Cs₃TlF₆, [mp-1211553](#) for

LuCrO₅, [mp-545399](#) for NaNiIO₆, [mp-1522253](#) for NaSrSnBiO₆, [mp-1220399](#) for NdGaAu₂, [mp-1100068](#) for Sr₂CoO₄, [mp-775149](#) for TiS₂O₈. Remarkably, these new phases were also not reported in the recent GNoME effort (which focused on compositions not included in MP). The WBM dataset does, however, report slightly more stable structures for NaNiIO₆ and NdGaAu₂. It also includes additional structures to ours for Cs₃BiF₆ and Cs₃TlF₆, which are less stable than the ones we found.

5.8 Bulk and Shear Moduli

Work in this section was led by [Matthew Kuner](#) at the University of California, Berkeley, USA. I helped write the text, contributed usage advice, implemented the [ElasticityCalc](#) property calculator (compatible with any [ASE Calculator](#)) together with Shyue Ping Ong in [MatCalc](#) [329] which was used to obtain bulk and shear modulus predictions, and the plotting functions to generate [Figure 40](#) implemented in [pymatviz](#) [313].

Being able to predict elastic properties such as bulk and shear modulus is relevant for many industrial applications including coating, cutting and polishing as well as turbine and other high-temperature and high-wear engine materials [241], [330]. Like phonons, elasticity depends on the second derivative of energy with respect to strain, again testing a model’s ability to precisely capture PES curvature.

To test MACE-MP in this regard, we benchmarked against the elastic properties of over 10 k MP materials by calculating the Voigt-Reuss-Hill average [331], [332], [333] bulk modulus and shear modulus as derived from MACE-MP stress-strain relations.

All 10 k structures are bulk crystals at equilibrium. While none of the DFT deformation calculations contained in MP are present in the MPtrj training set used for MACE-MP, and the deformations we applied are not identical given MACE-MP relaxes to a somewhat different ground state than the original MP structure, the purpose of this analysis is not to assess extrapolation performance but the fidelity of the PES for predicting downstream observables that also constitute application-relevant material properties.

The initial structures used for these calculations were the PBE-relaxed [101] configurations from MP which were first re-relaxed using MACE-MP and then deformed. Specifically, a total of 4 strain magnitudes were used along six independent strain modes (in Voigt notation):

$$\epsilon = [\epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{44}, \epsilon_{55}, \epsilon_{66}] \quad (46)$$

where the strain values we chose are

$$\begin{aligned} \epsilon_{11} = \epsilon_{22} = \epsilon_{33} &\in \{\pm 0.01, \pm 0.005\} \\ \epsilon_{44} = \epsilon_{55} = \epsilon_{66} &\in \{\pm 0.06, \pm 0.03\} \end{aligned} \quad (47)$$

These calculations were performed using the [MatCalc elasticity module](#) [329]. Moreover, to filter out likely unphysical DFT predictions, elastic properties from MP were excluded from

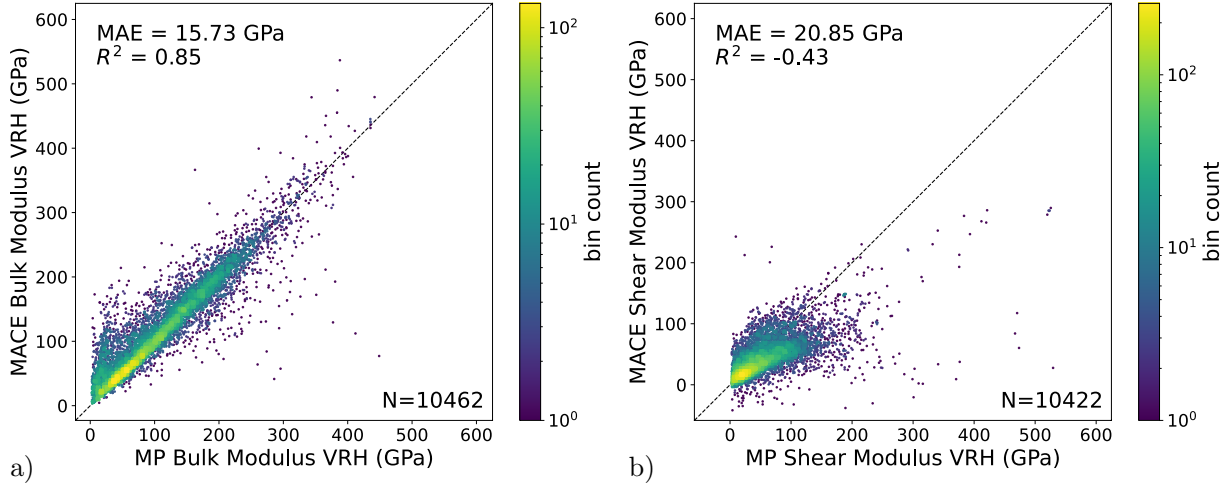


Figure 40: MACE-MP vs MP bulk (a) and shear (b) moduli for ~ 10 k MP materials. For bulk moduli K_{VRH} , a single point for [mp-721759](#) (face-centered cubic nitrogen) was excluded, as MACE-MP predicted an unphysically high bulk modulus (>600 GPa). For shear moduli G_{VRH} , 26 points were excluded from this plot as MACE-MP predictions were unphysically high (>600 GPa) or low (<-50 GPa). This includes [mp-2458](#), [mp-10030](#), [mp-13174](#), [mp-14549](#), [mp-30083](#), [mp-570354](#), [mp-631377](#), [mp-631475](#), [mp-631524](#), [mp-972442](#), [mp-976419](#), [mp-980204](#), [mp-989573](#), [mp-999086](#), [mp-999263](#), [mp-1002105](#), [mp-1002117](#), [mp-1002165](#), [mp-1002206](#), [mp-1002207](#), [mp-1008669](#), [mp-1009019](#), [mp-1009485](#), [mp-1009731](#), [mp-1077353](#), [mp-1428312](#).

this analysis if the DFT average bulk K_{VRH} or shear G_{VRH} modulus are less than -50 GPa or greater than 600 GPa.

$$-50 \text{ GPa} < (K_{\text{VRH}} \text{ or } G_{\text{VRH}}) < 600 \text{ GPa} \quad (48)$$

Results comparing MP and MACE-MP bulk moduli with MAE of 15.73 GPa and R^2 of 0.85 are shown in [Figure 40a](#)). This compares favorably to the R^2 value of 0.757 reported for MACE-MP [\[68\]](#). [Figure 40b](#)) reveals that MACE-MP struggles to predict shear properties, likely due to a lack of sheared structures in MP.

5.9 Quantifying PES Softening in Universal ML Potentials

Work in this section was led by [Bowen Deng](#) at University of California, Berkeley, USA and a preprint published separately from other sections in this chapter in [\[9\]](#), currently under review at npj Computational Materials. I wrote and ran the code for the phonon analysis, helped draft and edit the manuscript, and advised the MACE-MP analysis in other sections of the paper. Text and figures adapted with permission from B. Deng *et al.*, [\[9\]](#).

All simulation and prediction tasks involving high-energy configurations in [\[8\]](#) and [\[9\]](#) reveal a consistent misalignment of DFT and ML potential energy surfaces (PES) across all 3 MLFFs we tested (CHGNet, MACE-MP, M3GNet). This so-called PES softening is characterized by weaker

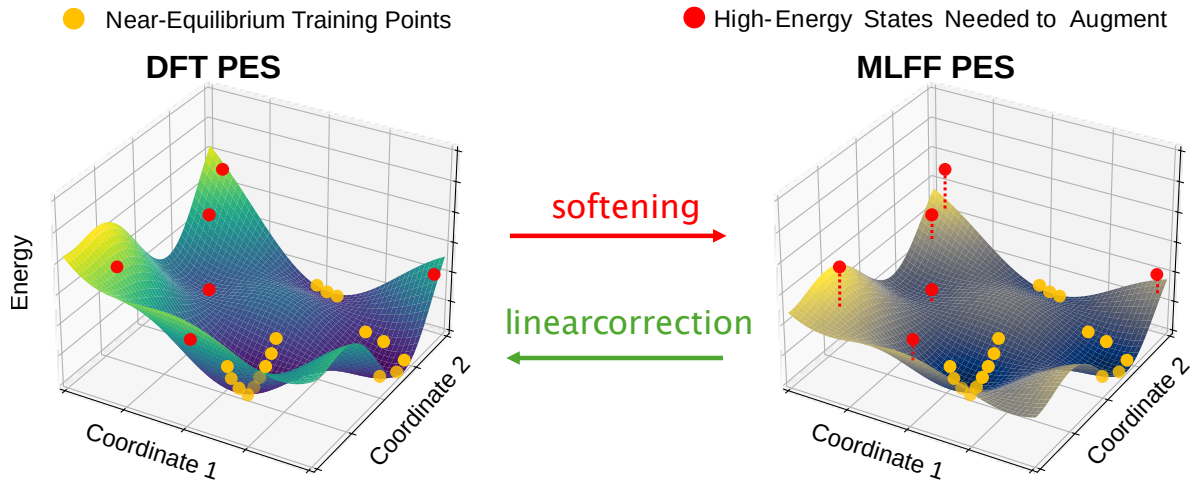


Figure 41: **PES softening occurs due to 2 biases: preferential sampling of low-energy configurations in current DFT pre-training datasets and bias towards PES smoothness built into ML potentials.** The two coordinate axes illustrate arbitrary dimensions in atomic configuration space and the height of the surface the energy of the corresponding structure. Ideally, the DFT PES (left) and ML PES (right) should match. In practice, we consistently observe (across CHGNet, MACE-MP, M3GNet) that the ML PES only accurately describes the well-sampled dark blue regions populated by near-equilibrium states in MPtrj (indicated by orange points). The ML PES exhibits significant systematic errors in high-energy, far-from-equilibrium yellow-colored regions (indicated by red points) due to lack of nearby training data. ML potentials are designed to smoothly vary output with respect to changes in input which incurs a bias towards flat energy surfaces. This bias causes energies and forces in unsampled high-energy regions of configuration space to be systematically underpredicted. Since PES softening errors are systematic, they can be efficiently rectified by a linear correction fitted to minimal additional training data. In fact, a single additional training label at any of the red points can fit a scalar energy multiplier that rescales the gradients of the PES, on average eliminating the systematic error. Figure 43a) demonstrates this works in practice, including across diverse chemical spaces (Figure 43b)).

curvature of the ML compared to the DFT PES, resulting in the systematic underprediction of energies and forces in far-from-equilibrium regions, as illustrated in Figure 41 and quantified through the PES softening scale c_S shown in Figure 42. It shows the tilted distribution of ML-predicted vs DFT forces for the specific compound $\text{Li}_2\text{B}_3\text{PO}_8$ (Figure 42a)) and the distribution of softening scales for M3GNet, CHGNet and MACE-MP across various chemical systems for 1000 randomly sampled WBM compounds (Figure 42b)).

Figure 43 shows that a single DFT label can correct the systematic softening in CHGNet, reducing the far-equilibrium force MAE from $0.19 \text{ eV}/\text{\AA}$ to $0.166 \text{ eV}/\text{\AA}$. Fine-tuning the model with 10 DFT labels further decreases the force MAE to $0.155 \text{ eV}/\text{\AA}$.

As further evidence to establish the issue of PES softening in current MLFFs, Figure 44b) plots the relative underestimation of DFT vs ML Ω_{max} , the highest phonon band frequency

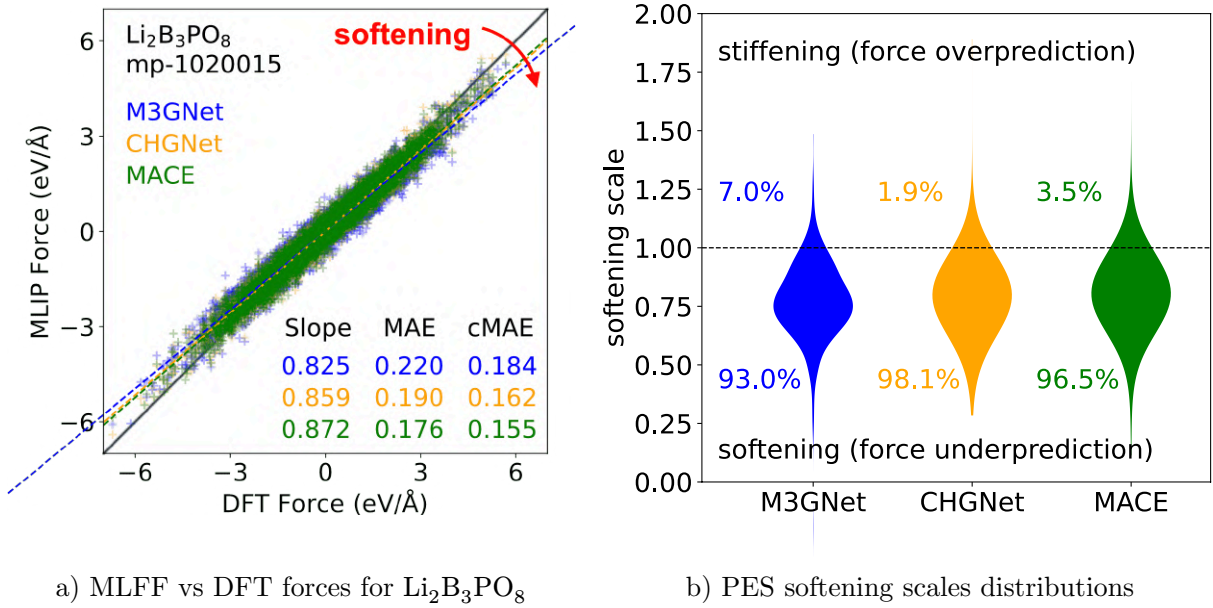


Figure 42: **Linear correction with a single training label solves PES softening.** (a) MLFF vs DFT forces for $\text{Li}_2\text{B}_3\text{PO}_8$ (mp-1020015) high-energy states sampled from MD at 1,000 K. The tilted distribution of forces from the diagonal means systematic softening in the PES from all 3 MLFFs. We call the slope c_S of a linear fit the PES softening scale, where $c_S < 1$ are softened forces and $c_S > 1$ (unobserved) would be stiffened forces. cMAE is the corrected MAE obtained by rescaling forces such that $c_S = 1$ (i.e. rotating the force distribution back to parity). (b) Distribution of softening scales of 1000 compounds sampled from the WBM dataset, showing the PES softening behavior is universal across various chemical systems.

of a crystal.¹¹ Excluding materials with a low relative error of $\pm 5\%$, MACE-MP’s strongly underestimates $\Omega_{\max}$ by more than 5% in 52% of cases whereas it overestimates it by more than 5% in only 1%. For CHGNet (52% vs 1%) and M3GNet (84% vs 2%), this imbalance is even more pronounced. MACE-MP has a noticeably more peaked error distribution with a mode that falls into the green-shaded area. As a result, 58% of MACE-MP predictions are within $\pm 5\%$ relative error compared to 47% for CHGNet and 14% for M3GNet.

Figure 36b) shows $\Omega_{\min}$, the lowest phonon frequency at any q -point. (Unlike $\Omega_{\max}^{\text{ML}}/\Omega_{\max}^{\text{PBE}}$, plotting the ratio of $\Omega_{\min}^{\text{ML}}/\Omega_{\min}^{\text{PBE}}$ does not make sense as frequently $\Omega_{\min} = 0$.)

To quantify the magnitude of PES softening observed in different universal potentials, we define a softening scale parameter c_S , which is determined by sampling $\text{Li}_2\text{B}_3\text{PO}_8$ (mp-1020015) off-equilibrium states from high-temperature MD at 1,000 K followed by running static calculations to get the interatomic forces on each atom with both DFT and the ML potential in question. Figure 42a) shows a parity plot of MLFF vs DFT forces from these off-equilibrium states.

¹¹Frequency underpredictions in Figure 44 may be partially due to comparing to PBEsol reference while all models are trained on PBE data which tends to have softer phonons. Comparison to actual PBE reference data is underway and will allow quantifying the degree of systematic error in this analysis which is expected to be minor given the small difference between PBEsol and PBE phonons relative to the significant ML softening.

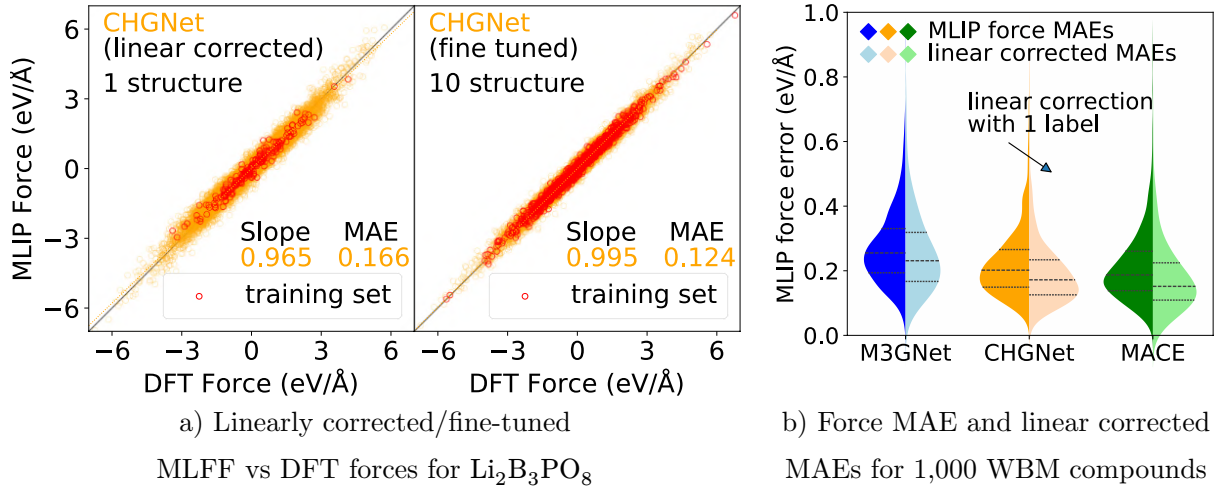
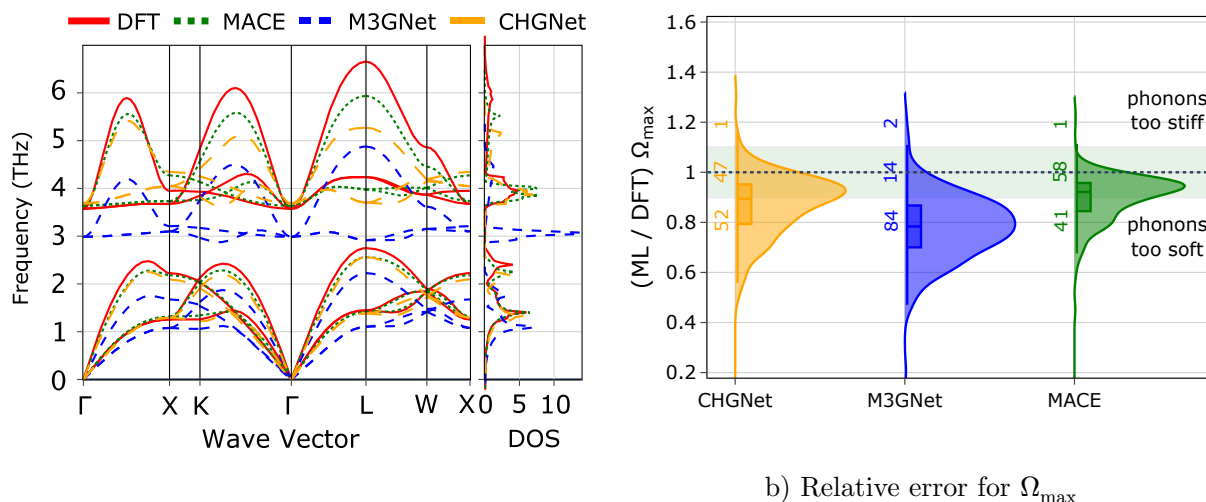


Figure 43: **Empirically observed PES softening scales c_S for different models and chemical systems.** (a) DFT forces vs CHGNet predictions for the same [mp-1020015](#) high-energy states as in [Figure 42a](#)) but *after* fine-tuning. Training force labels are plotted in red, and test force labels excluded from training in orange. The left plot shows a linear correction with a single DFT label that corrects the systematic softening and reduces force MAE from 0.19 eV/Å to 0.166 eV/Å. The right plot shows a more realistic fine-tuning scenario where *all* model parameters are updated with 10 DFT labels, further decreasing the force MAE to 0.155 eV/Å. While fine-tuning with 10 labels brings the slope closer to 1, even correcting with a single label should *on average* eliminate the systematic error. (b) Distribution of force MAEs and linear corrected MAEs (cMAEs) for 1,000 WBM compounds. All 3 error distributions are shifted closer to zero following the single-label linear correction. Dashed and dotted lines show mean and quartiles, respectively.

Systematic PES softening is characterized by a clockwise tilt of predicted forces from the diagonal parity line, which corresponds to systematic force underpredictions. A linear regression can be performed to extract the slope of the distribution, where the slope indicates the extent of the systematic error, and is therefore defined as the PES softening scale c_S . $c_S = 1$ corresponds to the MLIP’s force distribution aligning with the diagonal, which indicates the curvature of the MLIP-PES systematically agrees with DFT with only local random noise. $c_S > 1$ which is observed for all models tested here (CHGNet, MACE-MP and M3GNet), indicates the MLIP PES is overly smooth, resulting in low forces.

To investigate how broadly PES softening affects different chemistries, we randomly sampled 1000 unique chemical systems from the WBM dataset [1] not included in the MP pre-training set of any of the models to ensure testing under OOD conditions. We sampled 10 off-equilibrium structures from each MD trajectory at $T = 1,000$ K to calculate static DFT forces [334]. Comparing the force predictions from MLIPs and DFT shows a clear underprediction trend. [Figure 42b](#)) shows the distribution of PES softening scales c_S across the 1000 WBM OOD chemistries. We find c_S smaller than 1 for 93% of materials with M3GNet, 98% for CHGNet, and 97% for MACE-MP, which demonstrates the softening behavior is consistent across diverse chemical spaces and current universal ML potentials. GNoME similarly exhibits pronounced



a) Phonon dispersion relation and DOS of CsF

Figure 44: **A soft PES causes soft phonons.** (a) The phonon dispersion relation and DOS of CsF (mp-1784) calculated with DFT and MLFFs. This material is a typical example where all models underpredict the vibrational frequencies, most notably M3GNet by 27 % compared to CHGNet’s 19 % and MACE-MP’s 11 %. (b) The relative (ML / DFT) error for $\Omega_{\max}$, similar to Figure 36a) which shows the signed error (ML - DFT). Again the error distribution of MACE-MP is more sharply peaked and the mode lies within the green-shaded area indicating $\pm 5\%$ error.

softening when trained on the M3GNet dataset, as shown in [44, Figs. S34 – S37]. The softening effect is somewhat mitigated but still present after training on the full GNoME dataset of 89 M structures. These examples show that PES softening is prevalent across various models and datasets. A systematic analysis is needed to quantify and correct this error.

While similar signs of softening due to choice of training data were previously observed [335], [336], less attention was dedicated to a systematic examination. [335] reported a distribution shift from training CGCNN [123] to predict DFT energies of ICSD structures taken as is from the database without relaxation.

5.9.1 Super data-efficient fine-tuning

The systematic MLIP PES flatness appears as a tilted distribution of forces in the parity plot as shown in Figure 42a). Intuitively, one can easily rotate the distribution back to the diagonal to eliminate systematic errors and reset the softening scale to 1. In Figure 42a), the bottom-left table with columns Slope, MAE, cMAE report the linear fitted slope from the actual distribution (colored dash lines), the corresponding actual MAE and the corrected cMAE if all force predictions are uniformly divided by the slope to rotate predictions until slope and parity align. The cMAEs are 12 to 16% lower than the actual MAEs, indicating that $\frac{1}{7}$ to $\frac{1}{8}$ of MLFF force errors are systematic and can be easily corrected:

$$\begin{array}{ccccc}
\text{CHGNet : 0.22} & \xrightarrow{-16\%} & 0.18 & \text{MACE-MP : 0.19} & \xrightarrow{-15\%} & 0.16 & \text{M3GNet : 0.18} & \xrightarrow{-12\%} & 0.16 \\
& & & \text{MAE (eV/\AA)} & \xrightarrow{\text{percent change}} & \text{cMAE (eV/\AA)} & & &
\end{array} \quad (49)$$

Mathematically, rotating the force distribution is equivalent to multiplying a scalar to every force value, which can be realized by multiplying the MLIP-predicted energy $\text{MLFF}(\{\mathbf{r}_i\}, \{C_i\})$ by the scalar slope c_S .

$$E^{\text{corr}} = c_S \cdot \text{MLFF}(\{\mathbf{r}_i\}, \{C_i\}) = c_S \cdot E, \quad \mathbf{f}_i^{\text{corr}} = -\frac{\partial E^{\text{corr}}}{\partial \mathbf{r}_i} = c_S \cdot \mathbf{f}_i. \quad (50)$$

The above formulation is equivalent to fine-tuning an MLIP with all model weights fixed except for a final linear layer, which modifies only the scalar parameter c_S in eq. (50). Since only a scalar parameter is modified, a single label (1 force component) suffices for training. Fine-tuning typically involves hundreds to thousands of additional labels. This observation of remedying PES softness with a minute amount of fine-tuning data elucidates that softness is a data issue rather than a model issue. Figure 43b) shows the off-equilibrium force MAE and theoretical cMAE for the 1000 WBM structures. The distribution of decreased MAEs therefore provides a theoretical lower bound for the expected performance gain of fine-tuning MLFFs. This lower bound can in turn be cheaply and quickly approximated by fitting a final-layer linear correction from a single training label.

Our results suggest a theoretical explanation for the commonly observed data-efficient performance boost achieved by fine-tuning MLFFs compared to training MLIPs from scratch with the same data [337]. The data efficiency arises from the observation that a considerable part of the MAEs in pre-trained MLFFs are systematic and thus easily fixed with small amounts of data.

5.10 Limitations and Next Steps

MACE-MP (and to a slightly lesser extent CHGNet and M3GNet) approaching DFT accuracy and stable MD across such a wide range of atomic configurations covering most of the periodic table is a landmark achievement for a single interatomic potential. Yet many limitations in the current version remain. Foremost, the MPtrj training set was generated at the PBE-level of theory [101], which requires Hubbard U corrections to improve electronic correlations for particular transition metal–anion or metal–transition metal combinations. This introduces inconsistencies in the PES that again must be posthoc compensated [36], introducing a source of confusion for the model since two materials with similar chemistries may now have inconsistent energies if one contains a U -corrected element-pair and the other does not. PBE is also well-known to lack long-range dispersive forces which therefore MACE-MP likewise fails to model, requiring the addition of posthoc dispersion force corrections such as D3 [338]. Recent developments in DFT around the r2SCAN [201], [339], [340] meta-GGA functional are beginning to

supersede PBE by achieving significantly improved accuracy at acceptable increase in computational cost [341]. With future computing power, hybrid functionals [111], [342] or the random phase approximation [343] may become affordable enough to generate fine-tuning datasets and improve upon this even further, albeit at huge cost by today’s standards. Refitting the model to a more modern functional will significantly increase its predictive power and simultaneously reduce the need for system-dependent corrections, making it more universally applicable and closer to a plug-and-play tool that can be used even by non-experts.

Despite the success stories in Chapter 5 and even more in [8], MACE-MP is limited to a 12 Å receptive field afforded by two steps of message passing each with graph cutoff $r_C = 6$ Å. It therefore cannot incorporate truly long-range interactions acting over distances greater than 12 Å. Some observables, particularly in the context of dilute solutions and at interfaces, cannot be calculated with a medium-range model like this. Moreover, MACE-MP has no concept of magnetic or spin degrees of freedom. This is an obvious improvement for future model iterations as several approaches have already been published for incorporating explicit electrostatic interactions [181], [183], [344], [345], as well as spin degrees of freedom [67], [346], [347].

In the case of low-pressure simulations of ethanol [8, sec. A.16] we noticed these small quantitative deviations can lead to qualitative errors by shifting the temperature or pressure of a phase transition, which then result in unphysical equilibrium phases at ambient conditions. Another failure mode that popped up in two different test cases (random structure search [8, sec. A.14] and high-pressure hydrogen [8, sec. A.21]) are atoms at very close approach causing sudden extreme drops in energy, resulting in the respective simulations being trapped in unphysical erroneously low-energy geometries.

Both these failure modes are not specific to any particularly challenging chemical system but are direct shortcomings of the training data. As such, they should be straightforward to address by extending the existing training data to lower and higher pressures [348] and concatenating pressure as a global state variable onto the input. This will place stronger constraints on the model’s behavior in diverse pressure regimes. Likewise, a future iteration of the training set could be augmented using repulsive pair potentials [349] such as ZBL [328] to generate very-short atom pair–force labels to constrain the model a more physical PES at sub-angstrom distances. The current training set contains no examples of such close approach, leaving a physically agnostic statistical model with no guidance on what to predict.

In summary, promising avenues to improve the model’s accuracy include

Increasing model size More free parameters means lifting the expressivity ceiling. Unlikely to yield sizeable improvements given MACE-MP’s current size of 4.7 M parameters is already 10x that of CHGNet which performs worse but not by a large margin.

Removing inconsistencies As mentioned above, having trained only PBE-level data which require posthoc U corrections for some materials may confuse the model.

It also has an incomplete picture of physical forces given GGA’s lack of long-range dispersion forces [338], [350].

Training set augmentation Increasing the number and variety of atomic configurations for training is likely to unlock by far the biggest gains in accuracy and versatility given the performance of proprietary models such as GNoME and PFP which trained on 1 – 2 orders of magnitude larger datasets than MPtrj.

Efforts to alleviate current limitations around data inconsistency and size are ongoing. We started an open source MatPES collaboration with Shyue Ping Ong’s group mid-last year aiming to create one of the largest and most diverse r2SCAN-level datasets specifically designed for ML potential training. However, even in its current form, MACE-MP serves as a good starting point for both data- and compute-efficient fine-tuning. How much additional data is needed for such refinement depends on the required accuracy and degree of departure from the training set [351]. In many cases, we expect a fine-tuning set of hundreds of examples to be sufficient. Transfer-learning approaches can also enable these models to fit higher-quality reference data, often requiring an order of magnitude less high-quality data than the original training set [130], [309], [352].

While it is not yet wise to solely rely on ML potentials for any prediction without further validation [323], we are optimistic that training sets of significantly increased fidelity, diversity and size will be sufficient to achieve quantitative agreement with the target level of theory across chemical space. In that case, future versions of MACE-MP or other foundational force fields may truly provide a universal model for carrying out materials simulations.

Chapter 6

Summary

The projects in this thesis collectively explored the current boundaries of ML-accelerated materials discovery and simulation. ML capabilities have advanced at an impressive pace in the last few years with significant improvements across model architecture, training data, usability and reproducibility. Many models now come as versioned, easily installable Python packages¹². ML potentials covering most of the periodic table in particular have undergone a flurry of innovation and reached a critical juncture. Models like MACE-MP, CHGNet, M3GNet, SevenNet, PFP, GNoME and Equiformer v2¹³, among others can perform various forms of stability and property prediction as well as drive MD for challenging chemical processes over unprecedented length and time scales [353]. To put this into perspective, as recently as 2020, the ability of ML to accelerate materials discovery was still a topic of debate [10] and rightfully so given results at the time. MLFFs are now accurate and versatile enough to be serious contenders to compete with more established simulation methods on a variety of tasks. Even more importantly, their linear scaling with system size makes new kinds of complex, long-duration simulations possible for the first time.

6.1 Matbench Discovery

Matbench Discovery established machine-learned force fields as the state-of-the-art method for ML-guided materials discovery. By demonstrating their superiority over energy-only models such as MEGNet [125], ALIGNN [168], CGCNN [123], BOWSR [212] and Wrenformer [126], Matbench Discovery catalyzed the creation of CHGNet [67] and especially MACE-MP [8], two of the most accurate and versatile open-source ML potentials to date. Force models can chart a path through atomic configuration space approaching the DFT-relaxed structure from where a more reliable final energy prediction can be made. Matbench Discovery proved a highly useful tool for the initial validation and performance assessment of both models. Our findings include discovery acceleration factors (DAFs) compared to random selection greater than 3 across the whole WBM test set (Table 2), and 5.6 when considering only the 10k most stable predictions from each model (Table 4) (with 6.5 the maximum possible DAF on our test set). Our results

¹²`pip install chgnet mace-torch m3gnet` is all it takes to install the 3 currently best open models on Matbench Discovery.

¹³We and Zackary Ulissi’s group are each working on submitting Equiformer v1 and v2 trained on MPtrj to Matbench Discovery. Our work on v2 has stalled recently though due to persistent NaN predictions appearing during training in deeply nested `e3nn` which proved resistant to Philipp Benner’s debugging efforts.

advocate for the integration of both ML geometry optimization and relaxed energy prediction into future database expansion efforts, as filtering out ML-unstable structures strongly reduces the number of DFT calculations required to reach a desired level of convex hull exploration. Model errors as a function of training set size in log-log scale yield roughly straight lines even up to $\mathcal{O}(10^7)$ training samples [44, figs. 1e, S14–S21], [225, fig. 2], suggesting that further decreases in MLFF errors are readily unlocked with unbiased training data augmentation. We observe the same trends in Matbench Discovery from proprietary models like GNoME and PFP which significantly outperform all open models (see Table 11) having training on ~ 2 orders of magnitude more data ($\mathcal{O}(10^8)$ structures) than the canonical Matbench Discovery training set (1.6 M structures). Model architecture is likely to play an insignificant role here, as both PFP (TeaNet architecture [65]) and GNoME (NequIP architecture [80]) are standard GNNs devoid of unusual architectural features. Active learning loops where new training data is added based on model uncertainty or error estimates are likely to be even more effective than unbiased data increase. This prior was recently confirmed by MatterSim [354, table S2] which outperforms GNoME on most Matbench Discovery metrics using $\sim 5\times$ less training structures (~ 17 M vs ~ 89 M) highlighting the importance of optimizing for structural diversity.

It is worth emphasizing at this point that we controlled as best we could for data leakage on industry models with incompatible training sets by confirming performance metrics are insensitive to testing on subsets of WBM data filtered to remove compositional overlap with proprietary training sets such as GNoME. That said, we are unable to monitor what happens behind closed doors. This makes a benchmark such as Matbench Discovery with a publicly available test set [1] reliant on an honor system, especially for submissions with closed source and/or weights where fine-tuning the model on the test set cannot be verified to yield substantial improvements in performance (the absence of which would strongly hint at data leakage and perhaps prior training on the test set). A future benchmark of similarly permissive design would benefit from a hidden test set unavailable to external model submissions as is the case e.g. for the OCP series of benchmarks [62], [82], [215], allowing for somewhat more confident evaluation of proprietary models.

Several well-established benchmarks including Matbench [60] and Jarvis [61] already provide a platform for direct architecture comparisons. Our objective with Matbench Discovery was not to reproduce the rigid benchmarks common in conventional ML domains like language and vision which in our opinion do not map well onto chemistry. The crucial difference in those domains is data abundance, making architectural and training procedure innovations the primary focus of research. ML for chemistry by contrast is data-starved. This makes innovative architecture-data combinations more likely to be the primary source of progress which in turn renders rigid-data benchmarks less conducive to advancing the field. As an example, interatomic potentials are trained on forces and stresses and can therefore leverage available ab initio training data to greater effect than energy-only models. This constitutes a genuine domain-

advantage of ML potentials and we want Matbench Discovery to reflect that. Thus, our objective to ask system-level rather than purely architecture-focused questions led to permissiveness as an overarching design principle and the opinionated choice to accommodate models trained on different datasets.

To elaborate on this point, a recent study found a substantial portion of high-throughput data is redundant when considering in-distribution (ID) performance [219]. By specifying a maximum training set and a fixed test set, Matbench Discovery provides a controlled environment in which future work could innovate on downsampling schemes that result in more efficient training or more consistent data while measuring the impact on performance. Such work would be able to evaluate the claims in [219] in the context of OOD performance. To some extent, the answer to this question popped out for free from Matbench Discovery by comparing CHGNet which was trained on MPtrj (1.6 M structures) and M3GNet trained on the MPF.2021.2.8 dataset (187.7 k structures), an almost 8.4x smaller subset of the allowed data. Given the architectural similarities between CHGNet and M3GNet, the leaderboard reveals a surprisingly small degradation in regression performance from CHGNet to M3GNet. The more task-relevant classification metrics by contrast exhibit a larger deterioration, perhaps calling into question the transferability of results in [219] to the OOD stability classification setting. Of course, a rigorous ablation study would be necessary to thoroughly substantiate such claims.

To summarize, we believe Matbench Discovery succeeded at providing a walled garden for probing system-level questions in the context of ML for materials discovery. We hope the greater control it affords end-users will help inspire novel research directions, particularly with regards to utilizing a wider variety of the huge amounts of available (but often inconsistent) ab initio reference data.

6.2 Foundation Models

By demonstrating the effectiveness of ML stability screening for high-throughput materials discovery in Chapter 3, we gained trust in the reliability and transferability of these models. This in turn encouraged us to invest the significant time and resources required to train models like CHGNet and MACE-MP. Ref. [8] (on which most of Chapter 5 is based) explores their abilities and limitations in great detail on a diverse set of tasks, finding MACE-MP to be capable of accurately simulating a surprising range of chemistries in solid, liquid, and gaseous phases. Despite being trained solely on bulk crystals, MACE-MP demonstrates surprising versatility, able to model molecules and proteins as well as catalytic processes and entire batteries. Such foundation models, pre-trained on large structural and property databases, have the potential to run near-DFT-accuracy simulations at a fraction of the cost. Even more excitingly, they can simulate systems orders of magnitude larger over longer timescales than possible with DFT. They also serve as excellent starting points for data-efficient fine-tuning to specific applications

as demonstrated in [Section 5.9](#). The emergent ability to extrapolate the interactions governing large complex systems, both inorganic, organic and metal-organic, from having trained solely on inorganic bulk crystals, as thoroughly demonstrated in [Chapter 5](#) and [\[8\]](#), is a remarkable scientific discovery in its own right. More work is needed to fully understand how this extrapolation ability arises and how it can be further enhanced.

Together, the insights gained in [Chapters 3 – 5](#) lay the groundwork for future advancements in dataset and interatomic potential development and bring us closer to the goal of accurate, efficient off-the-shelf ML-driven materials simulation accessible to force-field non-experts.

Chapter 7

Outlook

Given this work focused on the nascent and fast-moving field of GNN interatomic potentials, many questions remain unanswered. Directions for future work include:

7.1 PES Softening

The initial phonon analysis of MACE-MP which we later extended to CHGNet and M3GNet reveals a consistent trend of underpredicted phonon frequencies across models and diverse chemistries attributable to underestimated restoring forces on atomic displacements. Many of the 35 simulation tasks in [8] showed similar issues also originating from an unphysically flat potential energy surface.

Section 5.9 investigates the root cause of this PES softening and finds it to be a data problem rather than a model problem, originating from the biased sampling of current pre-training datasets that are largely distributed around energy minima of bulk solids. As a result of the bias for smoothness built into MLFFs, far-from-equilibrium atomic configurations have systematically underestimated energies and forces, causing quantitative and sometimes even qualitative errors in simulations of high-energy dynamic processes like ion migrations and phase transformation.

We show that PES softness can be effectively resolved by fine-tuning on a handful of far-equilibrium states, explaining the data-efficient performance boost frequently observed from fine-tuning, indicative of correcting systematic errors.

This raises the need for improved next-generation datasets for training foundational MLFFs. Despite significant efforts dedicated to model design and training strategies, less emphasis has been placed on constructing comprehensive and well-curated open-source materials datasets. Current generation foundation models rely on databases such as MP that were not originally generated with ML in mind, leaving significant room for improvement. Results throughout this thesis but Section 5.9 in particular stress the importance of increasing both chemical and configurational diversity in next-generation datasets to ensure better sampling of high-energy regions of the PES. This will be crucial to allow MLFFs to simulate a wider range of physical phenomena like high-temperature and/or pressure MD, phase transitions, and chemical reactions.

Increasing training data diversity raises questions about the impact of model size and expressivity on the ability to capture intricate PES details [306]. Excited states often contain high-order interactions resulting in non-linear behavior. The Lennard-Jones potential serves as a toy example of this where the close-approach region is dominated by r^{-12} repulsion while the equilibrium region is governed by the r^{-6} term. Given that higher-order polynomials require more expressivity to learn [355], model size may need to increase to accurately capture future diversified near- and far-equilibrium training sets. While existing results shine little light on the effect of model size on PES physicality, further studies should investigate the number of model parameters needed to describe a truly universal PES.

7.2 Probing Bias-Variance Tradeoff in GNNs

The bias-variance tradeoff is the widely observed empirical trend that increasing model complexity (usually quantified by the number of trainable parameters) reduces bias but increases variance. Bias is the amount of prior expectations baked into the model by assumptions made during the design that later prevent it from correctly fitting the underlying process that generated the observed data. Variance, on the other hand, measures randomness in the model’s predictions due to the sampled data points used for training (which necessarily comprise a subset of the true distribution to be inferred). The more complex a model, the more it tends to fit to random noise.

However, empirical evidence from large over-parameterized neural networks including GNNs challenges the bias-variance paradigm. As the size and depth of GNNs increase, the test error even on OOD samples sometimes decreases, in contradiction to the expected variance increase. Several studies have proposed theoretical explanations for this phenomenon known as double descent [356]. Ref. [355] establishes strict upper and lower bounds on model complexity of models with a small Lipschitz constant, i.e. models that learn smooth functions. Ref. [357] showed that (depending on the interplay between model size, depth, and training) double descent can occur for small Lipschitz models. They found that models that underwent double descent yield better robustness to adversarial attacks, an observation not limited to neural networks. Ref. [358] further demonstrated that increasing model parameters can improve training fit while simultaneously restricting the solution to a smaller L_2 -norm ball in weight space, thus improving stability and generalization. Prior work had linked this effect to posterior contraction in Bayesian deep learning [359], [360]. Ref. [361] showed that models insensitive to small changes in the training data have more tightly coupled train and test errors, such that larger models with lower train error can also have lower test error.

More research is needed to substantiate the preventability of bias-variance tradeoff in GNNs and to understand the precise conditions needed for double descent. Such findings would have critical implications for larger next-generation MLFFs that tie into the PES softening issue

discussed above. Finding reliable methods to simultaneously reduce bias and variance would guide the way to more reliable ML foundation models for materials chemistry.

7.3 Benchmarking Foundation Models

Progress in machine learning research is often driven by good benchmarks, the most famous example being the ImageNet classification task in computer vision [362]. The availability of challenging and informative standardized tasks spurs advances by establishing a goal post with which to measure progress over time and compare the merit of different ideas and methods. For molecular systems, the QM9 [363] and MD17 [364] data sets have been the go-to benchmark tasks. Matbench [60] which inspired Chapter 3 has been an important first step in establishing a standard suite of ML tasks for inorganic materials. It measures predictive accuracy across 13 regression and classification tasks spanning a range of data set sizes. However, its limitation to supervised learning lacks a pre-training dimension that allows transfer-learning benefits and the ability to use more of the available training labels (e.g. forces plus energies) to be reflected. It also lacks a simulation aspect, arguably the most versatile use for ML potentials in downstream research which thus more directly connects benchmark metrics to real-world utility for users. As a result, Matbench could make no distinction between energy-only and force models which proved to be a critical factor in Matbench Discovery performance. Finally, the most informative metric is often task-dependent. E.g. precision and enrichment in selecting candidate materials are more pertinent when quantifying acceleration of materials discovery than predictive accuracy.

While Matbench Discovery (Chapter 3) was able to address some of these limitations, it focused solely on stability prediction, thus moving into sharp relief the now-pressing lack of a more comprehensive standardized benchmark for ML foundation models that goes beyond materials discovery to cover the rich diversity of possible downstream applications other researchers will use them for. One reason such a benchmark is difficult to design is the lack of agreement on what the most important tasks and related performance metrics are. Ideally, the field could agree on a minimal set of tasks that yield a maximally discriminative set of metrics. It will require careful consideration, iteration and extended discussion between benchmark authors and stakeholders to reach a consensus on which evaluation metrics best reflect the wide variety of possible use cases for these models. Identifying an agreeable set of benchmark tasks and metrics will greatly facilitate securing sufficient buy-in from model authors to solicit external model submissions. This in turn lowers the workload on benchmark curators and aids long-term maintainability and relevance of the benchmark.

A good question to ask when designing such a benchmark is which metrics capture the ability to predict a multitude of downstream observables and are therefore informative to readers from many subfields? This question naturally leads to metrics that are highly sensitive to the physi-

quality of the PES as the root source of all downstream predictions. They should also quantify foundation model stability and accuracy across diverse chemistries and physical conditions, the ability to extrapolate to new systems, computational efficiency, and reward a model that delivers high-quality uncertainty estimates. Given the large datasets and considerable training costs involved, such a benchmark would necessarily constitute a major curation effort that is cost-prohibitive for most academic groups. Standardizing a consistent foundation model training set may need to come from a well-known and trusted incumbent to attract sufficient attention, acceptance and ultimately adoption. Given the huge effort required to create the high-quality large-scale datasets needed for such a benchmark, the data creation and benchmark design may end up being separate efforts as was the case for Matbench Discovery. The inability of many academic groups to train even a single foundation model due to the prohibitive cost (often requiring dozens of GPUs over multiple days, i.e. $\mathcal{O}(10^4)$ GPU hours which is only going to grow), let alone the tens of training runs needed to populate a benchmark leaderboard establishes a barrier to entry.

Given architecture is decidedly less important than training data, it stands to reason that to set themselves apart, future foundation models will largely innovate on the training set, creating a natural pressure away from a consistent benchmark evaluation. That said, the expected near-term proliferation of foundation models and resulting confusion among users about which one to choose will significantly increase the need for a standardized benchmark and likely exert sufficient pressure to cause its creation.

7.4 Commoditizing Foundation Models

To realize the full potential of ML foundation models, several challenges must be addressed:

- Extend **accessibility to non-experts** by improving ease of use and setup. This may involve creating user-friendly visual interfaces into which researchers drag and drop structure files, and select desired tasks (relaxation, MD, property prediction, etc.) and the tool handles the rest including warning the user if results are untrustworthy.
- Improving **interpretability** can help build trust and aid scientific insights. These models are powerful but opaque, making them potentially treacherous in the hands of non-experts. Techniques to identify which input features drive specific outputs [323] and **reliable uncertainty estimation** possibly through model committees will be important to build trust and achieve deployment into high-stakes applications.
- Training on **multiple modalities** is essential to expose models to the full diversity of chemical space. This includes surfaces, defects, molecules, metal-organic hybrid materials, bulk solids, amorphous phases, interfaces and multi-element alloys and 2D materials. Having seen such systems will be vital for modeling numerous processes including catalysis and

electrochemical processes (e.g. water splitting), ab- and adsorption of gases and molecules, deposition of heterostructures and 2d crystals, nanoparticle growth, and many more. Handling the increased complexity may require more sophisticated model architectures. It will certainly substantially increase training cost. However, the biggest challenge will likely be achieving data consistency across that many modalities so as not to confuse the model. The best demonstration to date that this can be done was recently published in [86].

- Increasing **data fidelity**. All current universal potentials are fitted to PBE-level data which has known accuracy limitations. Refitting to r2SCAN [340], [341] is a highly promising avenue to improve accuracy and reduce the need for posthoc corrections.

7.5 Discovering Pareto Optimal Materials

Chapter 4 has shown our inability to reliably predict band gaps significantly degrades the hit rate of identifying new high-performance dielectric materials. The key failure mode stems from false insulators – metals predicted to be gapped. While we observed decreasing band gap error by increasing the training set to 319.6 k samples from combining MP and WBM, metals misclassified as insulators remained a critical cause for wasted DFPT calculations. The band gap is a non-local property of the electronic rather than ionic structure of a material, which most existing ML architectures have no notion of, making it particularly challenging to model with statistical methods. Consequently, alternative approaches for estimating the bandgap should be explored, such as predicting the electronic density of states (DOS) from which the bandgap can be extracted [295] or modeling the electronic structure directly [185]. A single DOS can be discretized into hundreds of training labels, increasing the amount of available reference data. More generally, effective architectures for predicting spectral properties would be valuable across a wide range of material science applications [8], [365].

Our dielectrics workflow would also benefit from using a top-performing MLFF like MACE-MP or CHGNet to enable pre-DFPT dynamic stability filtering. This would reduce the funnel’s false positive stable rate earlier and more cheaply (see Section 4.8). Thus the lessons learned from Matbench Discovery and especially Chapter 5 on data-driven thermodynamic and kinetic stability prediction tie directly into our dielectrics workflow, demonstrating how parts of the thesis inform each other. Specifically, Matbench Discovery informed the foundation model training and analysis in Chapter 5 which in turn will inform future iterations of our dielectrics workflow.

Identifying novel stable materials is just the first step in materials discovery; the ultimate goal is to predict and experimentally realize new functionality. Many of the most useful functional materials exhibit trade-offs between competing physical effects. For instance, good thermoelectric materials need to combine high electrical with low thermal conductivity. Good

dielectrics should be highly polarizable yet strong insulators. Although some studies have focused on optimizing a single functional property, such as hardness [212], [241], multi-objective optimization of competing properties is unavoidable in virtually all industrial applications. Tackling the multi-objective regime head-on will reveal the main shortcomings in current workflows, allowing us to better focus our efforts on resolving them.

7.6 Generative Modeling

The ultimate goal of computational materials design is a tool that given a set of target properties, suggests stable atomic configurations that can be made in a lab and satisfy the requested performance criteria. Initial efforts to develop generative diffusion models for inorganic crystals such as CDVAE [78], DiffCSP [75] and more recently MatterGen [76] have demonstrated encouraging potential to realize this inverse design paradigm. While we successfully employed a much simpler form of novel crystal generation using chemical-similarity-guided element substitution in Chapter 4, the hit rate (Table 6) of future implementations of the workflow presented in Figure 20 could benefit immensely from crystal generation successfully conditioned on band gap and dielectric constant, especially if such models can reduce or avoid the failure mode of metal-insulator misclassification. This would vastly reduce the number of expensive DFPT calculations required to identify a desired number of high- Φ_M materials, which in turn would speed up the discovery process and lower the barrier to entry.

We believe there’s been reason for skepticism on the viability of diffusion models for crystals in the past. However, MatterGen outperforming widely employed methods such as MLFF-assisted random structure search (RSS), element substitution, and ML-assisted screening provides compelling evidence that generative models will become an essential component of future materials design efforts. MatterGen’s step change in stability, uniqueness, and novelty of generated structures over prior work appears to derive principally from Alex-MP-20, a significantly larger training set, and two architectural improvements:

1. The joint diffusion for atom types, coordinates, and lattice (as proposed in DiffCSP [75]).
2. The introduction of adapter modules enabling fine-tuning for conditional generation across various properties.

Despite these advances, several challenges remain:

1. Diffusion models tend to generate low-symmetry $P1$ structures, particularly for larger crystals, symptomatic of a distribution mismatch with respect to crystals found in nature. Given this issue appears to have been partially resolved from CDVAE to MatterGen, further optimization of the denoising process and/or training set augmentation may suffice to fully resolve it.

2. Pioneering research on generative models dealt with molecules, i.e. non-periodic systems with fewer symmetry constraints typically consisting of a much smaller set of elements. While MatterGen and its predecessors have since succeeded at generalizing to inorganic crystals, further work is needed to enable the generation of catalytic surfaces, metal-organic frameworks and other large complex systems. This would open up highly impactful chemical reactions such as nitrogen fixation and carbon capture to the inverse design paradigm.
3. Another area for future development is the incorporation of non-scalar property constraints, such as band structures or X-ray diffraction (XRD) spectra. The ability to generate materials based on vectorial or even higher-order constraints would enable generative band engineering and the prediction of atomic structures from experimentally measured XRD spectra of unknown samples.
4. As with any subfield of ML, progress is spurred and can only be quantified in the presence of well-designed performance metrics and benchmarks to help identify areas for improvement and assess the quality of generated materials. The current lack of benchmarks for crystal generation makes model comparison difficult.
5. Finally, the ultimate test for a generative model is successful experimental synthesis and characterization which has yet to be demonstrated and will require closer collaboration with experimentalists. Until this is achieved, skeptics can rightly point out that this shortcoming calls into question the practical utility of generative models for materials.

7.7 Closing Remarks

From accelerating the stability prediction of hypothetical crystals, both thermodynamic and phononic, to guiding complex multi-step funnel-type discovery efforts, to enabling near-DFT-accuracy simulations across diverse chemistries, this thesis demonstrated how many areas of computational materials science are positively impacted by ML.

Looking ahead, improvements in high-quality open-source software, autonomous labs, declining computational costs, sophisticated error-correcting high-throughput DFT workflows [7], [69], [72], and rapid improvements in deep learning architectures are paving the way for the systematic design of atomic systems.

Close collaboration between materials experts (computational and experimental), ML researchers and robotics engineers will be essential to fully leverage the potent combination of theory, ML and automated labs without sacrificing scientific rigor. If successful, developing robotic experimentation for automated synthesis and characterization and integrating it with ML and theory guidance will dramatically accelerate and scale up materials research.

While industry efforts will likely drive rapid progress, the open-source community must step up to ensure that reproducibility and FAIRness (findable, accessible, interoperable, and reusable

data) are maintained in ML model development. An open community that freely shares new data and models will greatly improve the pace of progress and ensure we unlock their full potential.

Appendix

A.1 Code and Data Availability

A.1.1 Matbench Discovery

We actively solicit further model submissions to <https://github.com/janosh/matbench-discovery> where all code written for Matbench Discovery is publicly available and free to reuse under the MIT license.

We chose the latest Materials Project (MP) [12] database release (v2022.10.28 at time of writing) as the training set and the WBM dataset [1] available at <https://figshare.com/articles/dataset/22715158> as the test set for this benchmark. A snapshot of the entire uncleaned set of all 1.6 M ionic steps in MP v2022.10.28 including energies, forces, stresses and magnetic moments is available at <https://figshare.com/articles/dataset/22715158?file=43350447>.

A.1.2 Dielectrics

The MIT-licensed code for this work can be found at <https://github.com/janosh/dielectrics> and as a Zenodo archive at <https://doi.org/10.5281/zenodo.10456384>. Zenodo includes a complete dump of our DFPT dataset. Our live data is also publicly accessible through a MongoDB M2 Atlas instance with the schema of an `atomate tasks` collection. It can be queried free of charge and without registration using the read-only database credentials and example code snippet provided in the GitHub readme. This requires `pymongo` or any other MongoDB language driver. The query syntax will be familiar to users of the (legacy) Materials Project `MPRester` API. We used Materials Project data from the (v2020.09.08) database release and a cleaned version of the WBM dataset [1] available at <https://figshare.com/articles/dataset/22715158>.

A.1.3 MACE-MP

The code to train MACE-MP on `MPtrj` is available on GitHub at https://github.com/janosh/matbench-discovery/blob/71a5edc/models/mace/train_mace.py. The MACE source code was published to PyPI as the `mace-torch` package 9 days after publishing the model in [8] and can be installed with `pip install mace-torch`. The model weights for the small, medium and large versions of MACE-MP are available at <https://github.com/ACEsuit/mace-foundations>. Future versions of the model will also appear here as newer releases. They can be manually downloaded and will be auto-downloaded by the `mace-torch` package when requesting a given model for the first time. All code and data used in the phonon analysis (Section 5.4) can be found at <https://github.com/janosh/ffonons>. Various file artifacts from other parts of the model analysis can be found at <https://drive.google.com/drive/folders/1JQ-ry1RHvNliVg1Ut5>

OuyUxne51RHiT__, particularly in the `example-xyz-movie` folder. Hyperparameter tuning was recorded in the `Model Stats` spreadsheet.

A.1.4 Plotting

Most of the plots in this thesis were generated with `plotly` [366], some with `matplotlib` [367] and `seaborn` [368]. Many of the plot types in this thesis were turned into reusable, documented and tested functions in the `pymatviz` package [313] available at <https://github.com/janosh/pymatviz>. Thanks to [all contributors](#)!

A.2 MPtrj and WBM Exploratory Data Analysis

The MP/MPtrj and WBM datasets were heavily used throughout this thesis (as train and test sets respectively in [Chapters 3](#) and [5](#), and combined into a single training set in [Chapter 4](#)). To offer some high-level insights into the makeup of these datasets, this section offers an exploratory data analysis (EDA). [Figure 45](#) shows element distributions for structures in both data sets.

To show how frame selection from MP structure relaxation affected relative elemental abundance between MP relaxed structures and the snapshots in MPtrj, [Figure 46](#) shows element occurrence ratios between MPtrj and MP.

Believing MPtrj to be an influential dataset for the near-term continued development of universal interatomic potentials, we plot histograms showing the distributions of target values for energies, forces, stresses and magnetic moments in [Figure 47](#).

Similarly, [Figure 48](#) plots the distribution of WBM [1] formation energies and convex hull distances (with respect to the convex hull spanned by MP materials). [Figure 48b](#)) are the Matbench Discovery stability prediction targets. Formation energies were filtered to ± 5 eV/atom, discarding 22 structures to the right of the visible range with questionable formation energies between +5 and +80 eV/atom, as well as 502 suspicious structures left of the visible range with exactly -10 eV/atom formation energy. [Figure 48](#) not only gives insight into the nature of the dataset but comparing [Figure 48a](#)) and [Figure 48b](#)) reveals the increased difficulty of stability vs formation energy prediction arising from the much narrower distribution of convex hull distances. Even the filtered formation energies have a range of 10 eV/atom vs less than 1 eV/atom for hull distances. Accurate stability classification requires correctly ranking the subtle energy differences between chemically similar compounds in the same chemical system rather than comparing a single material with the reference energies of its constituent elements. DFT has been shown to significantly benefit from systematic error cancellation between similar chemistries when trying to identify the lowest-lying polymorph [369]. This beneficial cancellation has yet to be conclusively demonstrated for ML stability predictions. So far, only the lack thereof has been shown for a previous generation of models in [10] where they encountered a



Figure 45: Element occurrence in (a) MP, (b) WBM [1] and (c) MPtrj [67]. Values and color of periodic table tiles indicate the number of structures containing a given element in each dataset. The WBM test set is noticeably more chemically diverse than MP (and, by extension, MPtrj). Made with pymatviz [313].

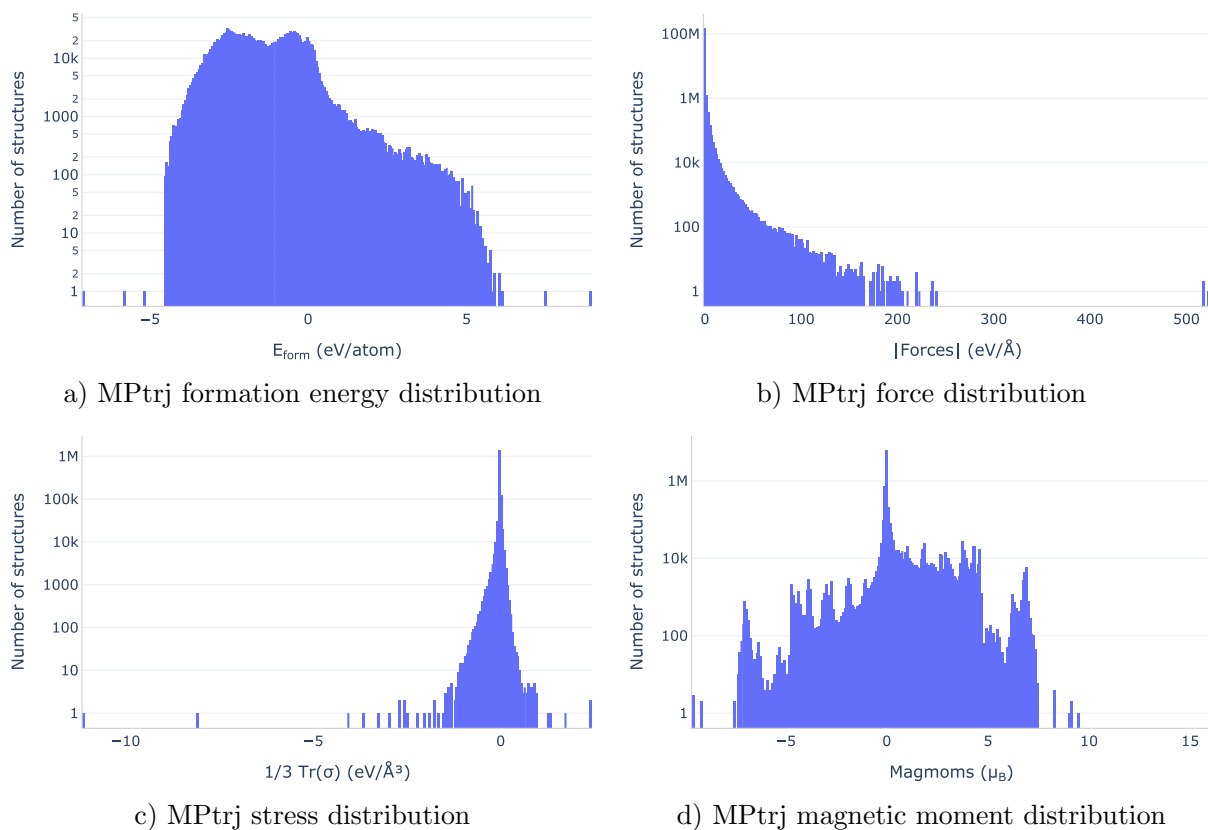


Figure 47: Distribution of energies, forces, stresses and magnetic moments MPtrj. The bimodality in the formation energy distribution is due to the MP anion correction scheme [116] which significantly lowers some formation energies, especially for oxides.

matminer [370] features for all initial structures as well as relaxed MP structures onto two UMAP dimensions in Figure 49.

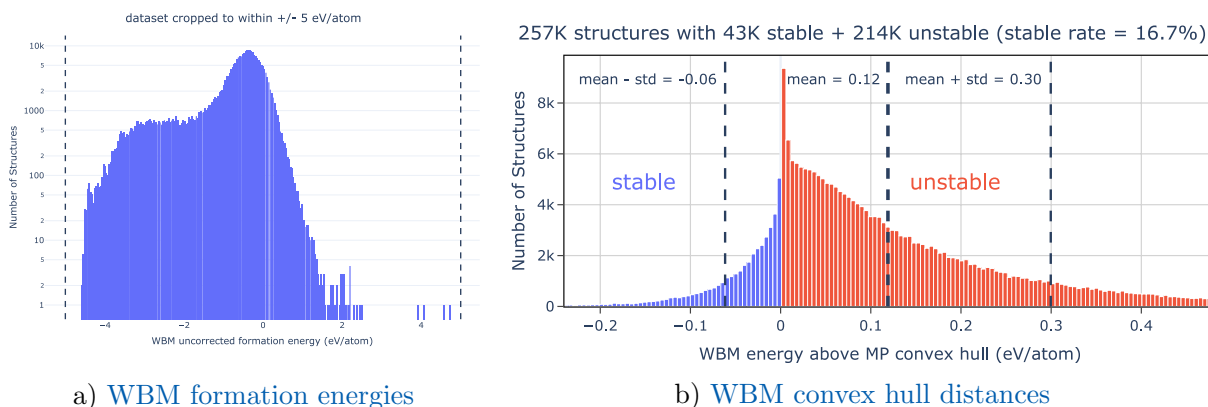


Figure 48: **Convex hull distances have a much narrower spread than formation energies, requiring more accurate energy predictions to correctly rank polymorphs by thermodynamic stability.** Distribution of formation energies (a) and convex hull distances (b) (both per atom) of the WBM test set [1]. In (a), we removed suspect calculations with formation energies outside the plotted range of ± 5 eV/atom.

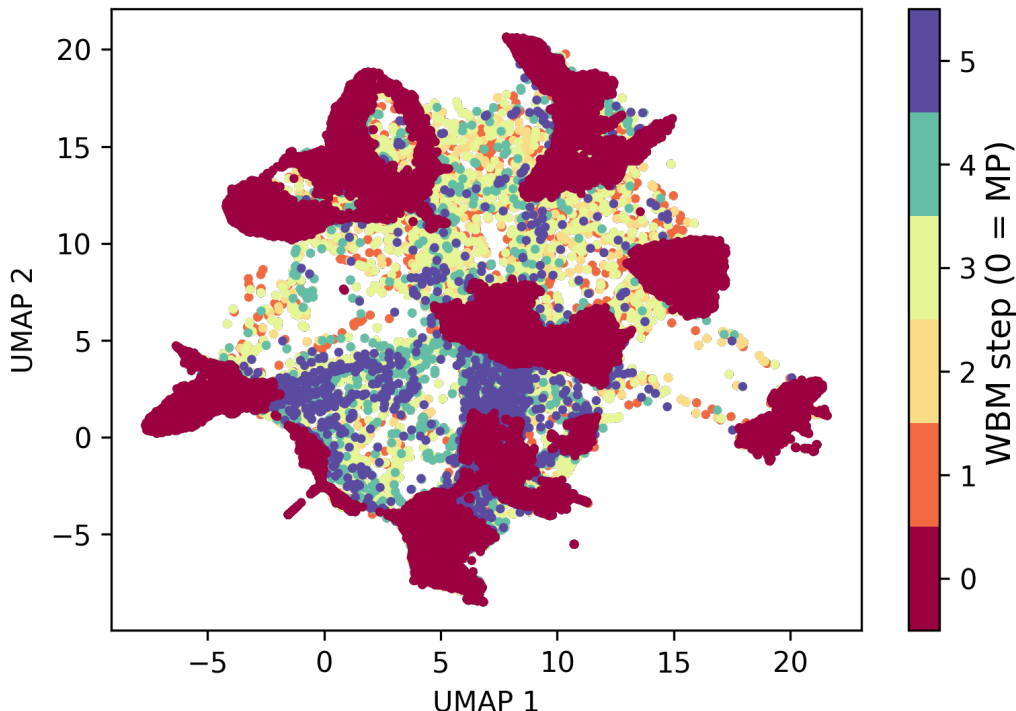


Figure 49: **WBM test set requires out-of-distribution extrapolation to perform well.** 2D UMAP projection of `matminer` [370] features computed for all initial structures in the WBM test set. Points are colored by element substitution step. Step 1 means a single element replacement applied to a source structure taken from MP. Step 2 underwent two successive element replacements and so on. Relaxed MP parent structures are marked as step 0. While the UMAP projection of MP and WBM structures are not completely disjoint, many areas of the WBM UMAP projection are not covered by MP, indicating our benchmark requires some degree of out-of-distribution extrapolation for models to perform well. See Figure 14 for further evidence of this.

Finally, Figure 50a) shows the elements-per-structure distribution of MP, MPtrj and WBM, normalized by dataset size. The mode of all three datasets is 3, but WBM’s share of ternary phases is noticeably more peaked than MP’s, which includes small numbers of unary and senary phases. The histogram in Figure 50b) shows the distribution of the number of sites in MPtrj structures. The inset displays the same histogram log-scaled y-axis as well as a cumulative line to show that 90% of MPtrj structures contain fewer than 70 sites.

We analyze the distribution of arity and number of atoms per structure in Figure 50. Compared to WBM, MP has a more even arity distribution, in particular with double the fraction of quaternary phases and a significant number of quinternaries which are almost absent from WBM. 3% of MP structures containing more than 5 elements (up to 9) are not shown in Figure 50a) for visual clarity. We also include MPtrj in this plot to highlight a slight drop in the relative abundance of quinternaries and higher phases in the sampled relaxation frames compared to MP ground states. This may be due to a poor choice of convergence criteria in

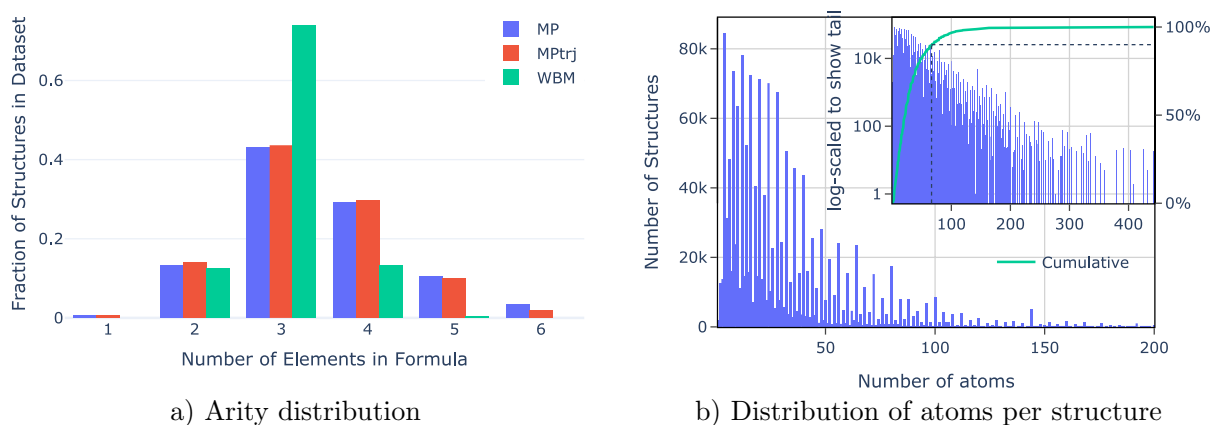


Figure 50: **(a)** Distribution of unique elements per structure in MP, MPtrj and WBM. The bar heights are normalized by the total number of structures in each dataset. WBM is dominated by ternary phases making up 74% of the dataset followed by about 13% each of binaries and quaternaries. **(b)** Histogram of number of atoms per structure. The inset shows the same distribution log-scaled to visualize the tail of large structures. The green cumulative line in the inset shows that 82% have less than 50 sites and 97% of structures in MPtrj have less than 100 atoms.

early MP relaxation workflows that scaled with the size of the structure (see `EDIFF_PER_ATOM` parameter in `pymatgen` VASP input sets), resulting in unconverged large structures with short relaxation trajectories entering the database. Assuming structures of higher arity correlate with larger structures, short relaxations would result in fewer frames of such structures selected for MPtrj.

Figure 51 shows the distribution of magnetic moments and forces for each element in MPtrj. Force distributions in Figure 51a) are truncated at $10 \text{ eV}/\text{\AA}$. Oxygen has the largest outliers with mean absolute forces of up to $160 \text{ eV}/\text{\AA}$. The y -axes in both figures are log-scaled to visualize the tails, revealing a small number of likely erroneous magnetic moments in MPtrj. For instance, Cr has a single-point calculation with a highly unphysical magnetic moment of $+17\mu_B$.

A.3 Proprietary Models on Matbench Discovery

We show the performance of proprietary models trained on incompatible training sets in Table 11. GNoME [44] from Google DeepMind and PFP [66] from Matlantis are the only two models in this category so far. GNoME has been [officially submitted](#) while vetting the [PFP results](#), testing for data leakage and preparing a preprint detailing the evaluation process including how to correct for PFP’s different elemental reference energies from having trained on diverse, incompatible DFT reference data is ongoing. GNoME [44] significantly outperforms all open models as well as PFP (based on preliminary results) across all regression and classification metrics. Importantly, DeepMind reported nearly identical metrics on a subset of WBM excluding all compositional overlap with the GNoME training data, thus measuring the

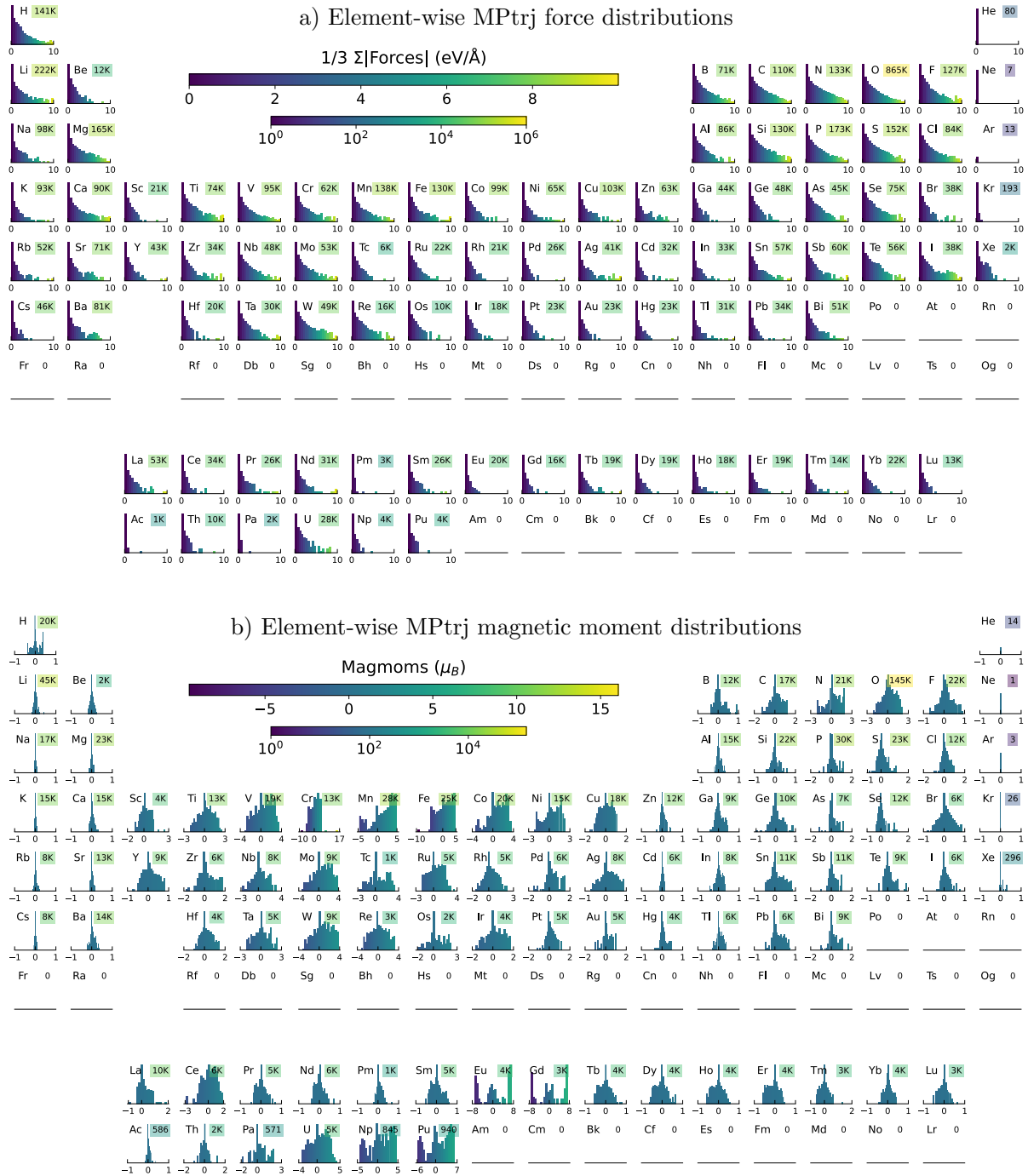


Figure 51: Distribution of forces (a) and magnetic moments (b) for each element in MPtrj. CHGNet and MACE-MP both use all forces in (a) as training targets while only CHGNet uses the absolute value of magnetic moments in (b). M3GNet used only a subset of forces in (a). The number in the top right corner of each element tile counts the number of target values for that element.

model’s generalization to unseen compositions. While not conclusive evidence, this suggests that data leakage is not the main driver of the model’s performance. GNoME’s impressive ability to benefit substantially from a much larger training set indicates that GNN expressiveness in general is far from exhausted. Despite the incompatible training data used (rendering GNoME

	F1 ↑	DAF ↑	Prec ↑	Acc ↑	TPR ↑	TNR ↑	MAE ↓	RMSE ↓	R ² ↑	Training Set	Model Params	Model Type	Targets	Date Added
MatterSim	0.859	5.646	0.863	0.957	0.856	0.975	0.026	0.080	0.812	17M	182.0M	UIP	EFS	2024-06-16
GNoME	0.829	5.523	0.844	0.955	0.814	0.972	0.035	0.085	0.785	6.0M (89.0M)	16.2M	UIP	EF	2024-02-03
SevenNet	0.724	4.252	0.650	0.904	0.818	0.919	0.048	0.092	0.750	145.9K (1.6M)	842.4K	UIP	EFS	2024-07-13
MACE	0.669	3.777	0.577	0.878	0.796	0.893	0.057	0.101	0.697	145.9K (1.6M)	4.7M	UIP	EFS	2023-07-14
CHGNet	0.613	3.361	0.514	0.851	0.758	0.868	0.063	0.103	0.689	145.9K (1.6M)	412.5K	UIP	EFSM	2023-03-03
M3GNet	0.569	2.882	0.441	0.813	0.803	0.813	0.075	0.118	0.585	62.8K (188.3K)	227.5K	UIP	EFS	2022-09-20
ALIGNN	0.567	3.206	0.490	0.841	0.672	0.872	0.093	0.154	0.297	155K	4.0M	GNN	E	2023-06-02
MEGNet	0.510	2.959	0.452	0.826	0.585	0.870	0.130	0.206	-0.248	133K	167.8K	GNN	E	2022-11-14
CGCNN	0.507	2.855	0.436	0.818	0.605	0.857	0.138	0.233	-0.603	155K	128.4K (N=10)	GNN	E	2022-12-28
CGCNN+P	0.500	2.563	0.392	0.786	0.693	0.803	0.113	0.182	0.019	155K	128.4K (N=10)	GNN	E	2023-02-03
Wrenformer	0.466	2.256	0.345	0.745	0.719	0.750	0.110	0.186	-0.018	155K	5.2M (N=10)	Transformer	E	2022-11-26
BOWSR	0.423	1.964	0.300	0.712	0.718	0.693	0.118	0.167	0.151	133K	167.8K	BO-GNN	E	2022-11-17
Voronoi RF	0.333	1.579	0.241	0.668	0.535	0.692	0.148	0.212	-0.329	155K	0.0	Fingerprint	E	2022-11-26
Dummy	0.185	1	0.154	0.687	0.232	0.769	0.124	0.184	0					

Table 11: **The proprietary GNoME [44] model outperforms all open models by a significant margin**, albeit having trained on a much larger, closed and non-admissible dataset. While it can’t be used by any but a select few researchers inside Google DeepMind, it shows what’s possible in terms of ML force field accuracy with a large and diverse enough dataset.

an unusual admission to a benchmark), we wanted to highlight this result as it signals to the community the viability of larger and more diverse training sets. Perhaps unsurprising given continuous improvements in model accuracy with increasing data size even up to internet-scale datasets have been consistently observed in other ML domains such as vision and language, it nonetheless de-risks scaling efforts for other ML force field authors to know the same holds in inorganic chemistry.

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